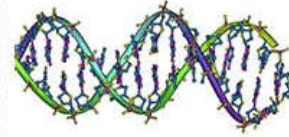
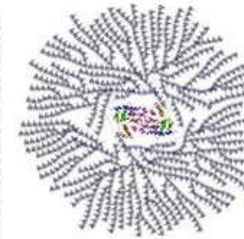


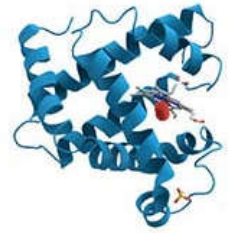
LIPIDS



NUCLEIC ACIDS



CARBOHYDRATES



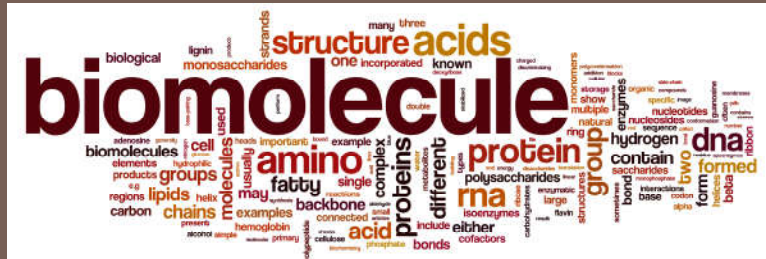
PROTEINS

Department of Medical Biochemistry

Module: Body Structure, Organization and Functions

Module code: BSOF 1011

Course: Biochemistry



December, 2022

Punctuality is very important

👉 Please.... Always remember

👉 *To behave as a mature adult*

Strictly, when in class

👉 Turn all cell phones off

👉 No...side talks

👉 Do not hesitate and ask to repeat unclear points

Come to class!!!

Try not to fall behind!!



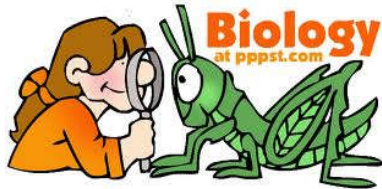
Reference books

- Harper illustrated biochemistry
- Lehninger 4th or 5th edition.
- Streyer 5th edition.
- Champ Lippincott illustrated review of biochemistry
- Marks-Basic Medical Biochemistry

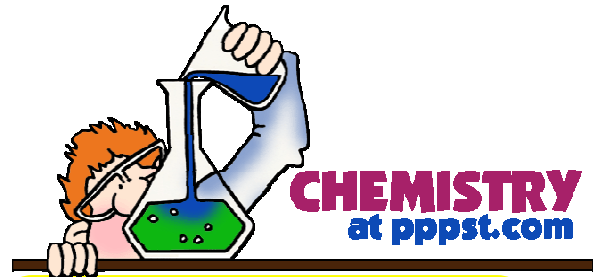
The word **'BIOCHEMISTRY'**-
means -**Chemistry of Living
beings or Chemical Basis of
Life.**

What is Biochemistry?

STUDY OF THE STRUCTURE, COMPOSITION, AND CHEMICAL REACTIONS OF MOLECULES IN LIVING SYSTEMS

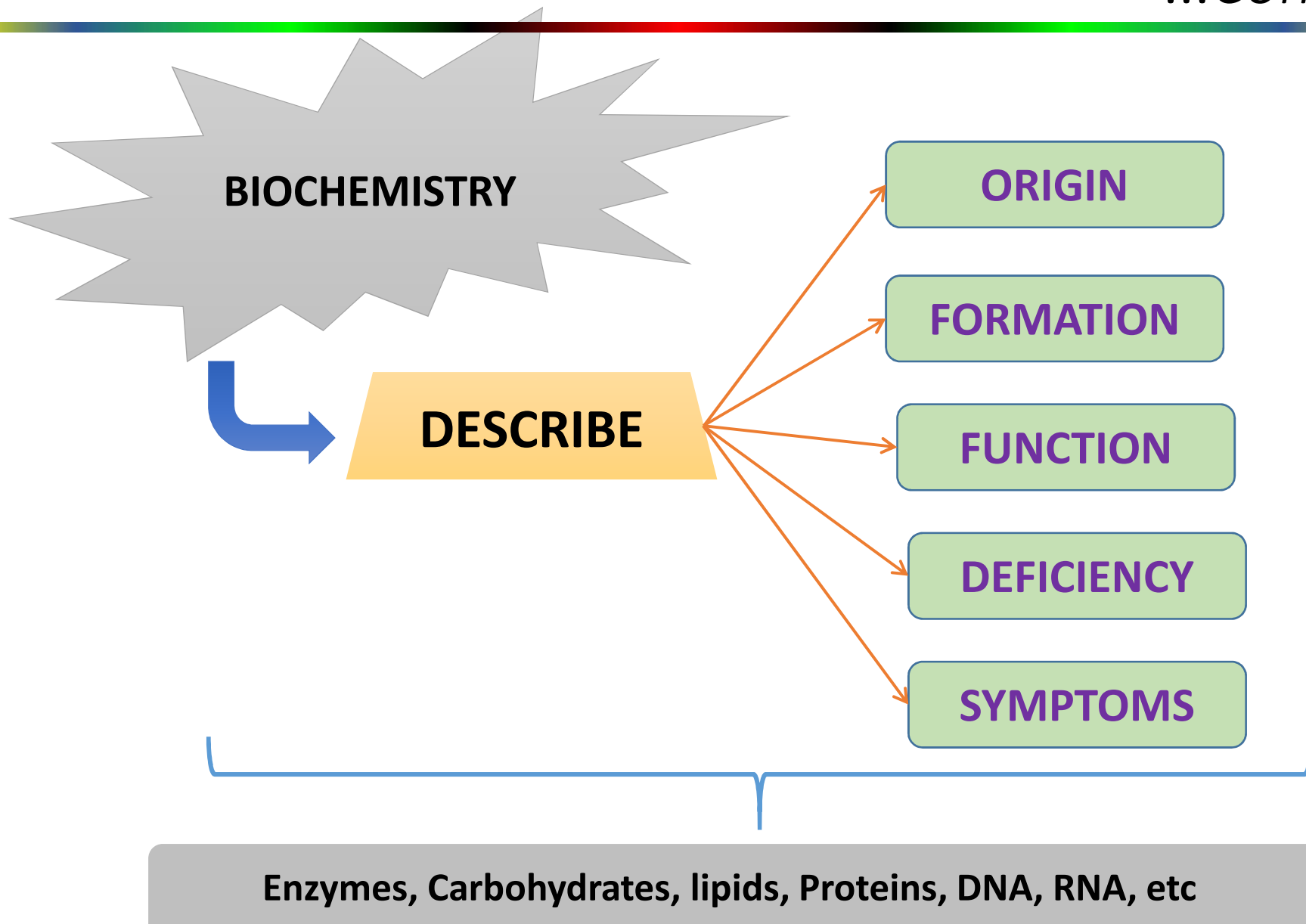


BIOLOGY



CHEMISTRY

BIOCHEMISTRY



Introduction to Biochemistry

- ♣ Biochemistry is a broad scientific discipline and an evolving Science.
- ♣ It is the study of the chemistry of life processes (Chemistry of life)
- ♣ It explains the complex processes of life from the smallest virus to the most complex and highly evolved human being.

- ✓ Basic principles of biochemistry are common to all living organism.
- ✓ Biochemistry emerged as a separate discipline when scientists began to study such topics of:
 - How living things obtain energy from food?
 - What is the chemical basis of heredity? And
 - What fundamental changes occur in disease?

Historical Development of Biochemistry

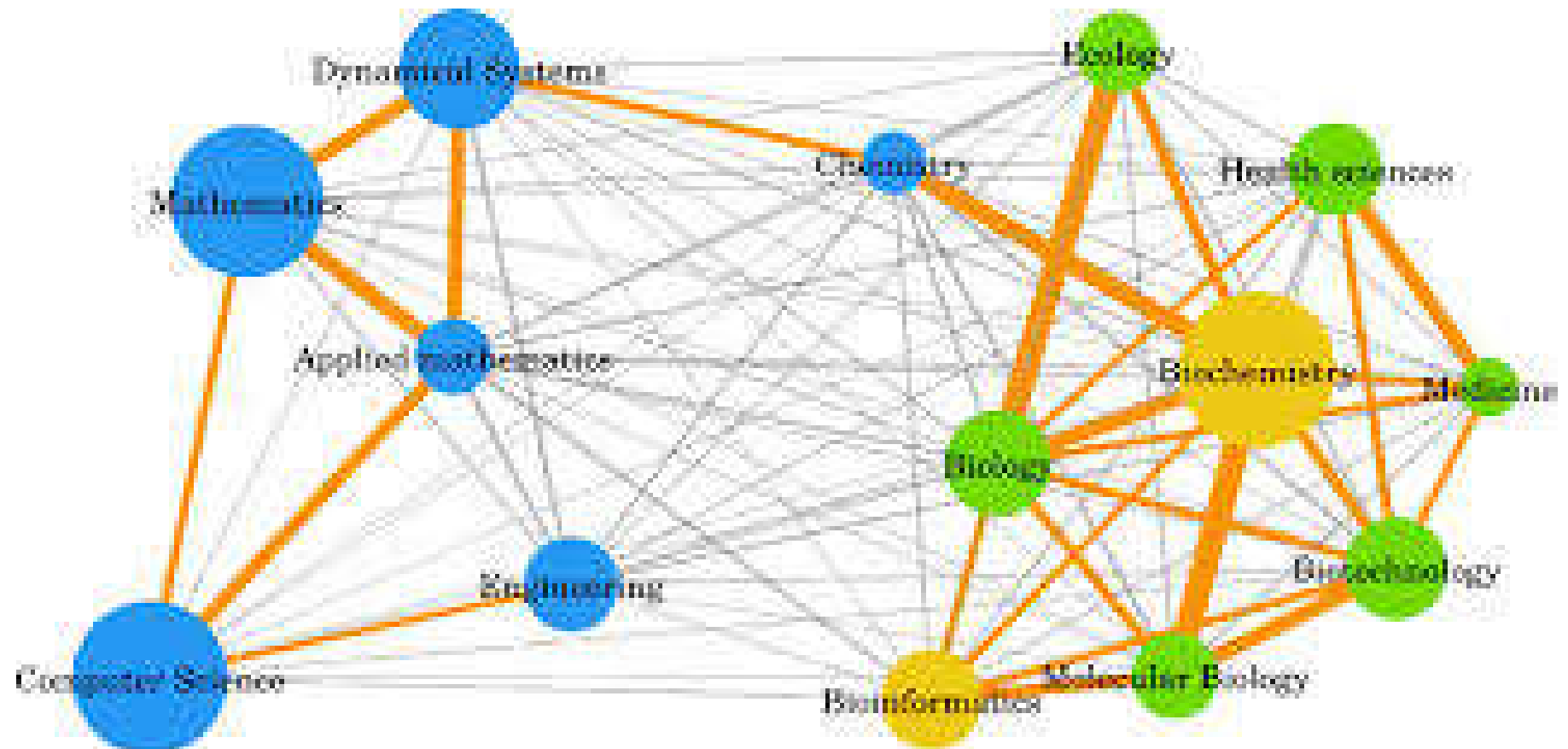
- ♥ Biochemistry emerged in the **late 18th and early 19th century.**
- ♥ The **term Biochemistry** was first introduced by the **German Chemist Carl Neuberg in 1903.**
- ♥ In the **1940s Clinical Biochemistry evolved, as an autonomous field.**

Scope of Biochemistry

- ✓ It is an interdisciplinary science that **integrates systematically** the principles of Molecular and Cell Biology, Chemistry, Genetic, Physiology
- ✓ In recent years the fusion of biochemistry, cell biology and microbiology to form molecular biology has led to spectacular advances in the **understanding and control of biological processes** in medicine, agriculture, pharmaceuticals and food industry.

- ② The knowledge of biochemistry can be **applied to solve problems** in medicine, dentistry, agriculture, forensics, anthropology, environmental sciences etc.
- ② The **causes of disease** and the **importance of proper nutrition** are fascinating topics in biochemistry.
- ② It also recognizes the fundamental nature of life at the molecular level and the ways in which species are related through **evolution from a common ancestor**.

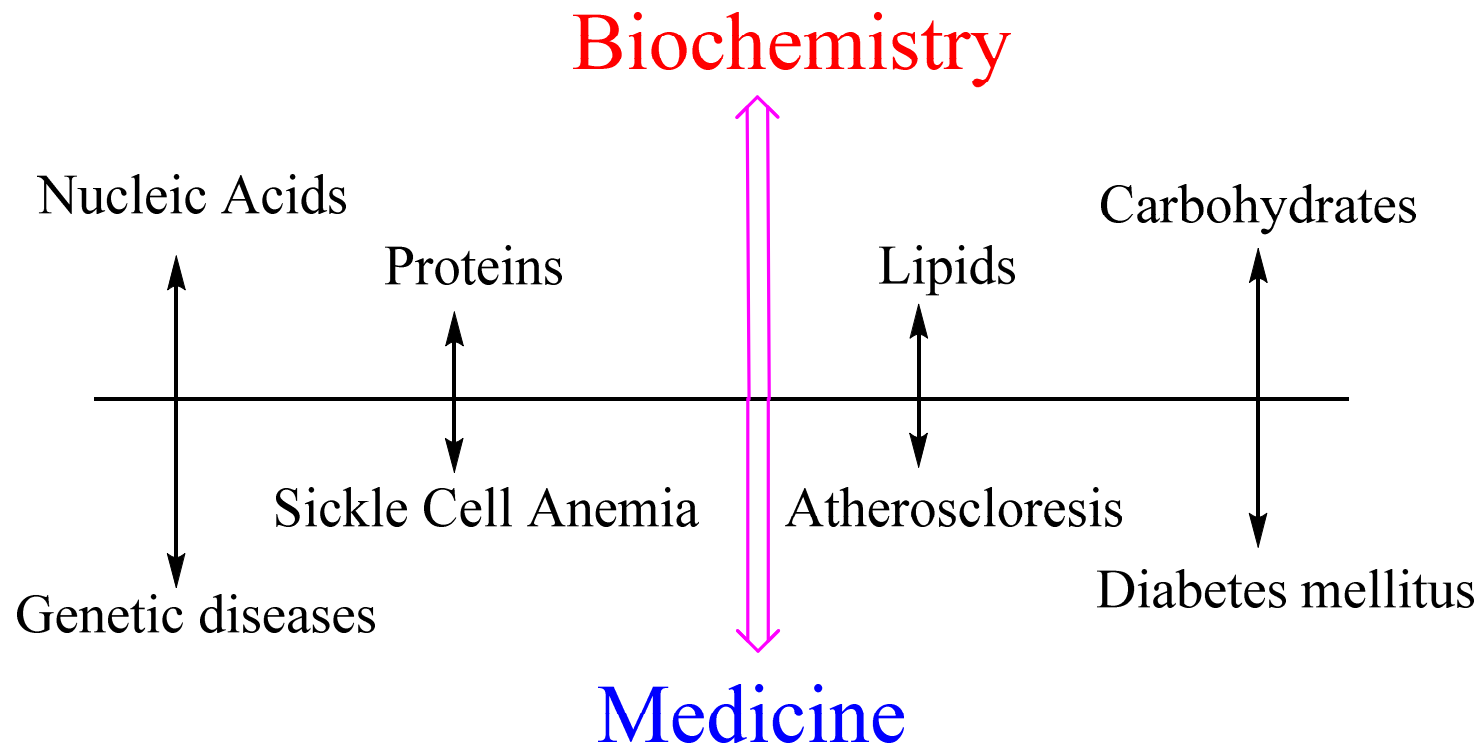
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Major Objectives of Biochemistry

- ✓ To understand **all the chemical processes** associated with **living cells at molecular level**.
- ✓ To do this Biochemists:
 - ❖ **Isolate** - the numerous molecules found in the cell
 - ❖ **Determine** - their structure
 - ❖ **Analyze** – how the function
- ✓ To understand **how life began**
- ✓ One of the goals of biochemists is to integrate a large body of knowledge into a molecular explanation of life and toward defining the basic reactions common to all cells

Most & perhaps all disease has a **biochemical basis.**



- ✓ Knowledge of the biochemical molecules has clarified our understanding of the **diseases and conversely, analyses of the diseases** is based on biochemical changes.

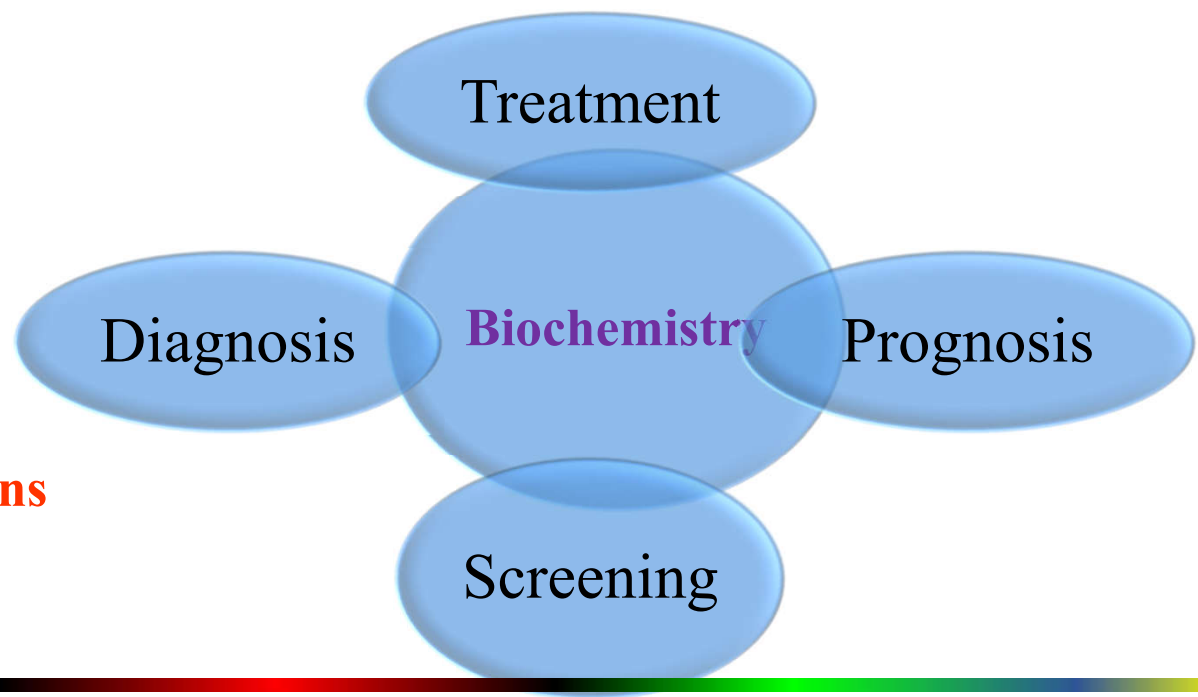
- Biochemical tests are used extensively in medicine and health either in disease that have a metabolic basis (e.g. DM) or in those in which the biochemical changes are a consequence of the disease (e.g. renal failure).
- Biochemical tests are used in diagnosis, prognosis, monitoring and screening

- **Medicine**

- **Agriculture**

- **Industrial applications**

- **Environmental applications**

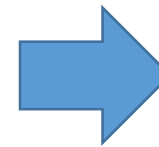


@ Significance to clinical medicine:

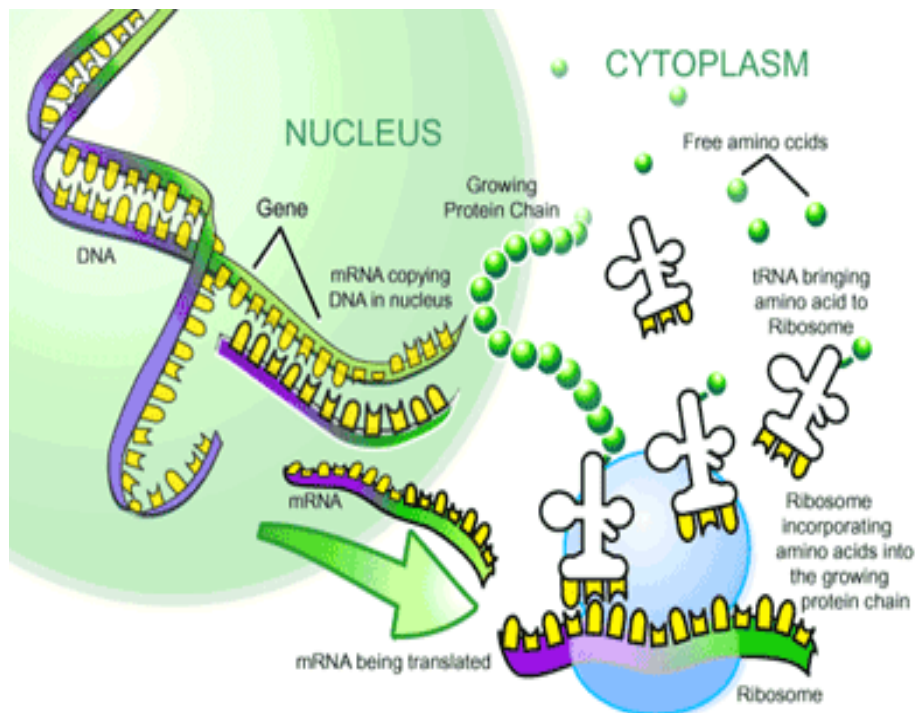
- To assess the functioning of body organs and systems.
 - ♥ For the state of normal functioning, values of the constituents such as Glucose, Urea, Uric acid in blood will be within certain limits.
 - ♥ In case of abnormal functioning, their values in blood and other body fluids will deviate from normal.
- To assess mechanisms of drug designing, synthesis, interaction and clearance from the body.
- For identification of inherited/inborn errors, metabolic abnormalities and ways of medication.
- To explain physiological disturbance and mechanisms of maintaining normal condition.

...Cont'd

Biochemistry helps to understand the biochemical changes and related physiological alteration in the body



The pathology of any disease is studied through biochemical changes



...Cont'd

In the present scenario many people rely in taking multivitamin and minerals for better health



The function and role of the vitamins is described only by biochemistry



Importance of Biochemistry in



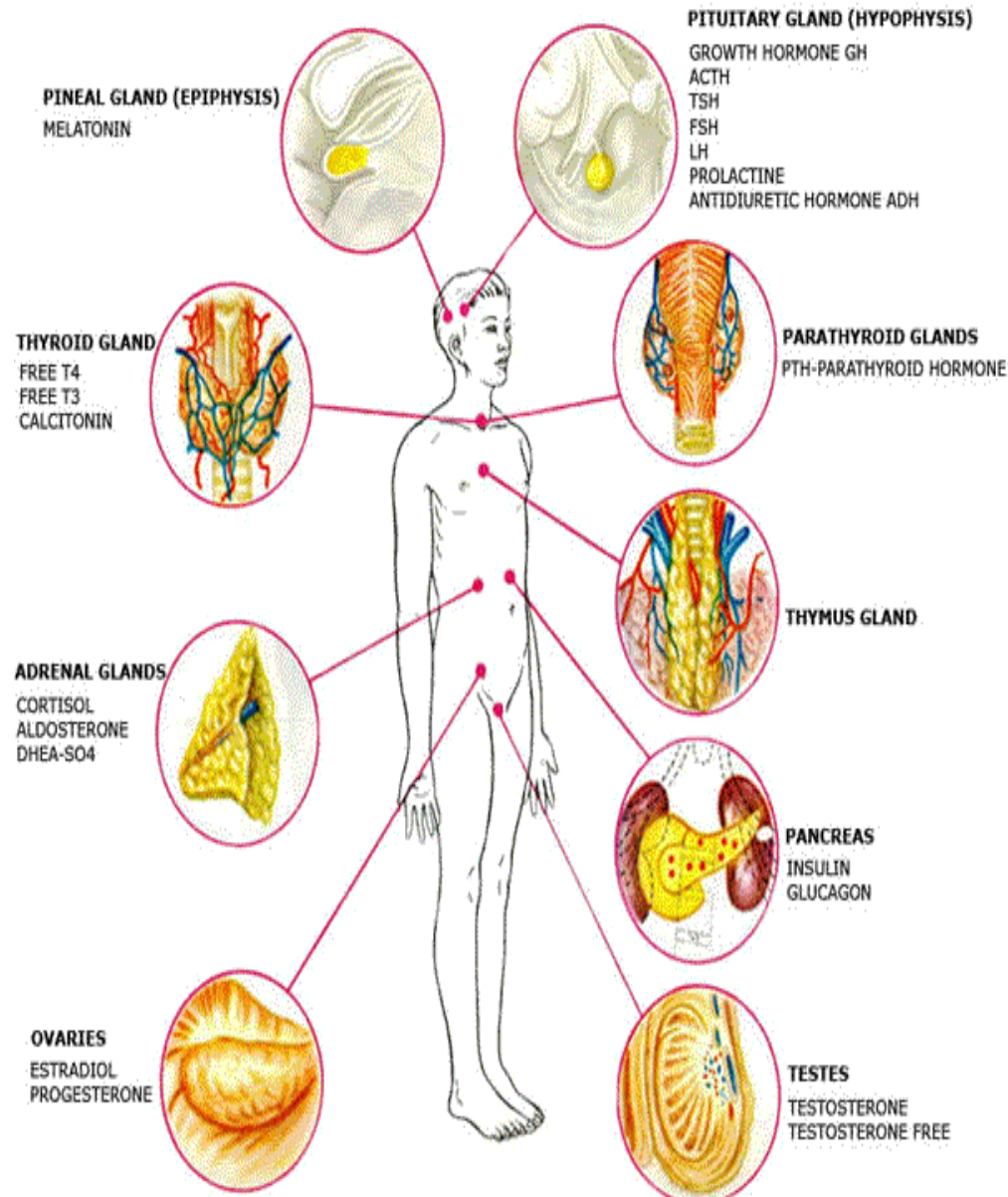
DRUGS

Gives an idea of its
constitution

Its change of degradation
with varying temperature

how drugs are metabolized
by biochemical reactions in
presence of enzymes

...Cont'd

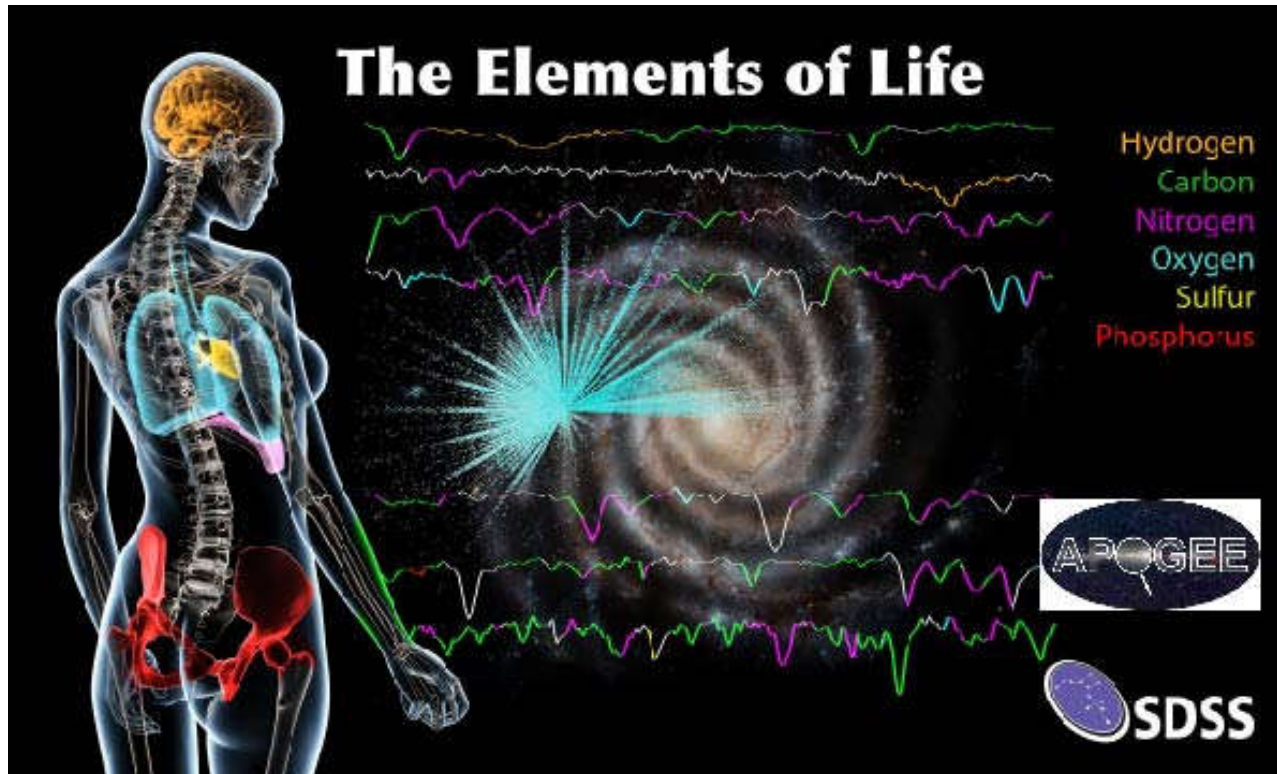


There are many disorders due to hormonal imbalance in specially women and children



The formation of hormones in the normal body function is taught in biochemistry by which the physician can understand the concerned problem during treatment

...Cont'd



➡ The human body on the left uses the same color coding to evoke the important role these elements play in different parts of our bodies, from oxygen in our lungs to phosphorous in our bones

- ➡ To say “we are stardust” may be a cliché, but it’s an undeniable fact that most of the essential elements of life are made in stars.
- ➡ The elements we measure include the atoms that make up 97% of the mass of the human body

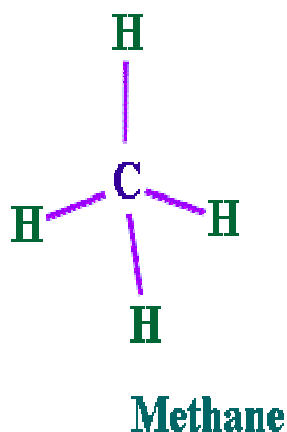
Chemical Constituents of Cells

- ☞ **Living matter** is composed of mainly *six elements*-carbon, hydrogen, oxygen, nitrogen, phosphorus and sulfur.
- ☞ **These elements together** constitute about 90% of the dry weight of the human body.
- ☞ Several other functionally important elements are also found in the cells.
- ☞ These include Ca, K, Na, Cl, Mg, Fe, Cu, Co, I, Zn, F, Mo and Se.

- ➡ Most abundant, essential for all organisms: C, N, O, P, S, H
- ➡ Less abundant, essential for all organisms : Na, Mg, K, Ca, Cl
- ➡ Trace levels, essential for all organism: Mn, Fe, Co, Cu, Zn
- ➡ Trace levels, essential for some organisms: V, Cr, Mo, B, Al, Ga, Sn, Si, As, Se, I,

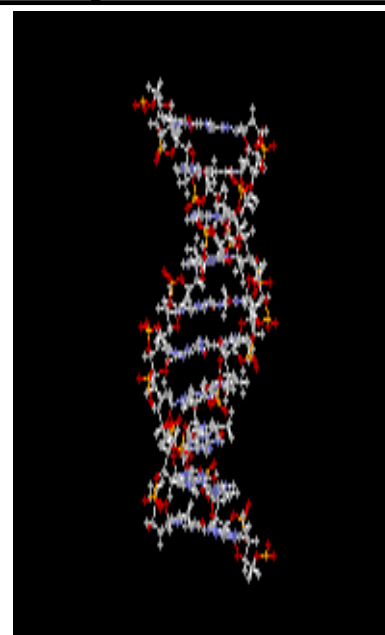
- ✓ Carbon can form immensely diverse compounds, from simple to complex.

Simple Molecule



Methane with 1 Carbon atom

Complex Molecule



DNA with tens of Billions of Carbon atoms

Carbon-a unique element of life

- ✓ **Carbon** is the **most predominant** and versatile element of life.
- ✓ It possesses as unique property to form **infinite** number of compounds.
- ✓ This is attributed to the ability of carbon to form stable covalent bonds and C-C chains of unlimited length.
- ✓ It is estimated that **about 90%** of **compounds** found in **living system** invariably contain **carbon**.

Important Functional Groups

Carbon-Oxygen groups



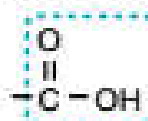
Alcohol



Aldehyde



Ketone



Carboxylic acid



Ether



Acid anhydride

Carbon-Sulfur groups



Sulfhydryl group

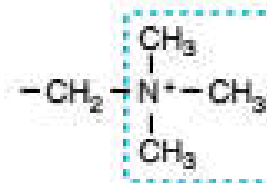


A disulfide

Carbon-Nitrogen groups



Amino group

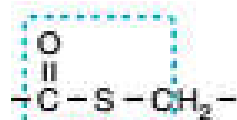


Quaternary amine

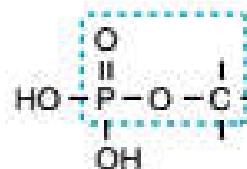
Esters and Amides



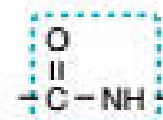
Ester



Thioester

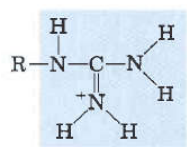
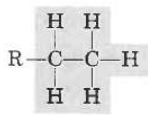
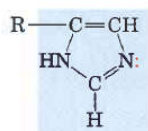
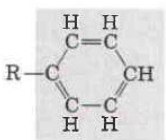


Phosphoester

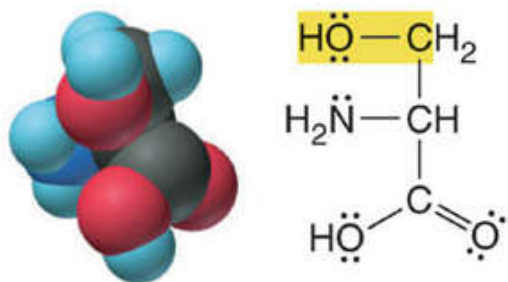


Amide

2. Major types of functional groups found in biochemical compounds of the human body.

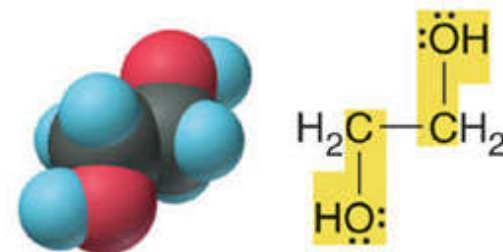
Methyl		Ether	R^1-O-R^2	Guanidinium	
Ethyl		Ester	$R^1-C(=O)-O-R^2$	Imidazole	
Phenyl		Acetyl	$R-O-C(=O)-CH_3$	Sulfhydryl	$R-S-H$
Carbonyl (aldehyde)	$R-C(=O)-H$	Anhydride (two carboxylic acids)	$R^1-C(=O)-O-C(=O)-R^2$	Disulfide	$R^1-S-S-R^2$
Carbonyl (ketone)	$R^1-C(=O)-R^2$	Amino (protonated)	$R-NH_3^+$	Thioester	$R^1-C(=O)-S-R^2$
Carboxyl	$R-C(=O)-O^-$	Amido	$R-C(=O)-NH_2$	Phosphoryl	$R-O-P(=O)(OH)_2$
Hydroxyl (alcohol)	$R-O-H$	Imine	$R^1-C=N-R^2$	Phosphoanhydride	$R^1-O-P(=O)(O^-)-O-P(=O)(O^-)-O-R^2$
Enol	$R-C(OH)=CH_2$	N-Substituted imine (Schiff base)	$R^1-C=N-R^2$	Mixed anhydride (carboxylic acid and phosphoric acid; also called acyl phosphate)	$R-C(=O)-O-P(=O)(OH)_2$

Some biomolecules with the alcohol functional group.



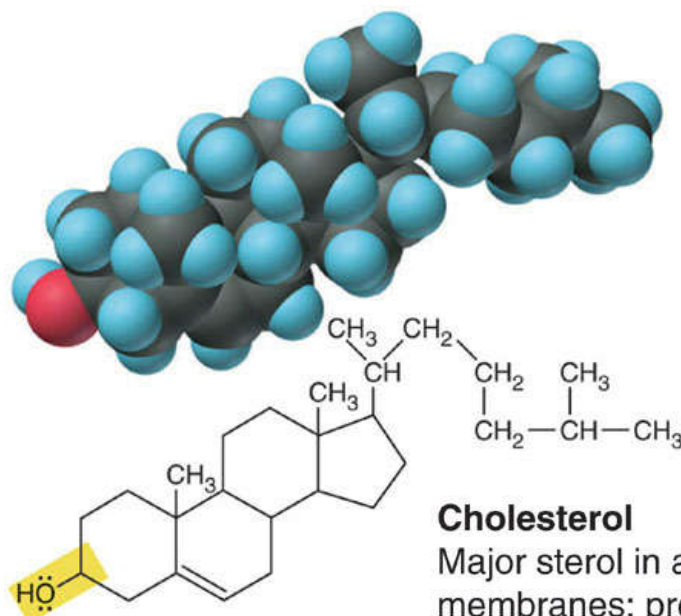
Serine

Amino acid found in most proteins



1,2-Ethandiol (ethylene glycol)

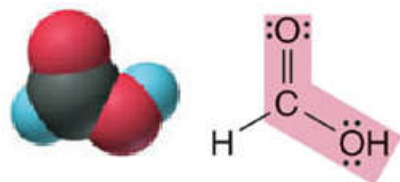
Main component of auto antifreeze



Cholesterol

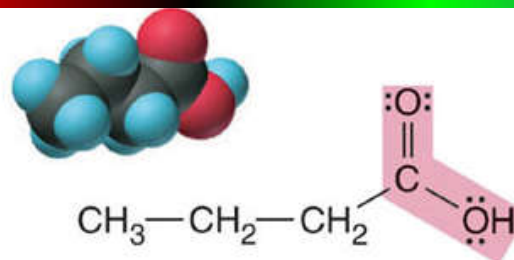
Major sterol in animals; essential for cell membranes; precursor of steroid hormones

Some molecules with the carboxylic acid functional group.



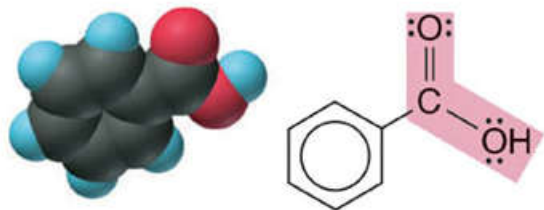
Methanoic acid (formic acid)

An irritating component of ant and bee stings



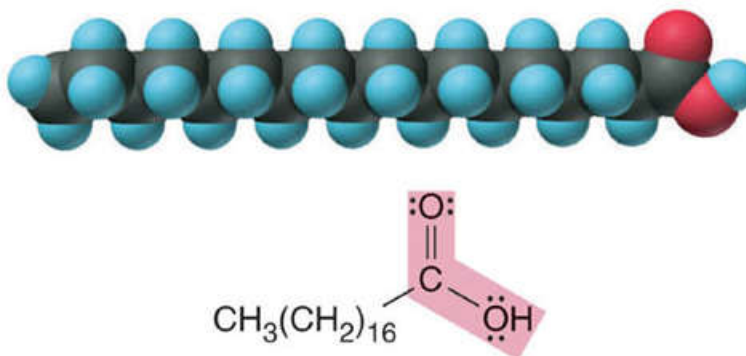
Butanoic acid (butyric acid)

Odor of rancid butter; suspected component of monkey sex attractant



Benzoic acid

Calorimetric standard; used in preserving food, dyeing fabric, curing tobacco



Octadecanoic acid (stearic acid)

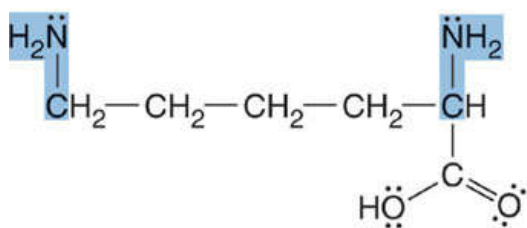
Found in animal fats; used in making candles and soaps

Some biomolecules with the amine functional group.

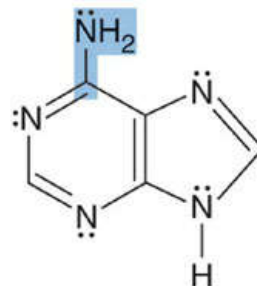


Lysine (1° amine)

amino acid found in proteins

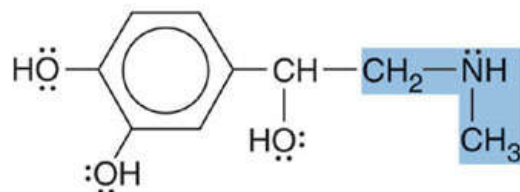
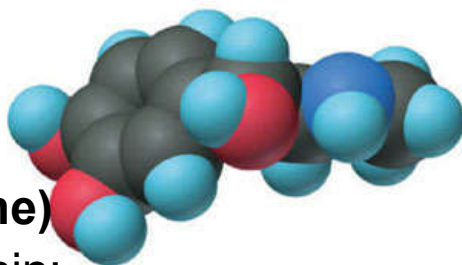


Adenine (1° amine)
component of nucleic acids



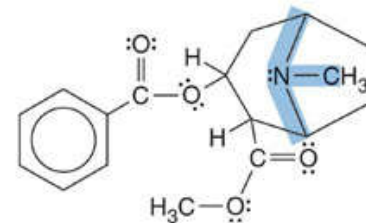
Epinephrine (adrenaline; 2° amine)

neurotransmitter in brain;
hormone released during stress

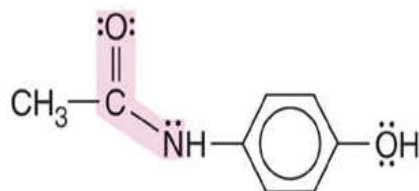


Cocaine (3° amine)

brain stimulant;
widely abused drug

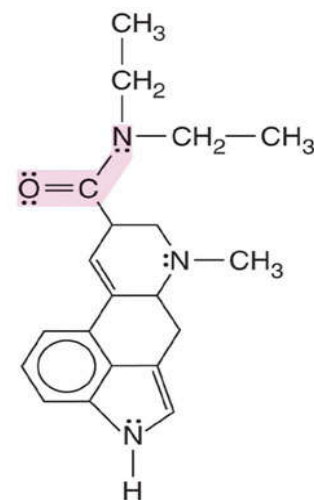
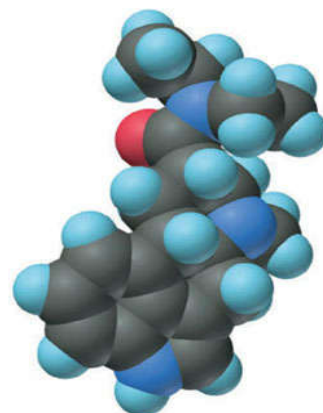


Some molecules with the amide functional group.



Acetaminophen

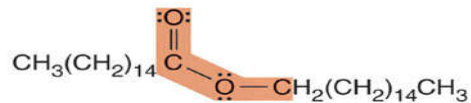
Active ingredient in nonaspirin pain relievers; used to make dyes and photographic chemicals



Lysergic acid diethylamide (LSD-25): A potent hallucinogen

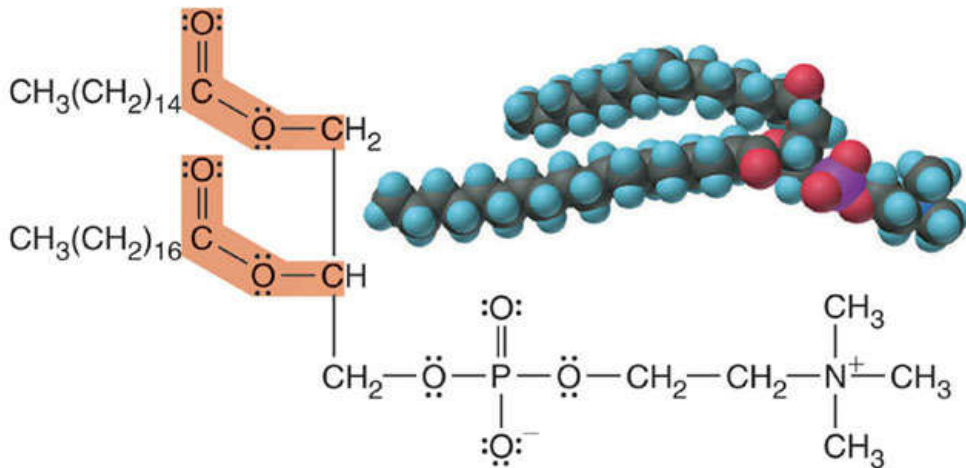
The **peptide bond**, which links amino acids in a protein, is an **amide group**.

Ester groups occur commonly in lipids, which are formed by the esterification of fatty acids.

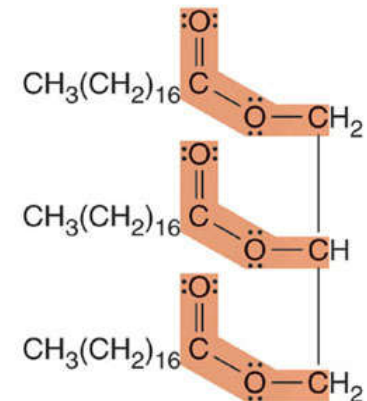
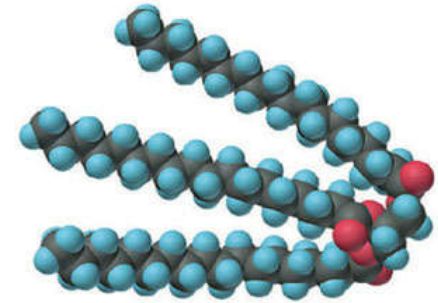


Cetyl palmitate

The most common lipid in whale blubber



Lecithin Phospholipid found in all cell membranes



Tristearin Typical dietary fat used as an energy store in animals

Sulfhydryl

STRUCTURE

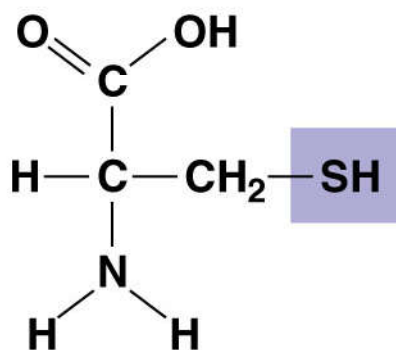


—SH

(may be
written HS—)

Thiols

EXAMPLE



Cysteine

- ➡ Two sulfhydryl groups can react, forming a covalent bond. This “cross-linking” helps stabilize protein structure.
- ➡ Cross-linking of cysteines in hair proteins maintains the curliness or straightness of hair. Straight hair can be “permanently” curled by shaping it around curlers and then breaking and re-forming the cross-linking bonds.

Chemical molecules of Life

- ☞ Life is **composed** of lifeless **chemical molecules**.
- ☞ A single cell of the bacterium, **Escherichia coli** contains about 6,000 different **organic compounds**.
- ☞ It is believed that man may contain about **100,000 different types of molecules** although only a few of them have been characterized.
- ☞ The organic compounds such as amino acids, nucleotides and monosaccharides serve as the monomeric units or building blocks of complex biomolecules-proteins, nucleic acids (DNA and RNA) and polysaccharides, respectively.
- ☞ As regards lipids, it may be noted that they are not biopolymers in a strict sense, but majority of them contain fatty acids.

Chemical Composition of Man

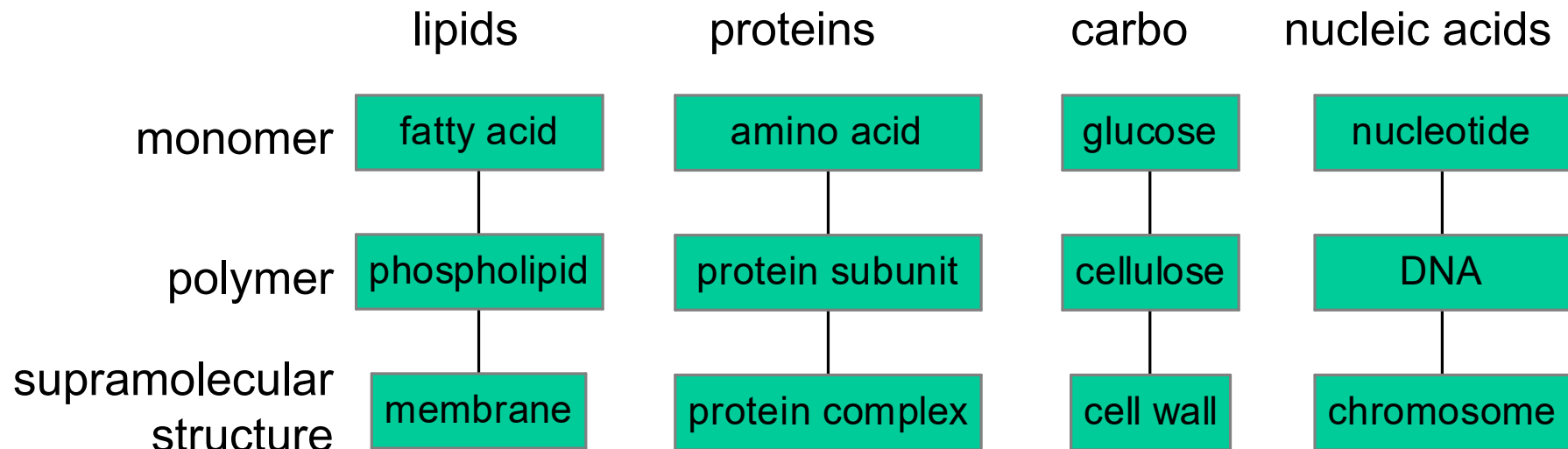
- ✓ **Water** is the solvent of life and contributes to more than 60% of the weight.
- ✓ This is followed by **protein** (mostly in muscle) and **lipid** (mostly in adipose tissue).
- ✓ The **carbohydrate** content is rather low which is in the form of glycogen.

Table: chemical composition of a normal man (weight 65 kg)

Constituent	Percent (%)	Weight(kg)
Water	61.6	40
Protein	17.0	11
Lipid	13.8	9
Carbohydrate	1.5	1
Minerals	6.1	4

Important Biomolecules are Polymers

- Monomers form polymers through condensations
- Polymers are broken down through hydrolysis.



... *Cont'd*

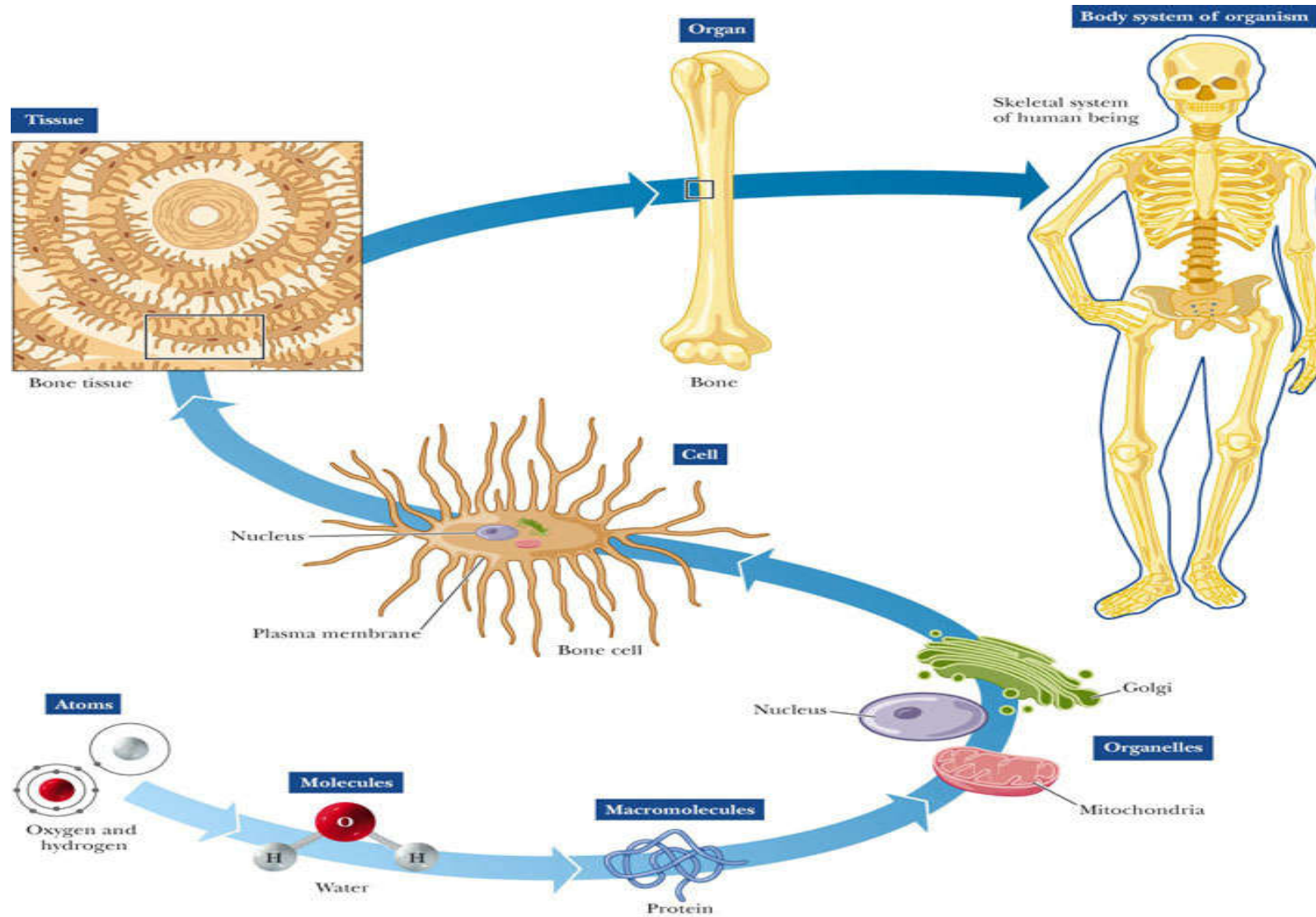
TABLE 1.1 The major complex biomolecules of cells

<i>Biomolecule</i>	<i>Building block (repeating unit)</i>	<i>Major functions</i>
1. Protein	Amino acids	Fundamental basis of structure and function of cell (static and dynamic functions).
2. Deoxyribonucleic acid (DNA)	Deoxyribonucleotides	Repository of hereditary information.
3. Ribonucleic acid (RNA)	Ribonucleotides	Essentially required for protein biosynthesis.
4. Polysaccharide (glycogen)	Monosaccharides (glucose)	Storage form of energy to meet short term demands.
5. Lipid	Fatty acids, glycerol	Storage form of energy to meet long term demands; structural components of membranes.

Hierarchical Organization of Organelles in Cells

- ♠ Living cells is composed of a very large marcomolecules such as proteins, nucleic acids, carbohydrate and lipids.
- ♠ The macromolecules form supramolecular assemblies (e.g membrane) which in turn organize into organelles, cells, tissues, organs and finally the whole organism.
- ♠ Cell is the most basic unit of life.
- ♠ Cells are the structural and functional units of all living organism.

Structural Hierarchy of an Organism



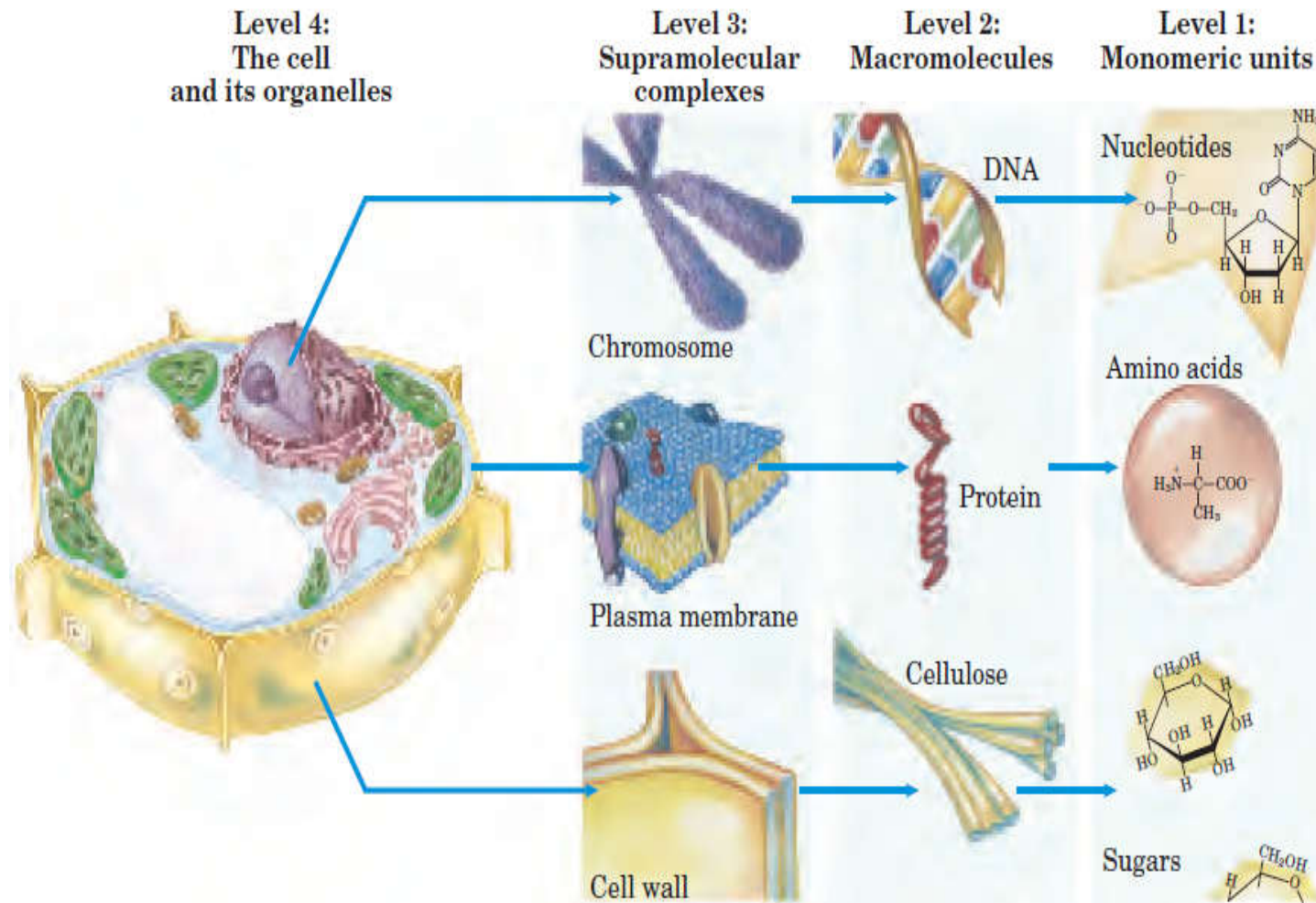


Fig. Structural hierarchy in the molecular organization of cells

Way To Live Life

- Be Balanced in all the life activities
- Work as per your priorities
- Try to adjust as per the need and condition of life
- Try your best to survive
- Live simple and natural Life
- Know your Do's and Don'ts of Life
- Do Right Judgements
- Work with Focus and Time Management
- Never Go Against the Nature
- Admire and Feel the Natures Life
- Think ,Thank and Praise the creator of Nature
- Imbibe Natural processes within ourselves
- Practice life like Natural processes



Thank You

National Aeronautics and Space Administration



Water, pH and Biological Buffer

Water

The Basis for Life

December, 2022

www.nasa.gov

Introduction to Water

➡ The human body is mostly water

☞ Lean muscle: 75% water

☞ Blood: 83% water

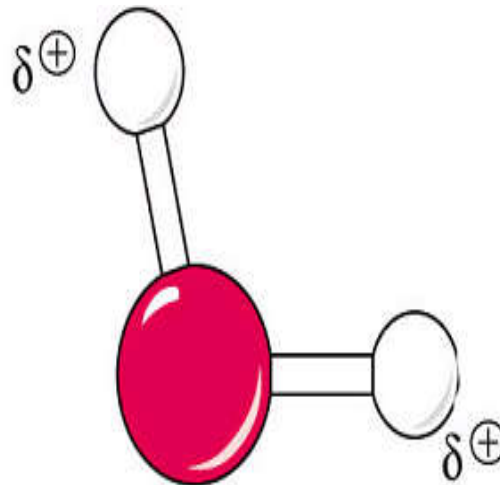
☞ Body fat: 25% water

☞ Bone: 22% water

➡ All living organisms require water to live

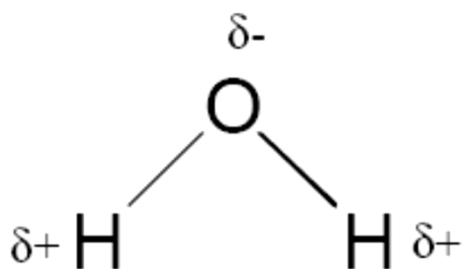
Properties of Water

- ❖ Very polar
- ❖ Oxygen is highly electronegative
- ❖ H-bond donor and acceptor
- ❖ High boiling point, melting point, heat of vaporization, surface tension

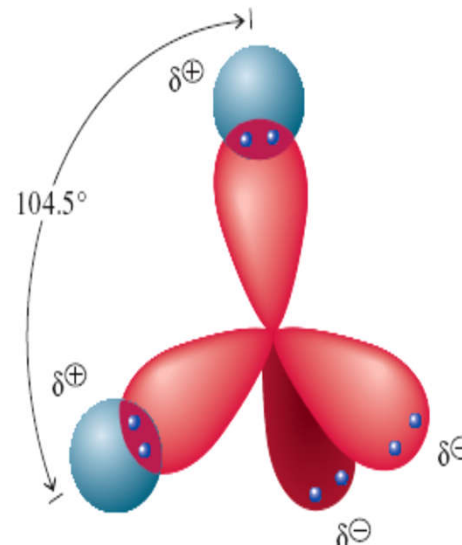


Water is Polar

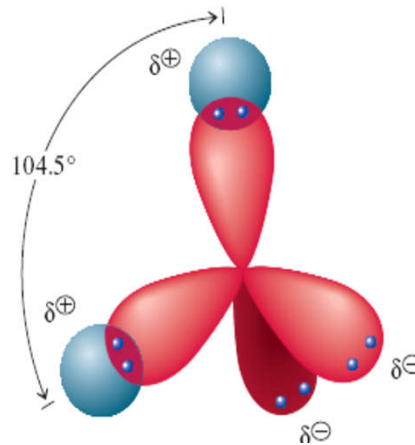
- ❑ Each -OH **bond** is polar
- ❑ The H₂O **molecule** is polar - due to its shape
- ❑ The H₂O molecule acts as a dipole (has a +ve and a -ve end)



↑
dipole
direction



- ➡ A water molecule is V-shaped and the angle between the two covalent bonds is 104.5° .



- ➡ Some important properties of water arise from its angle, shape and the intermolecular bonds that it can form.
- ➡ These interactions are important sources of structural stability in macromolecules and large cellular structures.

Water is an Excellent Solvent

- ➡ The physical properties of water combine to make it an excellent solvent.
- ➡ Water molecules are polar—this property has important consequences.
- ➡ In addition, water has an **intrinsic viscosity** that does not greatly impede the movement of dissolved molecules.
- ➡ Finally, water molecules themselves are **small compared** to some other solvents, such as ethanol and benzene.
- ➡ The small size of water molecules means that many of them can associate with solute particles to make them more soluble.

Ionic and Polar Substances Dissolve in Water

- ➡ Water can interact with and dissolve other polar compounds and compounds that ionize.
- ➡ Molecules that can dissociate to form ions are called **electrolytes**.
- ➡ Substances that readily dissolve in water are said to be **hydrophilic**, or water loving.

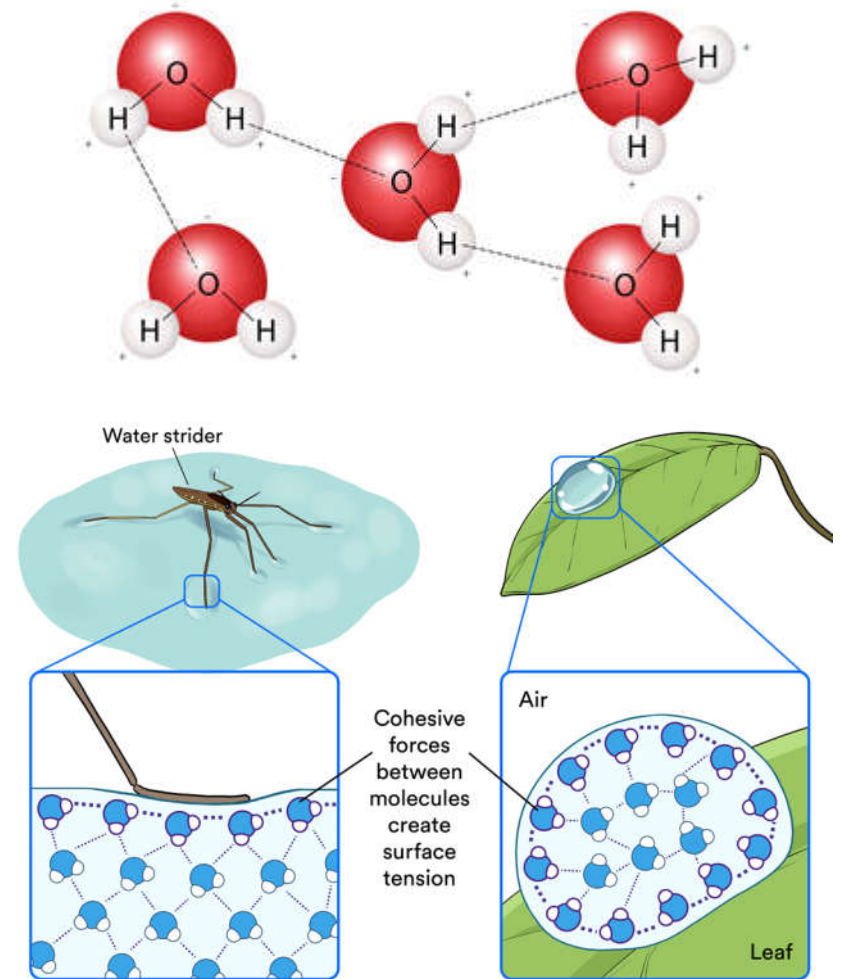
- ➡ Hydrocarbons and other non-polar substances have very low solubility in water because water molecules tend to interact with other water molecules rather than with nonpolar molecules.
- ➡ As a result, water molecules exclude nonpolar substances, forcing them to associate with each other.

Physical Properties of Water

- ✓ A molecular-level description of the following physical properties of water
 - ☞ Surface tension
 - ☞ Adhesion
 - ☞ Cohesion
 - ☞ Viscosity
 - ☞ Boiling and melting
 - ☞ Hardness and brittleness

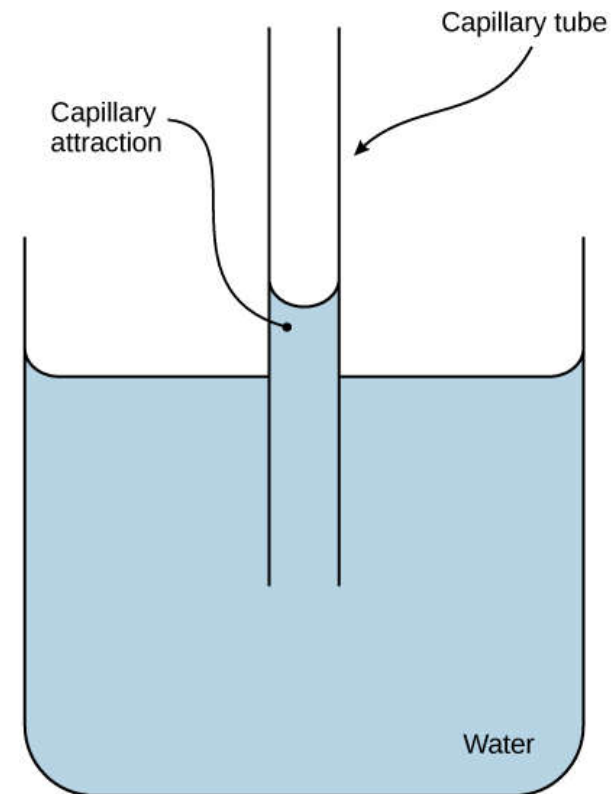
A. Surface Tension

- **Cohesion** - forces between like molecules in the liquid i.e intermolecular forces.
- Cohesion- water attracted to other water molecules because of polar properties. For water these are dominated by hydrogen bonding.
- Surface molecules experience a cohesive force tending to pull them into the bulk and to minimize surface area --> SURFACE TENSION



B. Adhesion

- ↑ Liquids interact with solid surfaces.
- ↑ If the adhesive forces are strong, the liquid ‘wets’ the solid surface.
- ↑ The shapes of droplets reflect the balance between adhesion and surface tension
- ↑ Adhesion allows water to move through very small pores or tubes against gravity (**capillary action**)



C. Viscosity

- ✚ Viscosity is the resistance to flow.
- ✚ High viscosity is associated with strong intermolecular forces
- harder for molecules to 'slide past each other' *E.g. water > octane, Paraffin oil > octane > pentane*
- ✚ The viscosity of water is fairly high - but not as high as might be expected based on the intermolecular forces argument
- ✚ Viscosity decreases with increasing temperature (for small molecules)

D. Heat Capacity

- ♣ Water has a higher heat capacity than many other liquids
- ♣ Water resists temperature change and absorb or release large amounts of heat energy with little change in temperature.
- ♣ Hydrogen bonding network in water can store more energy than a liquid with less extensive intermolecular forces;
- ♣ More energy is required to get the molecules to rotate in the liquid; as the liquid is heated it expands (at least above 4 °C) and energy is required to 'pull apart' the hydrogen bonds.

Weak interaction in aqueous systems

- Hydrogen bonds between water molecules provide the cohesive forces that make water a liquid at room temperature and favor the extreme ordering of molecules that is typical of crystalline water (ice).
- Polar biomolecules dissolve readily in water because they can replace water-water interactions with more energetically favorable water-solute interactions.
- In aqueous solutions non-polar molecules tend to cluster together.
- Hydrogen bonds and ionic, hydrophobic (Greek, "water-fearing") and van der Waals interactions are individually weak, but collectively they have a very significant influence on the three-dimensional structures of proteins, nucleic acids, polysaccharides, and membrane lipids.

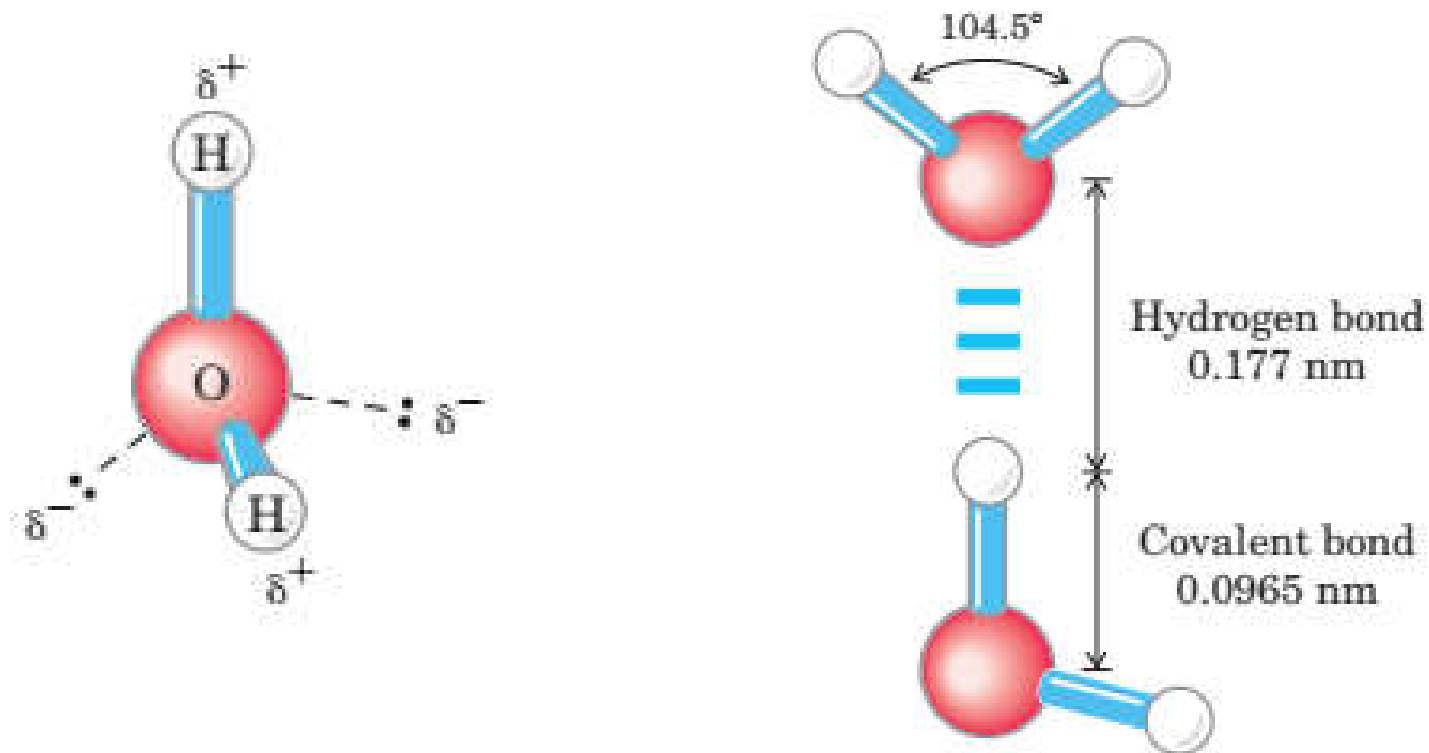
Hydrogen Bonding gives water its unusual Properties

- Water has a higher melting point, boiling point, and heat of vaporization than most other common solvents.
- These unusual properties are a consequence of attractions between adjacent water molecules that give liquid water great internal cohesion.
- A look at the electron structure of the H_2O molecule reveals the cause of these intermolecular attractions.
- Each hydrogen atom of a water molecule shares an electron pair with the central oxygen atom. The geometry of the molecule is dictated by the shapes of the outer electron orbitals of the oxygen atom.

Hydrogen Bonding gives water its unusual Properties

- The shared electrons are more often in the vicinity of the oxygen atom than of the hydrogen; since Oxygen is more electronegative.
- The result of this unequal electron sharing is two electric dipoles in the water molecule, one along each of the H-O bonds; each hydrogen bears a partial positive charge (δ^+) and the oxygen atom bears a partial negative charge equal in magnitude to the sum of the two partial positives ($2\delta^-$).

- As a result, there is an electrostatic attraction between the oxygen atom of one water molecule and the hydrogen of another, called a hydrogen bond.



... Cont'd

- ☞ Hydrogen bonds are **not unique to water**; readily form between an electronegative atom (the hydrogen acceptor, usually oxygen or nitrogen) and a hydrogen atom covalently bonded to another electronegative atom (the hydrogen donor) in the same or another molecule.

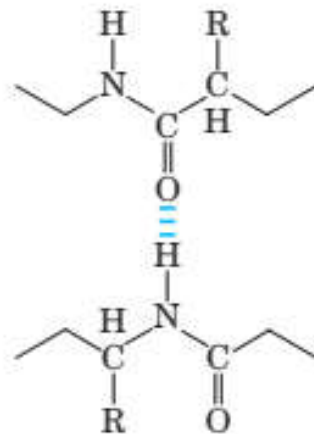
Between the hydroxyl group of an alcohol and water



Between the carbonyl group of a ketone and water



Between peptide groups in polypeptides



Between complementary bases of DNA

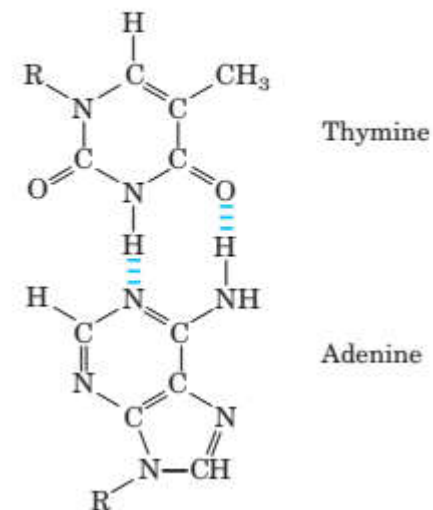


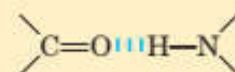
TABLE 2-5 Four Types of Noncovalent ("Weak") Interactions among Biomolecules in Aqueous Solvent

Hydrogen bonds

Between neutral groups



Between peptide bonds



Ionic interactions

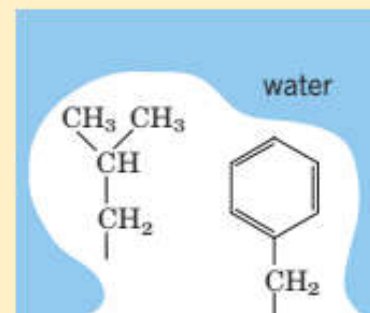
Attraction



Repulsion



Hydrophobic interactions



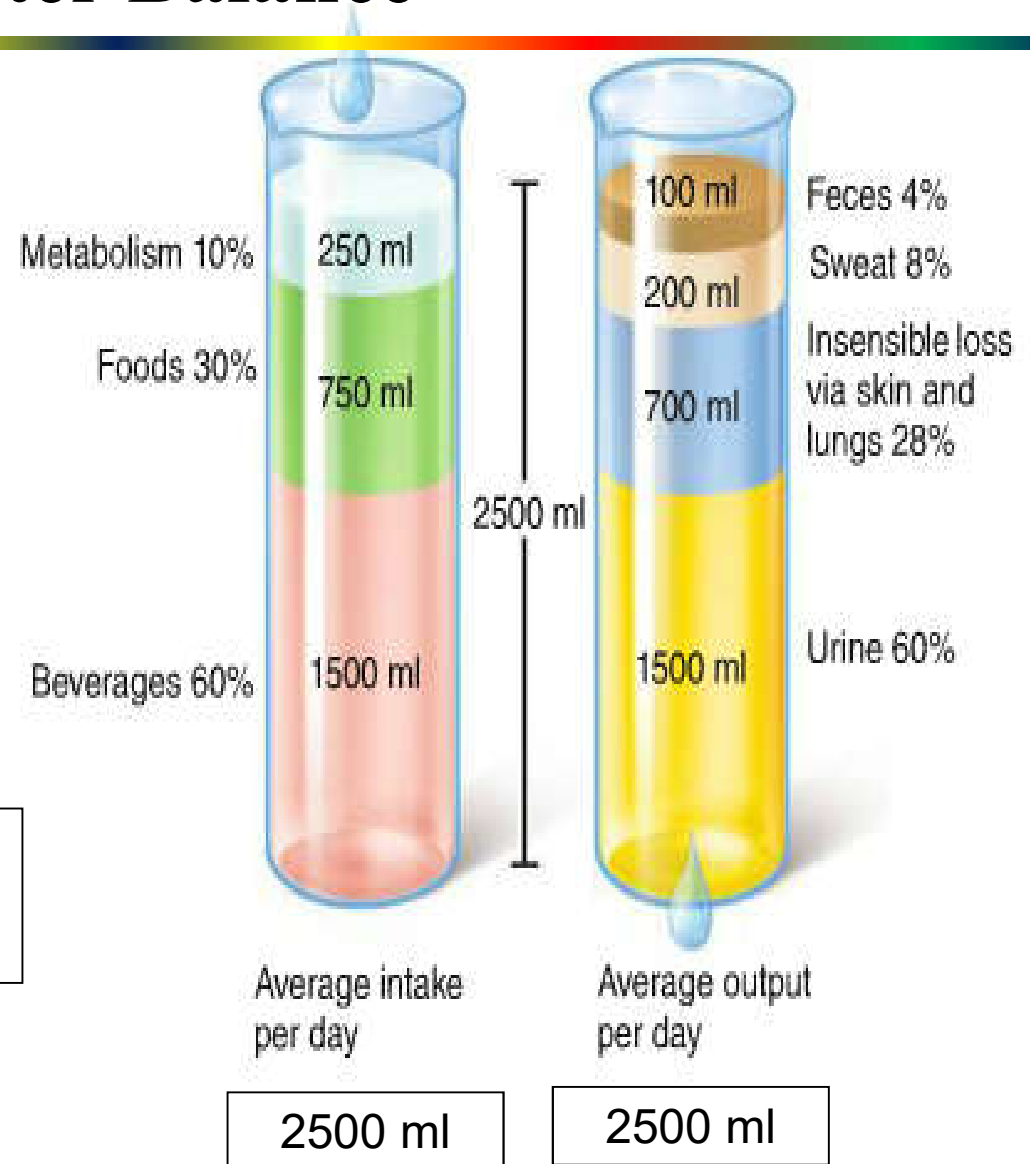
van der Waals interactions

Any two atoms in close proximity

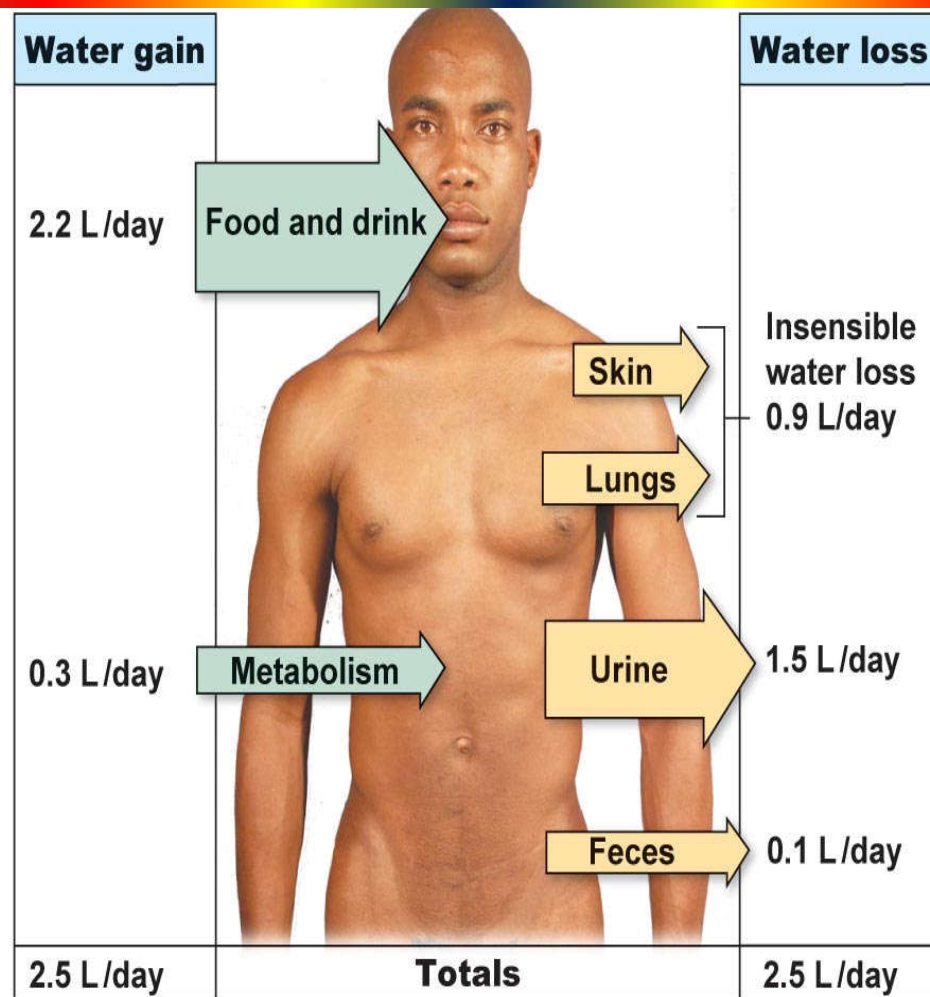
Water Balance

- ✓ There has to be a balance between the water intake and the water output
- ✓ To keep the composition of the cell water and extra cellular fluid constant.

Water intake must equal water output



... *Cont'd*



$$\begin{array}{l} \text{Intake} \\ 2.2 \text{ L/day} \end{array} + \begin{array}{l} \text{Metabolic production} \\ 0.3 \text{ L/day} \end{array} - \begin{array}{l} \text{Output} \\ 2.5 \text{ L/day} \end{array} = 0$$

Water Recommendation

✓ Water needs vary and it depends on:

☞ Type of **food** (i.e. salty food need more water)

☞ The **environment** temperature and humidity.

☞ The **activity level** of a person.

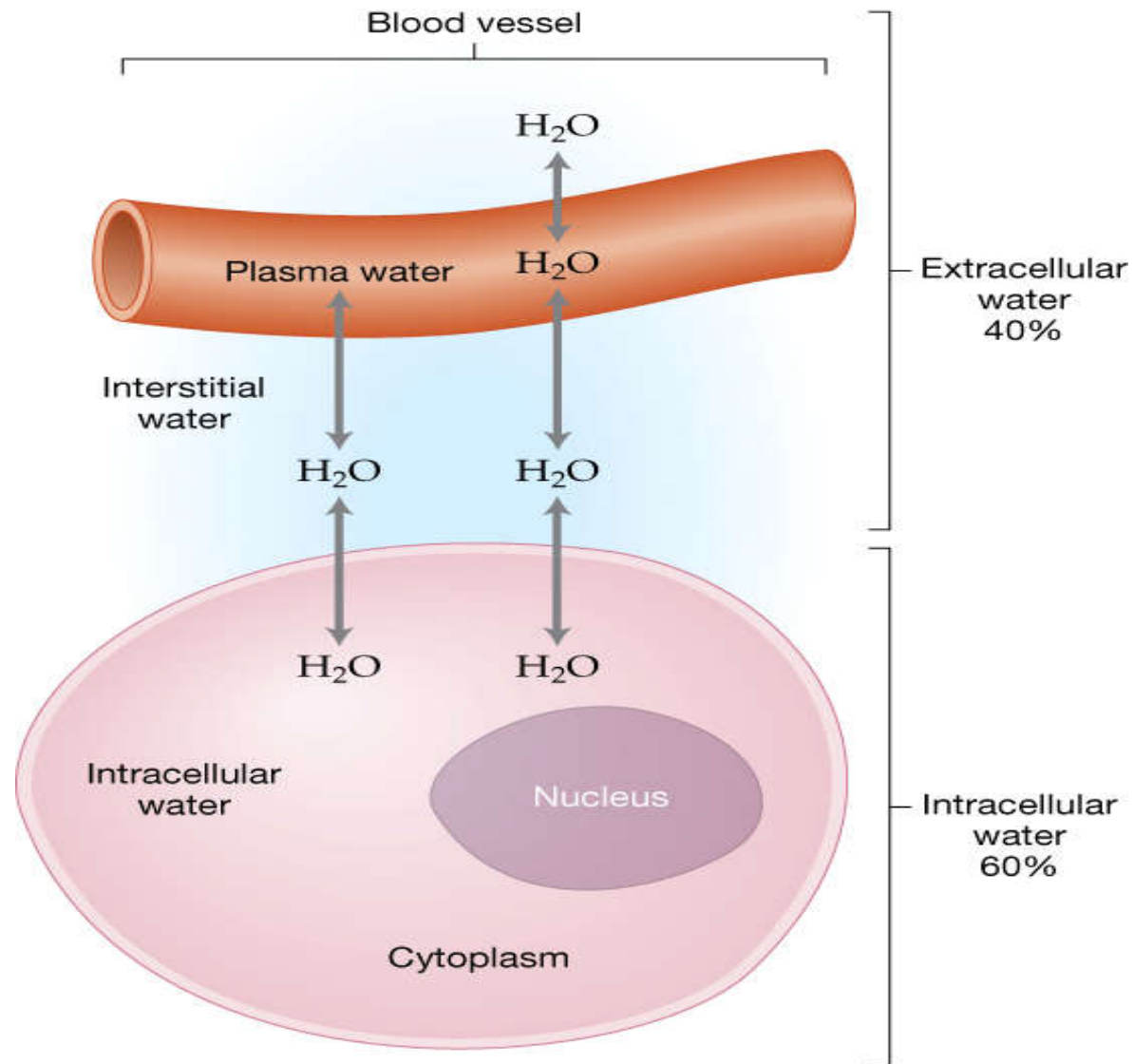
NB: both **alcohol** and **caffeine** act as diuretics causing the body to lose water

✓ **Diuretics** are medicines that help reduce the amount of water in the body

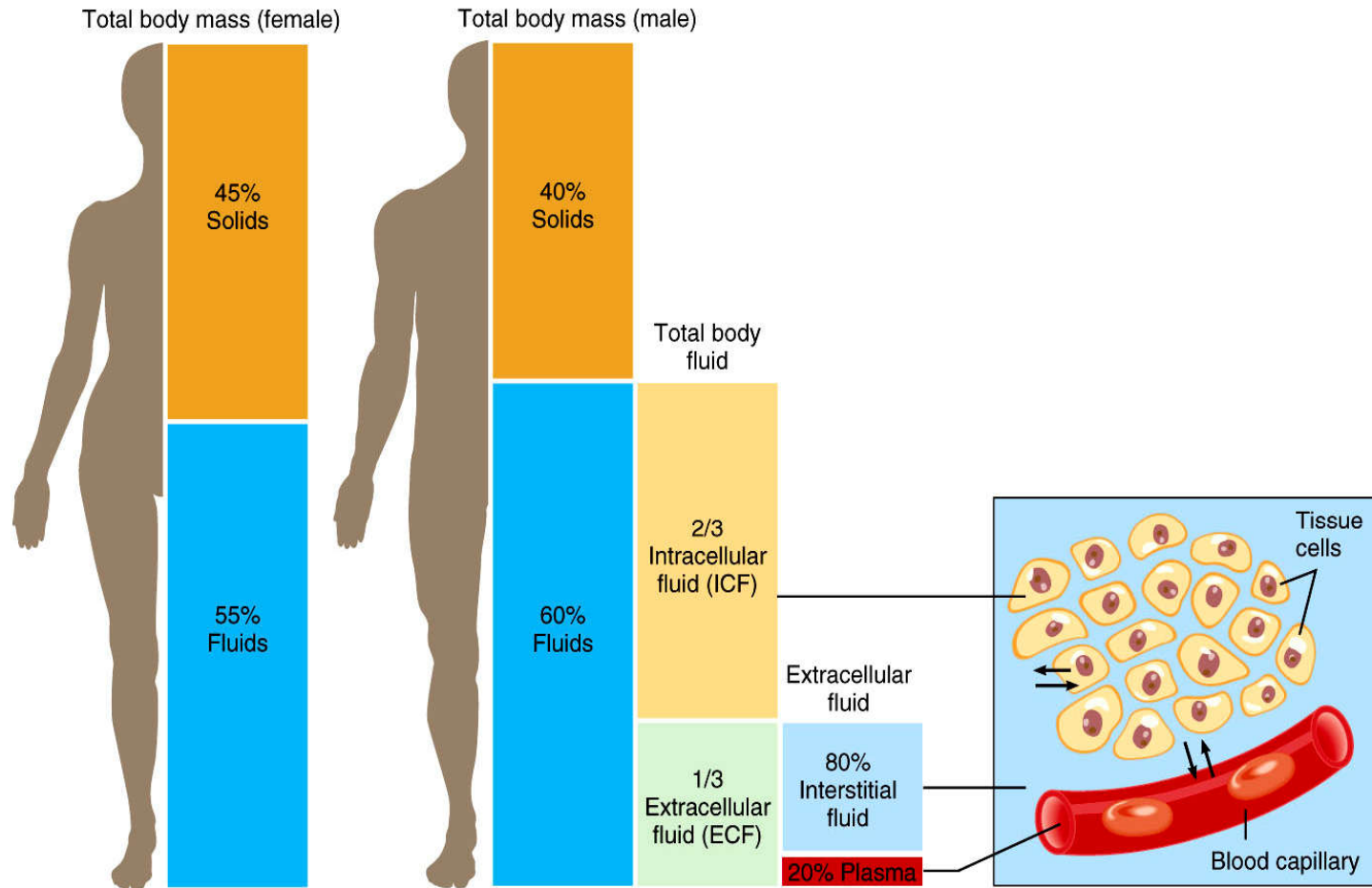
Water Distribution in Body

- ✓ All chemical reactions take place in aqueous medium in body, and all reactives are dissolved in body fluids.
 - ✓ Water occupies two main fluid compartments i.e Intracellular Fluid (ICF) & Extracellular Fluid (ECF).
 - ✓ The volumes of water for an average 70-kg person is 42 liter
 - ✓ Two-thirds of this water (28 L) is within body cells (ICF).
 - ✓ The remaining one-third (14 L) is ECF and among this:-
 - * 80 % (11 L) is interstitial fluid and
 - * 20 % (3 L) is plasma.
 - ✓ Both compartments are physically separated by the cellular plasma membrane.
-

... *Cont'd*



Body Fluid Compartments



(a) Distribution of body solids and fluids in an average lean, adult female and male

(b) Exchange of water among body fluid compartments

Figure 27.01 Tortora - PAP 12/e
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- **Intracellular Fluid** (60%) fluid inside all cells throughout body.
- **Extracellular Fluid- fluid outside cells**
 - **Interstitial fluid** (the water immediately outside cells, between and around cells) (30%)
 - **Plasma fluid** (the water inside blood vessels, but not in blood cells) (9%)
 - **Transcellular fluid** (the water enclosed in chambers lined by epithelial membranes) (1%)

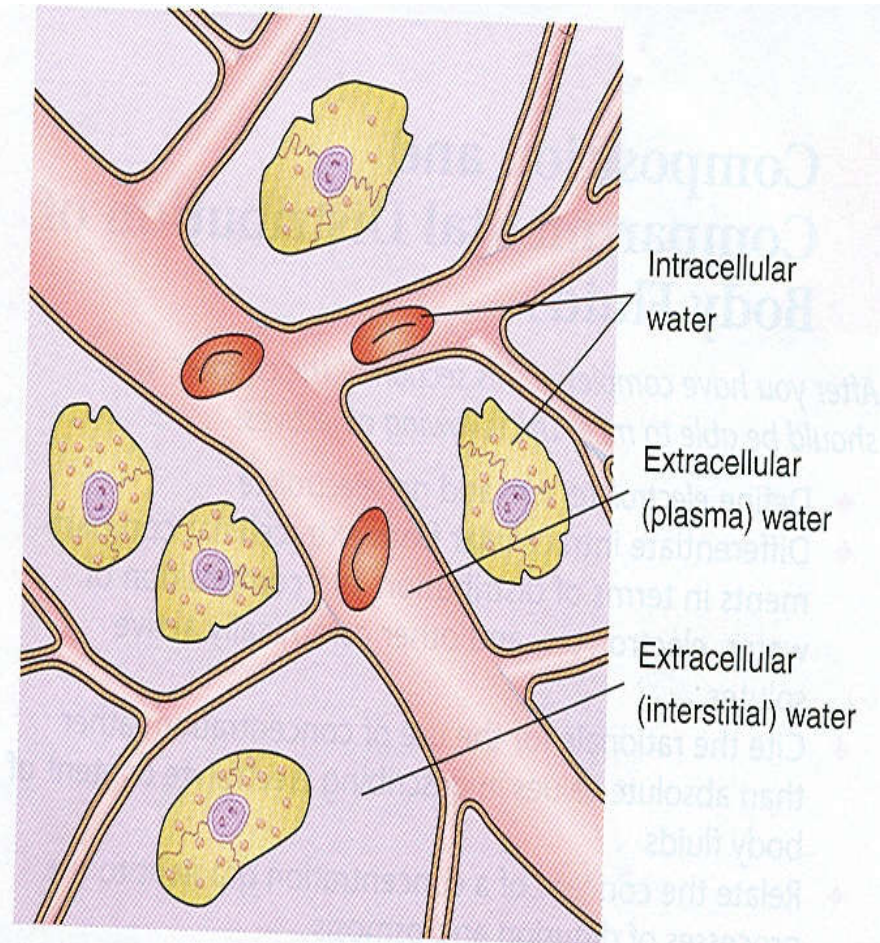


FIGURE 31-1 Distribution of body water. The extracellular space includes the vascular compartment and the interstitial spaces.

Eelectrolytes in Body Fluid

- ✓ Many compounds are carried into the cell by special carrying systems because of limited permeability of cellular membranes
- ✓ Cellular membranes are permeable to water and hydrophobic molecules but not permeable for ions and hydrophilic neutral molecules.
- ✓ These molecules are carried into cell by special carrier systems and pumps.
- ✓ The most effective pump in the cell membranes is Na^+/K^+ ATPase

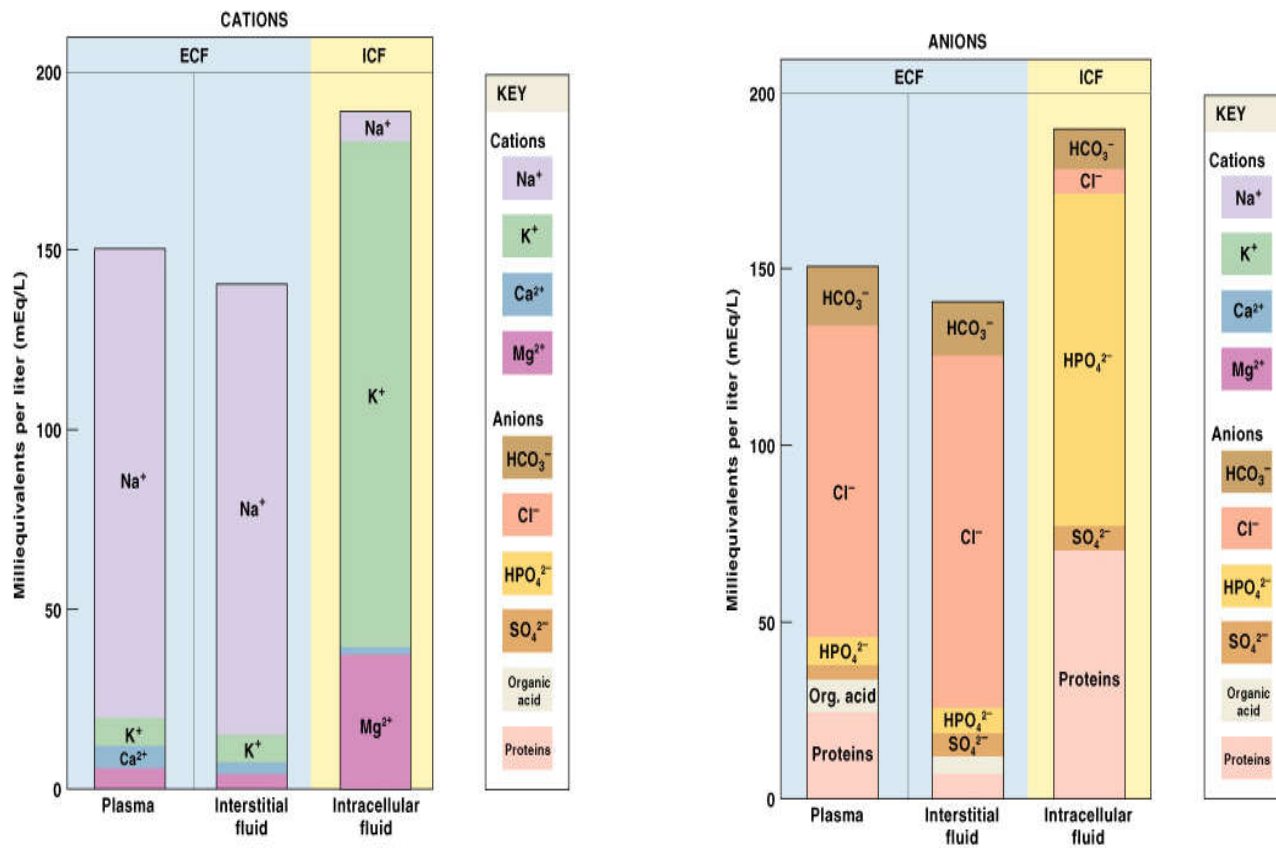
Eelectrolytes in Body Fluid

- ✓ Na^+/K^+ ATPase is responsible for setting and maintaining the intracellular concentrations of Na^+ and K^+ , and for generation of transmembrane electrical potential.
- ✓ Na^+/K^+ ATPase moves 3 Na^+ out of the cell for every 2 K^+ moves in.
- ✓ Na^+/K^+ pumps play major role in keeping K^+ high inside cells (ICF) and Na^+ high outside cell (ECF).
- ✓ The electrical potential is central for electrical signaling in neurons and the gradient of Na^+ is used to drive uphill co-transport of various solutes such as glucose, amino acids in a variety of cell types.

- ✓ Ions form when electrolytes dissolve and dissociate
- ✓ Four general functions of electrolyte
 - I. Control osmosis of water between body fluid compartments
 - II. Help maintain the acid-base balance
 - III. Carry electrical current
 - IV. Serve as cofactors

Cations and Anions in Body Fluids

- ✓ ECF - Prevalent cation is Na^+ and Prevalent anion is Cl^-
- ✓ ICF - Prevalent cation is K^+ and Prevalent anion is PO_4^{3-}



Edema

- ✓ Edema is the presence of excess fluid in the body tissues.
- ✓ Edema is often observed in children suffering from kwashiorkor, a nutritional disease caused by an inadequate intake of protein.
- ✓ Broadly, there are 2 types of edema:
 - I. Intracellular edema (less common)
 - II. Extracellular edema

Elephantiasis –
block of lymph return



© R. J. Heath-Chandrasekhar, TDR, WHO/SP, Photo Researchers

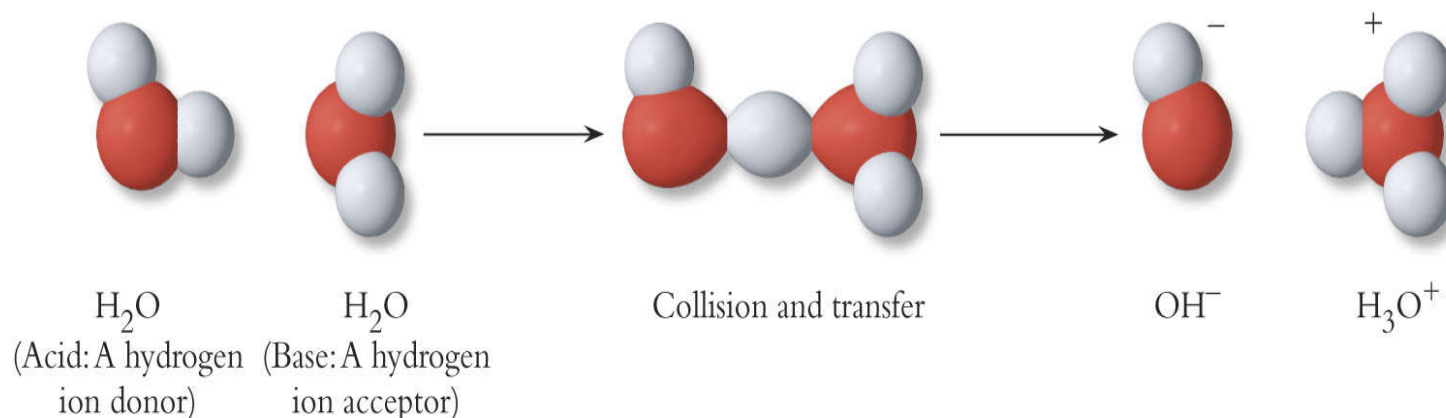
Vander's Human Physiology, 11th Ed.

Ionization of Water

- ➡ One of the important properties of water is its slight tendency to ionize.
- ➡ Pure water consists low concentration of hydronium ions (H_3O^+) and an equal concentration of hydroxide ions (OH^-).



- ➡ Hydronium ions are capable of donating a proton to another ion.
- ➡ The hydronium ion H^+ is a major source of protons in biochemical reactions.
- ➡ Hydroxide ions can accept a proton and be converted back into water molecules.



- ✓ Because the activity of pure water is 1, the equilibrium constant for this reaction is

$$K_{\text{eq}} = \frac{[\text{H}^+][\text{OH}^-]}{[\text{H}_2\text{O}]}$$

- Experimental measurements have determined that the concentration of each ion is $1.0 \times 10^{-7} \text{ M}$ at 25°C

- ⇒ In pure water at 25 °C, the concentration of water is 55.5 M.

$$K_{eq} = \frac{[H^+][OH^-]}{55.5 \text{ M}}$$

$$(55.5 \text{ M})(K_{eq}) = [H^+][OH^-] = K_w$$

- ⇒ This particular constant is called the **ion-product for water (K_w)**
- ⇒ The value for K_{eq} , determined by electrical-conductivity measurements of pure water, is $1.8 \times 10^{-16} \text{ M}$ at 25 °C.

$$\begin{aligned} K_w &= [H^+][OH^-] = (55.5 \text{ M})(1.8 \times 10^{-16} \text{ M}) \\ &= 1.0 \times 10^{-14} \text{ M}^2 \end{aligned}$$

- ⇒ When there are exactly equal concentrations of H^+ and OH^-

$$K_w = [H^+][OH^-] = [H^+]^2$$

$$[H^+] = \sqrt{K_w} = \sqrt{1 \times 10^{-14} \text{ M}^2}$$

$$[H^+] = [OH^-] = 10^{-7} \text{ M}$$

The pH Scale Designates the H⁺ and OH⁻ Concentrations

- ➡ The ion product of water, K_w, is the basis for the pH scale
- ➡ The term pH is defined by the expression:

$$\text{pH} = \log \frac{1}{[\text{H}^+]} = -\log [\text{H}^+]$$

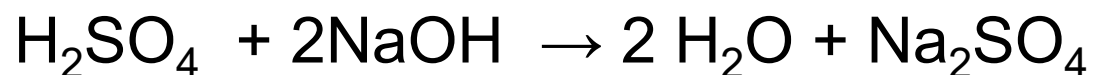
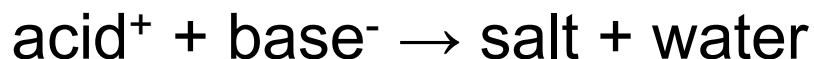
$$\text{pOH} = -\log [\text{OH}^-]$$

$$\text{pH} + \text{pOH} = 14$$

Acid-base theory

Arrhenius definition

- Arrhenius acids form hydrogen ions in aqueous solution with Arrhenius bases forming hydroxide ions.
- This definition assumes that water is the solvent.

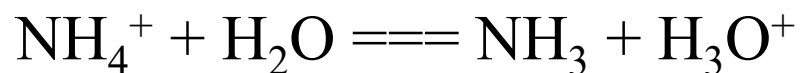
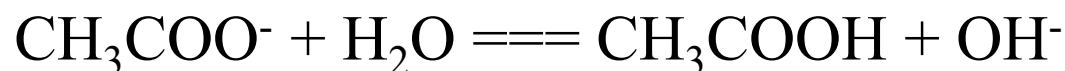
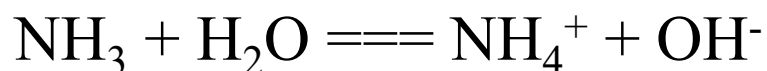
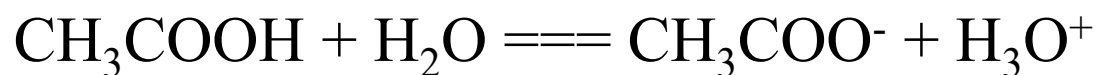


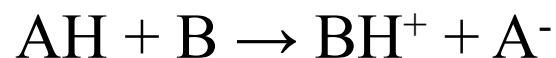
Acid-base theory

- **Brønsted and Lowry:** expanded upon the Arrhenius definitions, and defined **acids as proton donors** and **bases as proton acceptors**.
- They also introduced the concept of *conjugate acid-base pairs*.
 - ☞ For any solvent that can dissociate into a cation and an anion, the cation is the **acid**, and the **anion is the base**.
 - ☞ Any solute that causes an increase in the concentration of the **cation is an acid**, those that increase the concentration of the **anion are bases**.

... *Cont'd*

✓ **Brønsted-Lowry definition**-is based upon the idea of protonation of bases through the de-protonation of acids -- more commonly referred to as the ability of acids to "donate" hydrogen ions (H^+) or protons to bases, which "accept" them.

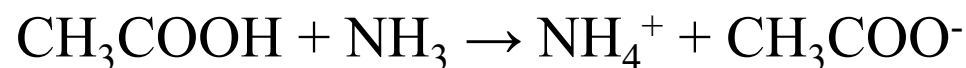




General formula for representing Brønsted-Lowry reactions.

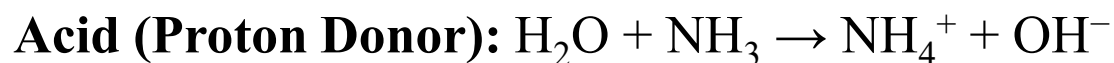
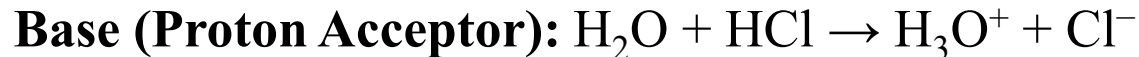


- Hydrochloric acid completely reacts with water to form the hydronium and chloride ions



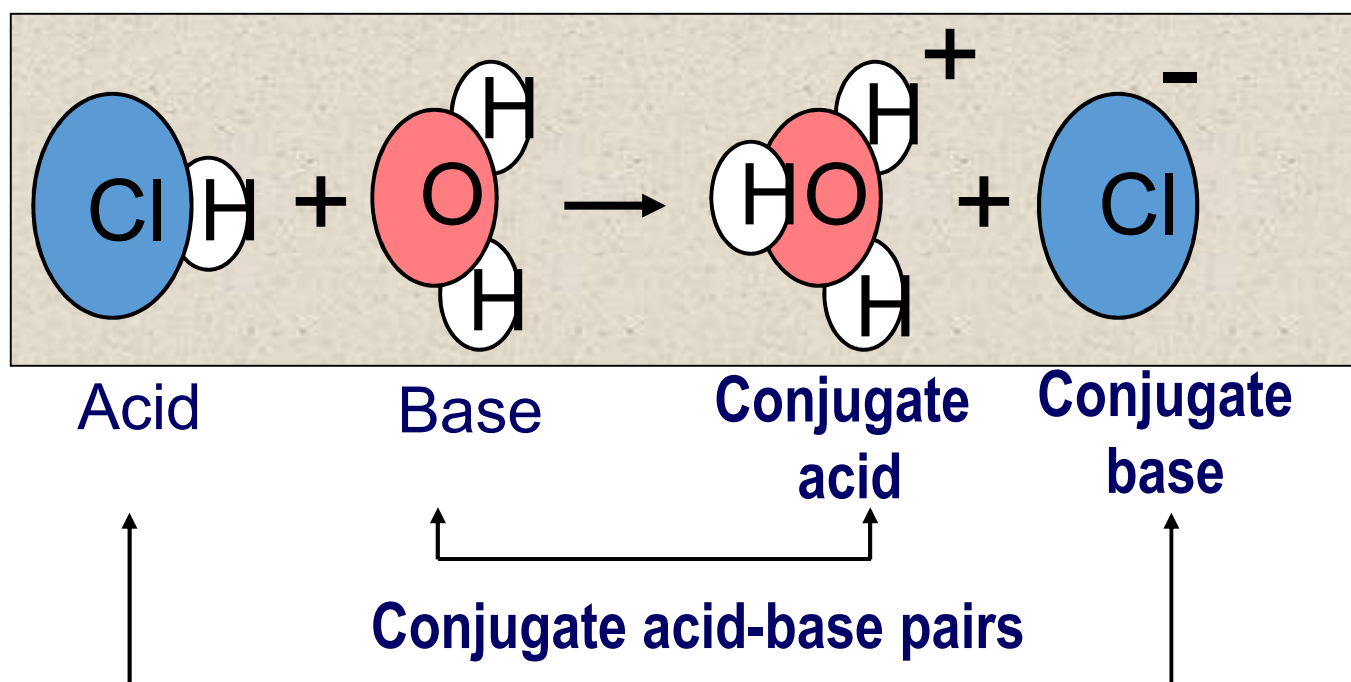
- ✓ Acetic acid reacts incompletely with ammonia, no hydronium ions being produced

- ✓ In differentiation from the [Arrhenius](#) definition, the Brønsted-Lowry definition postulates that for each acid, there is a conjugate acid and base or "*conjugate acid-base pair*" that is formed through a complete reaction, which also includes water, which is [amphoteric](#)



(Indeed, it can do both at once: $2\text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{OH}^-$)

- ✓ In this idea, the ionization of an acid by water is just one example of an acid-base reaction.



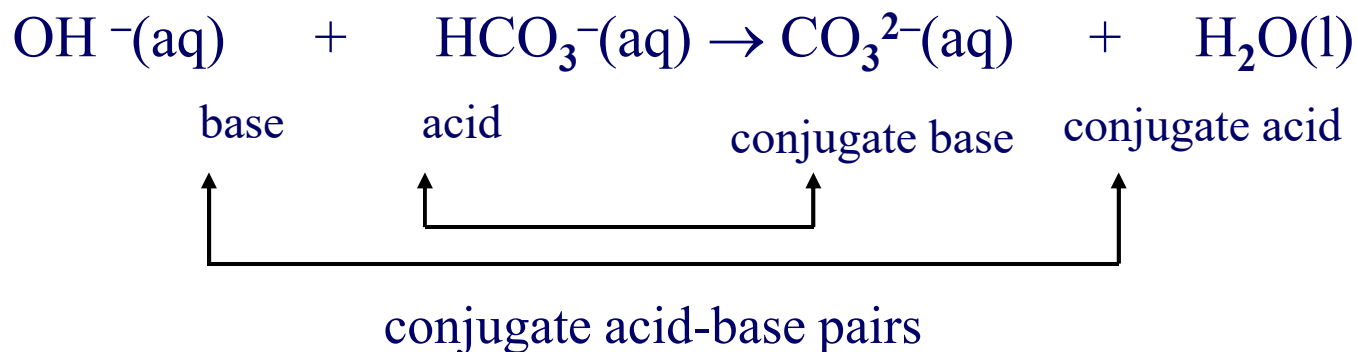
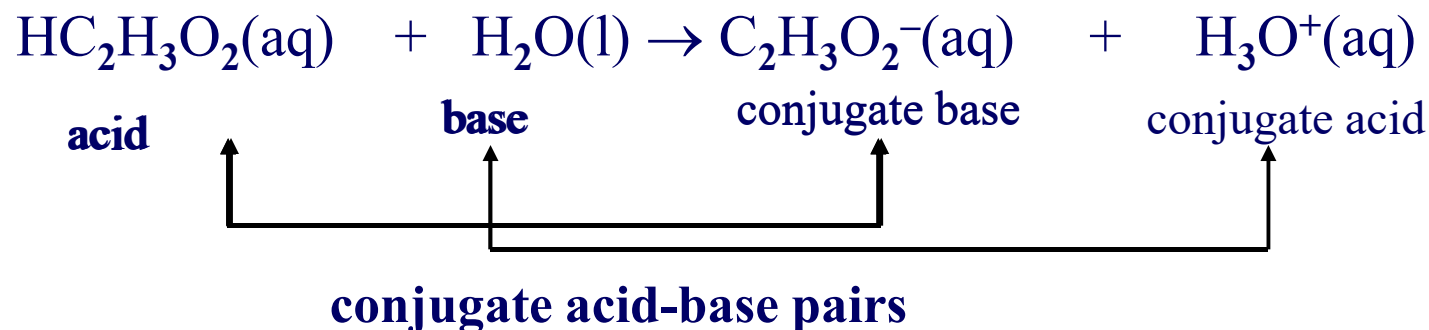
- ✓ Acids and bases are identified based on whether they donate or accept H^+ .
- ✓ “Conjugate” acids and bases are found on the products side of the equation.
- ✓ A conjugate base is the same as the starting acid minus H^+ .

... *Cont'd*

Acid (proton donor)	Base (proton acceptor)	Conjugate acid	Conjugate base
HA	$+ \text{H}_2\text{O}$ (water)	$\rightleftharpoons \text{H}_3\text{O}^+$ (Hydronium ion)	$+ \text{A}^-$
HA	$+ \text{NH}_3$ (Ammonia)	$\rightleftharpoons \text{NH}_4^+$ (Ammonium ion)	$+ \text{A}^-$

Practice Problems

✓ Identify the acid, base, conjugate acid, conjugate base, and conjugate acid-base pairs:

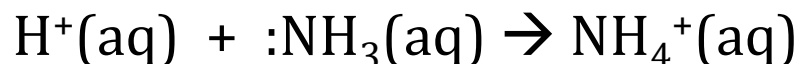


Acid-base Theory

Lewis Acids & Bases: A base is an electron-pair donor, and an acid is an electron-pair acceptor.



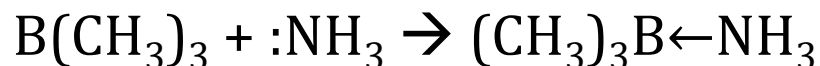
- This definition includes the “standard” Brønsted-Lowry acid-base reactions:



- It also includes the reactions of metal ions or atoms with ligands to form coordination compounds:



- In addition, electron-deficient compounds such as trivalent boron is categorized as a Lewis acid.



- There is the possibility of competing reaction pathways depending upon which reactants are present, and the relative energies of possible products.
- As a result, a compound such as water may serve as an acid, a base, an oxidizing agent or a reducing agent.

Lewis acids and bases

- *A Lewis acid is an electron acceptor*
- *A Lewis base is an electron donor*

Bronsted acids and bases

- *A Bronsted acid is a proton donor*
- *A Bronsted base is a proton acceptor*

Acid – Base Strength

- **Strong Acid-** Transfers all protons to water; completely ionizes; strong electrolyte; conjugate base is weaker and has negligible tendency to be protonated.
- **Weak Acid-** Fraction of protons transferred to water; partly ionized; weak electrolyte; conjugate base is stronger readily accepting protons from water.
 - ➔ As acid strength decreases, base strength increases.
 - ➔ The stronger the acid, the weaker its conjugate base
 - ➔ The weaker the acid, the stronger its conjugate base

... *Cont'd*

- ♠ When a strong acid is dissolved in water, it reacts with it to form H_3O^+ in the following reaction (where "HA" is a generic strong acid such as "HCl"):



- ♠ Any acid that is stronger than H_3O^+ reacts with H_2O to form H_3O^+ ; therefore, no acid can be stronger than H_3O^+ in H_2O .
- ♠ This is true with any solvent, although usually to a lesser extent than with water. Bases may be also leveled in solvents.

pK/ pKa

- Negative log of the ionization constant of an acid
- Strong acids would have a $pK < 3$
- Strong base would have a $pK > 9$
- acid ionization constant

pH

- Negative log of the hydrogen ion concentration
- $\text{pH} = \text{pK} + \log([\text{base}]/[\text{acid}])$
- Represents the hydrogen concentration

pKa

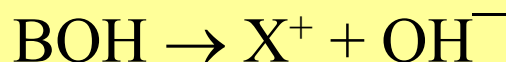
- ✓ When reactions are carried out with very highly dilute acids in water, the activity of water approaches unity.

... *Cont'd*

Acid - any substance which delivers hydrogen ion (H^+) to the solution.



Base - any substance which delivers hydroxide ion (OH^-) to the solution



$$K_{eq} = \frac{a_{A^-} a_{H^+}}{a_{AH} a_{H_2O}} \approx \frac{a_{A^-} a_{H^+}}{a_{AH}} \longrightarrow K_a = \frac{[A^-][H^+]}{[AH]}$$

$$pK_a = -\log(K_a) = \left\{ -\log\left(\frac{[A^-]}{[HA]}\right) + \log([H^+]) \right\}$$

$$pK_a = -\log(K_a) = -\log\left(\frac{[A^-]}{[HA]}\right) + pH$$

The pK_a is defined similar to the pH , $pK_a = -\log K_a$

The Henderson-Hasselbalch Equation

- ✓ The quantitative relationship between the pH of the solution and the concentration of a weak acid and its conjugate base is described by Henderson -Hasselbalch equation

$$\text{pH} = \text{pK}_a + \log_{10} \frac{[\text{A}^-]}{[\text{HA}]}$$

- ✓ **pKa** - the negative log of the dissociation constant, which is a measure of strength of an acid/base .
- ✓ **pKa** is defined as the pH at which the protonated and unprotonated species are present in equal concentrations.

- ✓ A low pK_a value indicates that the compound is acidic and will easily give up its proton to a base.
- ✓ At $pH < pka$, protonated acid form is more (it will give out its H^+ ions)
- ✓ At $pH > pka$, deprotonated base form is more in the solution (it will take H^+ ions to neutralize the pH).
- ✓ At $pKa = pH$, there is equal concentration of acid and its conjugate base.

pK_b (base-ionization constant)

For a base BOH,



$$K_b = \frac{[\text{B}^+][\text{OH}^-]}{[\text{BOH}]}$$

The pK_b is defined similar to pK_a

$$\text{p}K_b = -\log K_b, K_b = 10^{-\text{p}K_b}$$

Whatever you have learned for weak acids apply to weak bases

For acid, HA



$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

The pK_a is defined similar to the pH,

$$\text{p}K_a = -\log K_a, K_a = 10^{-\text{p}K_a}$$

The pH and pOH scales

- A convenient way to express acidity and basicity and pH is defined as:

$$\text{pH} = -\log[\text{H}_3\text{O}^+]$$

- In general, a lower case p before a symbol is read as ‘negative logarithm of’ the symbol

$$\text{pOH} = -\log[\text{OH}^-]$$

$$\text{pAg} = -\log[\text{Ag}^+]$$

and so forth

- ✓ If we know either $[\text{H}_3\text{O}^+]$ or $[\text{OH}^-]$, then we can calculate pH and pOH.

E.g. Calculate the pH of a solution in which the $[\text{H}_3\text{O}^+] = 0.030\text{M}$.

$$\text{pH} = -\log[\text{H}_3\text{O}^+]$$

$$\text{pH} = -\log(3.0 \times 10^{-2})$$

$$\text{pH} = 1.52$$

☞ The pH of a solution is 4.597. What is the concentration of H_3O^+ ?

$$\text{pH} = -\log[\text{H}_3\text{O}^+]$$

$$4.597 = -\log[\text{H}_3\text{O}^+]$$

$$\log[\text{H}_3\text{O}^+] = -4.597$$

$$[\text{H}_3\text{O}^+] = 10^{-4.597}$$

$$[\text{H}_3\text{O}^+] = 2.53 \times 10^{-5} \text{ M}$$

☞ A convenient relationship between pH and pOH may be derived for all dilute aqueous solutions at 25°C.

$$[\text{H}_3\text{O}^+][\text{OH}^-] = 1.0 \times 10^{-14}$$

- Taking the logarithm of both sides of this equation gives

$$\log[\text{H}_3\text{O}^+] + \log[\text{OH}^-] = -14.00$$

- Multiplying both sides of this equation by -1 gives

$$-\log[\text{H}_3\text{O}^+] + (-\log[\text{OH}^-]) = 14.00$$

- Or, in a slightly different form, we have

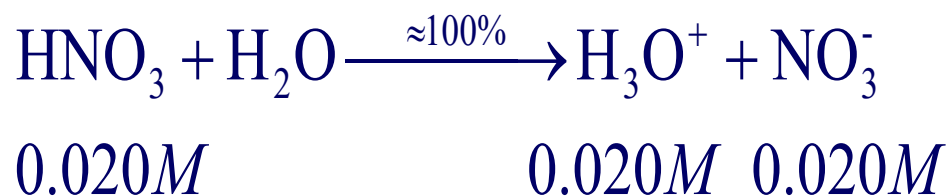
$$\text{pH} + \text{pOH} = 14.00$$

- Remember these two equations!!

$$[\text{H}_3\text{O}^+][\text{OH}^-] = 1.0 \times 10^{-14}$$

$$\text{pH} + \text{pOH} = 14.00$$

Q1. Calculate $[\text{H}_3\text{O}^+]$, pH, $[\text{OH}^-]$, and pOH for 0.020 M HNO_3 solution.



$$[\text{H}_3\text{O}^+] = 2.0 \times 10^{-2} M \quad \text{pH} = -\log(2.0 \times 10^{-2} M) = 1.70$$

$$[\text{H}_3\text{O}^+][\text{OH}^-] = 1.0 \times 10^{-14}$$

$$[\text{OH}^-] = \frac{1.0 \times 10^{-14}}{[\text{H}_3\text{O}^+]} = \frac{1.0 \times 10^{-14}}{2.0 \times 10^{-2}} = 5.0 \times 10^{-13} M$$

$$\text{pOH} = -\log[5.0 \times 10^{-13}] = 12.30$$

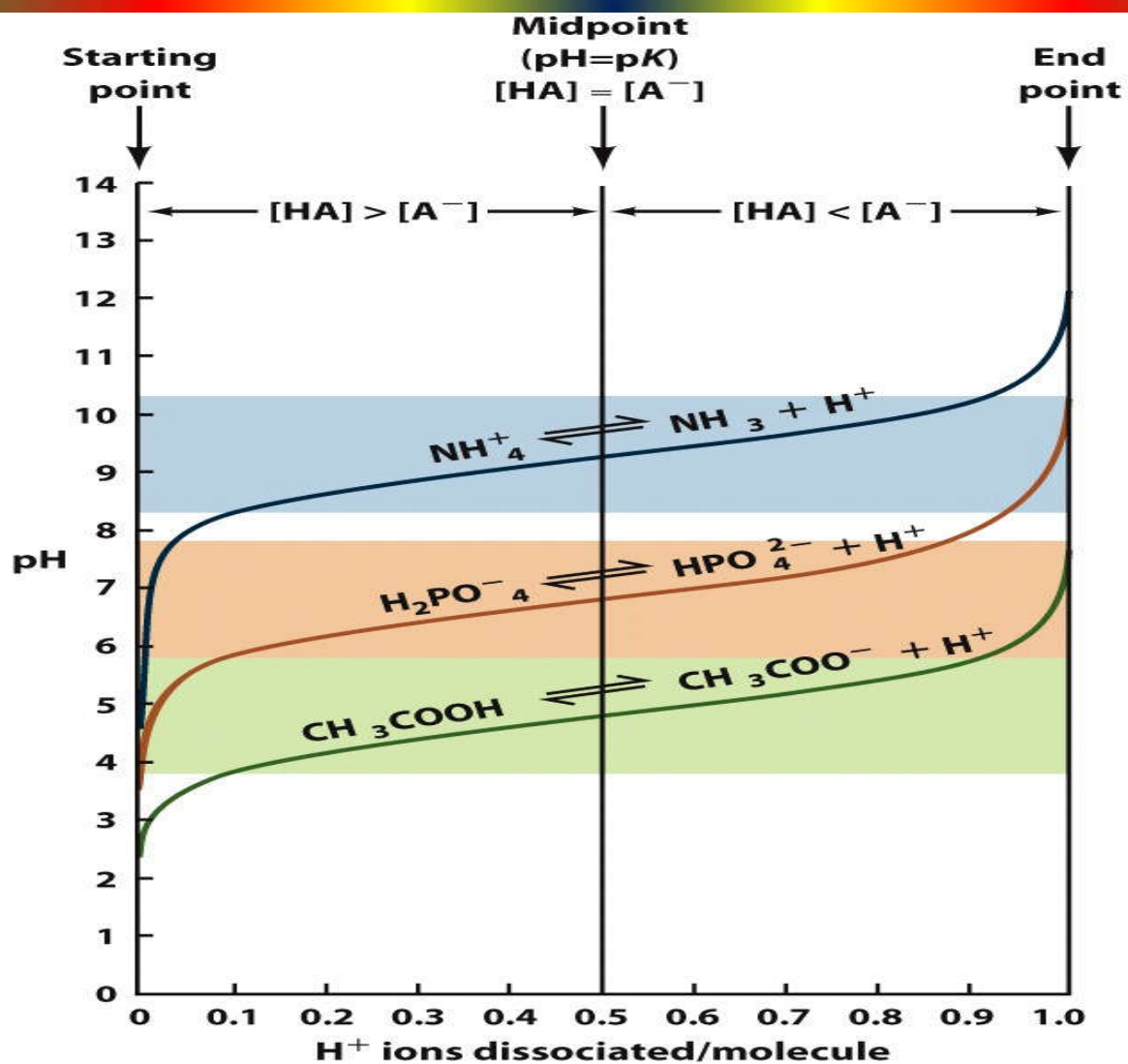
Solving Problems

- ① Calculate the pK_a of lactic acid, given that when the concentration of lactic acid is 0.010 M and the concentration of lactate is 0.087 M, the pH is 4.80.
- ② Calculate the pH of a mixture of 0.10 M acetic acid and 0.20 M sodium acetate. The pK_a of acetic acid is 4.76.
- ③ Calculate the ratio of the concentrations of acetate and acetic acid required in a buffer system of pH 5.30. The pK_a of acetic acid is 4.76

Buffer

- ✓ Buffer is solutions that resist changes in pH when acids and bases are added
- ✓ Most buffers consist of a weak acid and its conjugate base
- ✓ Effectiveness of a buffer is determined by:
 1. pK of buffering system
 2. pH of environment in which it is placed
- ✓ The buffering capacity is dependent on the concentration of the buffering agent.
- ✓ Buffering capacity is ± 1 pH unit from the pK
- ✓ Different compounds have different buffering ranges

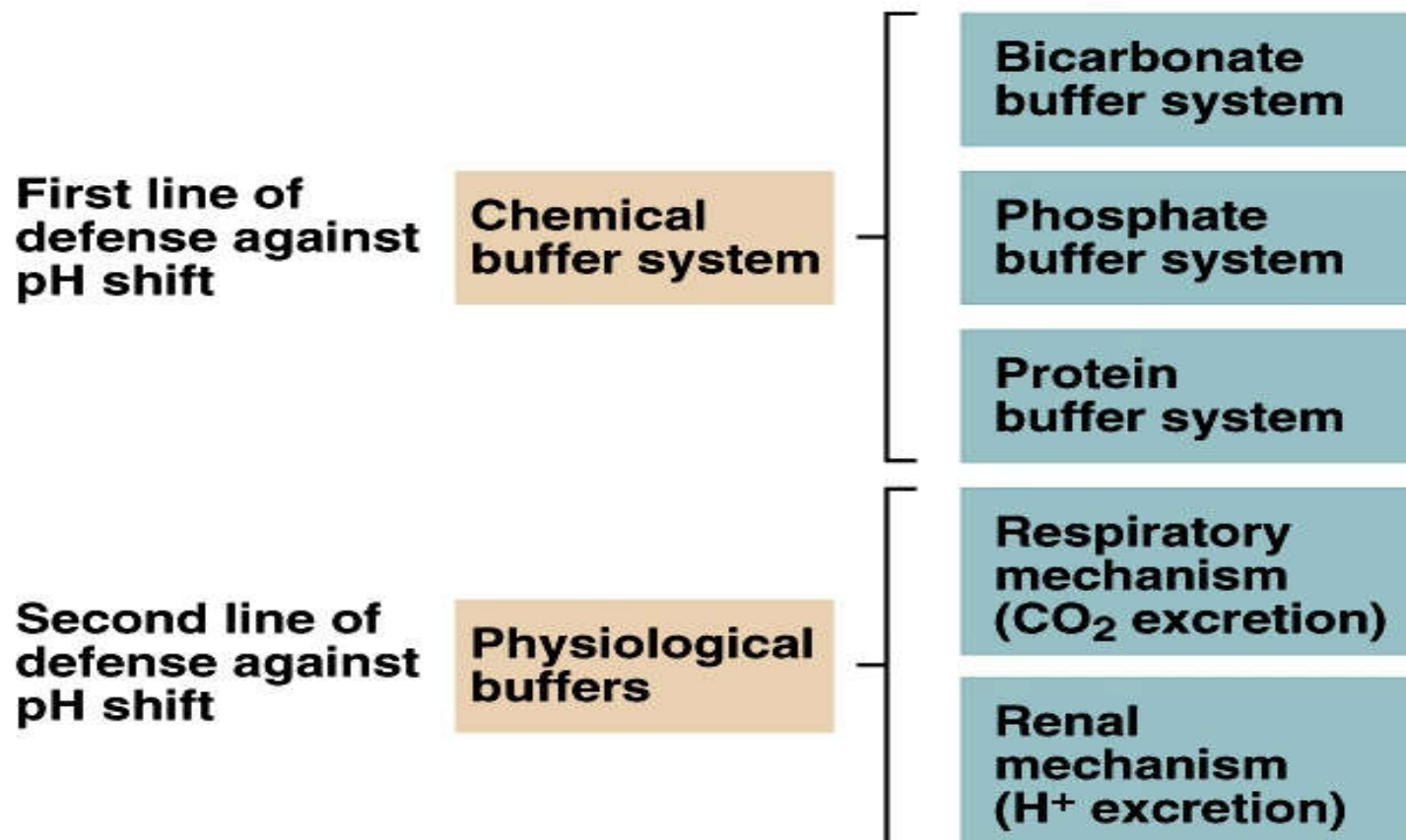
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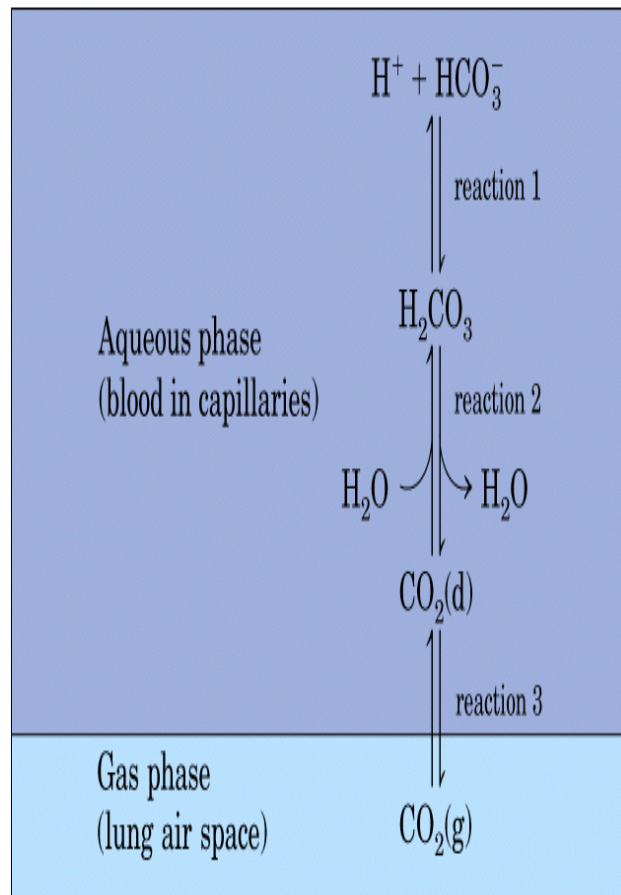
- ✓ Controlled by systems which maintain H^+ levels: Buffering Systems, Ventilation Rates, & Renal Function



Biocarbonate Buffer

- ✓ The most important buffer of the plasma is the bicarbonate/carbonic acid buffer system
- ✓ The ratio of base to acid ($\text{HCO}_3^-/\text{H}_2\text{CO}_3$) is nearly 20/1 in plasma under physiological conditions
- ✓ This buffer system is more complex than others, because carbonic acid (H_2CO_3) is formed from dissolved CO_2 which produced in tissues and diffused to plasma.

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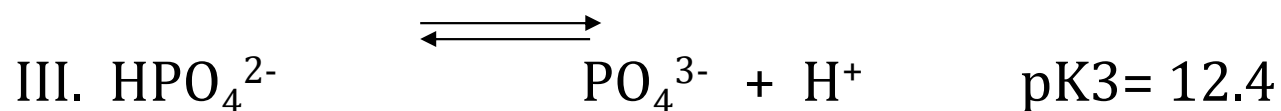


- ✓ This reaction is slow in plasma but in erythrocytes, **carbonic anhydrase** increases the rate of this reaction.
- ✓ When hydrogen ion concentration increases in plasma, HCO_3^- ions bind H^+ forming H_2CO_3 .
- ✓ H_2CO_3 is converted to $\text{CO}_2 + \text{H}_2\text{O}$
- ✓ CO_2 is released to atmosphere by lungs
- ✓ The normal values at $\text{pH} = 7.40$; $[\text{HCO}_3^-] = 24 \text{ mM}$; $[\text{CO}_2] = 1.2 \text{ mM}$.
- ✓ CO_2 balance is controlled by the lungs, HCO_3^- by the kidneys.

The Phosphate Buffer System

✓ Phosphate buffer system is most effective in intracellular medium, especially in kidneys.

✓ Phosphoric acid has 3 ionization steps:



✓ Among the 3 ionization steps, $\text{H}_2\text{PO}_4^- / \text{HPO}_4^{2-}$ is a good buffer because of its pKa value (6.8) which is close to physiological pH (7.4).

✓ $\text{HPO}_4^{2-} / \text{H}_2\text{PO}_4^- = 4$ at the pH (7.4).

- ➡ Phosphate buffer system is not effective in plasma, because phosphate ion concentrations are low.
- ➡ However it is important in the excretion of acids in the urine.
- ➡ H^+ secreted into the tubular lumen by the $\text{Na}^+ - \text{K}^+$ exchanger react with HPO_4^{2-} to form H_2PO_4^- .
- ➡ Some organic phosphates (2,3 diphosphoglycerate in erythrocytes) has also buffering capacity.

The Protein Buffer Systems

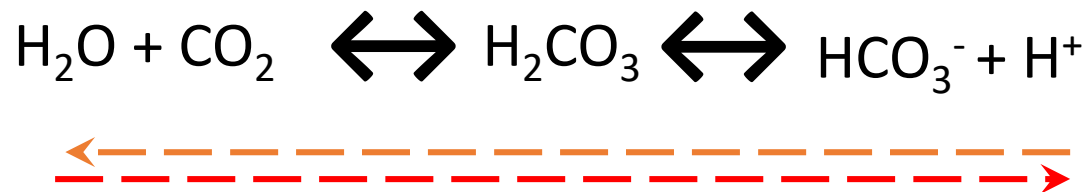
- ✓ In proteins, ionizable R groups (COOH groups of aspartate and glutamate, NH₂ groups of lysine and histidine) and N-terminale α -NH₂ groups of some amino acids are responsible for buffering.
- ✓ Buffering effect of proteins is low in plasma
- ✓ Proteins are much more effective buffers in intracellular medium.
- ✓ Proteins, especially albumin, account for the 95% of the non-bicarbonate buffer value of the plasma.
- ✓ The most important buffer groups of proteins at physiological pH range are the imidazole groups of Histidine which has a pKa value of 6.5
- ✓ Each albumin molecule contains 16 Histidine

The Hemoglobin Buffer Systems

- ✓ Hemoglobin (Hb) is a protein which carries O_2 to tissues and CO_2 from tissues to lungs and is an effective buffer.
- ✓ The most important buffer groups of Hb are histidines.
- ✓ Each globin chain contains 9 histidine.
- ✓ 95% of CO_2 which is released from tissues to plasma is diffused into erythrocytes.
- ✓ In erythrocytes, carbonic anhydrase constitutes H_2CO_3 from CO_2 and H_2O and then HCO_3^- and H^+ are released by the ionization of H_2CO_3 .
- ✓ Released protons take part in the formation of salt bridges between globin chains of Hb, and lead the change in the conformation of Hb molecule in tissue capillaries.

Lung & Effect on pH Balance

- ✓ It's all about CO₂ and the bicarbonate buffering system.
- ✓ Increased ventilation rate causes removal of CO₂ and H₂O.

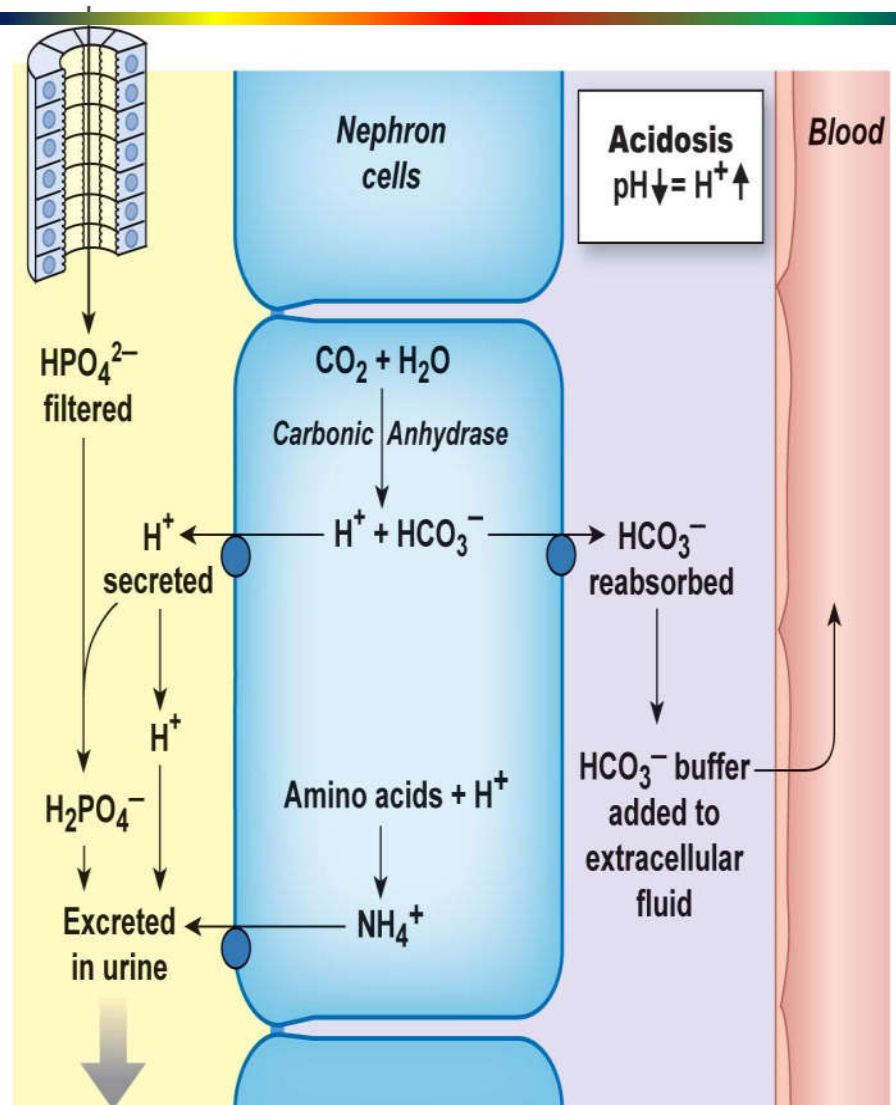


- ✓ Hyperventilation drives the reaction to the left causing removal of H⁺, pH goes up
- ✓ Hypoventilation drives the reaction to the right, causing additional H⁺, pH goes down

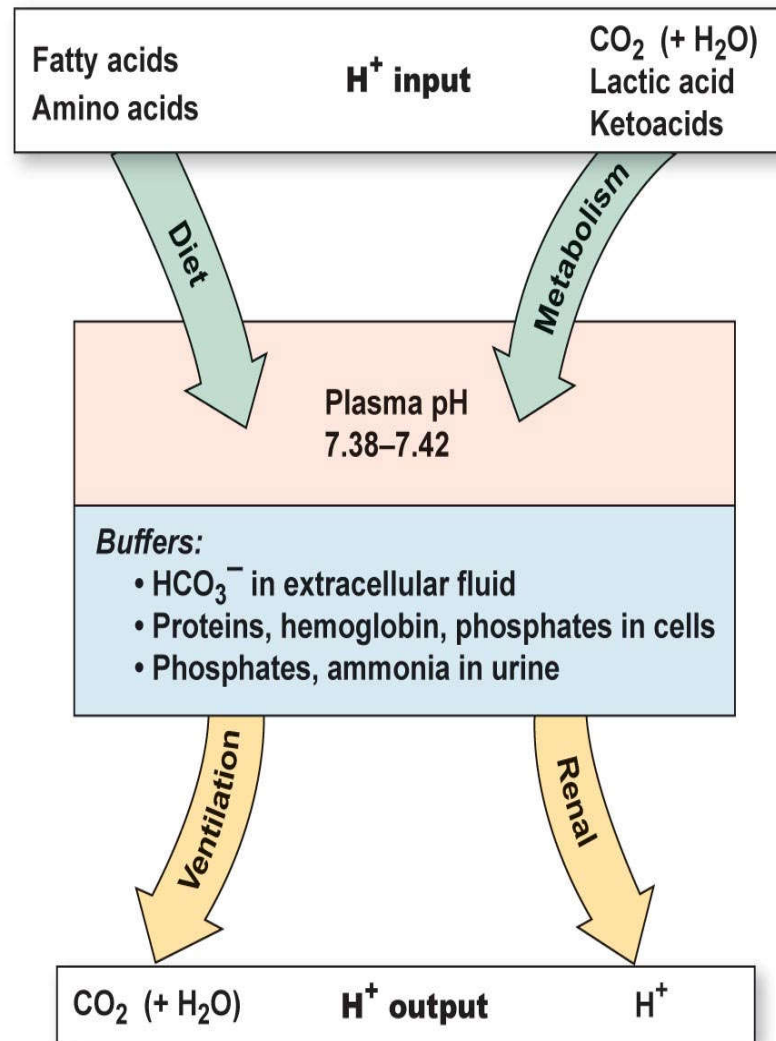
Renal Function

Through the aspects of tubular secretion and reabsorption:

- ✓ Bicarbonate (HCO_3^-) is produced and reabsorbed, acting as a buffer, stabilizing pH
- ✓ H^+ is capable of being secreted and excreted, reducing its concentration and causing pH to go up.
- ✓ Production of ammonia and excretion of NH_4^+



Overview of H^+ inputs and outputs as well as controlling mechanisms



Disorders of Acid and Base Balance

Acidosis

- Occurs when the blood pH is below 7.38
- Causes may be respiratory or metabolic (kidney):

1. **Respiratory acidosis:**

- ✓ Characterized by elevation of PCO_2 levels in arterial blood causes the pH to drop due to decreased movement of CO_2 from lungs to the air
- ✓ **Cause:** emphysema, pulmonary edema, medullary injury, airway obstruction, disorders of the muscle
- ✓ **Effects:** Kidneys increase secretion of H^+ ions and absorption of HCO_3^- ions (metabolic compensation)
- ✓ **Treatments:** increase exhalation of CO_2 , IV of HCO_3^- - artificial respiration, suction, and ventilation therapy.

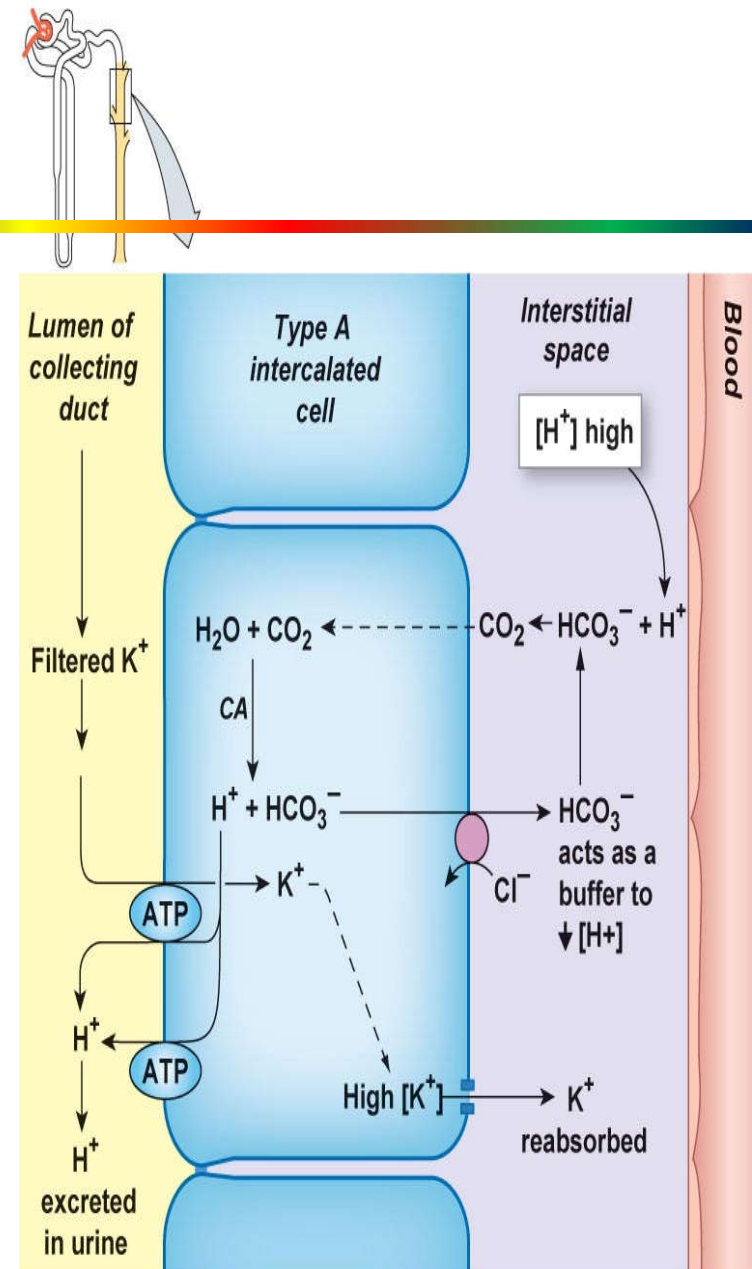
2. Metabolic Acidosis:

- ✓ blood concentration of HCO_3^- drops ($\downarrow$ 22mEq/L)
- ✓ because it is a buffer, the blood will lose its ability to pick up H^+ and will become more acidic (lower pH).

Causes

- Loss of HCO_3^- (diarrhea or renal failure)
- Accumulation of acid (e.g. ketosis)
- Kidney's failure to excrete H^+ from metabolism of dietary protein

Effects: Causes hyperventilation (respiratory compensation)



Alkalosis

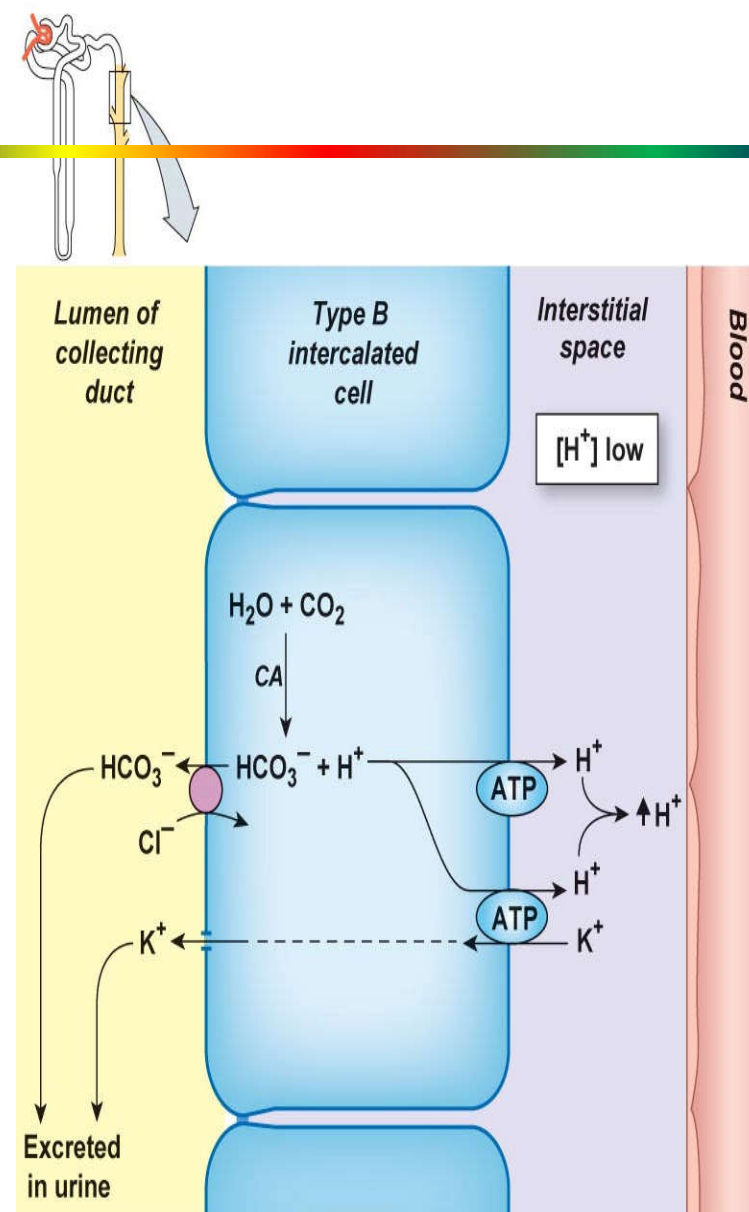
- ✓ occurs when the blood pH is above 7.45
- ✓ *Causes may be respiratory or metabolic (kidney):*

1. Respiratory alkalosis:

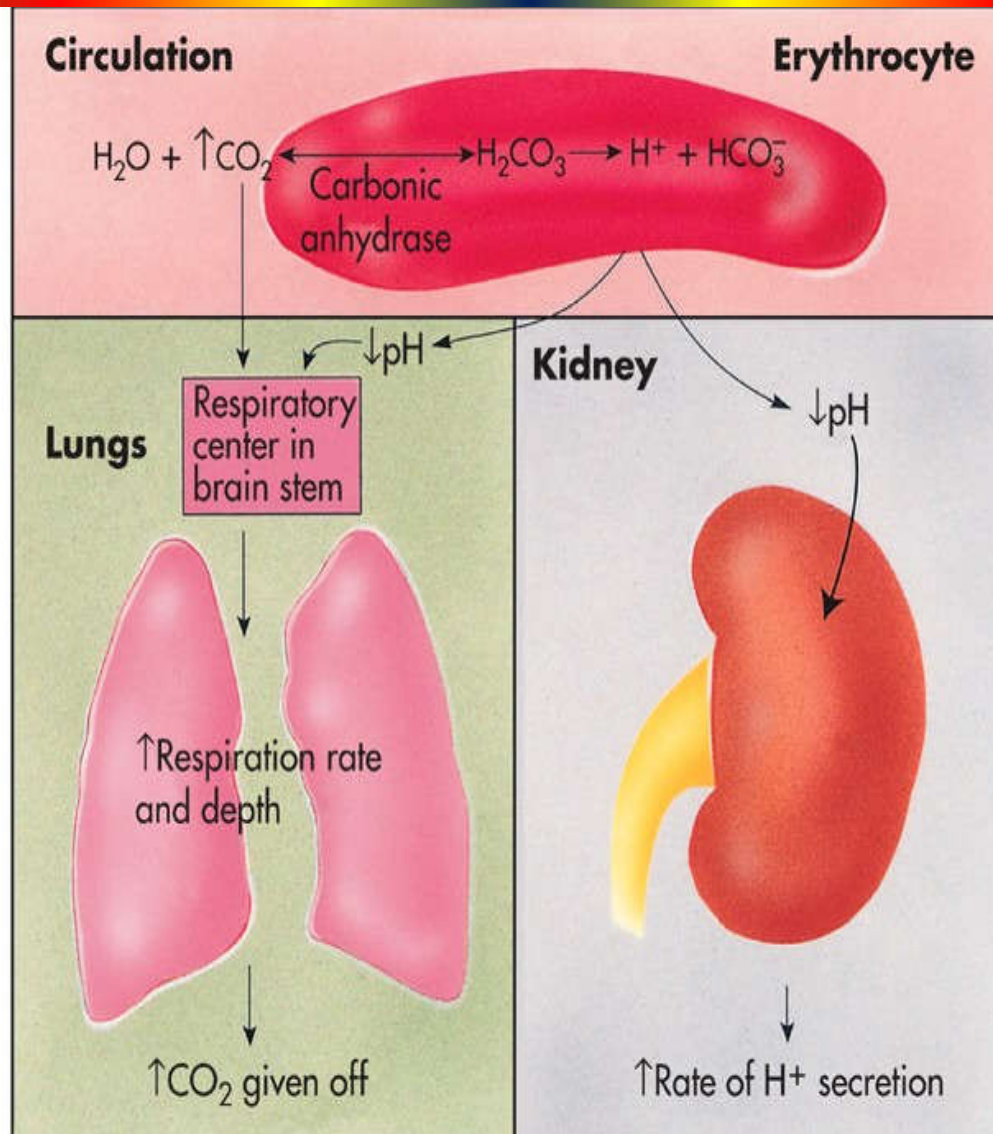
- ✓ *Characterized by decreased PCO_2 levels in arterial blood causes the pH to rise due to increased movement of CO_2 from lungs to the air*
- ✓ **Cause:** Hyperventilation due to voluntary behavior, oxygen deficiency at high altitudes, pulmonary dysfunction, stroke, anxiety
- ✓ **Effects:** Kidneys stop excreting H^+ , and reabsorbing bicarbonate creates a metabolic compensation
- ✓ **Treatments:** Re-breathe CO_2 , treat underlying behavior, reduce altitude.

2. Metabolic alkalosis

- ✓ blood HCO_3^- concentration increases ($\uparrow 26 \text{ mEq/L}$).
- ✓ Cause
 - due to loss of acid not related to respirations e.g. vomiting – most common, diuretics, endocrine problems
 - Overconsumption of antacids and severe dehydration
- ✓ Effects: causes hypoventilation



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Acid Base Disorders

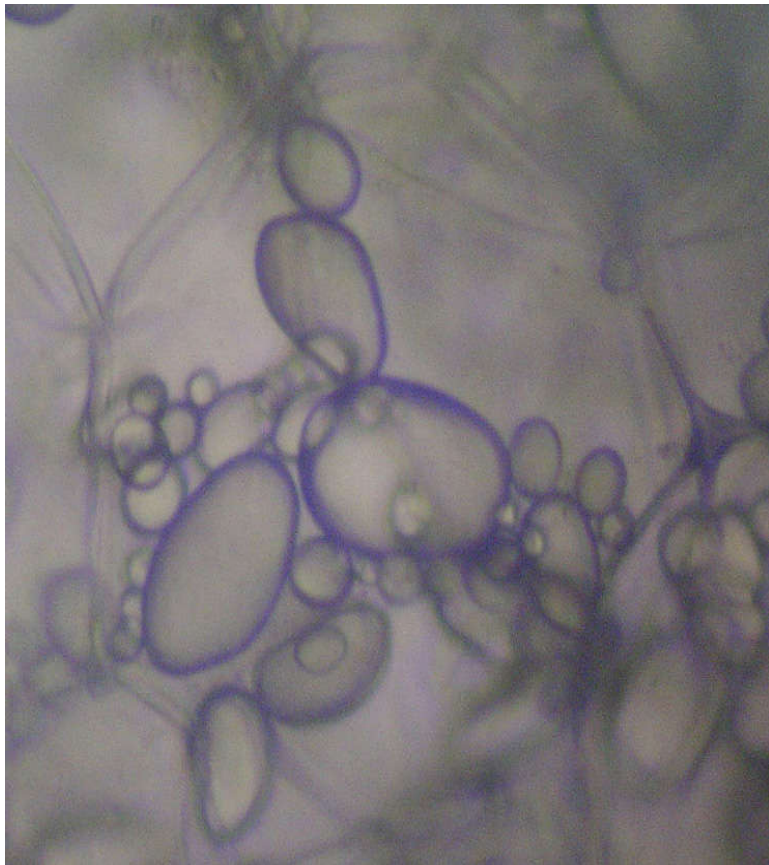
Disorder	pH	[H ⁺]	Primary disturbance	Secondary response
Metabolic acidosis	↓	↑	↓ [HCO ₃ ⁻]	↓ pCO ₂
Metabolic alkalosis	↑	↓	↑ [HCO ₃ ⁻]	↑ pCO ₂
Respiratory acidosis	↓	↑	↑ pCO ₂	↑ [HCO ₃ ⁻]
Respiratory alkalosis	↑	↓	↓ pCO ₂	↓ [HCO ₃ ⁻]



“Life is a struggle,
not against sin,
not against money power..
but against **Hydrogen ions**”

H.L. Mencken

Carbohydrates



Starch compartments in potato cells



Glycogen stored in liver cells (red)



Cellulose in plant cell walls (outer blue layer)

Introduction to Carbohydrate

- ➡ Carbohydrates are the most abundant macromolecules in nature.
- ➡ The general molecular formula of carbohydrate is $(CH_2O)_n$ where $n > 3$.
- ➡ Chemically, they contain the elements C, H and O.
- ➡ Carbohydrates are polyhydroxy aldehydes (aldose) or ketones (ketose) or compounds which give these substances on hydrolysis.

Medical and biological Function of Carbohydrate

- ✓ Variety of important functions in living systems:
 - ➔ Nutritional (energy storage, source, metabolic intermediates)
 - ➔ Structural component of cell membrane and nerve tissue
 - ➔ Components of skin, connective tissue, tendons, cartilage, bone and fiber.
 - ➔ Components of nucleic acids and blood group substances
 - ➔ Derivative of carbohydrates are drugs. For example, a glycoside ouabain is used in clinical medicine. Streptomycin an antibiotic is a glycoside.

- ➡ Aminosugars, derivatives of carbohydrates are components of antibiotics like erythromycin and carbomycin.
- ➡ Ascorbic acid, a derivative of carbohydrate is a water-soluble vitamin.
- ➡ Bacterial invasion involves hydrolysis of mucopolysaccharides.
- ➡ Survival of Antarctic fish in icy environment is due to presence of anti-freeze **glycoproteins** in their blood.
- ➡ Informational (molecular recognition, cell-cell communication)
- ➡ Osmotic pressure regulation

Classification and Structure of Carbohydrate

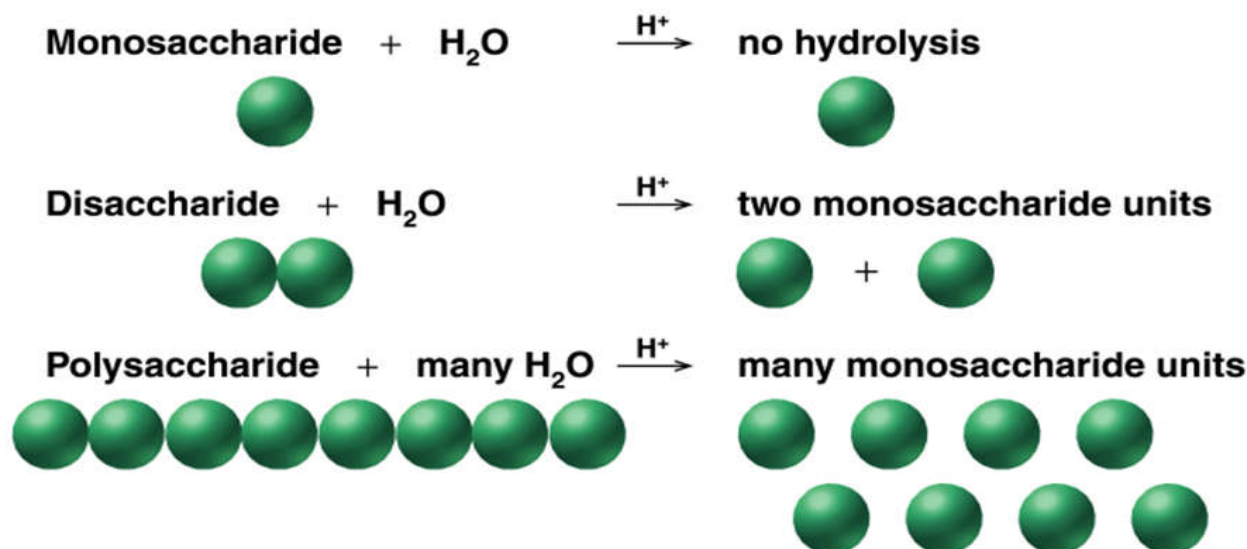
✓ There are four major classes of carbohydrates based on number of carbon chains present

I. Monosaccharides (simple sugar)

II. Disaccharides (2- monosaccharide units)

III. Oligosaccharides (3-10 monosaccharide units).

IV. Polysaccharides (>10 monosaccharide units).



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Monosaccharides

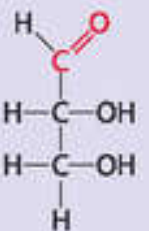
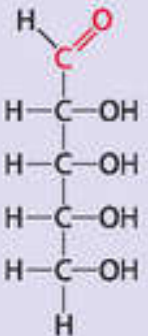
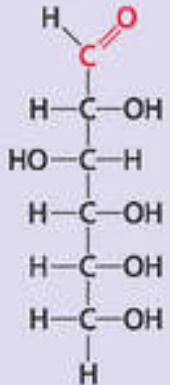
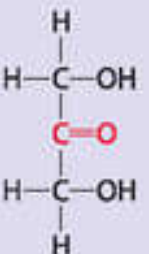
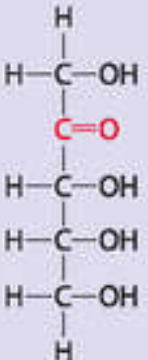
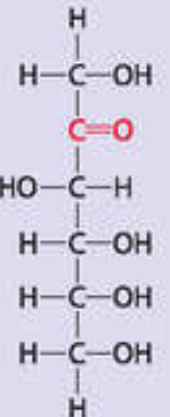
- They are also called simple sugars (one sugar unit).
- can not be hydrolyzed to small compounds
- They contain three to nine carbon atom.
- usually colourless, water-soluble and crystalline solids.
- Include D-glucose, fructose, ribose and deoxyribose.
- 2 types of monosaccharide structures: Aldoses and ketoses

<u>Generic names</u>	<u>Examples</u>
3 Carbons: trioses	Glyceraldehyde
4 Carbons: tetroses	Erythrose
5 Carbons: pentoses	Ribose
6 Carbons: hexoses	Glucose
7 Carbons: heptoses	Sedoheptulose
9 Carbons: nonoses	Neuraminic acid

Monosaccharides

➡ **Aldoses** (e.g., glucose)
have an **aldehyde** group
at one end.

➡ **Ketoses** (e.g., fructose)
have a **keto** group,
usually at C2.

	Triose sugars	Pentose sugars	Hexose sugars
ALDOSES	 Glyceraldehyde	 Ribose	 Glucose
KETOSES	 Dihydroxyacetone	 Ribulose	 Fructose

Isomers and Epimers

- ✓ **Isomers** are molecules that have the same molecular formula, but have a different arrangement of the atoms in space.
- ✓ Examples: Glucose, Fructose, Galactose, Mannose are isomers having the same chemical formula of $C_6H_{12}O_6$

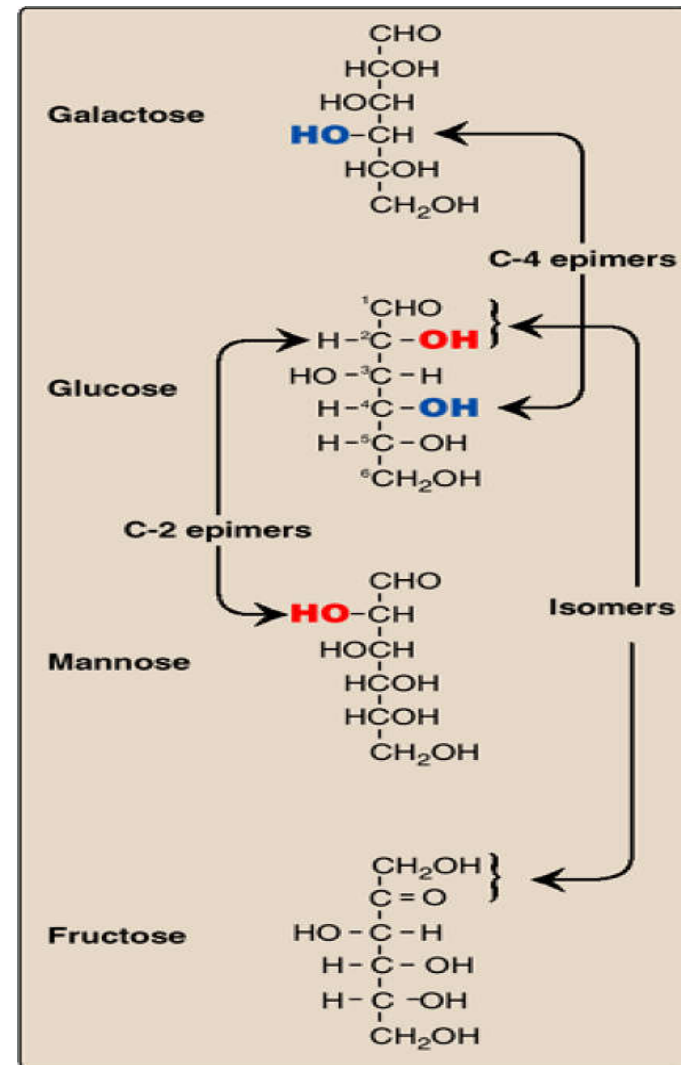
Epimers

✓ **Epimers** - stereoisomers that differ only in configuration about one chiral center.

✓ Examples of epimers :

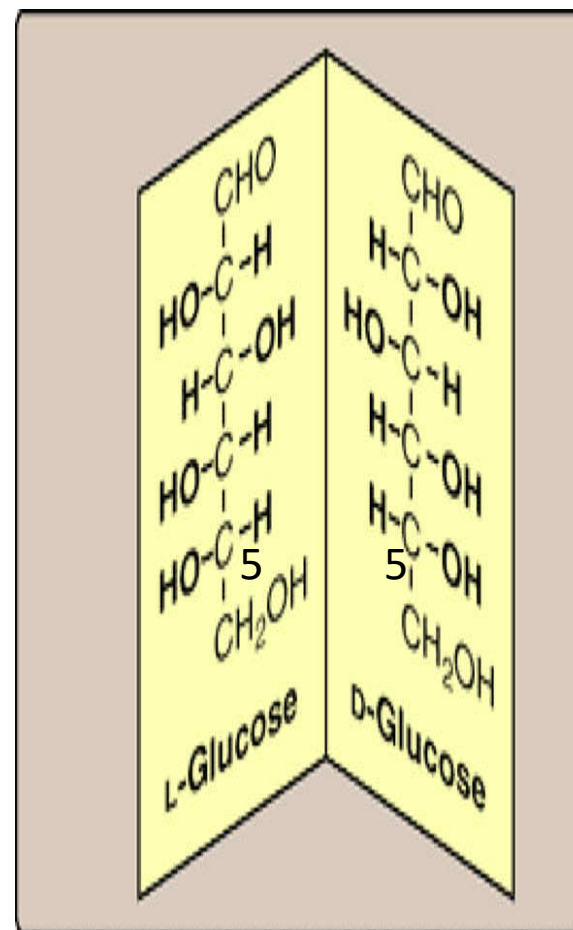
❖ D-glucose & D-mannose
(epimeric at C2)

❖ D-glucose & D-galactose
(epimeric at C4)



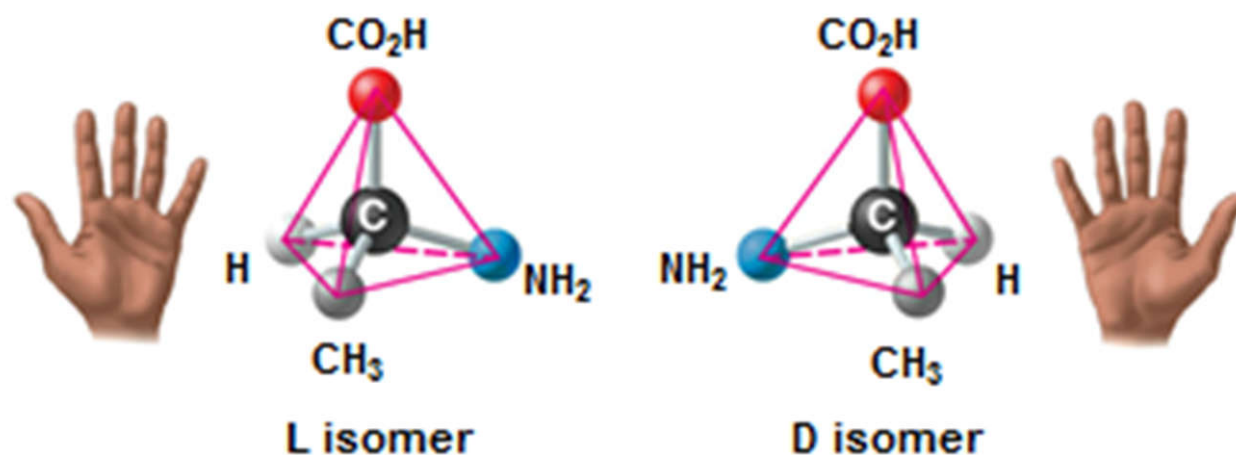
D and L Stereoisomers

- The two members of the pair are designated as D and L forms.
- In D form the OH group at C₅ on the asymmetric carbon is on the right.
- In L form the OH group at C₅ is on the left side.
- D-glucose and L-glucose are enantiomers:
- Non-superimposable *complete* mirror image (differ in configuration at every chiral center).
- Most sugars have D-configuration. E.g. D-glucose



- ➡ Enantiomers are important in the pharmaceutical industry
- ➡ Two enantiomers of a drug may have different effects
- ➡ Usually only one isomer is biologically active
- ➡ Differing effects of enantiomers demonstrate that organisms are sensitive to even subtle variations in molecules

(c) Enantiomers



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Note the Mirror Imaging

Drug	Condition	Effective Enantiomer	Ineffective Enantiomer
Ibuprofen	Pain; inflammation	 <i>S</i> -Ibuprofen	 <i>R</i> -Ibuprofen
Albuterol	Asthma	 <i>R</i> -Albuterol	 <i>S</i> -Albuterol

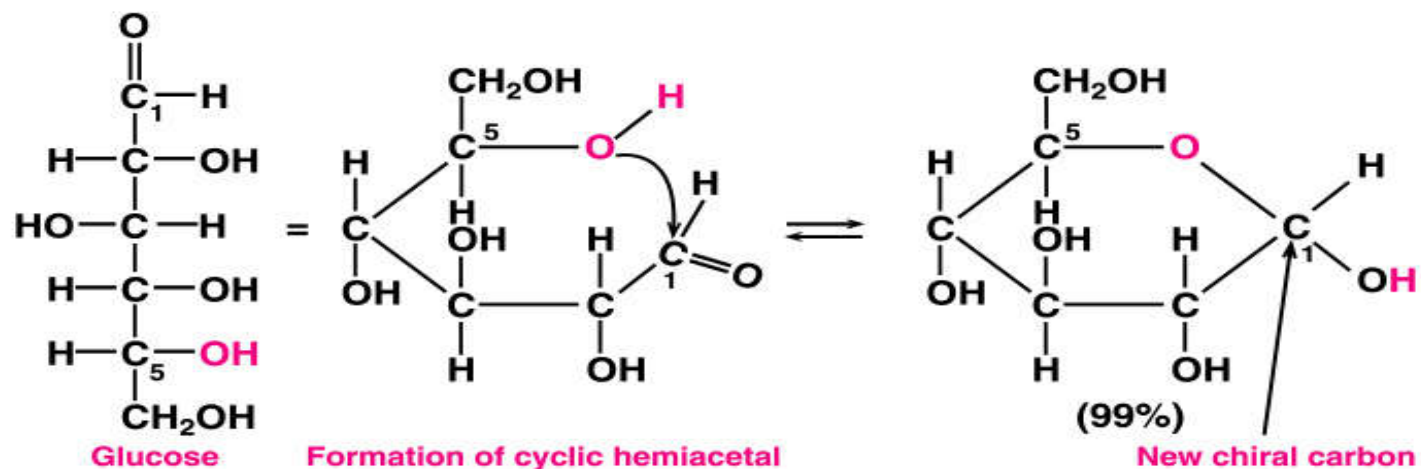
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➡ The symbol R comes from the Latin rectus for right, and S from the Latin sinister for left

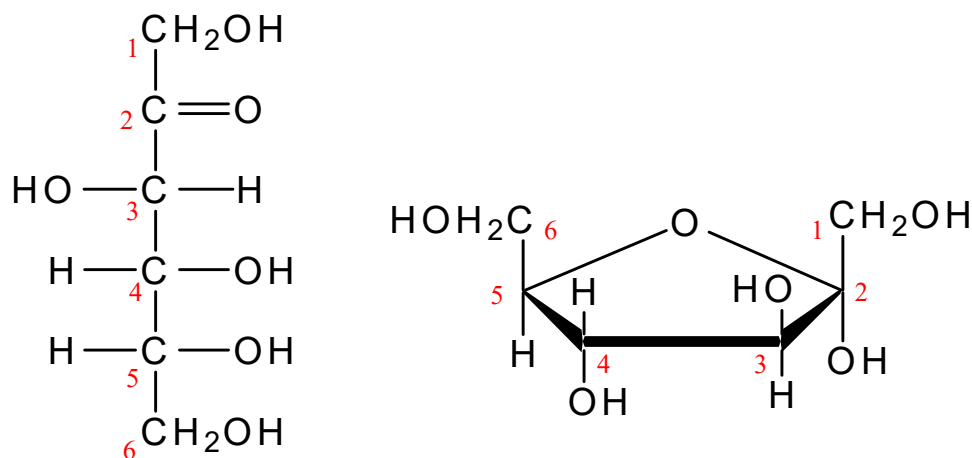
Cyclization of Monosaccharides

- ➡ Less than 1% of CHO exist in an open chain form.
- ➡ But, they predominantly found in **ring form**.
- ➡ Involving reaction of C-5 OH group with the C-1 aldehyde group or C-2 of keto group.
- ➡ Six membered ring structures are called **Pyranoses**.
- ➡ Five membered ring structures are called **Furanoses**.

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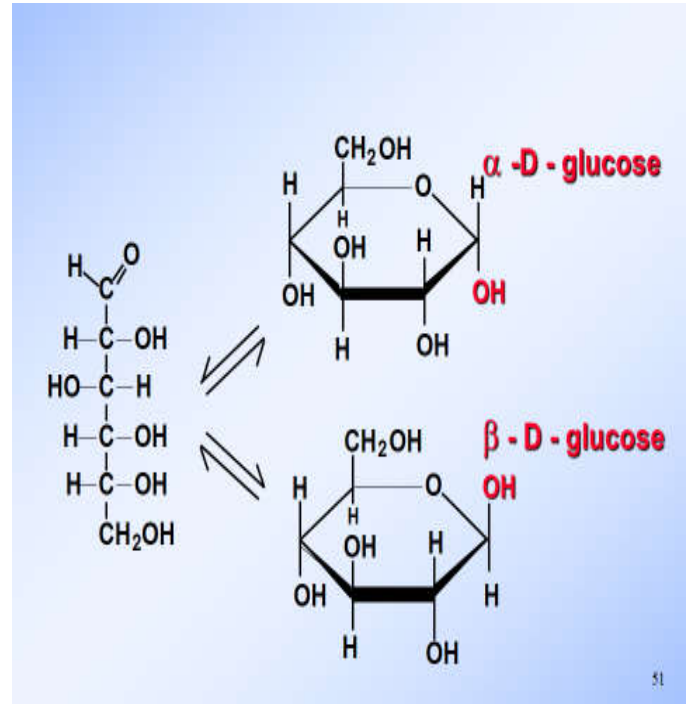


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Anomeric carbon

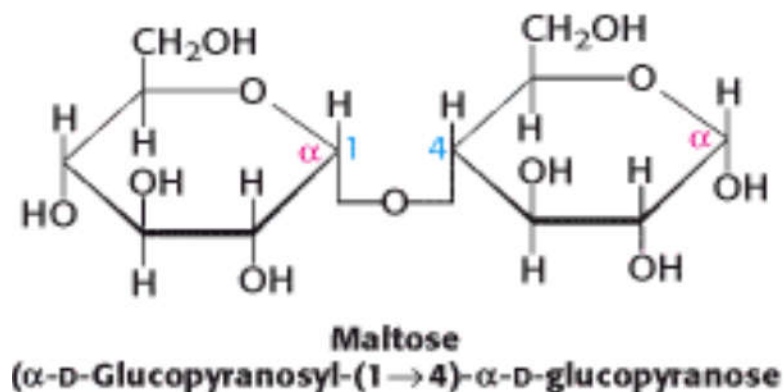
- ✓ The carbonyl carbon after cyclization becomes the anomeric carbon.
- ✓ This creates α and β configuration.
- ✓ α - OH group is down compared to CH_2OH (trans).
- ✓ β - OH group is up compared to CH_2OH (cis).
- ✓ The α and β forms are in equilibrium so one form can convert to the other - **mutarotation**.



Reducing Sugar

- ✓ Sugars in which the oxygen of the anomeric carbon is free and not attached to any other structure, such sugars can act as reducing agents and are called reducing sugars.

E.g. lactose and maltose



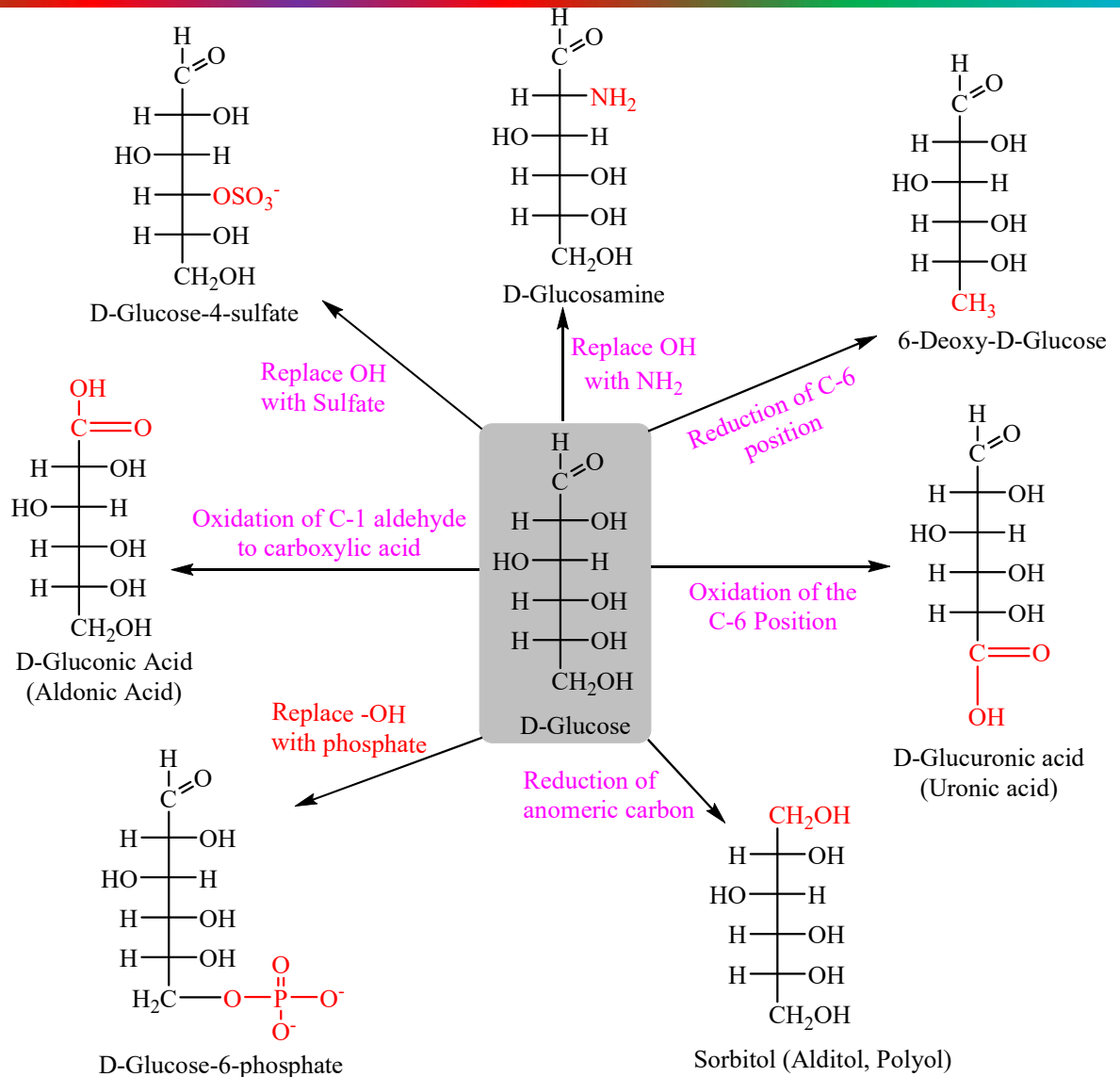
- ➡ In lab many tests are employed to identify the reducing action of sugars. These include Benedict's test, Fehlings test.

Source of Monosaccharides

- ➔ Glucose is found in several fruit juice blood of humans and in honey.
- ➔ Galactose is a part of lactose.
- ➔ Fructose is found in several fruit juices and honey.
- ➔ Commonly glucose is referred as dextrose.
- ➔ All monosaccharides containing free aldehyde or keto group reduces ions like Cu^{2+} under alkaline conditions.

Derivatives of Glucose

➡ Includes sugar acids, sugar alcohols, deoxy sugars, sugar phosphates and amino sugars.



Sugar acids

- ➡ Oxidation of sugars produces aldonic acids. E.g, oxidation of glucose produces glucuronic acid or uronic acid.
- ➡ Uronic acids are components of mucopolysaccharides and required for detoxification.
- ➡ Vitamin C or ascorbic acid is also sugar acid

Sugar alcohols

- ➡ Reduction of aldose and keto groups of sugar produces polyhydroxy alcohols or polyols.
- ➡ The alcohols formed from glucose, galactose and fructose are sorbitol, galactitol and sorbitol, respectively.

Deoxy sugars

☞ Those sugars in which oxygen of a hydroxyl group is removed leaving hydrogen.

Examples:

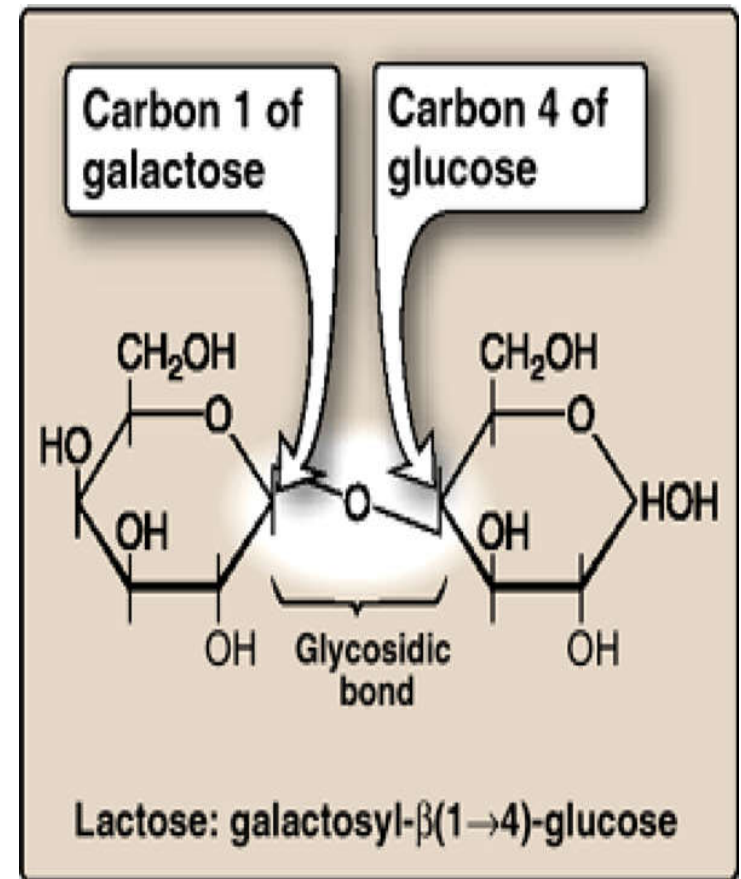
- ◆ Deoxyribose -present in nucleic acids.
- ◆ Fucose is another deoxy sugar present in blood group substances.

Amino sugars

- ☞ Those sugars in which an amino group is substituted for a hydroxyl group.
- ☞ D-glucosamine is an example for an amino sugar.
- ☞ Amino sugars are components of mucopolysaccharides, and antibiotics.

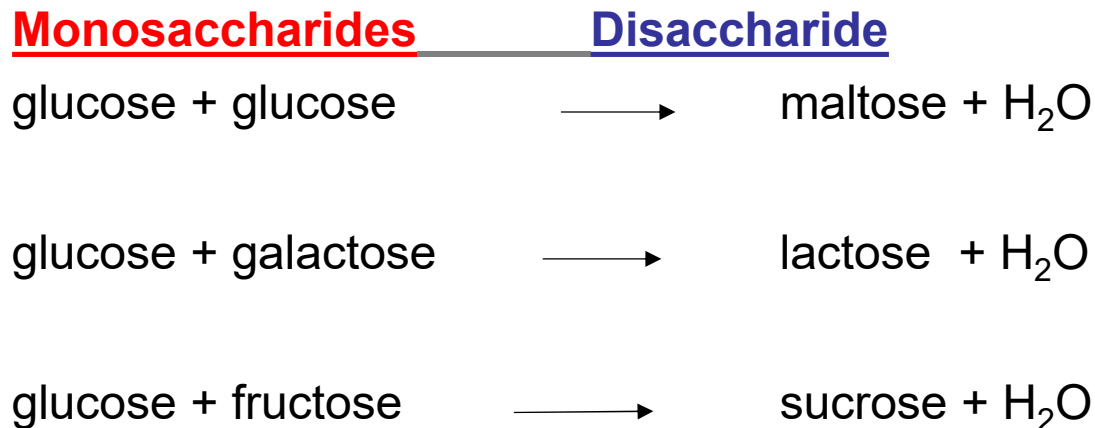
Joining of Monosaccharides

- ➔ Monosaccharides can be joined to form disaccharides, oligosaccharides, and polysaccharides.
- ➔ The bonds that link sugars are called *glycosidic bonds*.
- ➔ Glycosidic bond is formed when the hydroxyl group on one of the sugars reacts with the anomeric carbon on the second sugar.
- ➔ These are formed by enzymes known as *glycosyltransferases*.



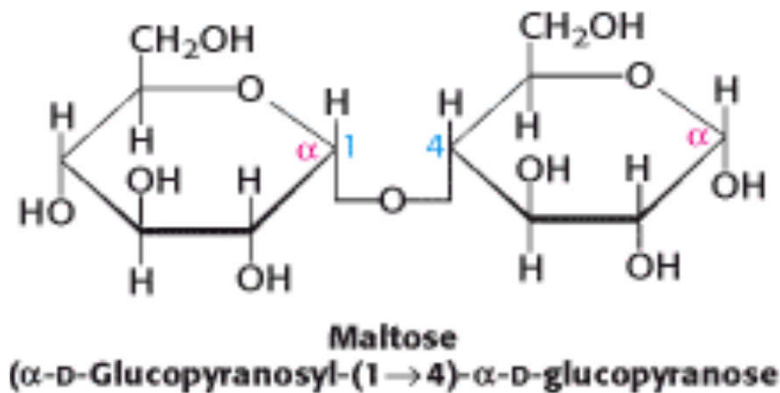
Disaccharides

- ✓ A disaccharide consists of two monosaccharides joined by glycosidic bond.
- ✓ Glycosidic bond is formed when the hydroxyl group on one of the sugars reacts with the anomeric carbon on the second sugar.
- ✓ Biologically important disaccharides are sucrose, maltose, and Lactose.



Maltose

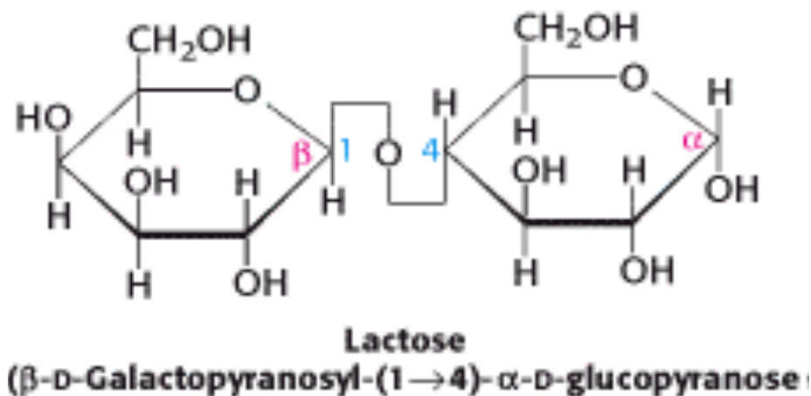
- ✓ Contains two D glucose residues joined by α -(1,4) glycosidic linkage.



- ✓ The major degradative product of Starch.
- ✓ It is hydrolyzed into two molecules of D- glucose by the intestinal enzyme maltase, which is specific for the α - (1, 4) glycosidic bond.

Lactose

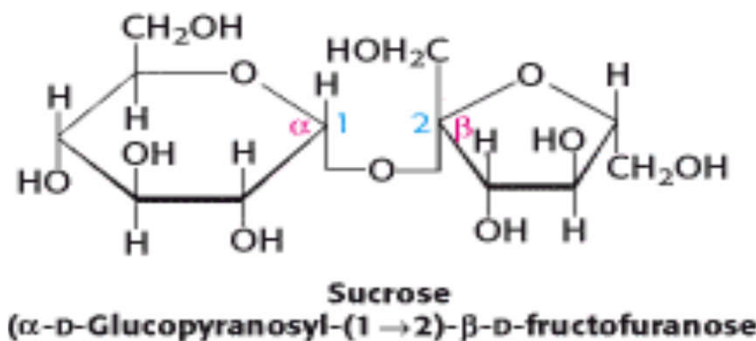
- ✓ Lactose is a disaccharide of β -D - galactose and β -D-glucose which are linked by β -(1,4) glycosidic linkage.



- ✓ It acts as a reducing substance since it has a free carbonyl group on the glucose.
- ✓ It is found exclusively in milk of mammals (Milk sugar).

Sucrose (Cane sugar)

- ✓ Sucrose is a disaccharide of α -D-glucose and β -D-fructose.
- ✓ Formed by α -(1,2) glycosidic linkage



- ✓ Since the anomeric carbons of both its component monosaccharide units are linked to each other, sucrose is non reducing sugar.

Other Disaccharides

Isomaltose

- ➡ It contains two glucose units and formed by $\alpha(1\rightarrow6)$ Glycosidic linkage.
- ➡ Isomaltose is present in glycogen, amylopectin and dextran.

Cellobiose

- ➡ It also contains two glucose units and joined in $\beta(1\rightarrow4)$ linkage.
- ➡ It is formed from cellulose.

Trehalose

- ➡ It contains two glucose units and formed by $\alpha(1\rightarrow1)$ glycosidic linkage.
- ➡ it is a non-reducing disaccharide.
- ➡ It is a major sugar of insect hemo lymph. In fungi it serve as reserve food material.

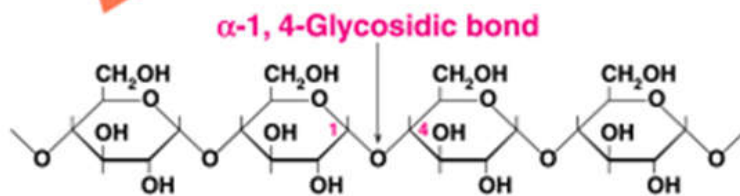
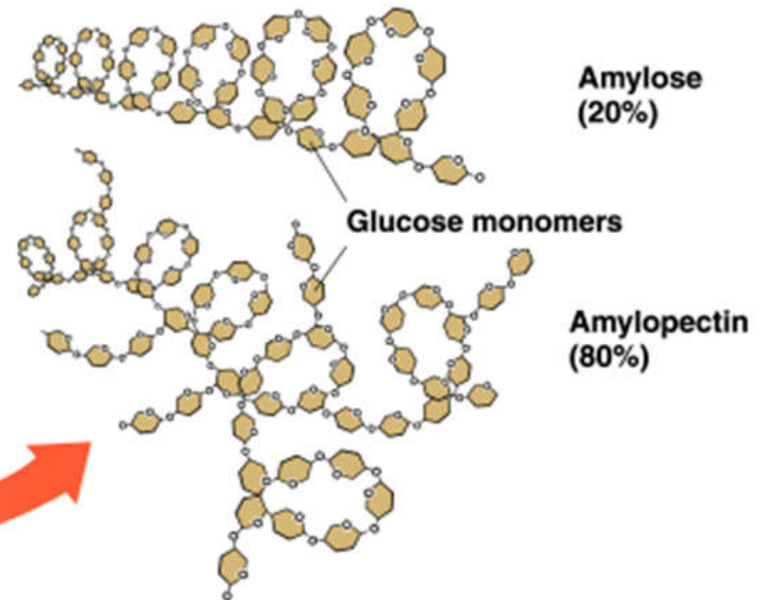
Polysaccharides

- ➔ Most of the carbohydrates found in nature occur in the form of high molecular polymers called polysaccharides.
- ➔ They are polymers of monosaccharides.
- ➔ They contain more than ten monosaccharide units.
- ➔ The monosaccharides are joined together by **glycosidic linkage**.
- ➔ There are two types of polysaccharides .
 1. **Homopolysaccharides** - contain only one type of monosaccharide building blocks. Example Starch, glycogen, Cellulose and dextrans.
 2. **Heteropolysaccharides** - contain two or more different kinds monosaccharide. E.g. GAGs, Proteoglycan, Glycoprotein

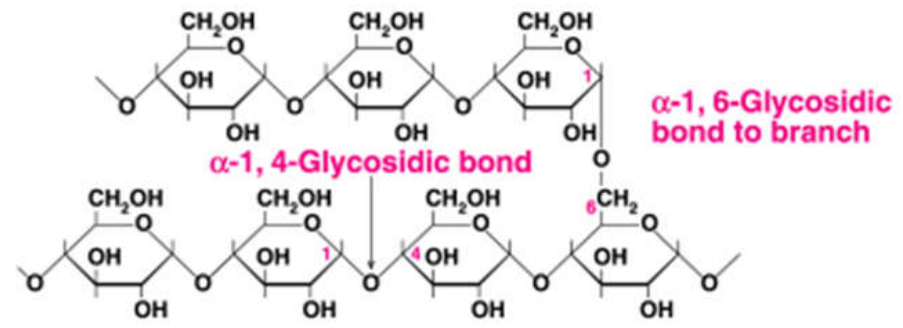
Starch

- ✓ Storage polysaccharide in plant cells.
- ✓ Starch consists of two polymeric units made of glucose called Amylose and Amylopectin but they differ in molecular architecture.
- ✓ **Amylose** is unbranched with 250 to 300 D-Glucose units linked by α -(1, 4) linkages
- ✓ **Amylopectin** consists of long branched glucose units with higher molecular weight.
- ✓ Glucose units in amylopectin are joined by α -(1,4) glycosidic linkage as in amylose , but the branch points of amylopectin are α - (1,6) linkages.
- ✓ The branch points repeat about every 20 to 30 (1-4) linkages.

Structures of Amylose and Amylopectin



(a) Unbranched chain of amylose

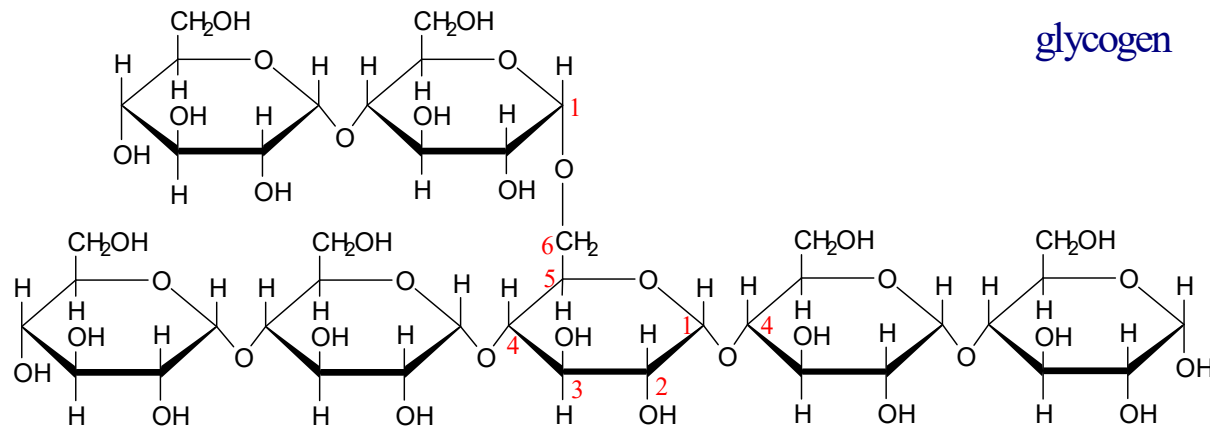


(b) Branched-chain of amylopectin

Timberlake, *General, Organic, and Biological Chemistry*. Copyright © Pearson Education Inc., publishing as Benjamin Cummings

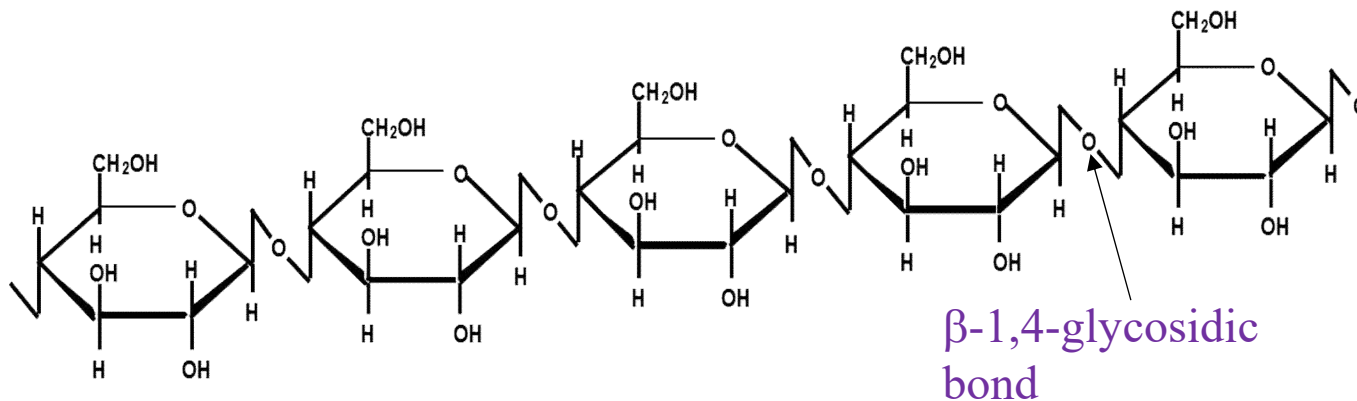
Glycogen

- ✓ The glucose storage device in animals
- ✓ Stored in liver and muscles as granules.
- ✓ Branched polymer of $\alpha(1-4)$ linked glucose residues with $\alpha(1-6)$ linked branches
- ✓ Similar to amylopectin but even more highly branched and more compact
- ✓ Branching increase H_2O -solubility



Cellulose

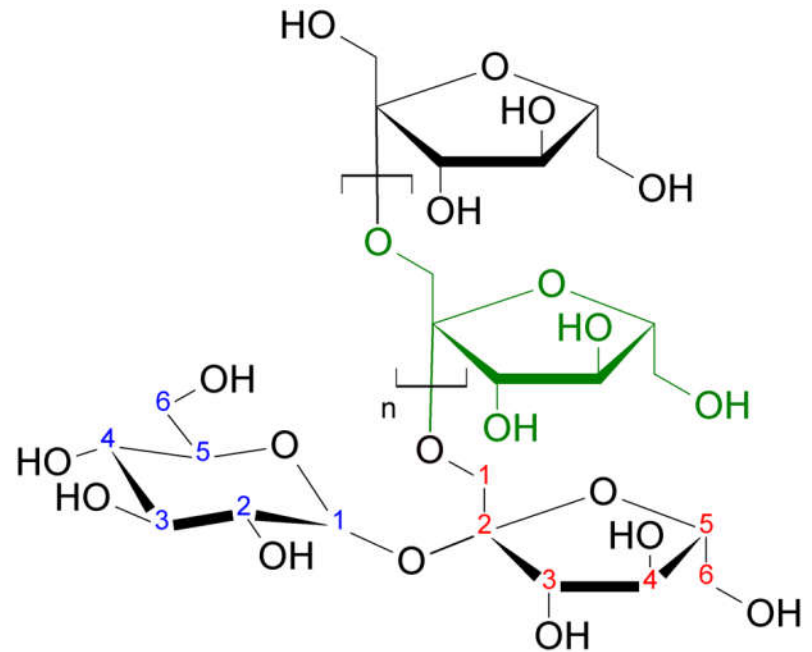
- ✓ The most abundant structural polysaccharide in plants.
- ✓ It is a linear unbranched homopolysaccharide of D- glucose units connected by β -(1, 4) glycosidic bonds.



- ✓ It cannot be digested by humans because humans cannot break down β -(1,4) glycosidic bonds due to lack of cellulase enzyme.

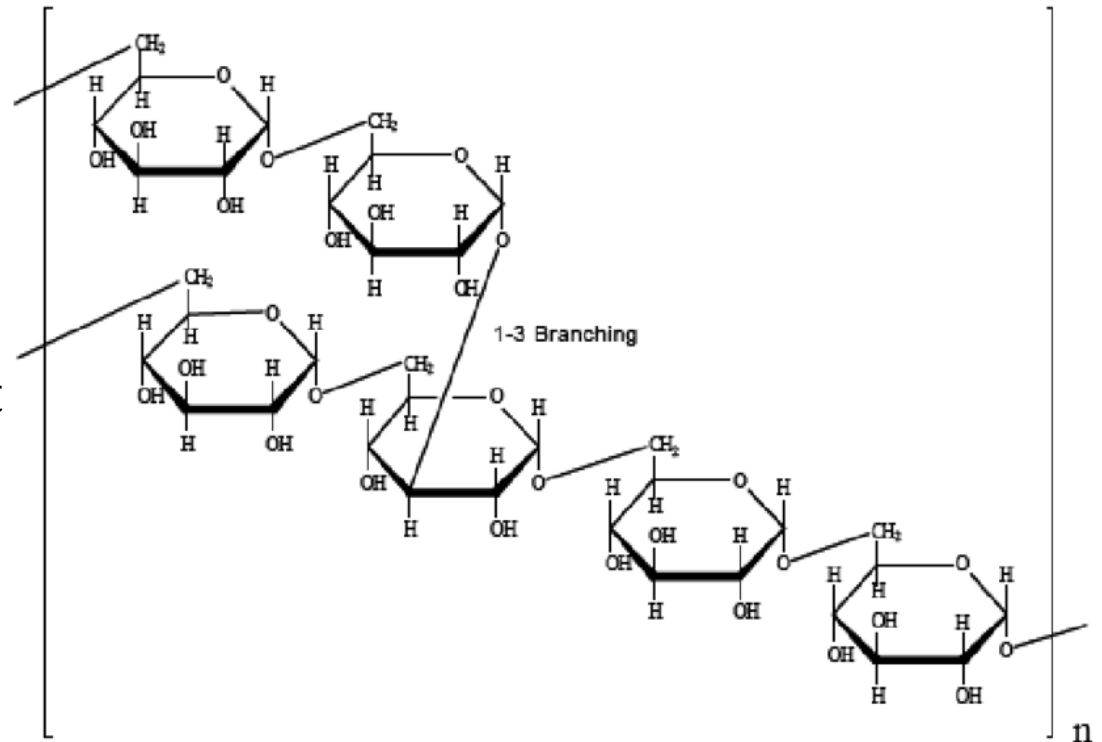
Inulin

- ✓ It is a polysaccharide composed of fructose.
- ✓ $\beta(1 \rightarrow 2)$ glycosidic linkage is present between fructose units.
- ✓ It is present in tubers of chicory, dhalia and in the bulb of onion and garlic.
- ✓ Inulin is used to determine glomerular filtration of kidney.



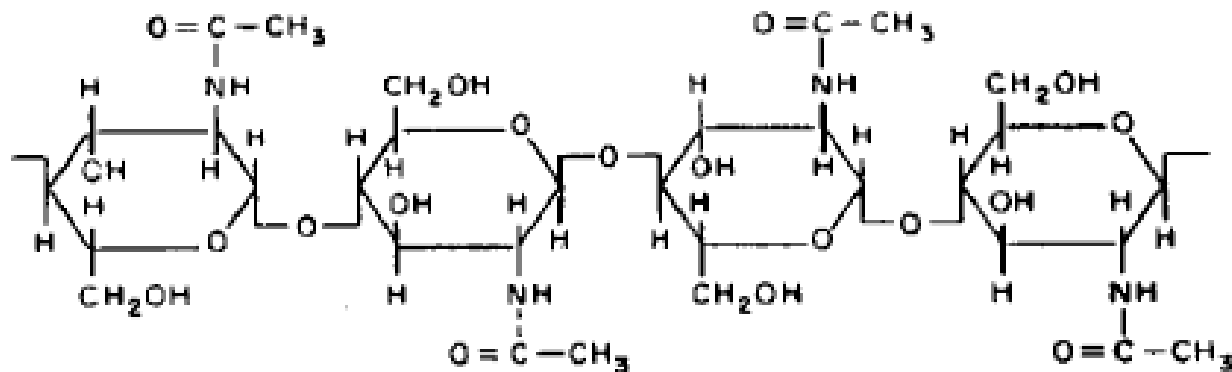
Dextran

- ➔ It has structure similar to amylopectin.
- ➔ In the linear part, glucose units are linked by $\alpha(1\rightarrow6)$ glycosidic bond and $\alpha(1\rightarrow3)$ glycosidic linkage is present between glucose unit at branch points.
- ➔ It is polysaccharide present in bacteria.
- ➔ Dental plaque is due to dextran synthesized from sucrose by oral bacteria.



Chitin

- ➡ Chitin is a structural homopolymer found in the exoskeletons of insects
- ➡ Chitin is a linear polymer similar to cellulose.
- ➡ It is made up of β -(1-4) linked N-acetylglucosamine (GlcNAc) residues rather than glucose residues.

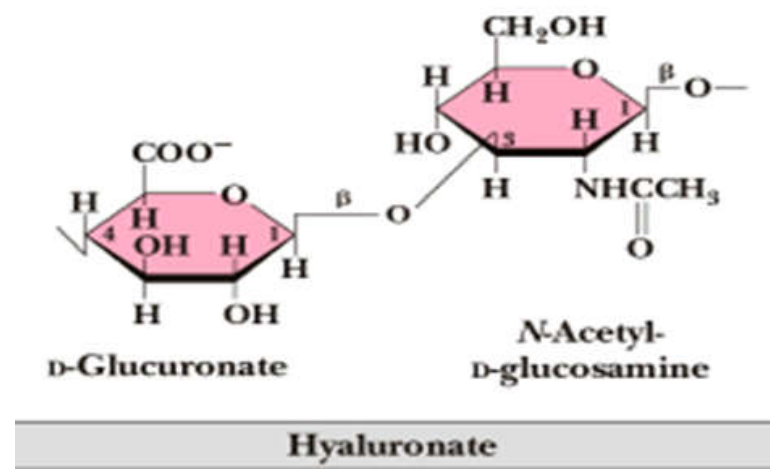


Heteropolysaccharide

- ➔ They are also called mucopolysaccharides or glycosaminoglycans
- ➔ These molecules contain a repeating disaccharide unit.
- ➔ The disaccharide units contain either of two modified sugars D-N-acetylgalactosamine (GalNAc) or D-N-acetylglucosamine (GlcNAc) and a uronic acid such as glucuronate or iduronate.
- ➔ They are components of extracellular matrix of bone, cartilage and tendons.
- ➔ Mucopolysaccharides also function as lubricants and shock absorbers.
- ➔ The complex of mucopolysaccharide and protein is called proteoglycan.
- ➔ The important mucopolysaccharides include hyaluronic acid, chondroitin 4-sulfate, heparin, dermatan sulfate and keratin sulfate.

Hyaluronic acid

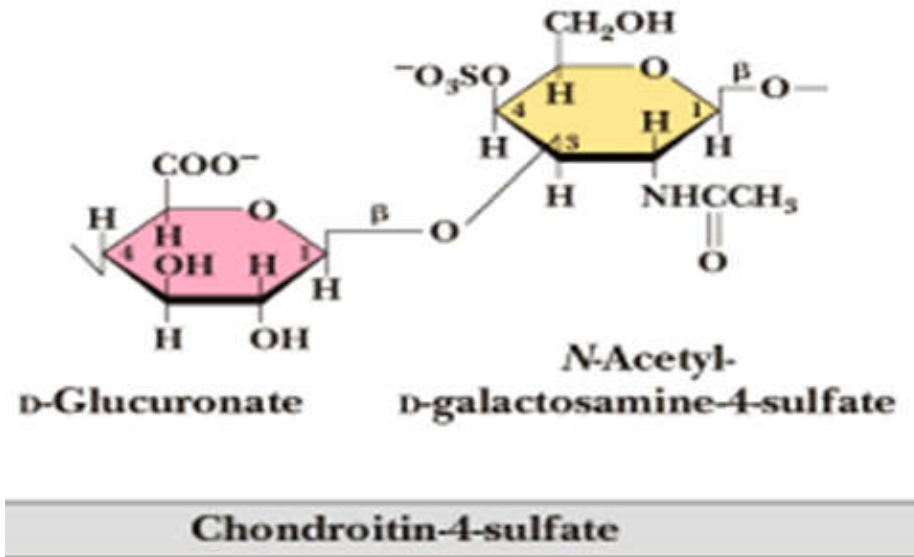
- ➔ Composed of D-glucuronic acid and N-acetyl –D-glucosamine.
- ➔ These two molecules form disaccharide units held together by β (1 $\rightarrow$ 3) glycosidic bond
- ➔ It is present in synovial fluid and function as lubricant and shock absorbent in joints.
- ➔ It also presents in skin, loose connective tissue, umbilical cord and ovum and vitreous body of eye



- ➡ **Hyaluronidase** breaks (β 1 $\rightarrow$ 4 linkages) hyaluronic acid and other GAG.
- ➡ This enzyme is present in high concentration in testes, seminal fluid, and in certain snake and insect venoms.
- ➡ **Hyaluronidase** of semen has an important role in fertilization to clears the gel (hyaluronic acid) around the ovum allowing a better penetration of sperm into the ovum.
- ➡ Hyaluronidase of bacteria helps their invasion into the animal tissues.
- ➡ Hyaluronic acid is involved in wound healing (repair).

Chondroitin Sulfate: Present in connective tissue

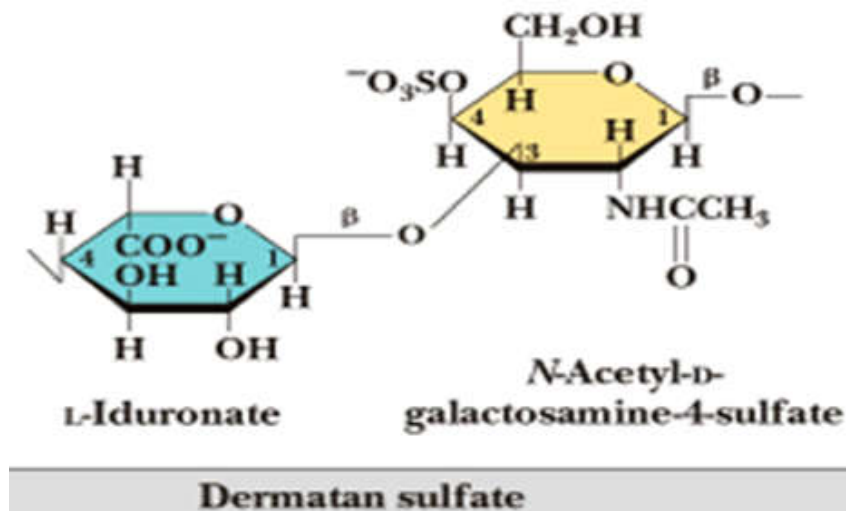
- ➔ a major constituent of bone, cartilage, tendons, heart, valves, skin, cornea etc.)
- ➔ Structurally, it is comparable with hyaluronic acid.
- ➔ It consists of repeating disaccharide units composed of D-glucuronic acid and N-acetyl D-galactosamine 4-sulfate.



- ➔ In chondroitin-4-sulfate, 4th carbon atom of N-acetyl galactosamine is sulfated where as in chondroitin-6-sulfate the 6th carbon is sulfated

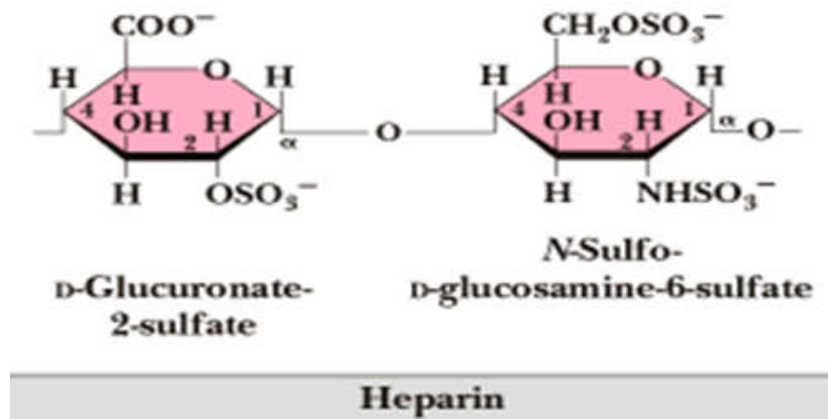
Dermatan Sulfate

- ➡ The name dermatan sulfate is derived from the fact that this compound mostly occurs in the skin, cornea and bone.
- ➡ It is structurally related to chondroitin 4-sulfate.
- ➡ The only difference is an inversion in the configuration around C5 of D-glucuronic acid to form L-iduronic acid.
- ➡ It has a role in **corneal transparency maintenance**



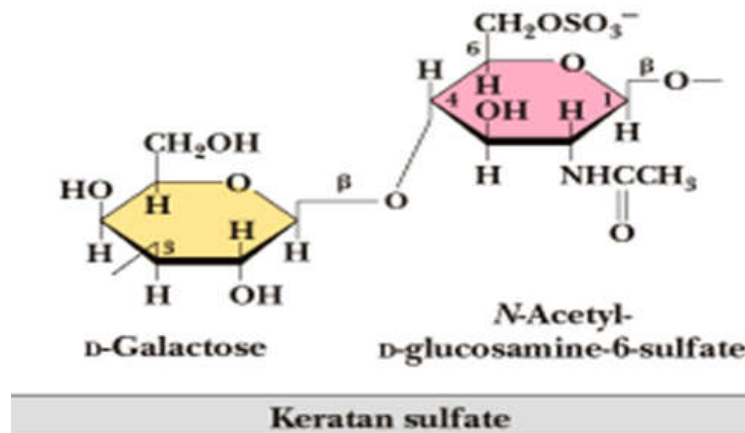
Heparin

- ✓ The repeating disaccharide unit consist of glucosamine and either iduronic acid or glucuronic acid.
 - ✓ Majority of uronic acids are iduronic acids. Further amino groups of glucosamine is **sulfated**.
 - ✓ Heparin is anti-coagulant (prevents blood clotting) in blood.
 - ✓ It is produced by mast cells present in the arteries, liver, lung and skin.
- ➔ Unlike other glycosaminoglycans, heparin is an **intracellular component**
 - ➔ Heparin helps in the release of the enzyme **lipoprotein lipase** which causes **clearing the turbidity of lipemic plasma**.



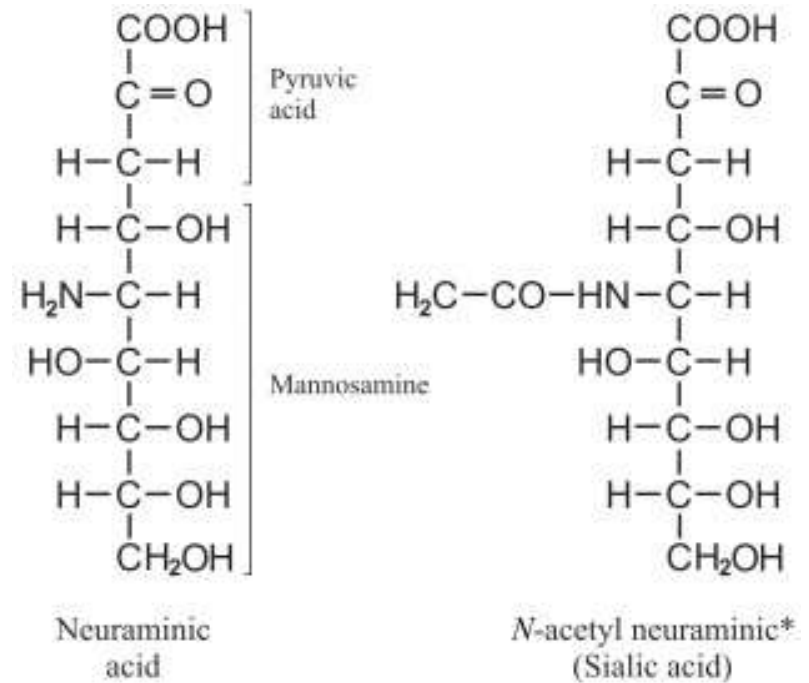
Keratan sulfates I and II

- ➔ The repeating disaccharide consist of galactose and N-acetyl glucosamine sulfate.
- ➔ Type I and II have different attachments to protein.
- ➔ They are components of cartilage, cornea and loose connective tissue.
- ➔ Keratan sulfate I is important for corneal transparency



Sialic Acids

- ➔ Sialic acids are acyl derivatives of neuraminic acid.
- ➔ Neuraminic acid is a nine-carbon sugar consisting of mannosamine and pyruvate.
- ➔ The amino group of mannosamine of neuraminic acid is acetylated.
- ➔ N-acetyl neuraminic acid (NANA) is an example for sialic acid
- ➔ Sialic acid is an important constituent of glycolipids present in cell membrane and nervous tissue



Important carbohydrates

Carbohydrate	Composition	Where found
<i>Monosaccharides</i>		
Glucose	$C_6H_{12}O_6$	Fruit juices
Fructose	$C_6H_{12}O_6$	Fruit juices, honey
Galactose	$C_6H_{12}O_6$	Not found naturally in its free form
<i>Disaccharides</i>		
	<i>Formed from:</i>	
Maltose	Glucose + glucose	Germinating grain
Sucrose	Glucose + fructose	Sugarcane, sugar beet
Lactose	Glucose + galactose	Milk
<i>Polysaccharides</i>		
	<i>Polymer of:</i>	
Glycogen	Glucose	Energy store in animals—in the liver and muscles
Starch	Glucose	Energy store in plants—abundant in wheat, potatoes, corn etc.
Cellulose	Glucose	Plant fibre—in plant cell walls

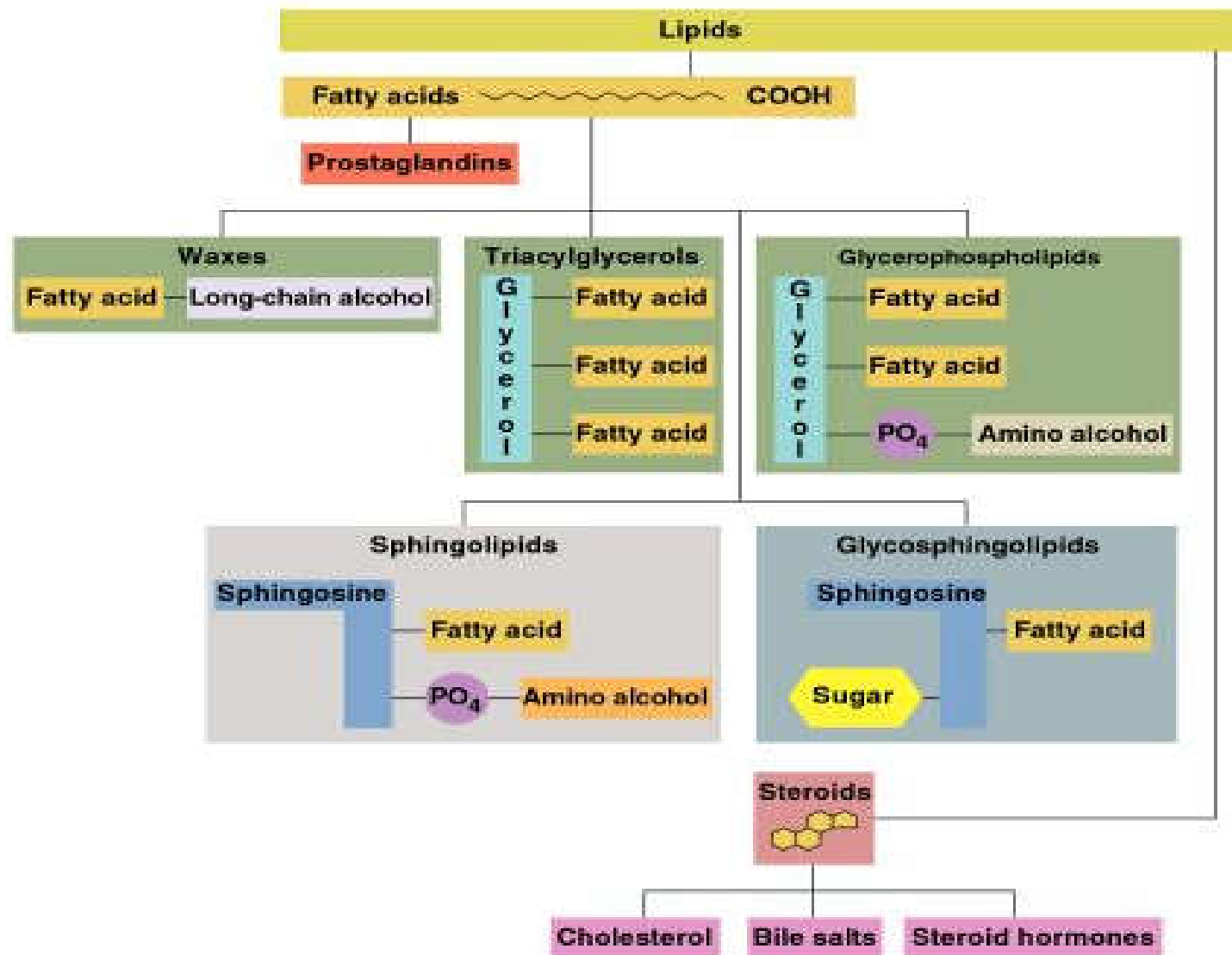


Introduction to Lipids

Introduction

- ✍ **Lipids** are diverse in form and are defined by solubility in non-polar solvents and insoluble in water
- ✍ Lipids are the group of greasy organic compounds, which are soluble in organic solvents like chloroform, ether and benzene but insoluble in water.
- ✍ Represent highly reduced forms of carbon and, upon oxidation in metabolism, yield large amounts of energy. It yields 9.7 cal/gm of energy while carbohydrates yield 4 cal/ gm
- ✍ They are fats, waxes, compound lipids, steroids, fatty acids and fat soluble vitamins.
- ✍ Chemical structure: glycerol + fatty acids.
- ✍ **Lipids have both non-polar** (hydrophobic) and polar (hydrophilic) ends.

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Biological Functions of Lipids

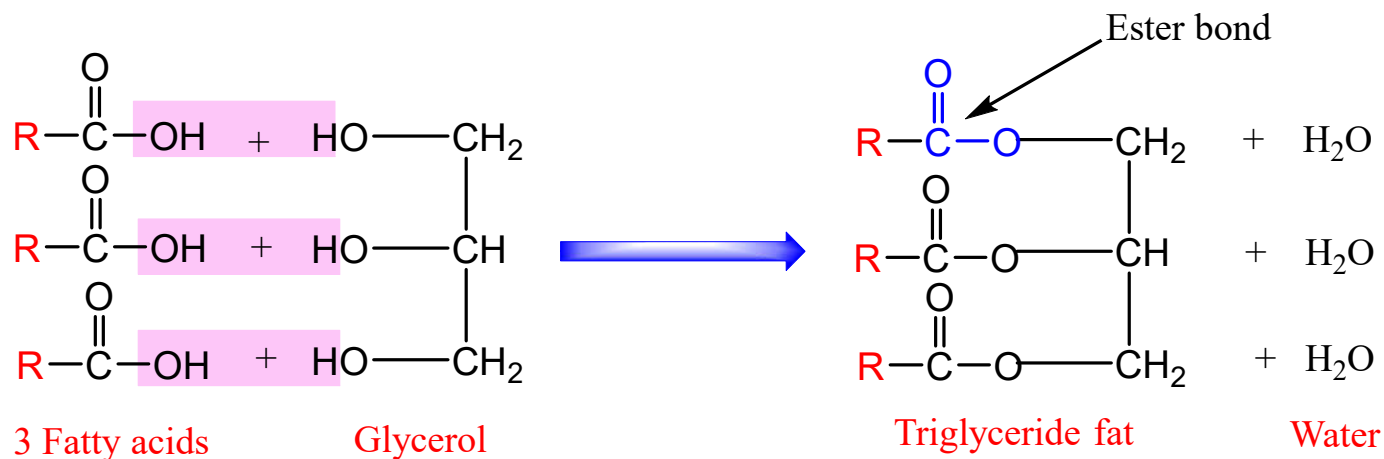
- ➔ Major source of energy for the body
- ➔ Provide energy storage in adipocytes
- ➔ As structural components of cell membranes
- ➔ Serve as a critical intercellular messengers
- ➔ Acts as a carrier of lipid-soluble vitamins into cells of small intestine
- ➔ Serve as thermal insulators against cold.
- ➔ Serve as precursors for hormones (steroid hormones).
- ➔ Fat around kidney serve as padding against injury
- ➔ Fats serve as surfactants by reducing surface tension

Structure and Classification of Lipids

- ✓ Lipids can be divided into five categories, on the basis of lipid function
 1. Energy-storage lipids (triacylglycerols)
 2. Membrane lipids (e.g. phospholipids)
 3. Emulsification lipids (bile acids)
 4. Messenger lipids (e.g. steroid hormones)
 5. Protective-coating lipids (biological waxes)
- Based on the composition lipids are classified into:
 - ① Simple lipids
 - ② Compound lipids and
 - ③ Derived lipids

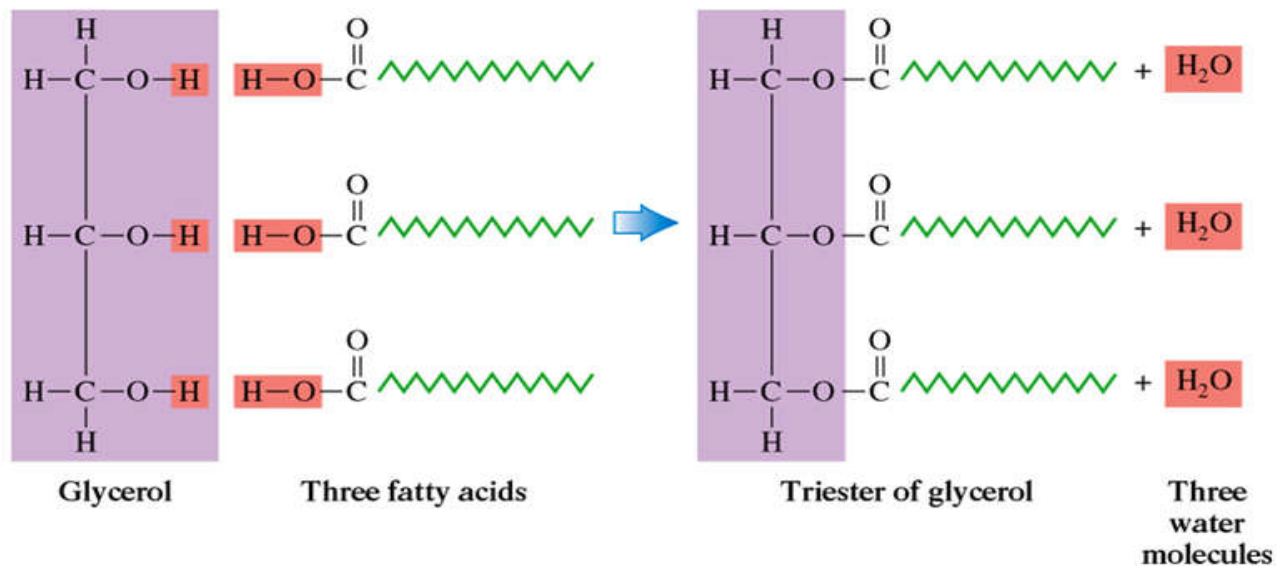
Simple Lipids

- They are esters of fatty acids with alcohols.
- Fats and waxes are example for simple lipids
- Fats are esters of fatty acids with glycerol
- Fats are also called as triglycerides or triacylglycerols



Triacylglycerols (Triglycerides)

- ✓ They are the major form of storage fatty acid.
- ✓ Triacylglycerols are tri-fatty acid esters of glycerol Eg. Triolein, Tripalmitate.
- ✓ Mainly found in adipocytes (fat cells).
- ✓ produce twice as much energy as that of carbohydrates per gram.



Triacylglycerols can be classified as fats or oils

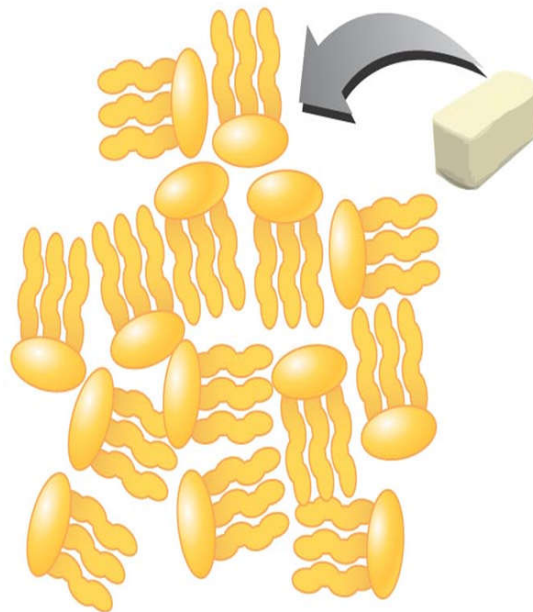
Fats

- ☞ Solid at room temperature
- ☞ Contain mainly saturated fatty acids
- ☞ Mostly obtained from animals

Oils

- ☞ Liquid at room temperature
- ☞ Contain mainly unsaturated fatty acids
- ☞ Obtain from plants

a Saturated fatty acids



b Unsaturated fatty acids



Properties of Triglycerides

- ① **Hydrolysis**: form fatty acids and glycerol by lipases
- ② **Saponification**: The alkali hydrolysis of triacylglycerols to produce glycerol and soaps
- ③ **Rancidity**: is the term used for deterioration of fats and oil resulting in unpleasant taste and smell.
 - ➔ Fats containing unsaturated fatty acids are more susceptible to rancidity.
 - ➔ When fats and oils are exposed to air moisture, light or bacteria, rancidity occurs.
 - ➔ The substances which prevent the occurrence of oxidative rancidity are called as **antioxidant** such as tocopherol (Vit. E), hydroquinone and α neptho

Hydrogenation of Triglyceride

- ✓ It converts unsaturated fatty acids of triglycerides into saturated fatty acids
- ✓ Since hydrogenation converts liquid fat to solid fat, it is also called as hardening
- ✓ Saturated fats tightly packed, clog arteries as **atherosclerosis**
- ✓ When unsaturated vegetable oils are hydrogenated to form more saturated oils (as in margarine), some of the cis fatty acids are isomerized to trans fatty acids.
- ✓ Trans fats may contribute to heart disease and some cancers

Waxes

- ➡ These are the esters of fatty acids usually long chains of fatty acids with alcohol other than glycerol.
- ➡ The alcohol may be aliphatic or alicyclic.
- ➡ Cetyl alcohol is most common in waxes.
- ➡ Bees wax is an ester of palmitic acid with myricyl alcohol
- ➡ Waxes serve as protective coatings on skin, fur and feathers of animals, birds and on fruit and leaves of plants.
- ➡ Waxes act as water barrier for animals, birds and insects.

Compound Lipids

- ➡ Compound lipids are esters of fatty acids with alcohol containing an additional group such as phosphate, nitrogenous base, carbohydrates, proteins, etc.
 - Phospholipids e.g. glycerophosphatides and Sphingomyelin
 - Nonphosphorylated Lipids e.g. glycolipids also called cerebrosides
 - Steroids e.g. cholesterol and hormones

Phospholipids

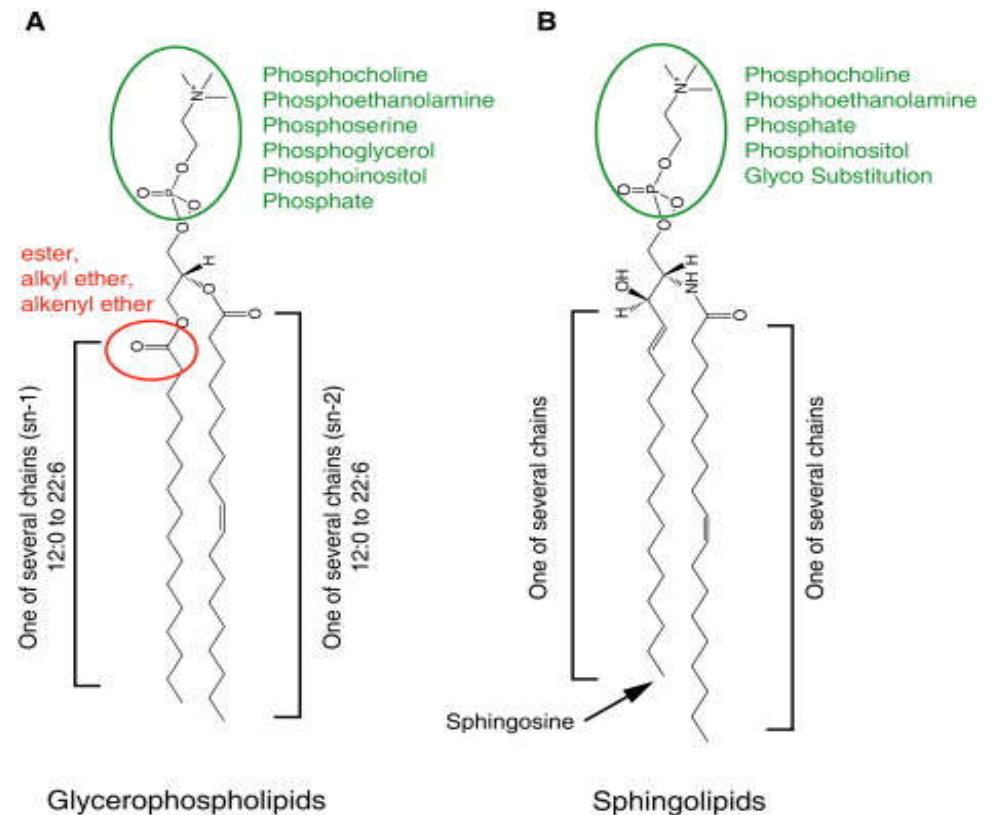
- ✓ Phospholipids are polar, ionic compounds composed of an alcohol attached by a phosphodiester bridge to either diacylglycerol or sphingosine.
- ✓ They are amphipathic in nature, each has a hydrophilic head (the phosphate group plus whatever alcohol is attached to it (serine, ethanolamine, and choline) and a long hydrophobic tail (containing fatty acids or fatty acid-derived hydrocarbons)
- ✓ Phospholipids are the predominant lipids of cell membranes.

Function of Phospholipid

- ♣ Membrane phospholipids also function as a **reservoir for intracellular messengers**, and, for some proteins, phospholipids serve as **anchors to cell membranes**.
- ♣ Nonmembrane-bound phospholipids serve additional functions in the body, for example as
 - ✓ **Components of lung surfactant** and
 - ✓ **Essential components of bile**, where their **detergent properties** aid in the **solubilization of cholesterol**.

Structure of Phospholipids

- ✓ There are two classes of phospholipids:
 - I. Those that have **glycerol** as a backbone called **Glycerophospholipids** and
 - II. Those that contain **sphingosine** called **sphingophospholipids**.
- ✓ Both classes are found as structural components of membranes, and both play a role in the generation of lipid-signaling molecules



Glycerophospholipids

- ✓ Glycerophospholipids are also called phosphoglycerides
- ✓ Glycerophospholipids constitute the major class of phospholipids
- ✓ They have a **glycerol** backbone.
- ✓ Glycerophospholipids are named as derivatives of the parent compound, phosphatidic acid according to the polar alcohol in the head group.
- ✓ Phosphatidic acid is the simplest phosphoglyceride, and is the precursor of the other members of this group.
- ✓ For example, Phosphatidylcholine and phosphatidylethanolamine have choline and ethanolamine as their polar head groups.

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Saturated fatty acid (e.g., palmitic acid) Unsaturated fatty acid (e.g., linoleic acid) Fatty acids Glycerol Head-group substituent			
Name of glycerophospholipid	Name of X — O	Formula of X	Net charge (at pH 7)
Phosphatidic acid	—		-2
Phosphatidylethanolamine	Ethanolamine		0
Phosphatidylcholine	Choline		0
Phosphatidylserine	Serine		-1
Phosphatidylglycerol	Glycerol		-1
Phosphatidylinositol 4,5-bisphosphate	<i>myo</i> -Inositol 4,5-bisphosphate		-4*
Cardiolipin	Phosphatidyl-glycerol		-2

➡ **Lecithins (Phosphatidyl choline)**

- ➡ Contain phospholipidic acid with choline as base.
- ➡ These are found in liver and sperm.
- ➡ It is secreted in alveoli of lungs.
- ➡ It is involved in the maintenance of shape of alveoli. It acts as surfactant in the lungs

➡ **Cephalins (Phosphatidyl ethanol amine)**

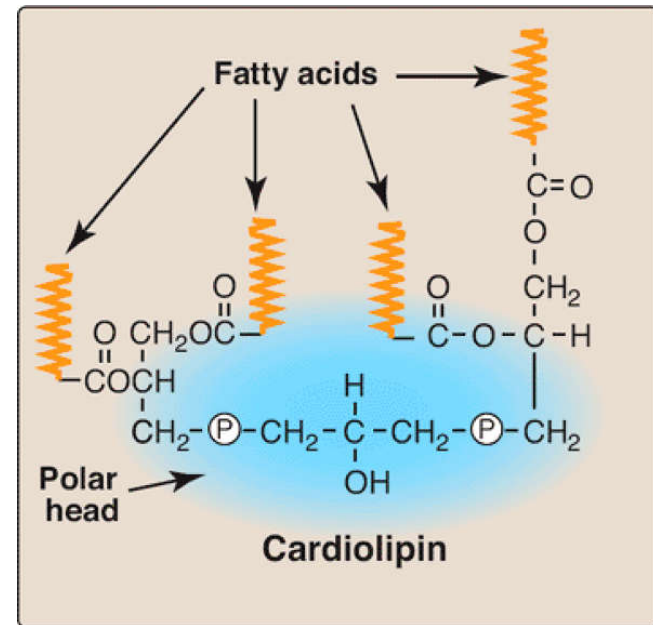
- ➡ Contains ethanolamine as base
- ➡ Found in milk and these are identical with thrombokinase which initiates the blood clotting.

➡ **Plasmalogens**

- ➡ When a fatty acid is attached by ether linkage at C1 of glycerol in glycerolphosphatide there occurs formation of plasmalogens.
- ➡ They are chemically phosphatidyl ethanolamine.
- ➡ These are found in liver and muscles.

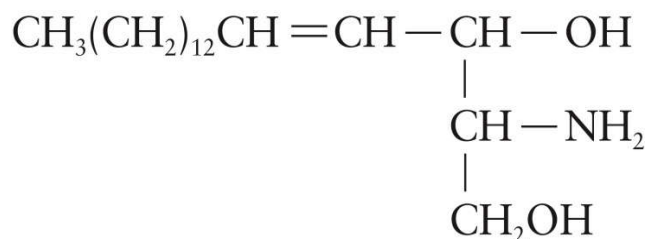
Cardiolipin (diphosphatidylglycerol)

- ✓ Composed of two molecules of phosphatidic acid connected by a molecule of glycerol.
- ✓ Required for the maintenance of certain respiratory complexes
- ✓ It is present in inner mitochondrial membrane.
- ✓ It is also present in heart muscle.
- ✓ It has immunological properties and used in the diagnosis of **syphilis**.



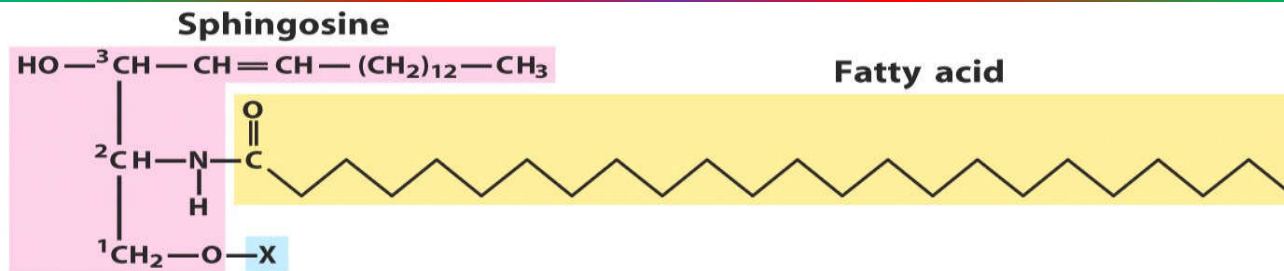
2. Sphingophospholipids

- ♪ Are similar to phospholipids, complex lipid found in cell membrane
- ♪ The backbone of sphingophospholipids is the **amino alcohol sphingosine**, rather than **glycerol**.

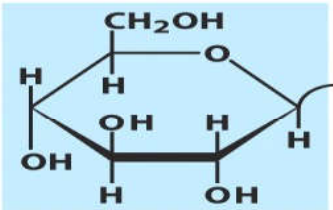


sphingosine

- ♪ Contain *sphingosine* instead of glycerol (a long-chain amino alcohol), a fatty acid, phosphate, and a small amino alcohol.
- ♪ A long-chain fatty acid is attached to the amino group of sphingosine through an **amide linkage**, producing a **ceramide**, which can also serve as a **precursor of glycolipids and sphingophospholipids**
- ♪ Have polar and nonpolar regions and are found in brain and nerve tissue, and in myelin sheath that surrounds nerves.



**Sphingolipid
(general
structure)**

Name of sphingolipid	Name of X	Formula of X
Ceramide	—	— H
Sphingomyelin	Phosphocholine	$ \begin{array}{c} \text{O} \\ \parallel \\ \text{P}-\text{O}-\text{CH}_2-\text{CH}_2-\text{N}^+(\text{CH}_3)_3 \\ \\ \text{O}^- \end{array} $
Neutral glycolipids Glucosylcerebroside	Glucose	
Lactosylceramide (a globoside)	Di-, tri-, or tetrasaccharide	
Ganglioside GM2	Complex oligosaccharide	

Sphingolipids

✓ There are three subclasses of sphingolipids, which are all derivatives of ceramide, but differ in their head groups.

I. Sphingomyelins

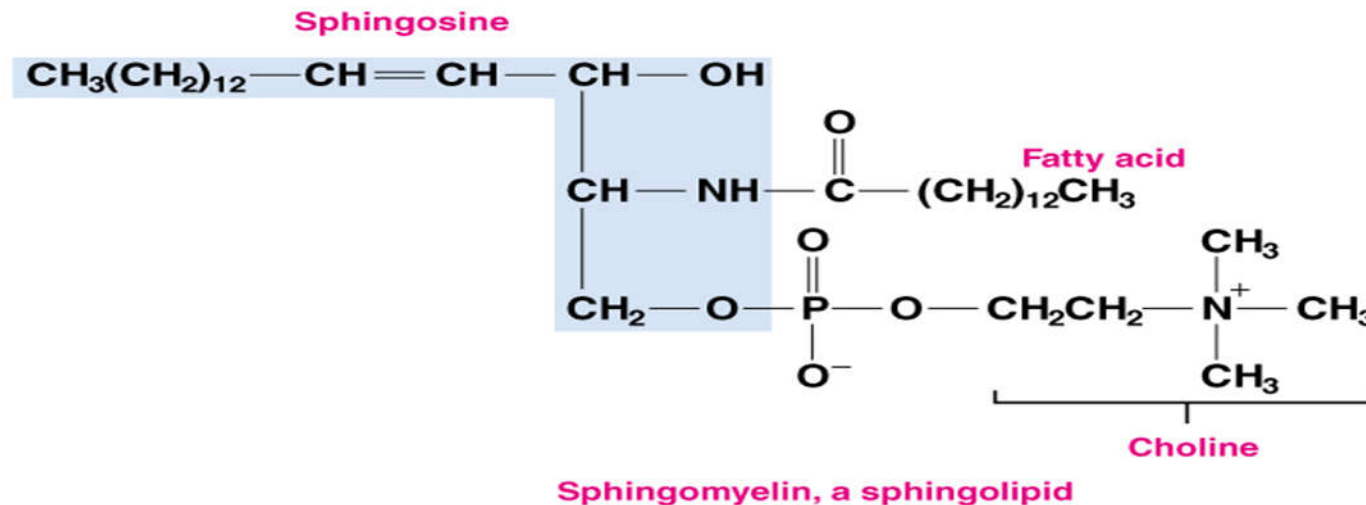
II. Glycosphingolipids

III. Gangliosides

Name of sphingolipid	Name of X—O	Formula of X
Ceramide	—	—H
Sphingomyelin	Phosphocholine	$\begin{array}{c} \text{O} \\ \parallel \\ \text{—P—O—CH}_2\text{—CH}_2\text{—N}^+(\text{CH}_3)_3 \\ \mid \\ \text{O}^- \end{array}$
Neutral glycolipids Glucosylcerebroside	Glucose	
Lactosylceramide (a globoside)	Di-, tri-, or tetrasaccharide	
Ganglioside GM2	Complex oligosaccharide	

Sphingomyelin

- ➔ Contain phosphocholine and phosphoethanolamine as their polar head groups.
- ➔ Structurally very similar to glycerophospholipids.
- ➔ Present in the plasma membranes of animal cells.
- ➔ Prominent in myelin, a membranous sheath that surrounds and insulates the axons of some neurons.

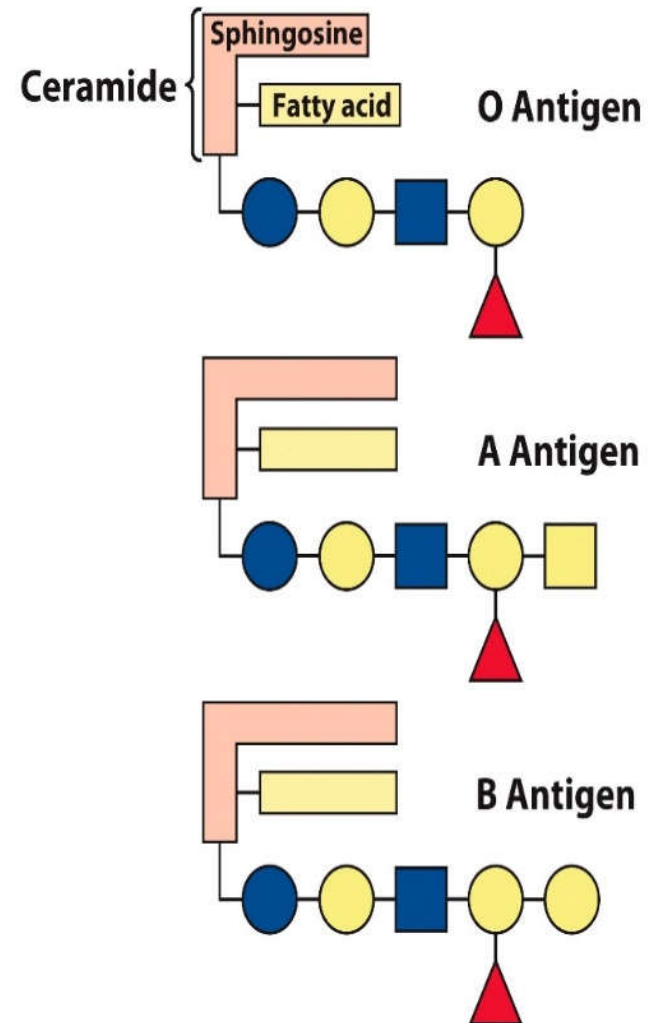


Glycosphingolipids

- ✓ These are neutral glycolipids that lack charges in the sugar head groups, and lack phosphates.
- ✓ Glycosphingolipids include the cerebrosides and globoside.
- ✓ **In cerebrosides** - a single sugar is linked to C-1 of the ceramide moiety.
- ✓ **In globosides** - two or more sugars usually D-glucose, D-galactose, or N-acetyl-D-galactosamine are linked to ceramide.
- ✓ The blood group antigens are members of the globoside class.

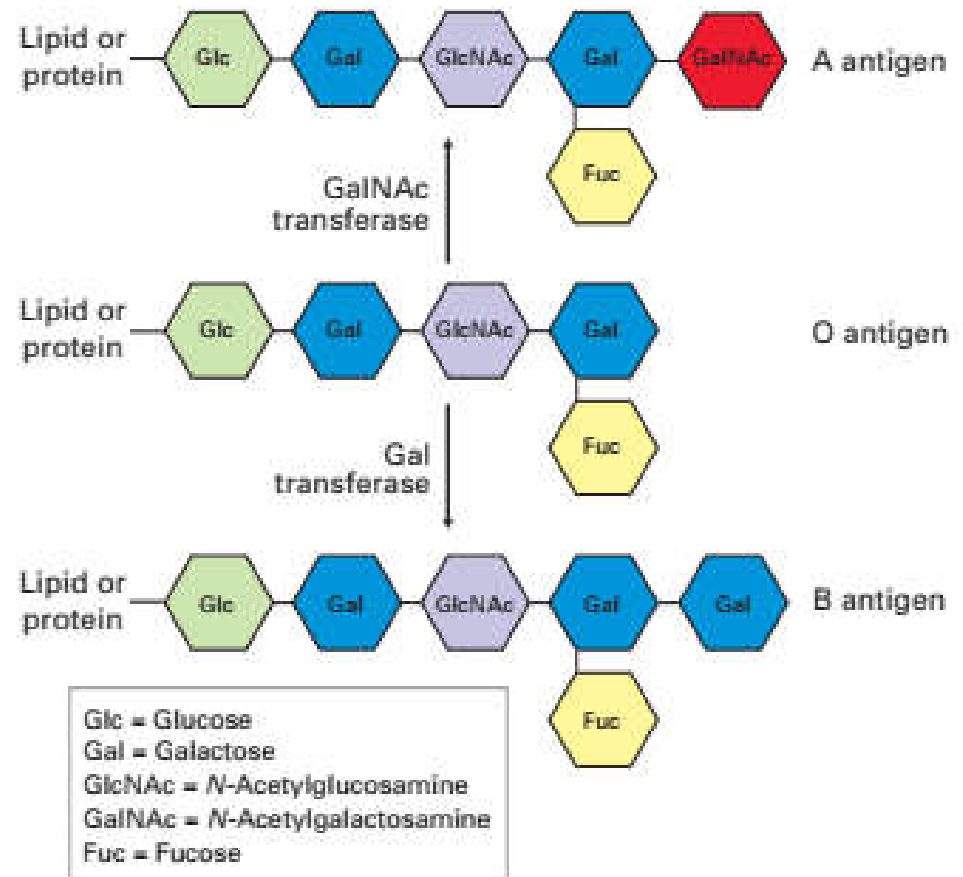
Blood Group Antigens

- ✓ The human blood groups (O, A, and B) are determined by the oligosaccharide head groups of globoside present in the outer leaflet of the red blood cell membrane.
- ✓ All three blood group antigens contain the same core of five sugars,
- ✓ But A and B antigens also contain an additional terminal N-acetylgalactosamine (yellow square) or galactose (yellow circle) residue, respectively.



Blood Group Antigens

- ➔ Blood antigens are oligosaccharide chains covalently attached to glycolipids or glycoproteins in the plasma membrane.
- ➔ The terminal oligosaccharide sugars distinguish the three antigens.
- ➔ The presence or absence of the glycosyltransferases that add galactose (Gal) or N-acetylgalactosamine (GalNAc) to O antigen determine a person's blood type.



Derived Lipids

☞ As the name implies they are formed from simple and compound lipids by hydrolysis.

☞ They includes:

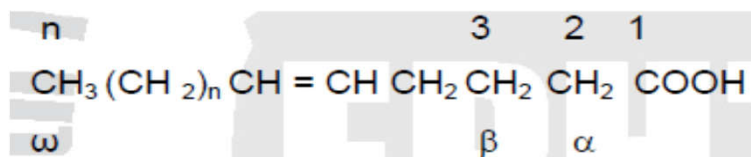
- Fatty acids,
- Steroids,
- Glycerol and
- Retinol

Fatty acids

- ✓ The simplest lipids are the **fatty acids**
- ✓ They are long unbranched straight-chain carboxylic acids.
- ✓ Fatty acids are long chain organic acids having one polar carboxyl group (head) and a non-polar hydrocarbon chain (tail).
- ✓ Most naturally occurring fatty acids have got even number of carbons.
- ✓ Fatty acids can be classified as:
 1. **Saturated** fatty acids have only single bonds
 2. **Monounsaturated** fatty acids have one double bond
 3. **Polyunsaturated** fatty acids have two or more double bonds

Fatty acid Nomenclature

- ✓ There are two systems of numbering the carbon atoms in a fatty acid.
- 1. Numbering starts from carboxyl carbon. The last carbon is the "n" carbon
- 2. The second carbon is the " α " and the third the " β " Carbon. The last carbon atom is omega.

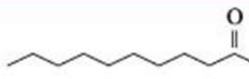
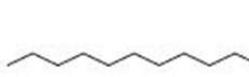
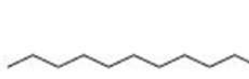
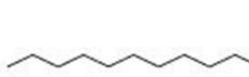
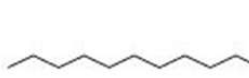
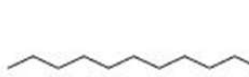


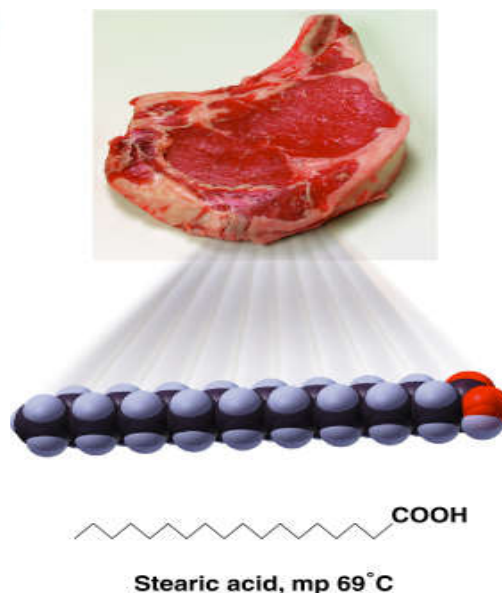
- ✓ α -linolenic acid - $18:3 \Delta^{9,12,15}$, - Fatty acid with 18 C-atoms and three double bonds at C-9, C-12, and C-15.

Saturated Fatty Acids

✓ Saturated fatty acids have:

- ❖ Molecules that fit closely together in a regular pattern
- ❖ Strong attractions (dispersion forces) between fatty acid chains
- ❖ High melting points that makes them solids at room temperature.

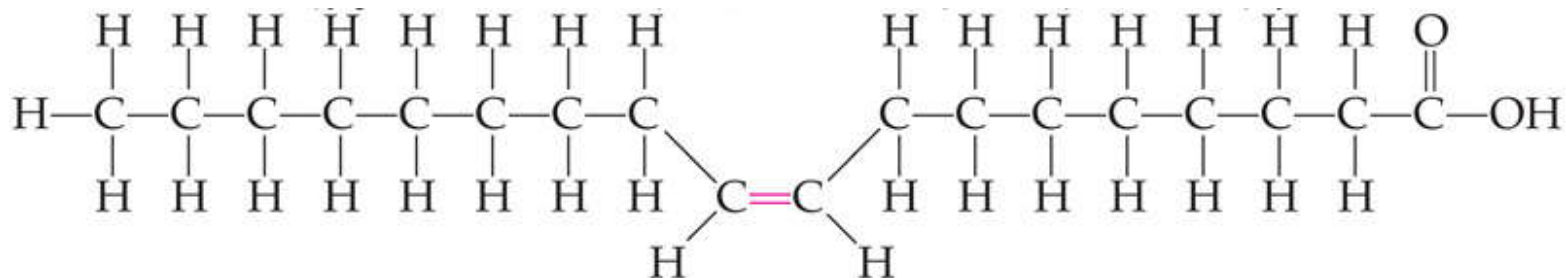
Name	Carbon Atoms	Structure	Melting Point (°C)	Source
Saturated Fatty Acids				
Capric acid	10		32	Saw palmetto
Lauric acid	12		43	Coconut
Myristic acid	14		54	Nutmeg
Palmitic acid	16		62	Palm
Stearic acid	18		69	Animal fat
Arachidic acid	20		76	Peanut oil, vegetable and fish oils



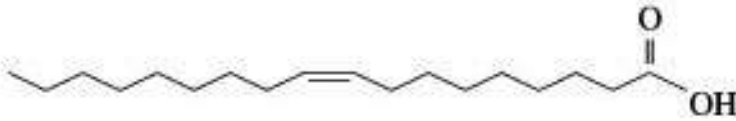
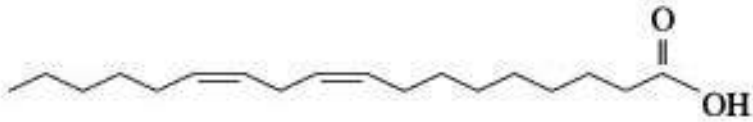
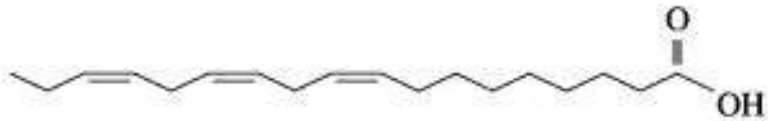
Unsaturated Fatty Acids

Unsaturated fatty acids have:

- ✓ Nonlinear chains that do not allow molecules to pack closely
- ✓ Weak attractions between fatty acid chains
- ✓ Low melting points and so are liquids at room temperature
- ✓ The double bond is normally in a *cis* configuration



Structures and Melting Points of Unsaturated Fatty Acids

Name	Carbon Atoms	Structure	Melting Point (°C)	Source
Monounsaturated Fatty Acids				
Palmitoleic acid	16		0	Butter
Oleic acid	18		13	Olives, corn
Polyunsaturated Fatty Acids				
Linoleic acid	18		-9	Soybean, safflower, sunflower
Linolenic acid	18		-17	Corn
Arachidonic acid	20		-50	Prostaglandins

Essential Fatty Acids

- ✓ Two fatty acids are dietary essentials in humans
 - ❖ linoleic acid -18:2 Δ (9,12)
 - ❖ α -linolenic acid -18:3 Δ (9,12,15)
 - ❖ arachidonic acid -20:4 Δ (5,8,11,14)
- ✓ Essential fatty acid deficiency can result in a scaly dermatitis, as well as visual and neurologic abnormalities.

Fatty acids with chain lengths of four to ten carbons are found in significant quantities in milk

Structural lipids and triacylglycerols contain primarily fatty acids of at least sixteen carbons.

COMMON NAME	STRUCTURE
Formic acid	1
Acetic acid	2:0
Propionic acid	3:0
Butyric acid	4:0
Capric acid	10:0
Palmitic acid	16:0
Palmitoleic acid	16:1(9)
Stearic acid	18:0
Oleic acid	18:1(9)
Linoleic acid	18:2(9,12)
α -Linolenic acid	18:3(9,12,15)
Arachidonic acid	20:4(5, 8,11,14)
Lignoceric acid	24:0
Nervonic acid	24:1(15)

Precursor of prostaglandins

Essential fatty acids

...Cont'd

Fatty acid trivial name	Systematic name	ω -series	Short name	Sources
Palmitoleic acid	Hexadeca - Δ^9 - <i>cis</i> -mono enoic acid or <i>Cis</i> -9-hexadecaenoic acid	ω -7	16:1, Δ^9	All animal fats
Oleic acid	Octadeca- Δ^9 - <i>cis</i> -enoic acid or <i>Cis</i> -9-octadecaenoic acid	ω -9	18:1, Δ^9	Many animal fats, vegetable oils
Linoleic acid	Octadeca- $\Delta^{9,12}$ -dienoic acid (all <i>cis</i>) or All <i>cis</i> -9,12-octadecadienoic acid	ω -6	18:2, $\Delta^{9,12}$	Vegetable oils like peanut, coconut, sun flower etc.
Linolenic acid	Octadeca- $\Delta^{9,12,15}$ -trienoic acid (all <i>cis</i>) or All <i>cis</i> -9,12,15-octadecatrienoic acid	ω -3	18:3, $\Delta^{9,12,15}$	Linseed oil, codliver oil, egg yolk
Arachidonic acid	Eicosa- $\Delta^{5,8,11,14}$ -tetra enoic acid (all <i>cis</i>) or All <i>cis</i> -5,8,11,14-eicosatetraenoic acid	ω -6	20:4, $\Delta^{5,8,11,14}$	Codliver oil and egg yolk

Functions of Essential Fatty Acids

- ① **They are used to maintain the fluidity of membrane and essential for skin integrity, normal growth and reproduction.**
- ② **They are required for the synthesis of Phospholipids, cholesterol ester, steroids, lipoproteins, prostaglandins, leukotrienes and thromboxanes.**
- ③ **They act as fat mobilizing agents in liver and protect liver from accumulating fats (fatty liver).**
- ④ **They are useful in the treatment of atherosclerosis by transporting blood cholesterol and lowering it and transporting triglycerides.**
- ⑤ **They have an important role in blood clotting (intrinsic factor).**
- ⑥ **Important role in health of the retina and vision.**
- ⑦ **They can be oxidized for energy production.**

Deficiency Manifestations

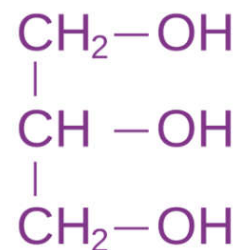
- ✍ Slow growth rate
- ✍ Skin lesions, hyperkeratosis eczema like dermatitis.
- ✍ Abnormalities in pregnancy
- ✍ Fatty liver accompanied by increased rates of fatty acids synthesis.
- ✍ Lessened resistance to stress.
- ✍ Kidney damage
- ✍ Degenerative changes in arterial wall

Glycerol

- ➡ It is a tri-hydric alcohol (i.e., containing three OH groups) and has the popular name glycerin.
- ➡ It is synthesized in the body from glucose.
- ➡ It has the properties of colorless viscous oily liquid with sweet taste.
- ➡ On esterification with fatty acids it gives: Mono, di, and triacylglycerol.

➡ Glycerol plus

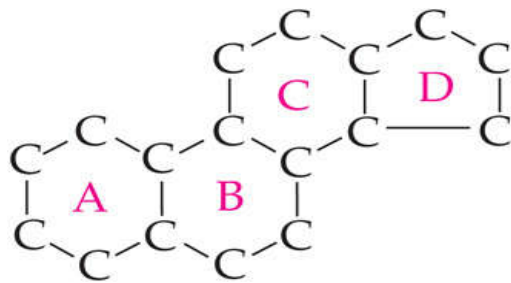
- ☞ 1 fatty acid = monoacylglycerol
- ☞ 2 fatty acids = diacylglycerol
- ☞ 3 fatty acids = triacylglycerol



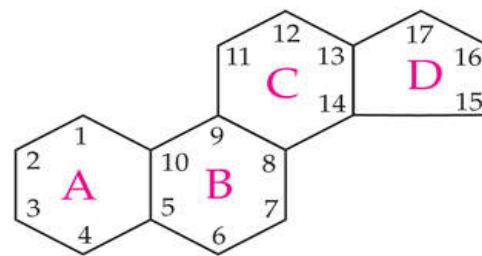
➤ **Medical application:** Nitroglycerine is used as vasodilator

Sterols

- ✍ Steroids are the compounds containing a cyclic steroid nucleus or 4 ring fused structure.
- ✍ Sterols are complex fat-soluble molecules, which are present in the plasma lipoproteins
- ✍ Sterols are synthesized from simple 5-carbon isoprene units.
- ✍ It includes steroid hormones, bile acids, cholesterol and lipoproteins.



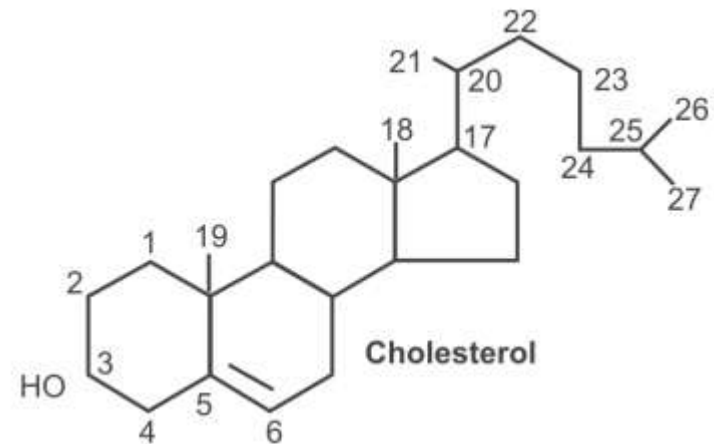
Carbon skeleton of the steroid nucleus



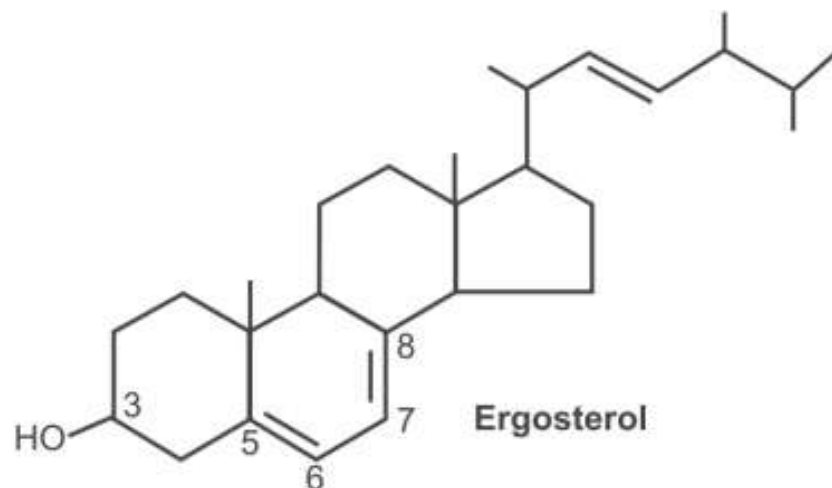
Steroid nucleus

Cholesterol

- ✓ Cholesterol is exclusively found in animals and is most abundant animal sterol.
- ✓ Cholesterol (Greek : chole bile) was first isolated from bile so, cholesterol means solid alcohol from bile.
- ✓ Cholesterol is one of the important non fatty acid lipid.
- ✓ Cholesterol have the following function:
 - ❖ Cell membranes, precursor to bile salts, sex hormones, vitamin D, and adrenocortical hormones
 - ❖ Have been linked to atherosclerosis



- ➔ **Ergosterol**— Found in plants and performs the function to produce ergocalciferol which is a compound containing vit D activity.
- ➔ Ergosterol is a provitamin of vitamin D found in yeast and plants
- ➔ Sitosterol Present in plants



Eicosanoids

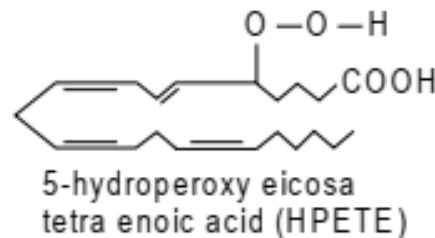
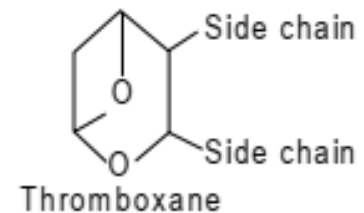
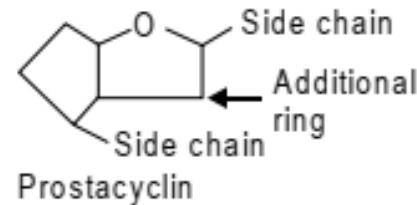
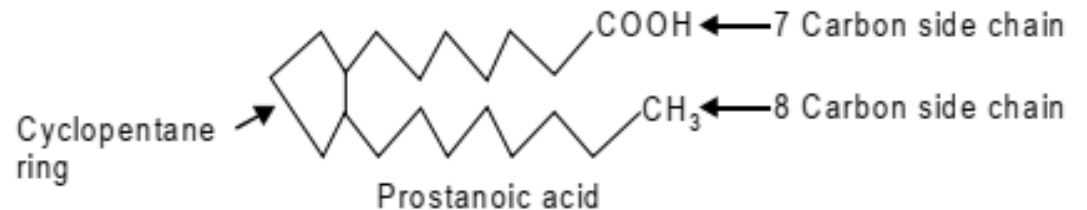
➡ They are derived from eicosapolyenoic fatty acid like arachidonic acid.

➡ They are prostanoids, leukotrienes (LTA) and lipoxins (LX).

➡ The prostanoids include prostaglandins (PG), prostacyclins (PGI) and thromboxanes (TXA).

➡ They function as local hormones.

➡ They act on several organs and produce physiological as well as pharmacological effects.



Prostaglandins

- ➡ Initially found in prostate gland they are named as prostaglandins.
- ➡ But later they are identified in all cells and tissues except erythrocytes.
- ➡ Chemically prostaglandins are derivatives of a 20 carbon prostanoic acid.
- ➡ Prostanoic acid is a cyclic compound with two side chains.
- ➡ The cyclic ring present in prostanoic acid is a cyclopentane ring.
- ➡ There are some six or more types of prostaglandins.

Prostacyclins

- ⇒ They contain a second five-numbered ring in addition to the one common to all prostaglandins.
- ⇒ Prostacyclins inhibit platelet aggregation

Thromboxane

- ⇒ They are identified first in thrombocytes.
- ⇒ They contain a six numbered heterocyclic oxane ring
- ⇒ Thromboxanes causes platelet aggregation and clot formation
- ⇒ Thromboxane A₂ regulates acquired immunity. It causes constriction of smooth muscle cells.

Leukotriens

- ➔ They are found in leukocytes.
- ➔ They are derivatives of arachidonic acid and contain no cyclic ring.
- ➔ HPETE derived from arachidonic acid serves as precursor for leukotriens
- ➔ Leukotriens are involved in the regulation of neutrophil and eosinophil function.
- ➔ They act as mediators of immediate hyper sensitivity reaction.
- ➔ The slow reacting substance of anaphylaxis (SRS-A) is a leukotriene

Plasma Membrane

Membrane is composed of:

A. *Lipids*

- ✓ Phospholipids
- ✓ Sterols

B. *Proteins*

- ✓ Integral
- ✓ Peripheral

C. *Carbohydrates*

- ✓ Glycolipids
- ✓ Glycoproteins

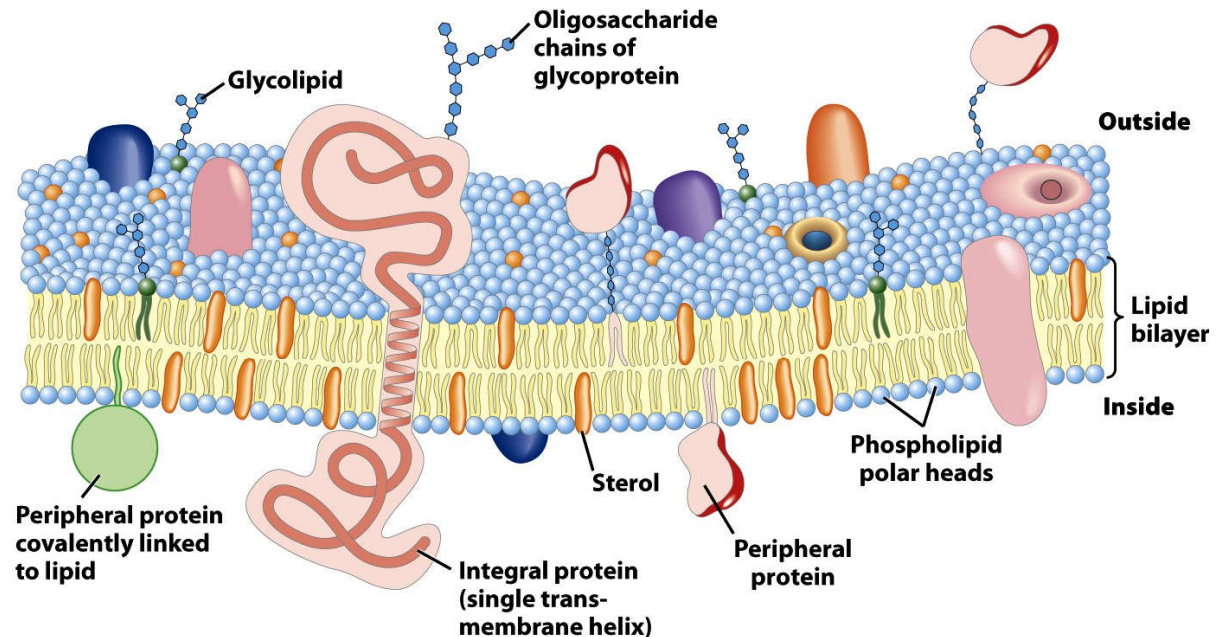
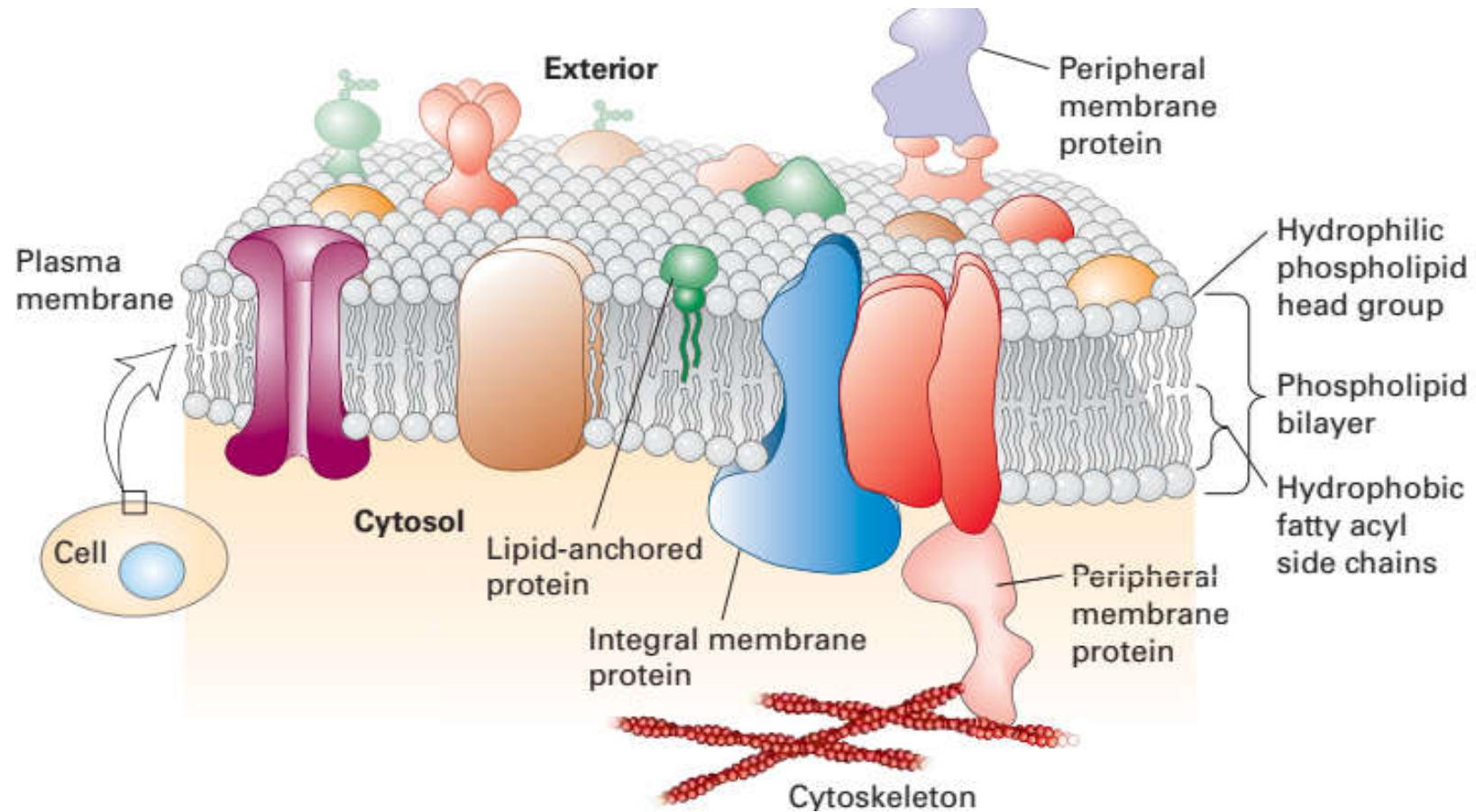


Figure 11-3
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Fluid Mosaic Model of Biomembrane



Plasma Membrane

- ➡ Composition of lipid, protein and carbohydrate varies from one membrane to another,
- ➡ Ratio of Lipid to Protein is not fixed in most membranes, varies widely from **4:1 to 1:4**, depending on the specific functions of the membrane,

TABLE 11-1 Major Components of Plasma Membranes in Various Organisms

	Components (% by weight)			Sterol type	Other lipids
	Protein	Phospholipid	Sterol		
Human myelin sheath	30	30	19	Cholesterol	Galactolipids, plasmalogens
Mouse liver	45	27	25	Cholesterol	—
Maize leaf	47	26	7	Sitosterol	Galactolipids
Yeast	52	7	4	Ergosterol	Triacylglycerols, steryl esters
Paramecium (ciliated protist)	56	40	4	Stigmasterol	—
<i>E. coli</i>	75	25	0	—	—

Note: Values do not add up to 100% in every case because there are components other than protein, phospholipids, and sterol; plants, for example, have high levels of glycolipids.

Table 11-1

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Evolution of Differences in Membrane Lipid Composition

- Variations in lipid composition of cell membranes of many species appear to be adaptations to specific environmental conditions
 - Ability to change the lipid compositions in response to temperature changes has evolved in organisms that live where temperatures vary
-

Membrane Proteins and Their Functions

- A membrane is a collage of different proteins, often grouped together, embedded in the fluid matrix of the lipid bilayer
 - Proteins determine most of the membrane's specific functions
 - Two basic types of membrane proteins:
 - ① Integral (Intrinsic) membrane proteins, and
 - ② Peripheral membrane proteins
-

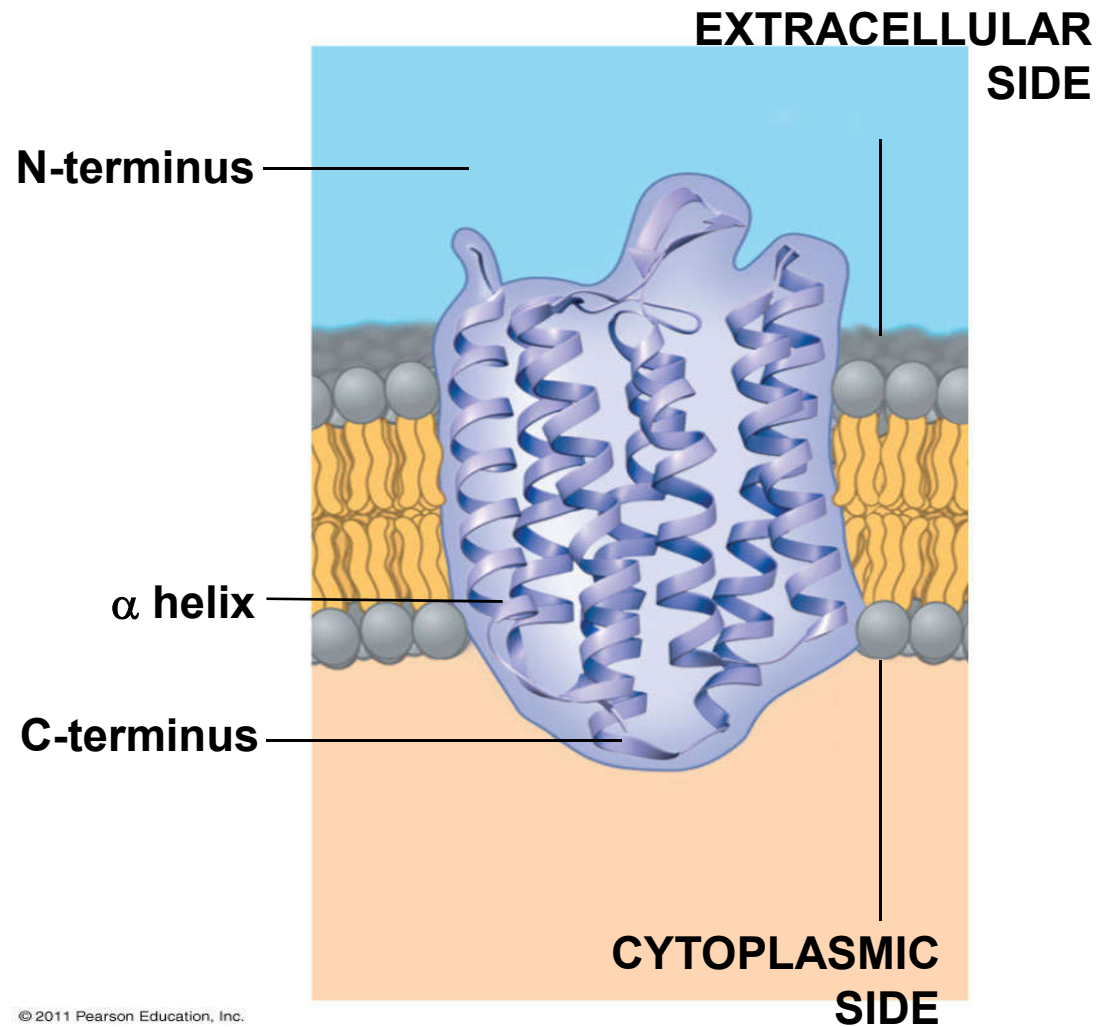
Peripheral Proteins

- ➡ **Peripheral proteins** are bound to the surface of the membrane.
 - ➡ Loosely bound by Hydrogen bonding or Electrostatic interactions to the exposed surface of integral proteins,
 - ➡ Readily be removed by washing the membranes with solutions of high ionic strength,
 - ➡ High ionic strength disrupts the Non-covalent ionic and Hydrogen bonds holding the proteins on the surface of the membrane
-

Integral Proteins

- ➡ **Integral proteins** tightly associated with hydrophobic core of lipid bilayer.
 - ➡ It interact with the hydrocarbon tails of lipids,
 - ➡ The hydrophobic regions of an integral protein consist of one or more stretches of nonpolar amino acids, often coiled into alpha helices.
 - ➡ Transmembrane proteins are integral proteins that are exposed on both sides of the membrane; they extend across the width of the membrane,
 - ➡ Integral proteins can be removed from the membrane only with organic solvents or detergents that disrupt the membrane structure,
-

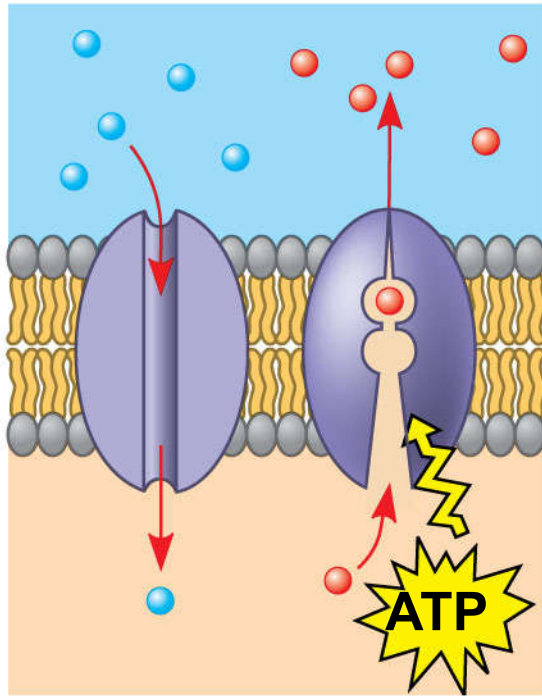
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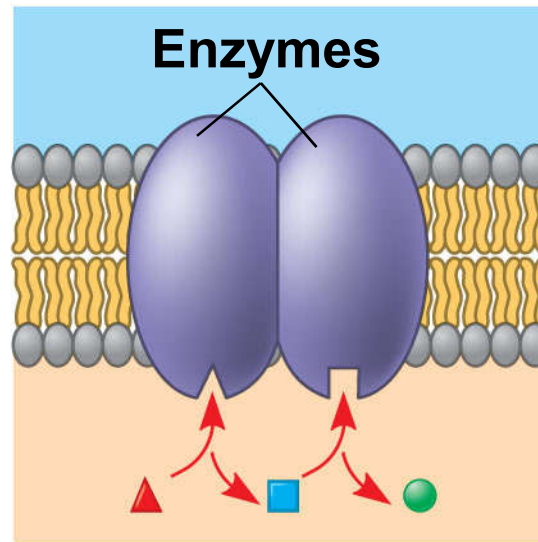
- Six major functions of membrane proteins
 - Transport
 - Enzymatic activity
 - Signal transduction
 - Cell-cell recognition
 - Intercellular joining
 - Attachment to the cytoskeleton and extracellular matrix (ECM)
-

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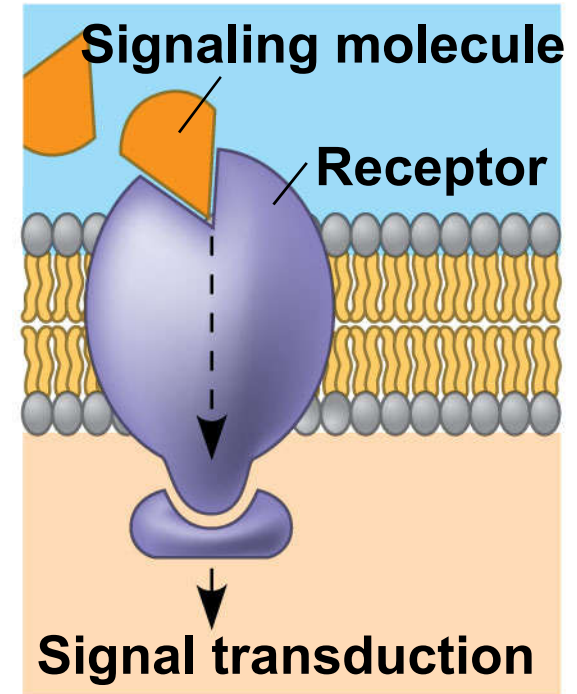


(a) Transport

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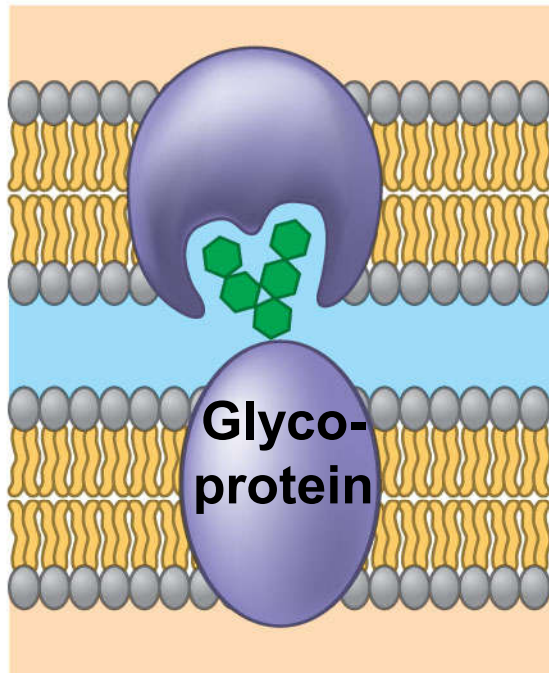


(b) Enzymatic activity

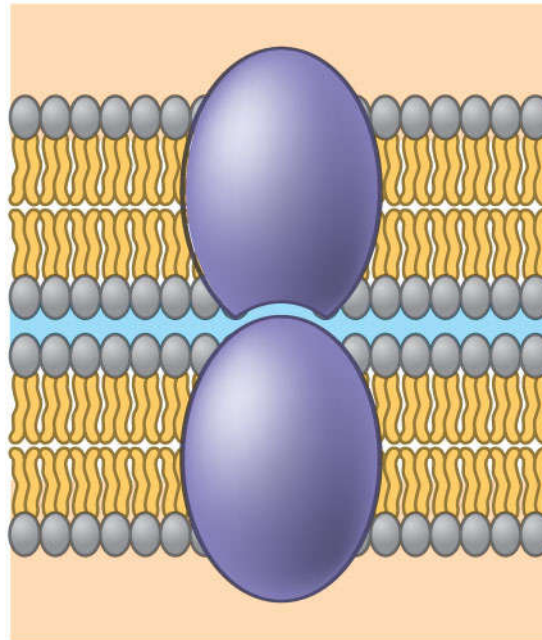


(c) Signal transduction

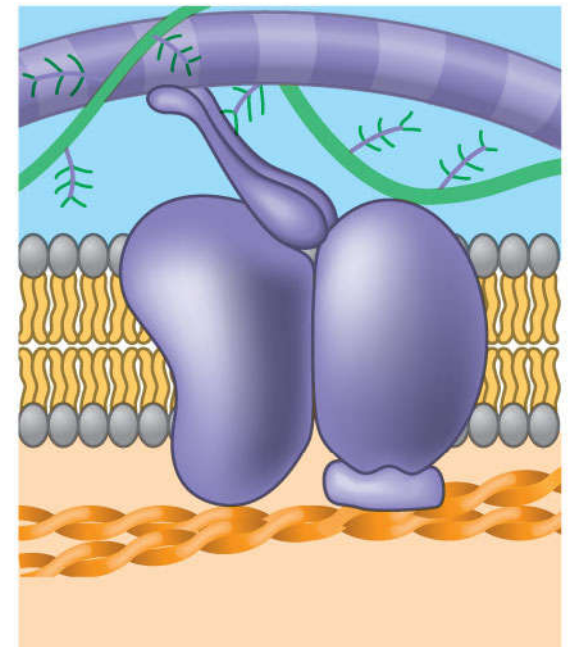
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(d) Cell-cell recognition



(e) Intercellular joining

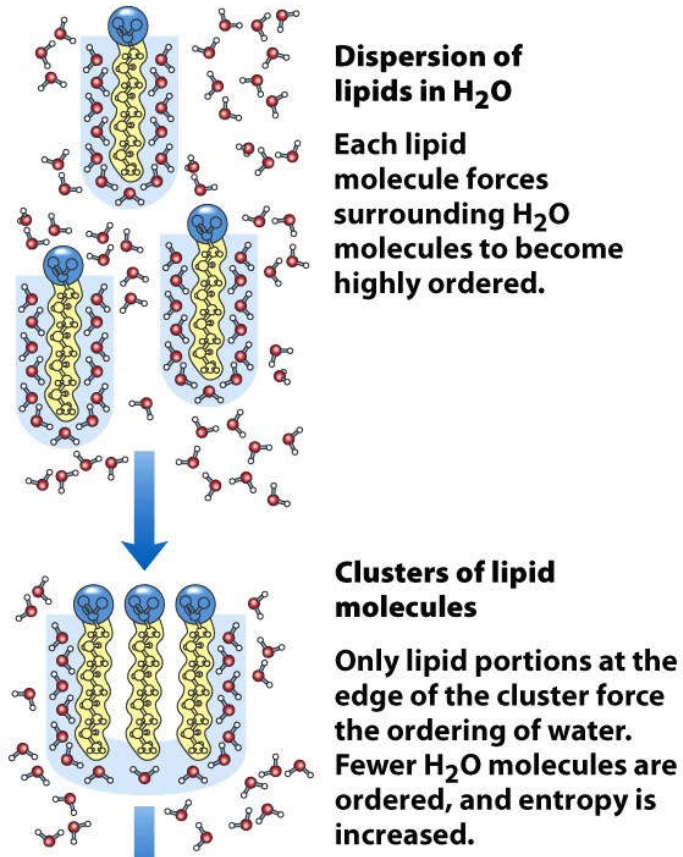


(f) Attachment to the cytoskeleton and extracellular matrix (ECM)

The Role of Membrane Carbohydrates in Cell-Cell Recognition

- Cells recognize each other by binding to surface molecules, often containing carbohydrates, on the extracellular surface of the plasma membrane
 - Membrane carbohydrates may be covalently bonded to lipids (forming **glycolipids**) or more commonly to proteins (forming **glycoproteins**)
 - Carbohydrates on the external side of the plasma membrane vary among species, individuals, and even cell types in an individual
-

Lipid Aggregates



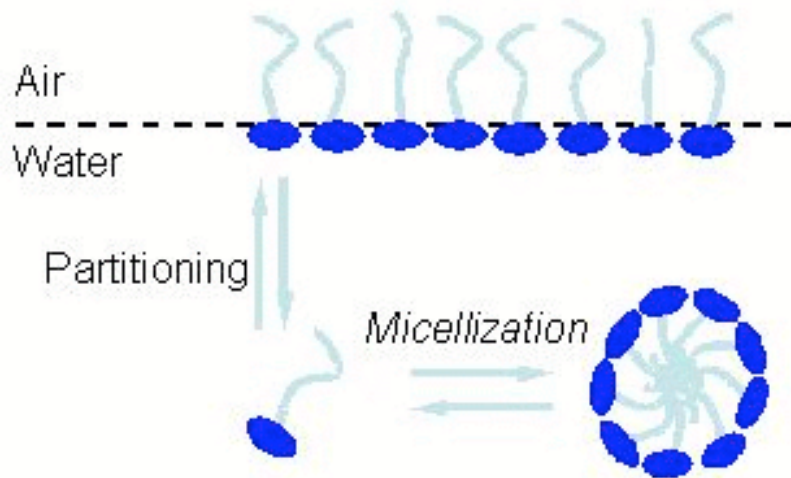
Lipids spontaneously aggregate in water as a result of the Hydrophobic Effect.

Figure 2-7b

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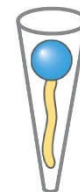
Lipid Aggregates



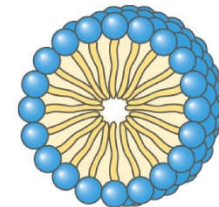
➡ Amphiphilic lipids form structures that solvate their head groups and keep their hydrophobic tails away from water.

➡ Above the critical micelle concentration, single-tailed lipids form micelles.

(a) Micelle

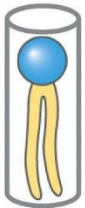


Individual units are wedge-shaped (cross section of head greater than that of side chain)

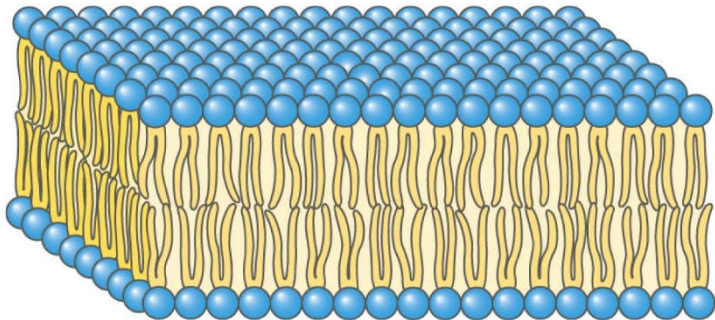


Lipid Aggregates

(b) Bilayer



Individual units are cylindrical
(cross section of head equals that
of side chain)

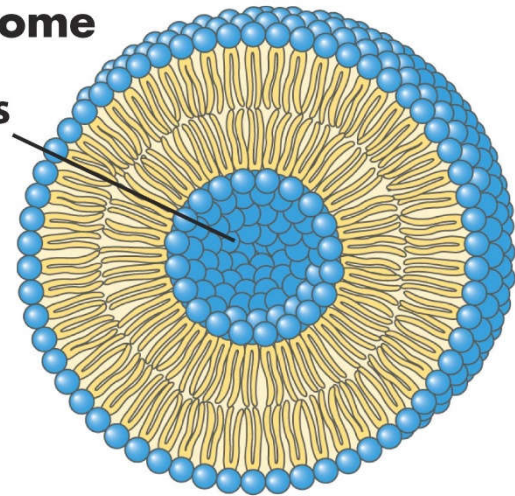


Bilayers can form vesicles
enclosing an aqueous cavity
(*liposomes*).

➡ Double-tailed lipids form
bilayers, the basis of cell
membranes.

(c) Liposome

Aqueous
cavity



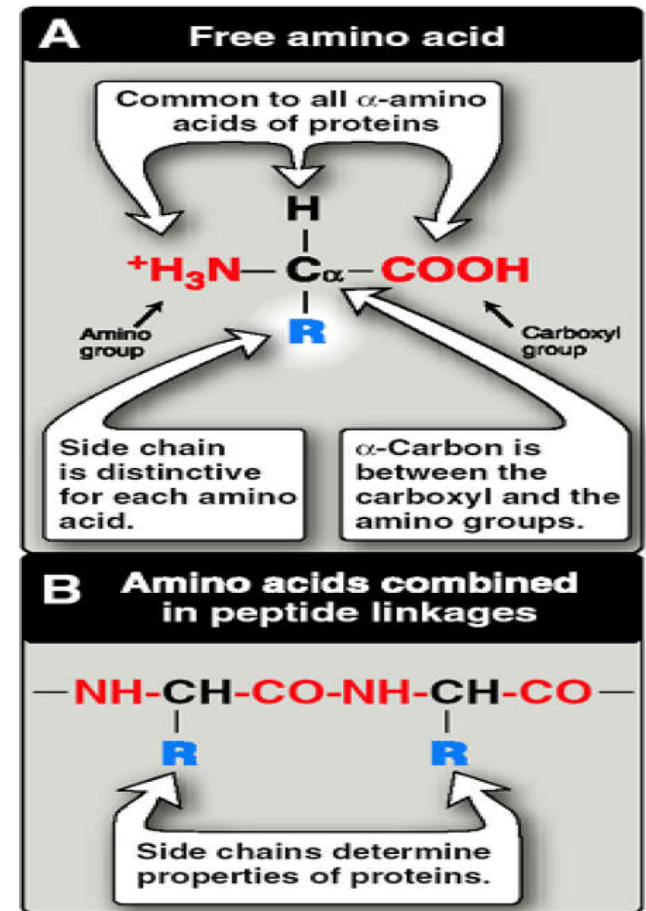
Amino acids and Protein

Introduction to Amino Acids

- Approximately 300 amino acids present in various animals, plants, and microbial systems, but **only 20 amino acids are coded by DNA to appear in proteins.**
- Cells produce **proteins** with strikingly **different properties and activities** by joining the same 20 amino acids in **many different combinations and sequences.**
- This indicates that the **properties of proteins are determined** by the **physical and chemical properties of their monomer units, the amino acids.**

Definition:

- Amino acids are the basic structural units of proteins consisting of:
 - **Amino group**, (-NH₂)
 - **Carboxyl** (-COOH) group
 - **Hydrogen** (H) atom and
 - a (variable) **distinctive** (R) group.
- All of the substituent's in amino acid are attached (bonded) to a **central α carbon** atom.
- This carbon atom is called α because it is bonded to the carboxyl (acidic) group.

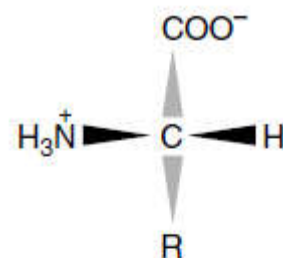


Stereochemistry (Optical activity)

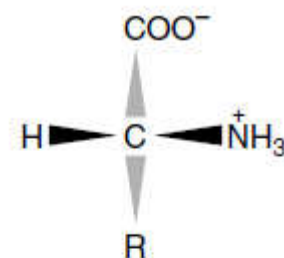
➡ Stereochemistry mainly emphasizes the configuration of amino acids at the α carbon atom, having either **D** or **L**- isomers.

➡ Out of the 20 amino acids, **proline** is not an α amino acid rather an α - **imino acid**.

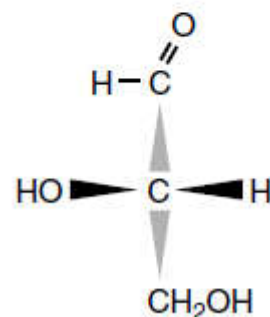
➡ Except for **glycine (achiral)**, all amino acids contain at least one **asymmetric carbon atom** (the α - carbon atom).



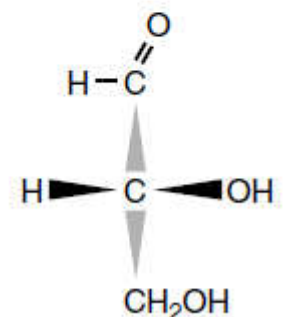
L-Amino Acid



D-Amino acid



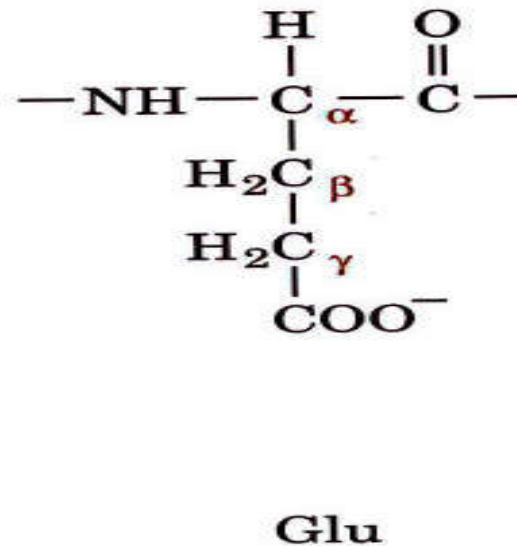
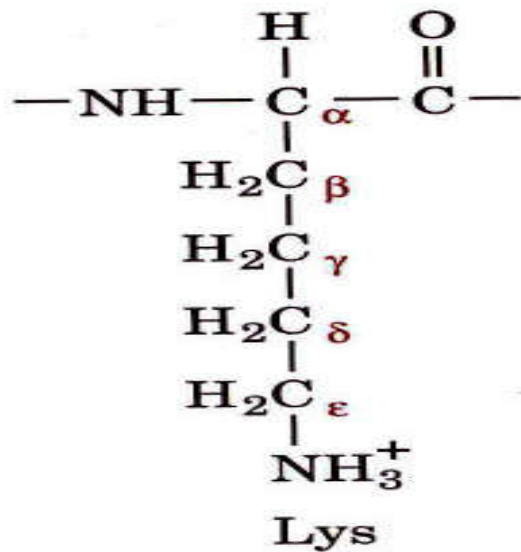
L-Glyceraldehyde



D-Glyceraldehyde

Nomenclature

- ✓ All amino acids have one and three letter codes
- ✓ The atoms of the side chain are assigned sequential letters in the Greek alphabet beginning with the carbon next to the **carboxyl group**



...Cont'd

TABLE 3.1 Names and Abbreviations of the Common Amino Acids

Amino Acid	Three-Letter Abbreviation	One-Letter Abbreviation
Alanine	Ala	A
Arginine	Arg	R
Asparagine	Asn	N
Aspartic acid	Asp	D
Cysteine	Cys	C
Glutamic acid	Glu	E
Glutamine	Gln	Q
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Leucine	Leu	L
Lysine	Lys	K
Methionine	Met	M
Phenylalanine	Phe	F
Proline	Pro	P
Serine	Ser	S
Threonine	Thr	T
Tryptophan	Trp	W
Tyrosine	Tyr	Y
Valine	Val	V

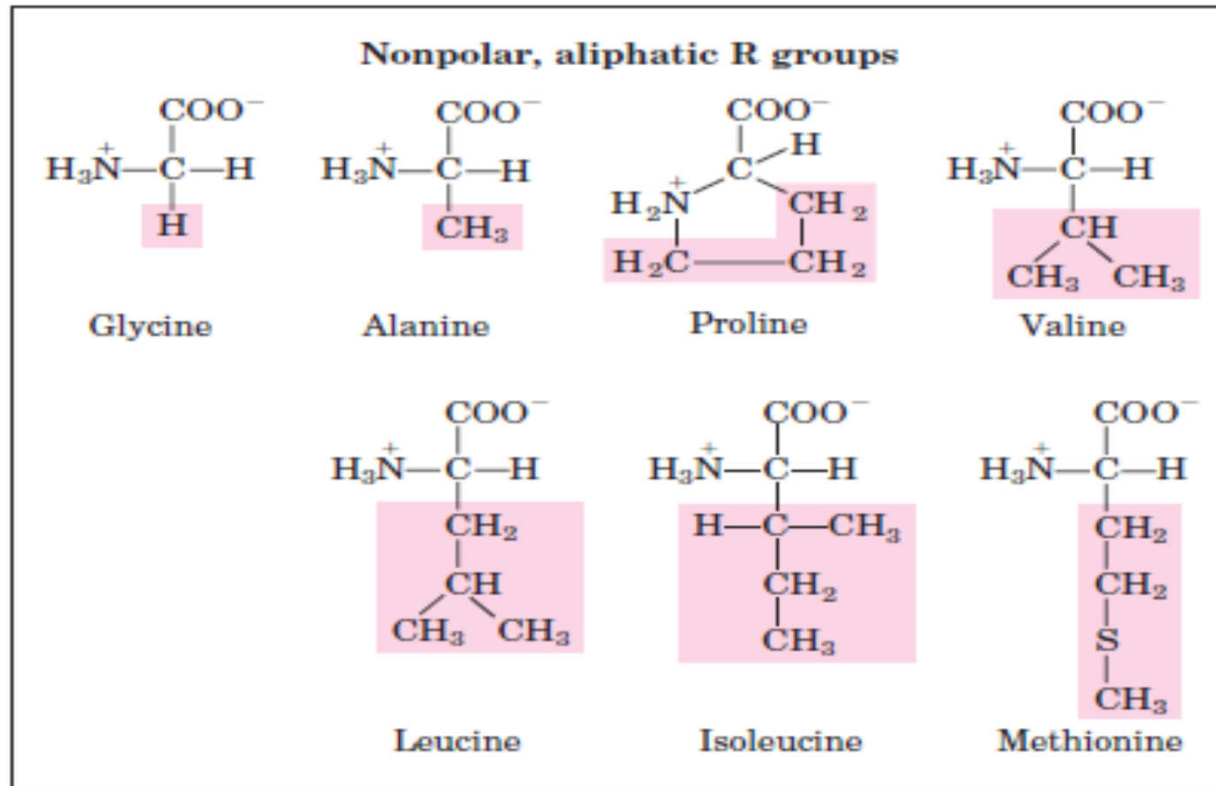
Classification of Amino Acids

- L-Amino acids are frequently grouped according to the **chemical nature of their side chains**.
- Common groupings of amino acids are **aliphatic, hydroxyl/sulfur, cyclic, aromatic, basic, acidic and acid amides**.

I. Structural Classification

- This classification is **based on the side chain radicals** (R-groups)

Nonpolar, Aliphatic R Groups

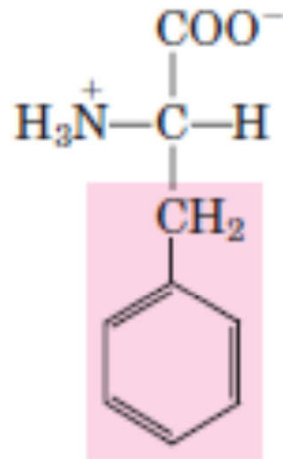


The R groups in this class of amino acids are nonpolar and hydrophobic.

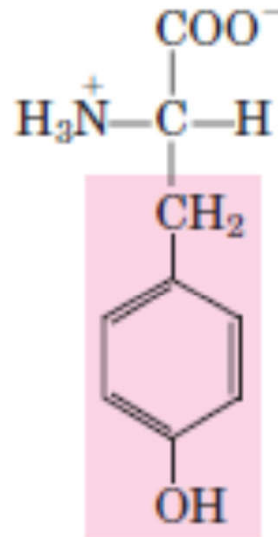
- The side chains of **alanine**, **valine**, **leucine**, and **isoleucine** tend to cluster together within proteins, stabilizing protein structure by means of hydrophobic interactions.
- **Glycine** has the **simplest structure**. Although it is formally nonpolar, its **very small side chain** makes **no real contribution to hydrophobic interactions**.
- **Methionine**, one of the two **sulfur-containing** amino acids, has a nonpolar **thioether group** in its side chain.
- **Proline** has an aliphatic side chain with a distinctive **cyclic structure**. The secondary amino (**imino**) **group** of proline residues is held in a rigid conformation that **reduces the structural flexibility of polypeptide** regions containing proline.

Aromatic R Groups

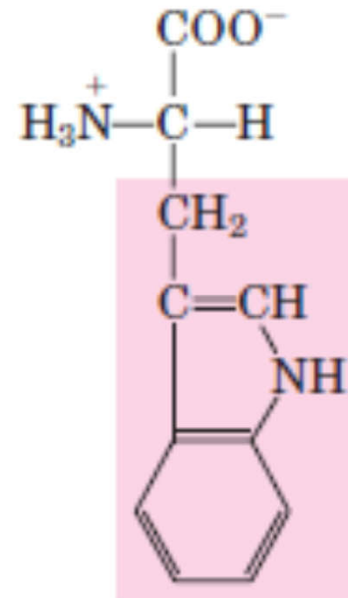
Aromatic R groups



Phenylalanine



Tyrosine

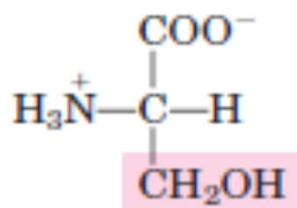


Tryptophan

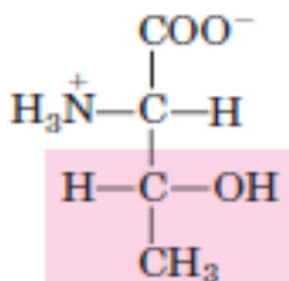
- ➡ **Phenylalanine, tyrosine, and tryptophan**, with their aromatic side chains, are relatively nonpolar (hydrophobic).
- ➡ All can participate in hydrophobic interactions.

Polar, Uncharged R Groups

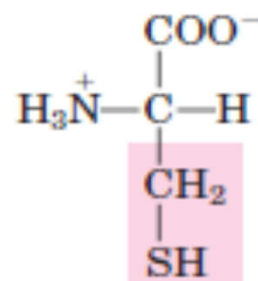
Polar, uncharged R groups



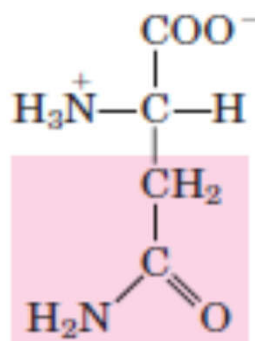
Serine



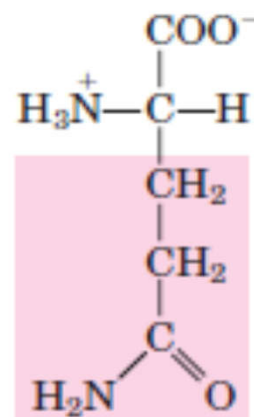
Threonine



Cysteine



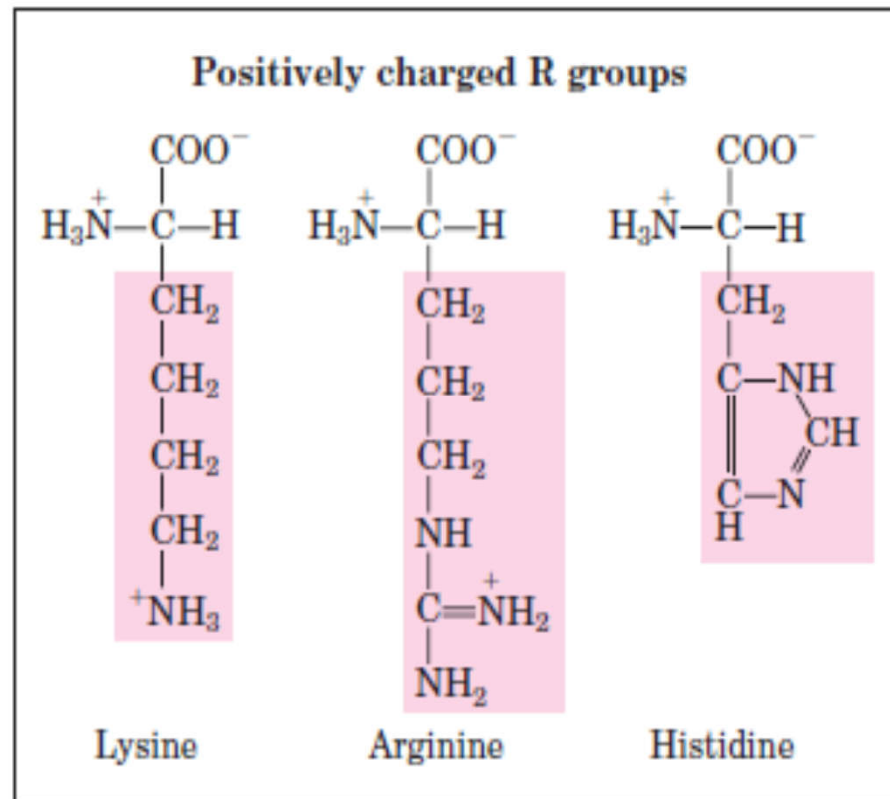
Asparagine



Glutamine

- The R groups of these amino acids are **more soluble in water**, or more hydrophilic, than those of the nonpolar amino acids, because:
 - they contain functional groups that form hydrogen bonds with water.
 - includes **serine, threonine, cysteine, asparagine, and glutamine.**
 - The polarity of :
 - **serine** and **threonine** is contributed by their **hydroxyl groups**;
 - **cysteine** by its **sulfhydryl group**; and
 - **asparagine** and **glutamine** by their **amide groups**.

Positively Charged (Basic) R Groups

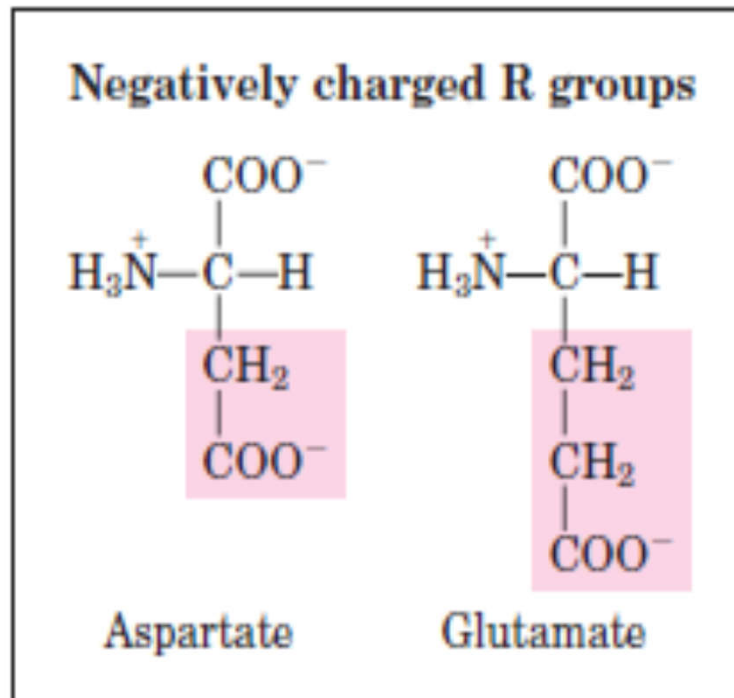


- ➡ The most hydrophilic R groups are those that are either positively or negatively charged.

- The amino acids in which the R groups have significant positive charge at pH 7.0 are:
 - **Lysine**, which has a **second primary amino group** at the **ε** position on its aliphatic chain;
 - **Arginine**, which has a positively charged **guanidino** group; and
 - **Histidine**, which has an **imidazole** group.
- ✓ Histidine is the only common amino acid having an ionizable side chain with a pK_a near neutrality.
- ✓ In many enzyme-catalyzed reactions, a His residue facilitates the reaction by serving as a **proton donor/acceptor**.

Negatively Charged (Acidic) R Groups

- The two amino acids having R groups with a net negative charge at pH 7.0 are:
 - **aspartate** and **glutamate**, each of which has a second carboxyl group.



II Classification based on Chemical Constitution

- ➡ **Small amino acids** – Glycine, Alanine
- ➡ **Branched amino acids** – Valine, Leucine, Isoleucine
- ➡ **Hydroxy amino acids** (-OH group) – Serine, Threonine
- ➡ **Sulfur amino acids** – Cysteine, Methionine
- ➡ **Aromatic amino acids** – Phenylalanine, Tyrosine, Tryptophan
- ➡ **Acidic amino acids and their derivatives** – Aspartate, Asparagine, Glutamate, Glutamine
- ➡ **Basic amino acids** – Lysine, Arginine, Histidine
- ➡ **Imino acid** - Proline

III. Biological or Physiological Classification

Based on the functional property of amino acids for the organism:

I. Essential

- ✓ Not synthesized in the body and
- ✓ Must be obtained from the diet to meet metabolic needs.
- ✓ There are nine essential amino acids.

II. Nonessential

- ✓ Not obtained from diet,
- ✓ Can be biosynthesized in adequate amounts within the organism.
- ✓ There are 11 nonessential amino acids

III. Conditionally essential

- ✓ can be synthesized within the organism but their synthesis is not in sufficient amounts due to illness or lack of necessary precursors
- ✓ Premature infants lack sufficient enzymes needed to create arginine

Essential Amino Acids

Histidine (His)^a

Isoleucine (Ile)

Leucine (Leu)

Lysine (Lys)

Methionine (Met)

Phenylalanine (Phe)

Threonine (Thr)

Tryptophan (Trp)

Valine (Val)

Nonessential Amino Acids

Alanine (Ala)

Arginine (Arg)^b

Asparagine (Asn)

Aspartic acid (Asp)

Cysteine (Cys)^b

Glutamic acid (Glu)

Glutamine (Gln)^b

Glycine (Gly)^b

Proline (Pro)^b

Serine (Ser)

Tyrosine (Tyr)^b

^a Histidine was once thought to be essential only for infants. It is now known that small amounts are also needed for adults.

^b These amino acids can be “conditionally essential” if there are either inadequate precursors or inadequate enzymes available to create these in the body.

III- Metabolic classification:

- According to metabolic or degradation products of amino acids they may be:

1- Ketogenic amino acids: give ketone bodies .

- ➡ **Lysine** and **Leucine** are the only pure ketogenic amino acids.

2- Mixed ketogenic and glucogenic amino acids:

- ➡ which give both ketonebodies and glucose.

- ➡ These are: isoleucine, phenyl alanine, tyrosine and tryptophan.

3- Glucogenic amino acids: give glucose.

- ➡ They include the rest of amino acids.

- ➡ These amino acids by catabolism yields products that enter in glycogen and glucose formation.

Rare Amino Acids or Unusual Amino Acids

These are the amino acids that are not found in proteins but play important roles in metabolism. **Examples**

- ① Ornithine, citrulline and arginino succinic acid are parts of urea cycle.
- ② β -alanine is part of co-enzyme A.
- ③ Taurine is part of bile acids.
- ④ γ -aminobutyric acid is a neurotransmitter.
- ⑤ Mono- and di-iodotyrosine are precursors of thyroxine.
- ⑥ Pantothenic acid is a water-soluble vitamin.
- ⑦ Homoserine is an intermediate of methionine catabolism.
- ⑧ Homocysteine is an intermediate of methionine catabolism

Physico-chemical properties of amino acids

1. Amino Acids act as Buffers
2. Amino Acids Carry a Net Charge of Zero at a Specific pH.
 - Zwitterions predominate at pH values between the pK_a values of amino and carboxyl group. At this point, the net charge is zero
3. Amino acids does not migrate in electric field
4. Amino acids have high melting points, generally over 200 °C.
5. Amino acids are more soluble in water than they are in ether, dichloromethane, and other common organic solvents.
6. Amino acids have much larger dipole moments than simple amines or simple acids.
7. Amino acids are less acidic than most carboxylic acids and less basic than most amines.

Amino Acids Can Act as Acids and Bases

- ➡ When an amino acid is dissolved in water, it exists in solution as the dipolar ion, or zwitterion (German for “hybrid ion”).
- ➡ At physiological pH, the amino group is protonated and the carboxylic acid group is deprotonated.
- ➡ They are amphoteric .
- ➡ **Isoelectric pH:** It is the pH at which the net charge of an amino acid is zero or when the number of positive charges are equal to number of negative charges.

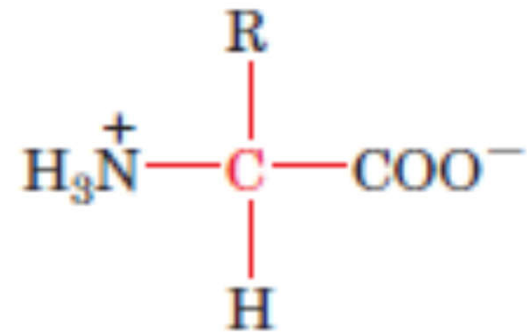


Figure: A dipolar amino acid.

Chemical properties of amino acids

1- Reactions due to COOH group:

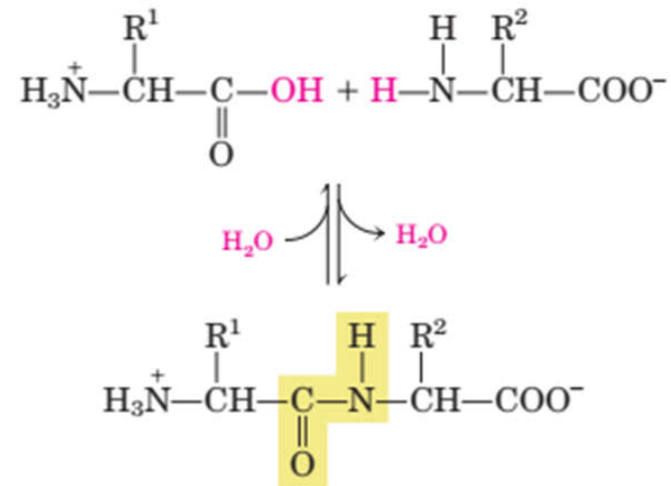
- ➡ Salt formation with alkalis, ester formation with alcohols, amide formation with amines and decarboxylation

2- Reactions due to NH₂ group:

- ➡ deamination and reaction with ninhydrin reagent.
- ➡ Ninhydrin reagent reacts with amino group of amino acid yielding blue colored product.
- ➡ The intensity of blue color indicates quantity of amino acids present.

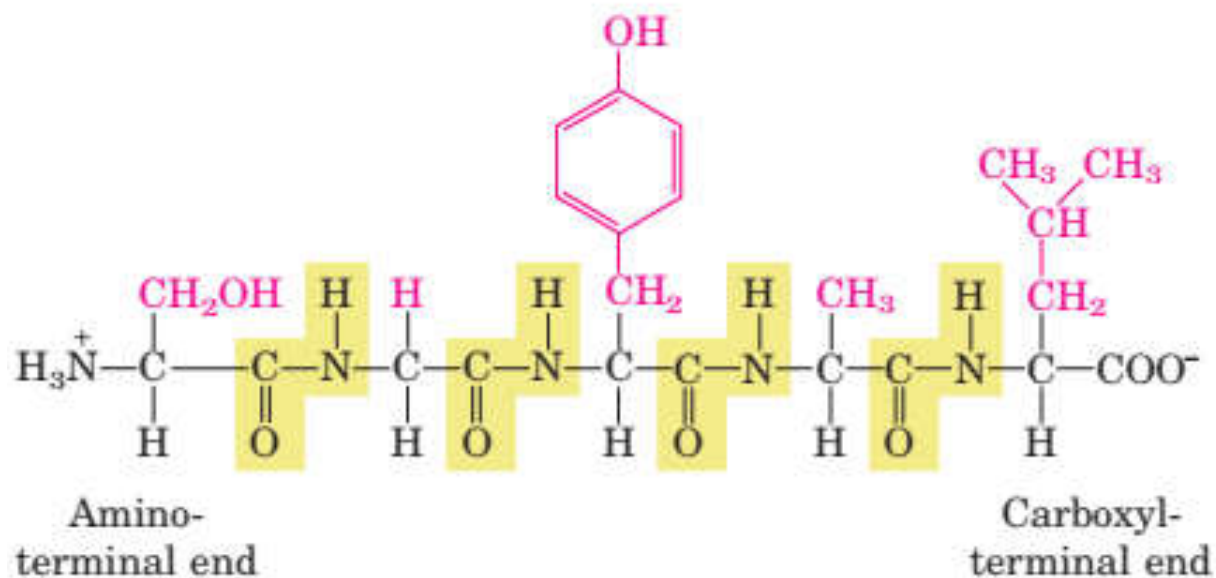
Peptides and Proteins

- ✓ Peptides consist of 2 or more amino acid residues linked by peptide bond.
- ✓ Proteins are macromolecules with a backbone formed by **polymerization of amino acids** in a **polyamide structure**.
- ✓ These amide bonds in protein, known as **peptide bonds** formed by linkage of **α - carboxyl group** of one amino acid with **α - amino groups** of the next amino acid.



Each peptide chain:

- ▶ starts on the left side by free amino group of the first amino acid enter in chain formation and termed (N- terminus).
- ▶ ends on the right side by free COOH group of the last amino acid and termed (C-terminus).



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- During the formation of a peptide bond, a molecule of water is eliminated as shown below:-

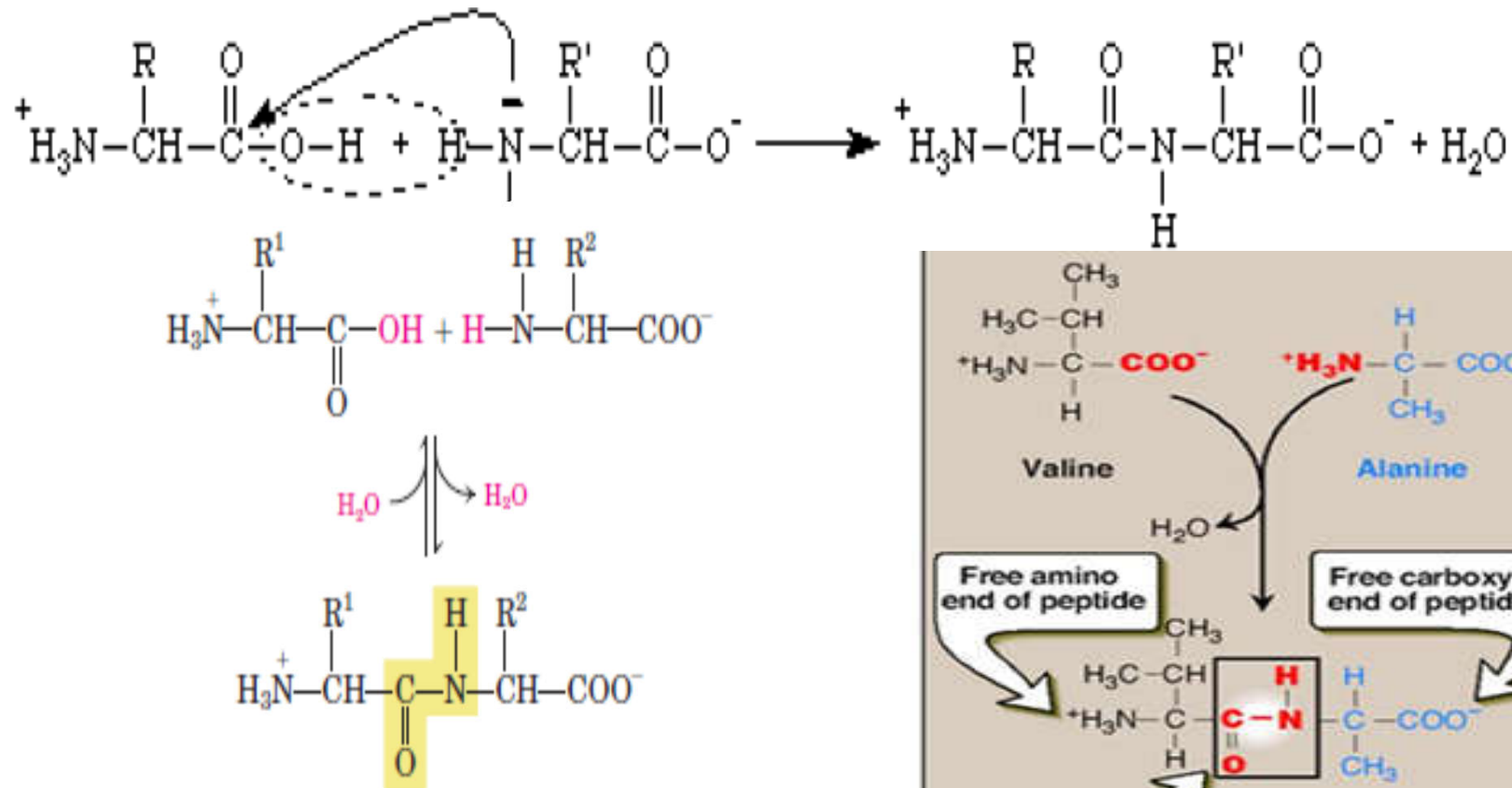
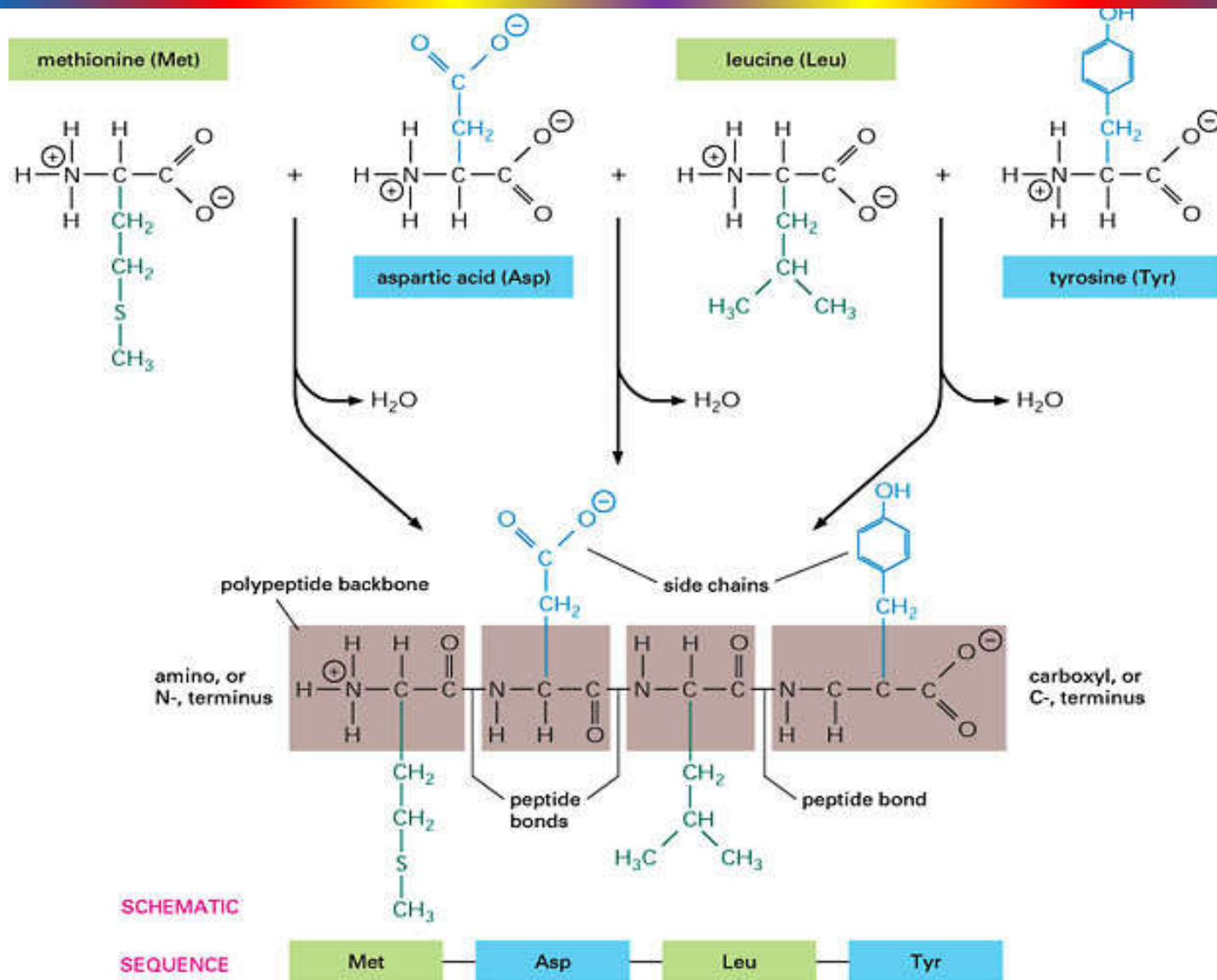


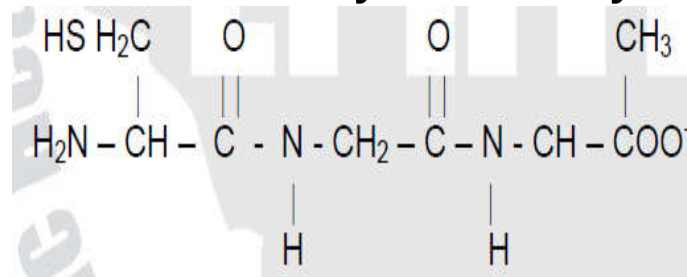
Fig : Peptide bond synthesis

...Cont'd



✓ A peptide chain consisting of:

1. **Dipeptide:** *two amino acids joined by one peptide bond*
2. **Tripeptide:** *Three amino acids linked by two peptide bonds (e. g*
Glutathione is formed from Cysteine, Glycine and Alanine.



- 3- **Octapeptides:** (*8 amino acids*): E.g.: Two hormones; oxytocin, vasopressin (ADH) and Angiotensin II.
- 4- **Polypeptides:** 10- 50 amino acids: e.g. Insulin hormone

PROTEINS

- The word protein is derived from Greek word, *proteious* meaning **primary**. So, proteins are the **major components** of any living organism.
- Proteins are natural substances with **high molecular weights** ranging from 5,000 to many millions.
- Besides **Carbon, Hydrogen and Oxygen**, they also contain **Nitrogen**, and sometimes, **Sulfur** and **Phosphorous**.
- Proteins are most important **constituent of cell membranes and cytoplasm**.
- Muscle and blood plasma also contain certain specific proteins.

- **Protein containing foods** are essential for living organism, because:
 - Protein is the most important biological molecules in **building up** and **maintenances** of the **structure of body**,
 - Giving as **much energy as carbohydrates** in the course of metabolism in the body.
- Proteins are the molecular instruments in which **genetic information is expressed**; **hormones, antibodies, transporters, muscle, the lense protein, antibiotics, mushroom poisons**, and a myriad of other substances having distinct biological activities are derived.

Function of Proteins

➡ Catalysis:

- ☞ enolase (in the glycolytic pathway)
- ☞ DNA polymerase (in DNA replication)

➡ Transport:

- ☞ hemoglobin (transports O₂ in the blood)
- ☞ lactose permease (transports lactose across the cell membrane)

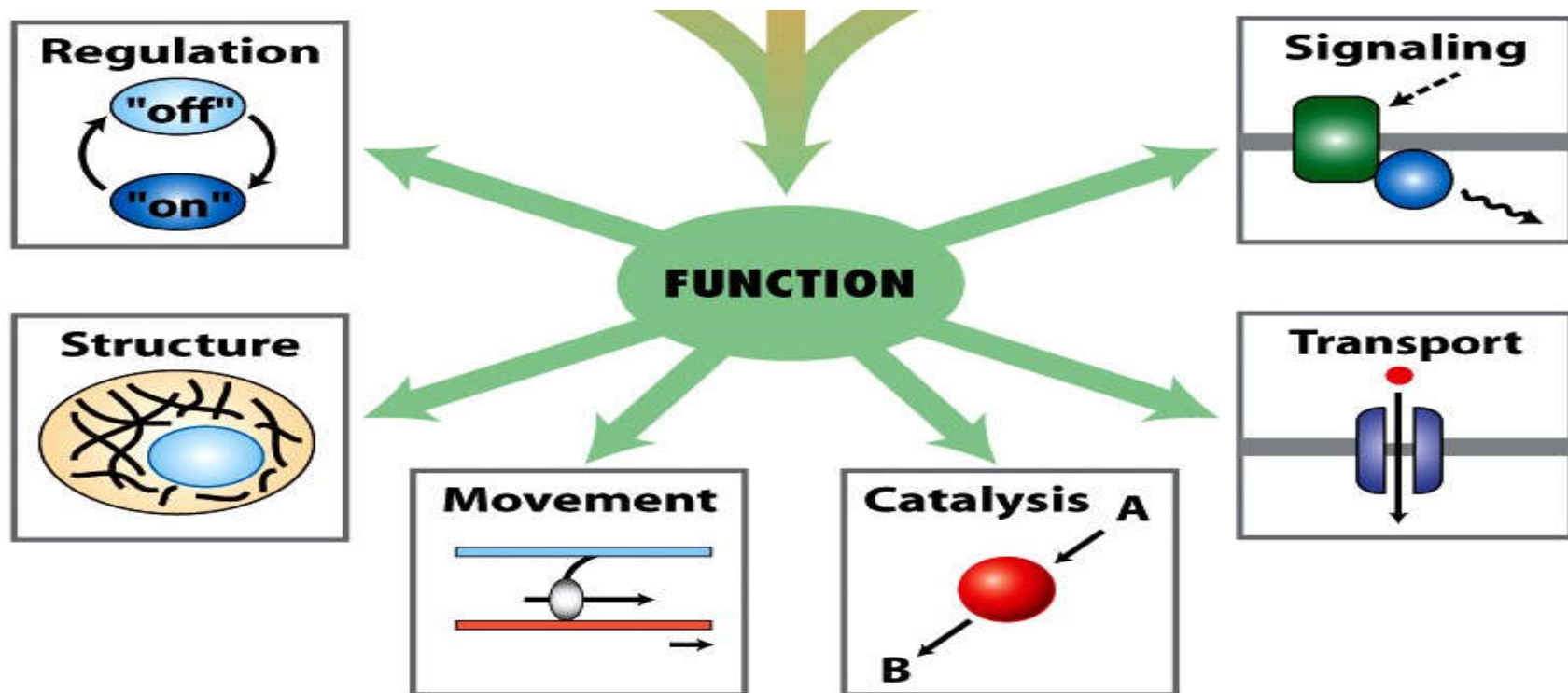
➡ Structure:

- ☞ collagen (connective tissue)
- ☞ keratin (hair, nails, feathers, horns)

➡ Motion:

- ☞ myosin (muscle tissue)
- ☞ actin (muscle tissue, cell motility)

...Cont'd



...Cont'd

Class of protein	example	Function example
Structural protein	Collagen, keratin	Strengthen tendons, skin , hair, nails
Enzymes	DNA polymerase	Replicates and repairs DNA
Transport protein	Hemoglobin	Transports oxygen to the cells
Contractile proteins	Actins, myosine	Cause contraction of muscles
Protective proteins	Antibodies	Complex with foreign proteins
Hormones	Insulin	Regulates glucose metabolism
Toxins	Snake venoms	Incapacitate prey

Classifications of Proteins

- **There is no universally accepted classification system.**
- **But , proteins may be classified** on the basis of their:
 - ☞ Composition,
 - ☞ Solubility,
 - ☞ Shape,
 - ☞ Biological function and
 - ☞ Three dimensional structure.

I. Composition

A. Simple proteins

- ✓ **Yields only amino acids and no other major organic or inorganic hydrolysis products**
 - ✓ E.g. Albumin, Trypsin, Chymotrypsin, pepsin, insulin, soyabean trypsin inhibitor and ribonuclease

B. Conjugated Proteins

- Yields amino acids and other organic and inorganic components.
 - **Nucleoprotein** (a protein containing Nuclei acids)
 - **Lipoprotein** (a protein containing lipids)
 - **Phosphoprotein** (a protein containing phosphorous)
 - **Metalloprotein** (a protein containing metal ions of Fe^{2+})
 - **Glycoprotein** (a protein containing carbohydrates)

II. Overall Shape

A. Fibrous proteins

- In these protein, the molecule are constituted by **several coiled cross-linked polypeptide chains**,
- They are **insoluble in water** and **highly resistant to enzyme digestion**.
- A few sub groups are:

1. Collagens:

- ✍ The major protein of the **connective tissue**,
- ✍ insoluble in water, acids or alkalis.
- ✍ But they are convertible to **water-soluble gelatin**, easily digestible by enzymes.

2. Elastins:

- ✓ Present in tendons, arteries and other elastic tissues,
- ✓ not convertible to gelatin.

3. Keratins:

- ✓ protein of hair, nails etc.

B. Globular proteins:

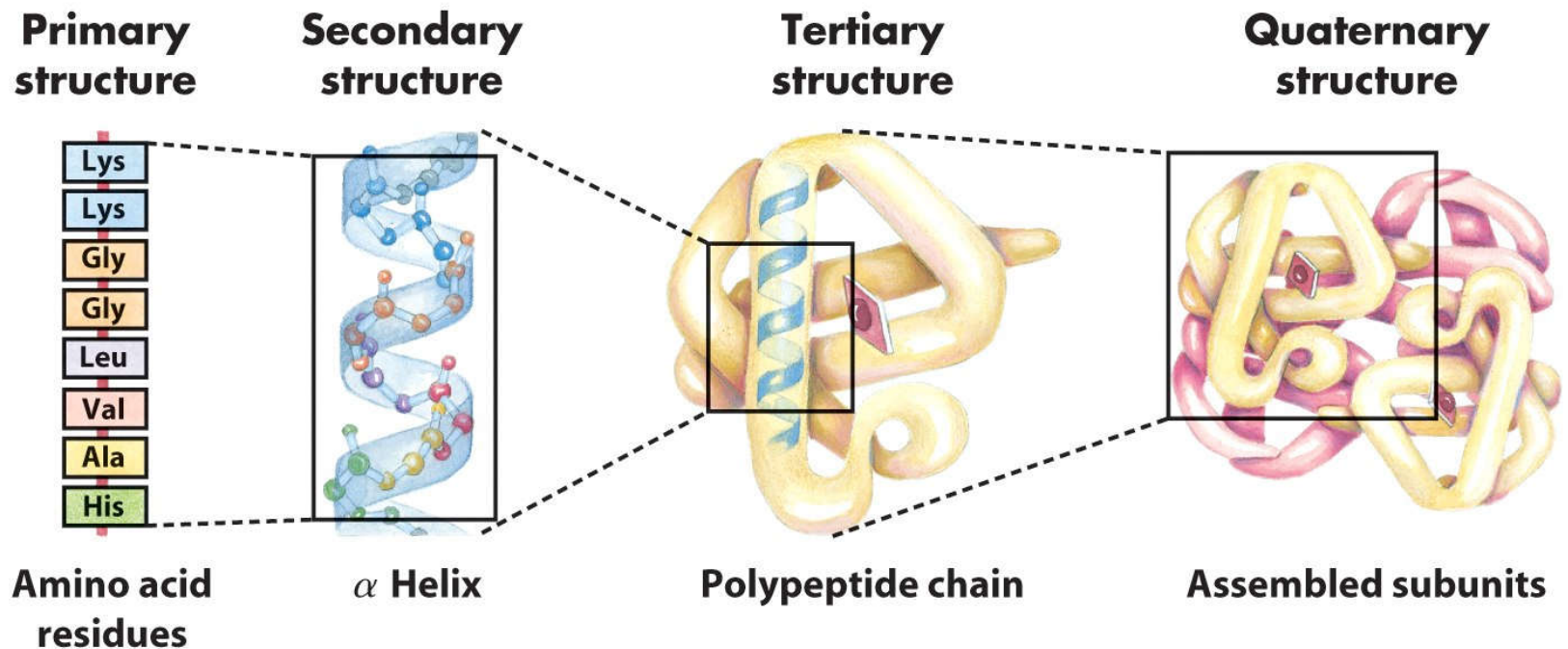
- ✍ Polypeptide chain(s) of these proteins are folded into compact globular (Spherical) shape.
- ✍ These are globular or ovoid in shape and soluble in water
- ✍ Constitute the enzymes, oxygen carrying proteins, hormone, Albumin, globulins and histones.

IV. On their Biological Functions:

- **Proteins** are **sometimes described** as the "**workhorses**" of the cell.
 - **Enzymes**: kinases, transaminases etc.
 - **Storage proteins**: myoglobin, ferritin
 - **Regulatory proteins**: peptide hormones, DNA binding proteins
 - **Structural protein**: collagen, proteoglycan
 - **Protective proteins**: blood clotting factors, Immunoglobins,
 - **Transport protein**: Hemoglobin, plasma lipoproteins
 - **Contractile or motile Proteins**: Actin, tubulin

Molecular Structures of Proteins

- ✓ Primary Structure
 - ✓ Secondary structure
 - ✓ Tertiary structure
 - ✓ Quaternary structure
- Spatial structure



I. Primary Structure

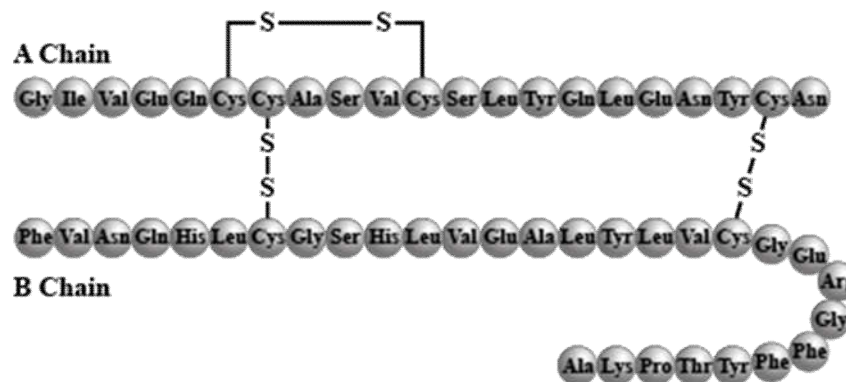
- ✓ The **primary structure** of proteins is defined as a linear sequence of amino acids joined together by peptide bonds.

Pro-Val-Thr-Gly-Lys-Cys-Glu

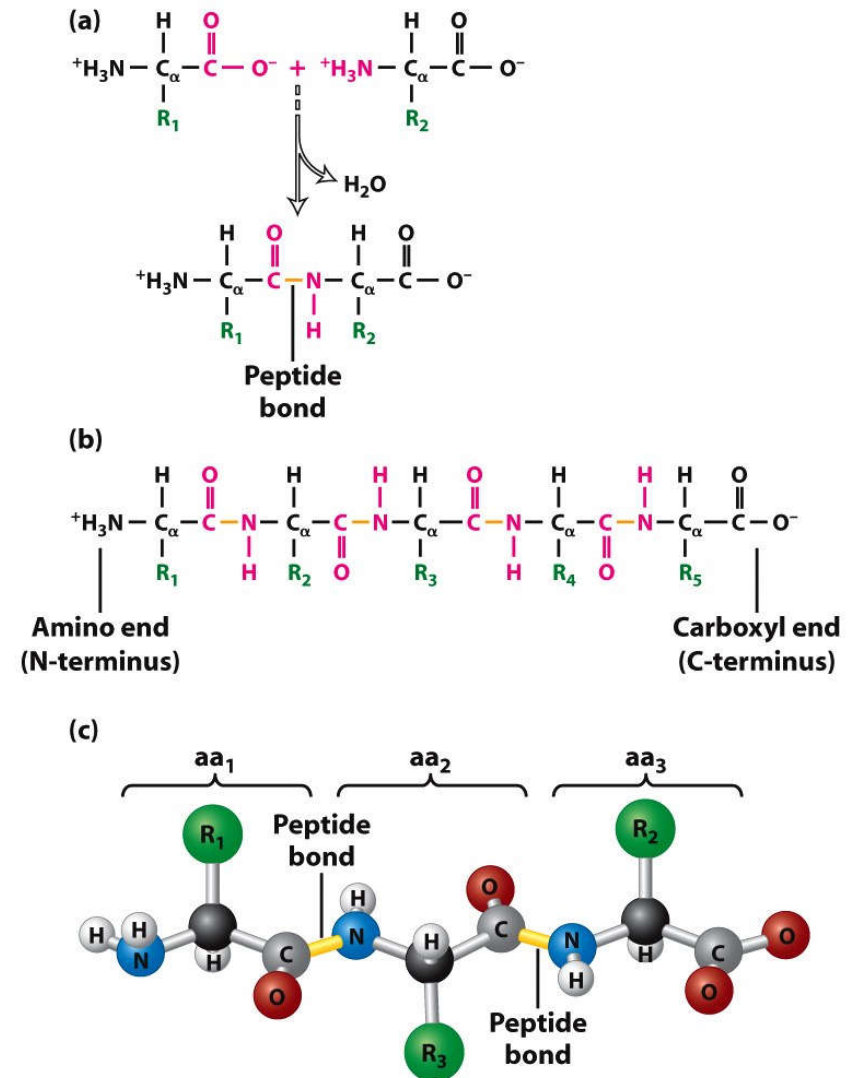
- ✓ **Peptide bonds** and **disulfide bonds** are responsible for maintaining the primary structure.
- ✓ The amino acid composition of a peptide chain has a profound effect on its physical and chemical properties of proteins.
- ✓ All the properties of the protein are determined, directly or indirectly, by the primary structure.

☞ The primary structure of protein is required to understand:

- Its structures and mechanisms of action
- Its biosynthesis including the post-translational modification of its structure
- Its relationship to other proteins.



Amino acids sequence of insulin



II. The Secondary Structure

- ✓ Folding of polypeptide chain along its long axis is called as secondary structure of protein.
- ✓ Folding of polypeptide chain can be ordered, disordered or random
- ✓ The secondary structure of a protein is a local spatial structure of a certain peptide segment,.
- ✓ Secondary structure is often referred as conformation.
- ✓ **H-bonds** are responsible for stabilizing the secondary structure.
- ✓ Four common types of secondary structure are:
 - ❖ α -helix
 - ❖ β -pleated sheet
 - ❖ β -turn
 - ❖ Random coil

A. α -Helix

- ✓ Is the most common polypeptide helices that are found in nature.
- ✓ It is a **spiral structure** consisting of a tightly packed and coiled polypeptide backbone core,
- ✓ The **side chains** of the amino acids extend **outward** from the central axis to avoid interfering sterically with each other.

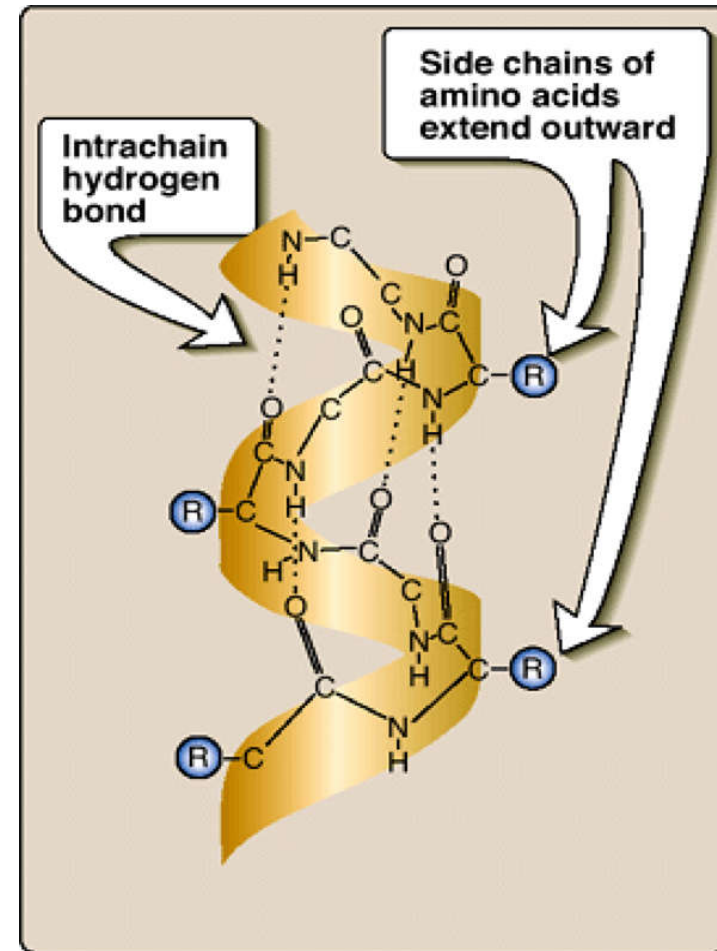
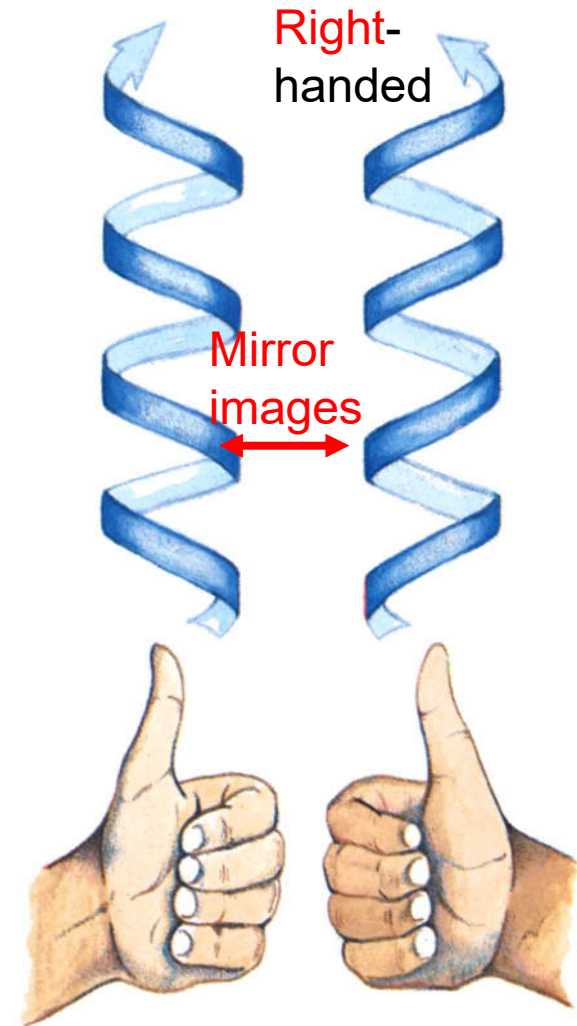


Fig. 2. α -Helix showing peptide backbone.

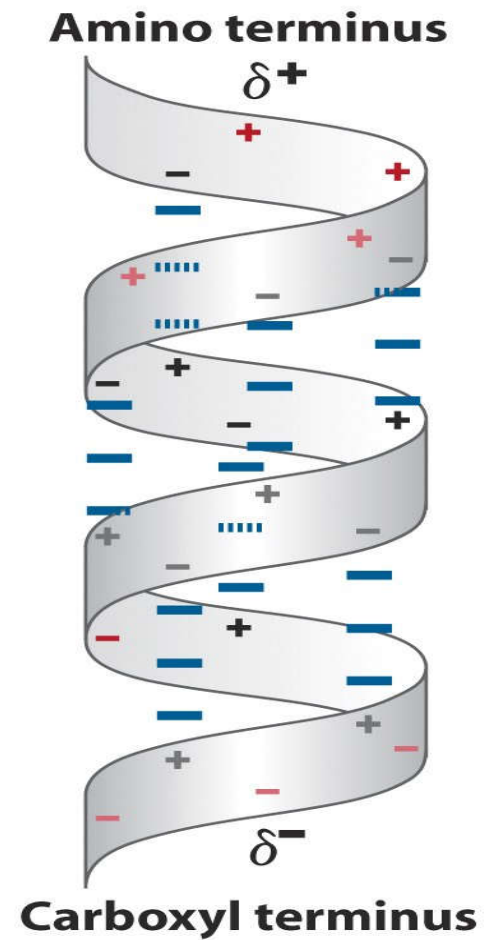
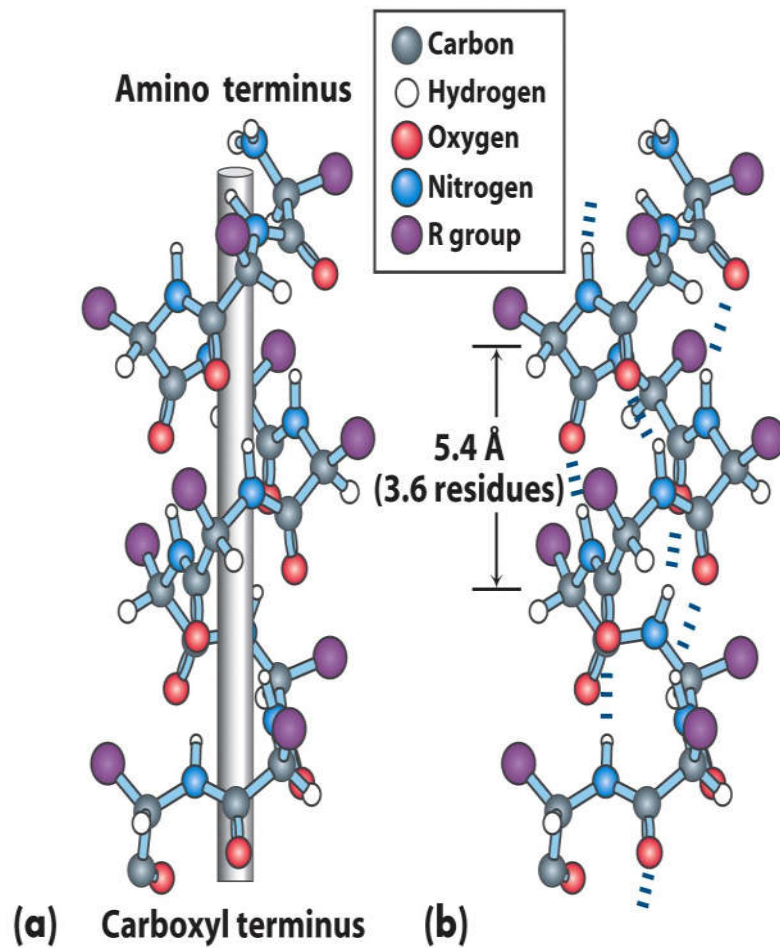
- ☞ Only extended right-handed α -helices are found in proteins.
- ☞ The tendency of a given segment of a polypeptide chain to fold up as an α helix depends on the identity and sequence of amino acid residues within the segment.
- ☞ Framework for structural proteins such as nails, hair and skin



1. Hydrogen bonds

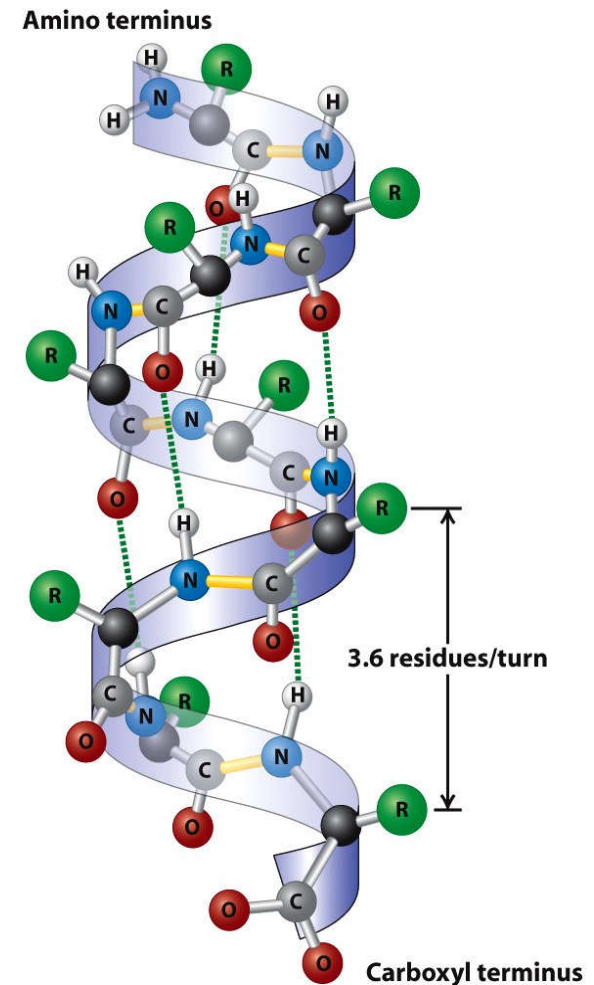
- ✓ **An α -helix is stabilized by extensive hydrogen bonding between the peptide-bond carbonyl oxygens and amide hydrogens that are part of the polypeptide backbone.**
- ✓ The hydrogen bonds extend up and are parallel to the spiral from the carbonyl oxygen of one peptide bond to the -NH- group of a peptide linkage four residues ahead in the polypeptide.
- ☞ Stabilized by H-bonds between $C=O(n)$ and $N-H(n+4)$. $[-N-H \cdots O=C-]$ (H-bonds occur between residues located in the n and $n + 4$ positions relative to one another).

...Cont'd



2. Amino acids per turn

- ✓ In the α helix, the backbone adopts a cylindrical spiral structure.
- ✓ Each turn of an α -helix contains 3.6 amino acids.
- ✓ Thus, amino acid residues spaced three or four residues apart in the primary sequence are spatially close together when folded in the α -helix.

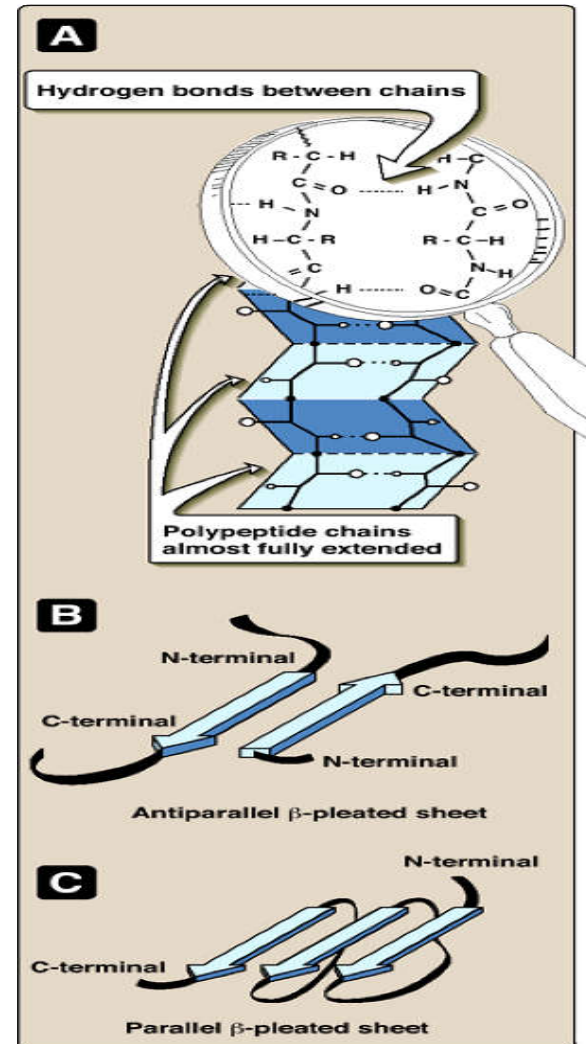


3. Amino acids that disrupt an α -helix

- ✓ Proline disrupts an α -helix because its secondary amino group is not geometrically compatible with the right-handed spiral of the α -helix.
- ✓ Instead, it inserts in the chain, which interferes with the smooth, helical kink structure.
- ✓ Large numbers of charged amino acids (E.g. glutamate, aspartate, histidine, lysine, or arginine) also disrupt the helix by forming ionic bonds, or by electrostatically repelling each other.
- ✓ Amino acids with bulky side chains, such as tryptophan, or amino acids, such as valine or isoleucine, that branch at the β -carbon can interfere with formation of the α -helix if they are present in large numbers. WHY?

β -Pleated Sheet

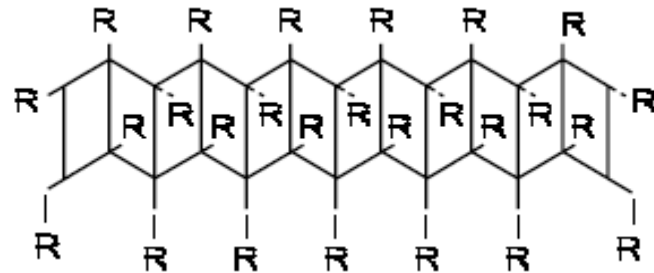
- ✓ Core of many proteins is the β sheet
- ✓ Form rigid structures with the H-bond
- ✓ In β -sheets the hydrogen bonds are perpendicular to the polypeptide backbone
- ✓ Can be of 2 types
- ✓ **Anti-parallel** – run in an opposite direction of its neighbor (B)
- ✓ **Parallel** – run in the same direction with longer looping sections between them (C)



Structure of a β -sheet

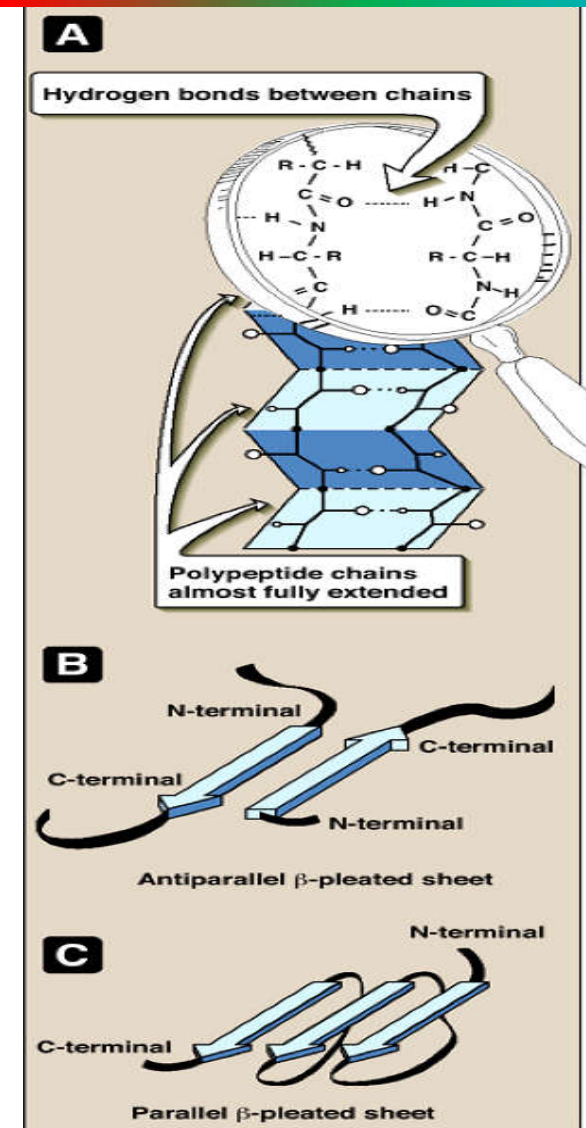
β -Pleated Sheet Features

- ✓ In β -pleated sheet, the polypeptide chain is **fully extended**.
- ✓ In β -pleated sheet, polypeptide chains line up **side by side to form sheet**. The **side chains are above or below the plane of the sheet**.



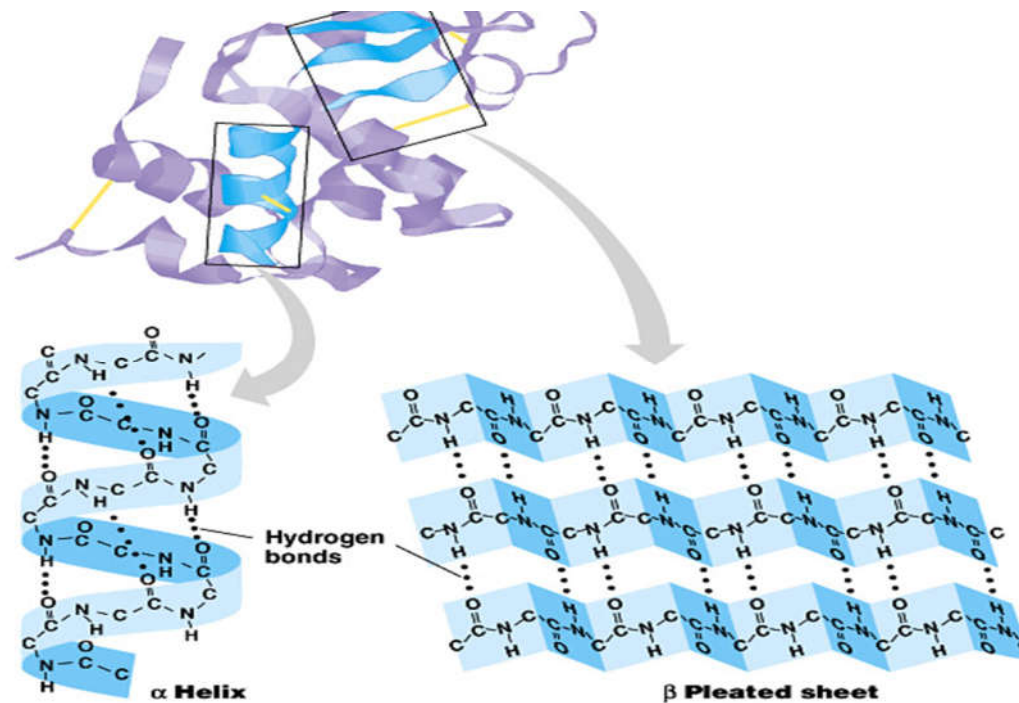
- ✓ From **2 to 5**, adjacent strands of polypeptides may combine and form **these structure**.
- ✓ When the adjacent polypeptide chains run in **same direction** (N to C terminus) the structure is termed as **parallel β -pleated sheet**.

- ➔ When the adjacent polypeptide chains run in **opposite** direction the structure is termed as **anti-parallel β -pleated sheet**.
- ➔ The **β -pleated** sheet is stabilized by **inter chain hydrogen bonds**.
- ➔ **β -keratin** contains **anti parallel β -pleated sheet**.
- ➔ Both parallel and anti-parallel β -pleated sheet occur in other proteins.
- ➔ **Amyloid protein** present in **Alzheimer's disease** has anti parallel β -pleated sheet. It accumulates in the CNS.



Comparison of a β -sheet and an α -helix

- Unlike the α -helix, β -sheets are composed of **two or more peptide chains** (β -strands), or segments of polypeptide chains, which are almost **fully extended**.
- **NB** : In β -sheets the hydrogen bonds are **perpendicular** to the polypeptide backbone.

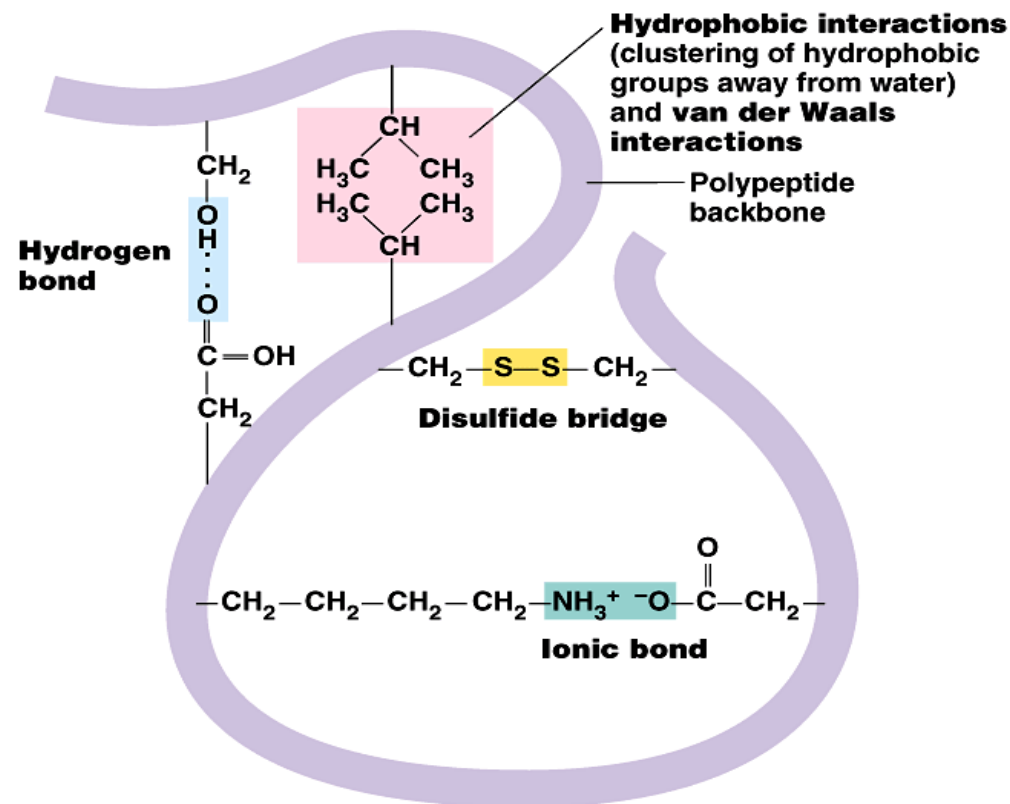


C. Tertiary Structure

- ➡ The overall 3D arrangement of all atoms in a protein
- ➡ It refers both to the folding of domains and to the final arrangement of domains in the polypeptide.
- ➡ **It is the 3D, folded and biologically active conformation of a protein.** e.g. Myoglobin
- ➡ **The structure** reflects the **overall shape of the molecule.**
- ➡ The term domain is used to represent the basic unit of protein structure.
- ➡ The amino acid sequence contains enough information to specify tertiary structure.

- ✓ Five types of interactions stabilize the protein tertiary structure.
 - ❖ Hydrophobic interaction
 - ❖ Ionic interaction
 - ❖ Hydrogen bond
 - ❖ Van der Waals interaction (e.g. proline)
 - ❖ Disulfide bond (e.g. **cysteine-cysteine** disulphide bridge)

1. The weak interactions include: **Hydrogen bonds** among polar side chains, **Ionic bonds** between charged R groups (basic and acidic amino acids), **Hydrophobic interactions** among hydrophobic (non polar) R groups.



2. **Strong covalent bonds** include **disulfide bridges**, that form between the sulfhydryl groups (SH) of cysteine monomers, stabilize the structure.

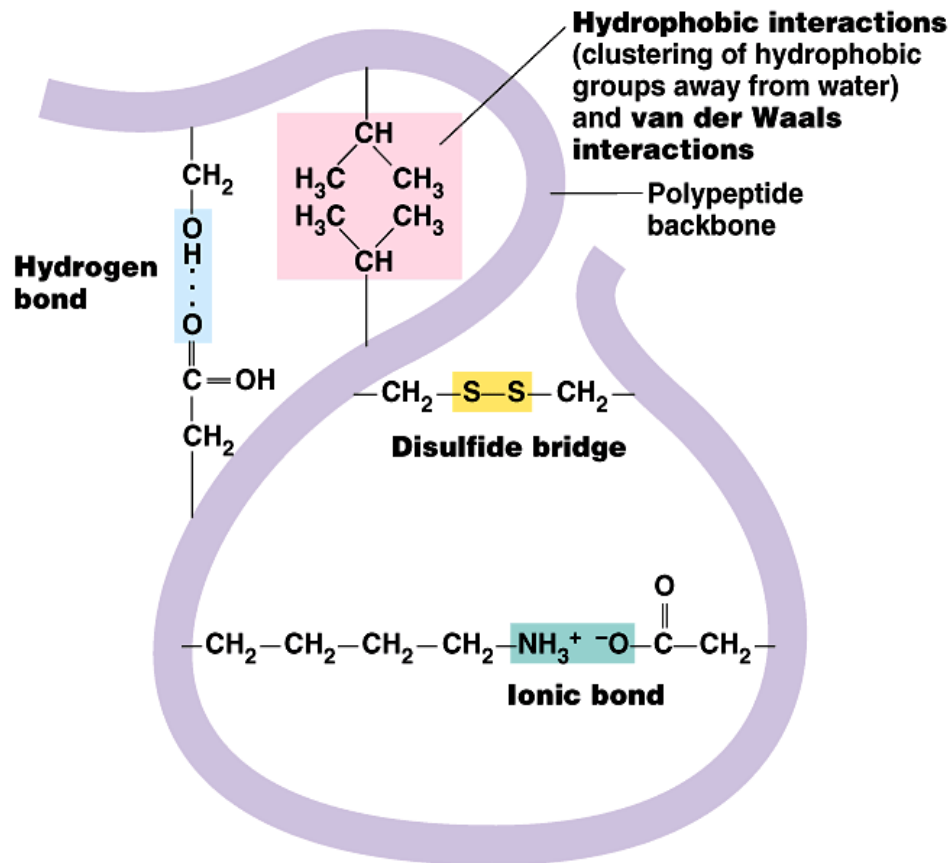


TABLE 4-2 Approximate Amounts of α Helix and β Conformation in Some Single-Chain Proteins

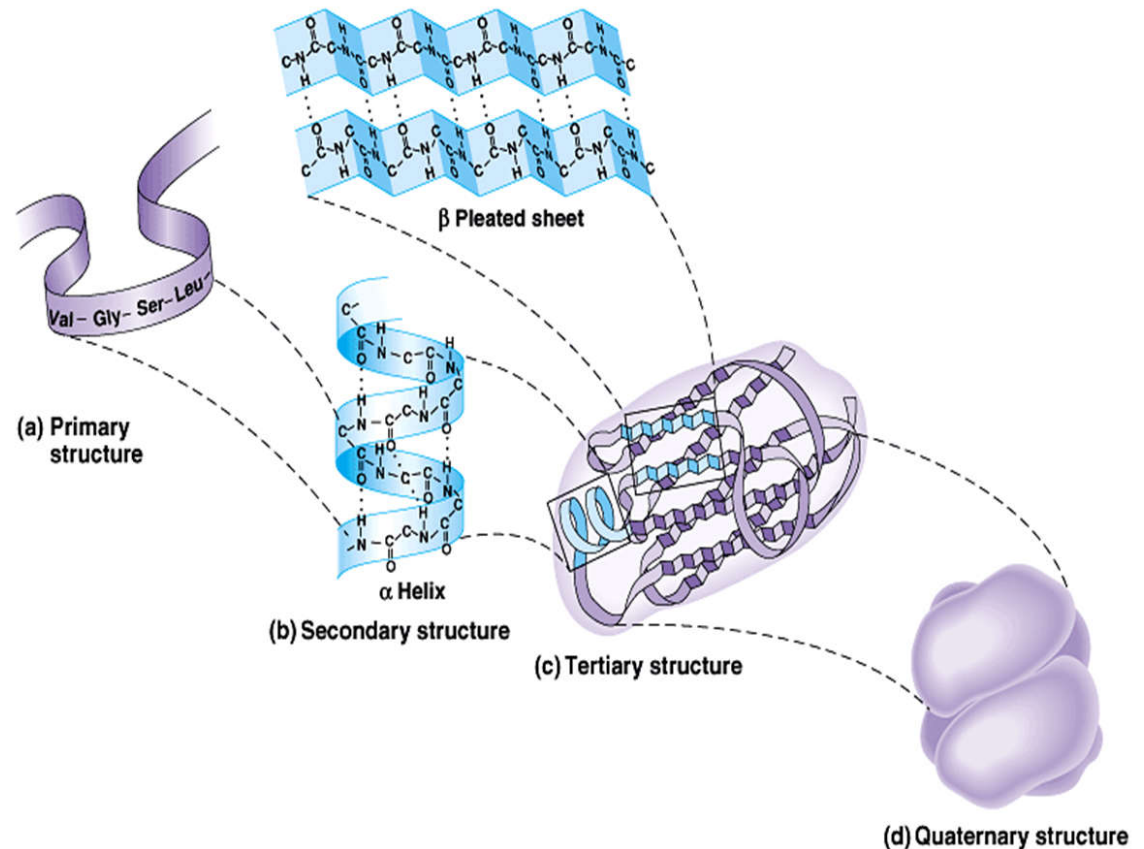
<i>Protein (total residues)</i>	<i>Residues (%)*</i>	
	<i>α Helix</i>	<i>β Conformation</i>
Chymotrypsin (247)	14	45
Ribonuclease (124)	26	35
Carboxypeptidase (307)	38	17
Cytochrome c (104)	39	0
Lysozyme (129)	40	12
Myoglobin (153)	78	0

Source: Data from Cantor, C.R. & Schimmel, P.R. (1980) *Biophysical Chemistry, Part I: The Conformation of Biological Macromolecules*, p. 100, W. H. Freeman and Company, New York.

D. Quaternary Structure

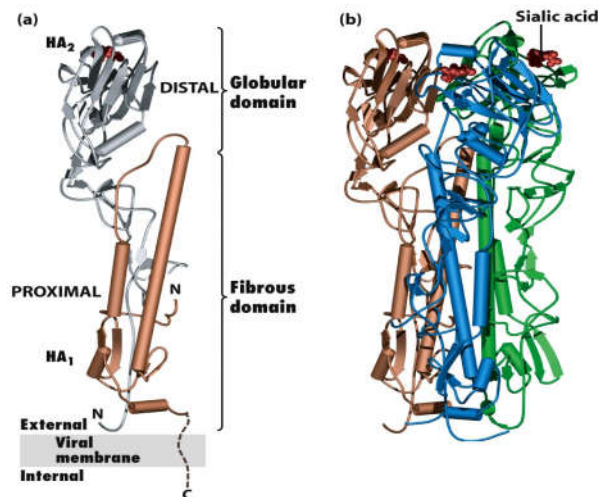
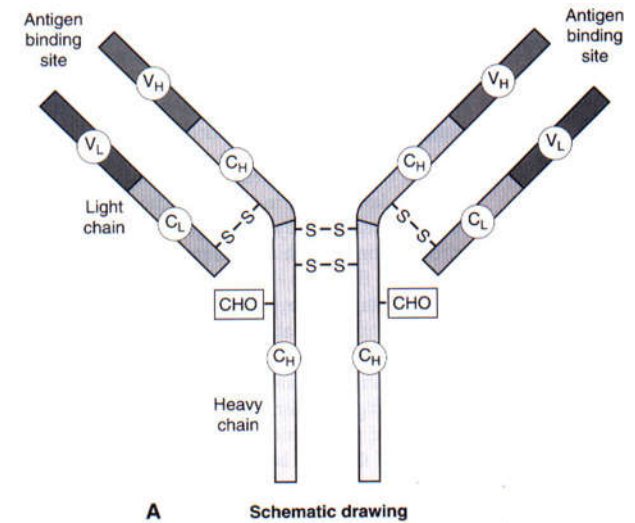
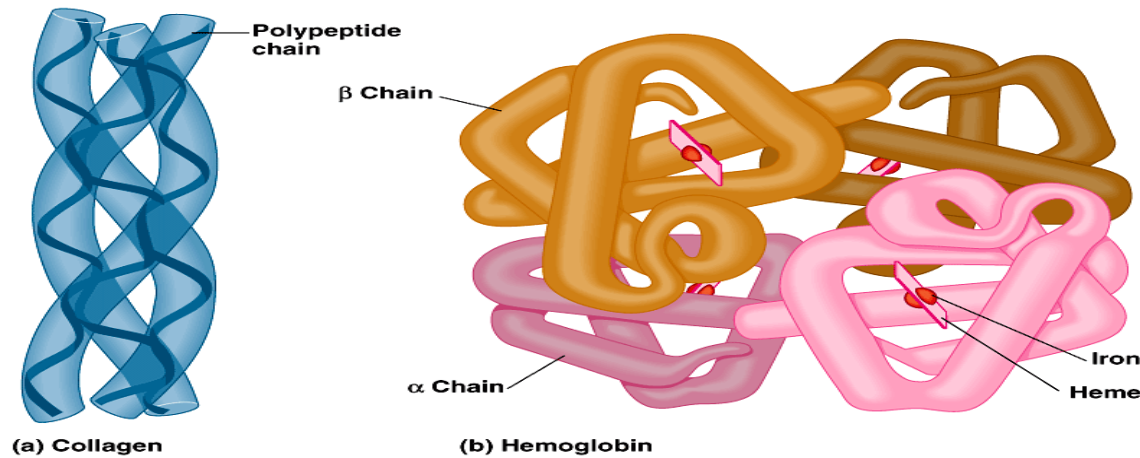
✓ Refers to a complex or an assembly of two or more separate peptide chains that are held together by non-covalent.

☞ Refers to the number of subunits, their relative positions, and contacts between the individual monomers in a multimeric protein.



...Cont'd

E.g. **Collagen** (trimeric), **Insulin** (dimeric) and **Hemoglobin** (tetrameric)



Trimetric hemagglutinin
surface protein of
influenza virus

Tests of Proteins

- The proteins give several reactions which are often useful to identify the nature of amino acids present in them

A. Biuret reaction:

- Biuret test is answered by compounds containing two or more CONH group or peptide bonds.
- The principle of Biuret test is conveniently used to detect presence of proteins in biological fluids (oxamide) $\text{CONH}_2 + \text{Biuret}$.

B. Ninhydrin Test:

- This test is used to detect presence of α - amino acids in a powder.
- It is positive for proteins, peptide including ammonia.
- The triketohydrantin hydrate forms a complex with amino or carboxylic group of amino acids or other primary amine developing a blue colour.

Protein Folding

- ✓ Proteins are truly amazing machines: before they do their work, they assemble themselves! This self-assembly is called "folding."
- ✓ The process by which a protein goes from being an unfolded polymer with no activity to a uniquely structured and active protein.
- ✓ Proceeds through a definite series of steps or a Pathway.
- ✓ A protein reaches a complex marginal stable 3D structure, providing unique environments and orientations of functional groups that give them their special functional properties.
- ✓ Protein misfolding has been implicated in many human diseases (Alzheimer's, Parkinson's)

👉 Two terms are used to describe a folding pattern

① Motif and

② Domain.

👉 A motif is a recognizable folding pattern involving two or more elements of secondary structure and the connections between them.

👉 Motifs can be simple, such as:

① β - α - β loop motif;

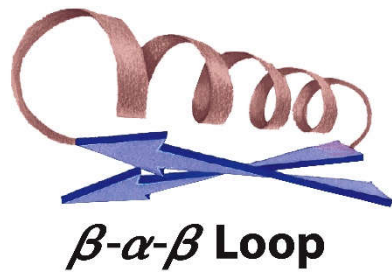
② β barrel motif .

👉 In some cases a single large motif may comprise the entire protein.

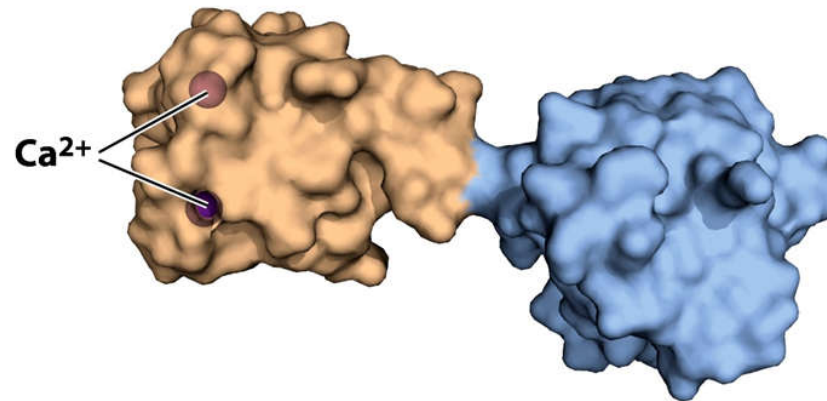
👉 Motif and fold can be used interchangeably, however the term fold is more commonly applied to describe more complex folding patterns.

👉 For example, the distinctive arrangement of eight α helices in myoglobin is found in all globins and is called the globin fold.

...Cont'd



- ♠ **Domains** are independently folding and functionally specialized tertiary structure units within a protein.
- ♠ Each domain can appear as a distinct globular lobe region as is true for the calcium binding protein of muscle, troponin C .



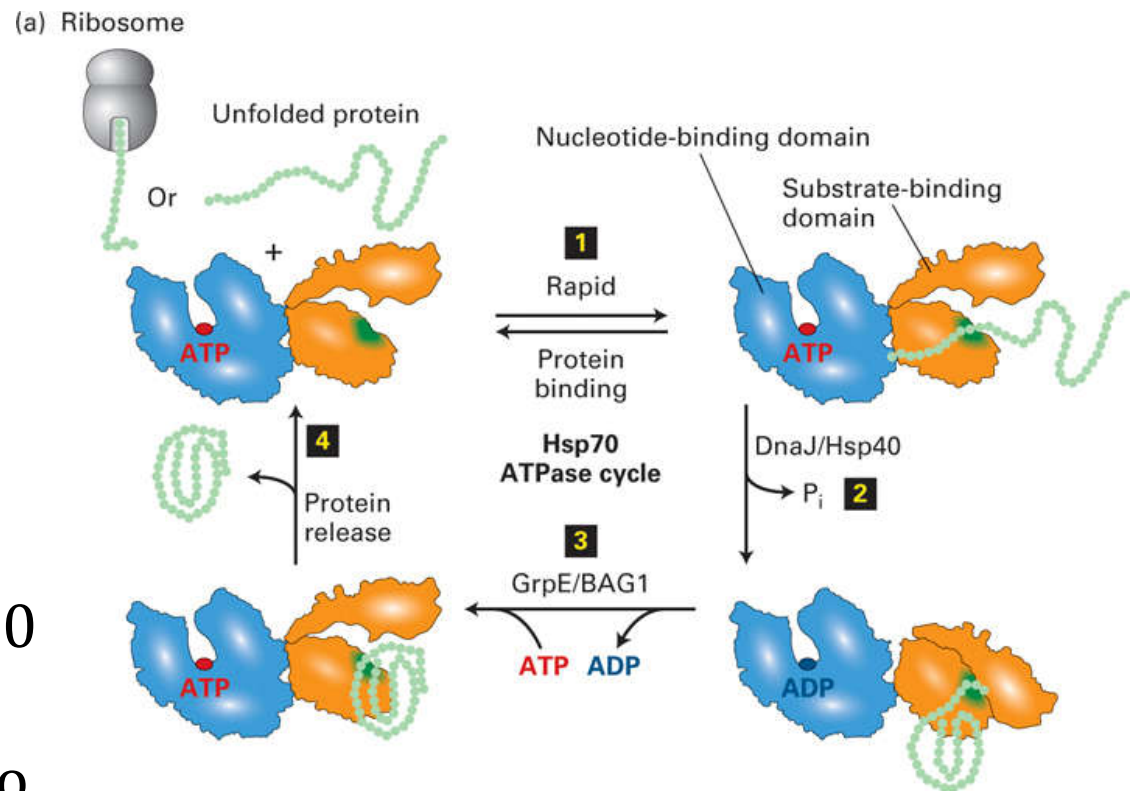
- ♠ The two or more domains of a protein often have different functions, which are coordinated in the overall function of the protein.
- ♠ Small proteins such as myoglobin usually only have one domain.

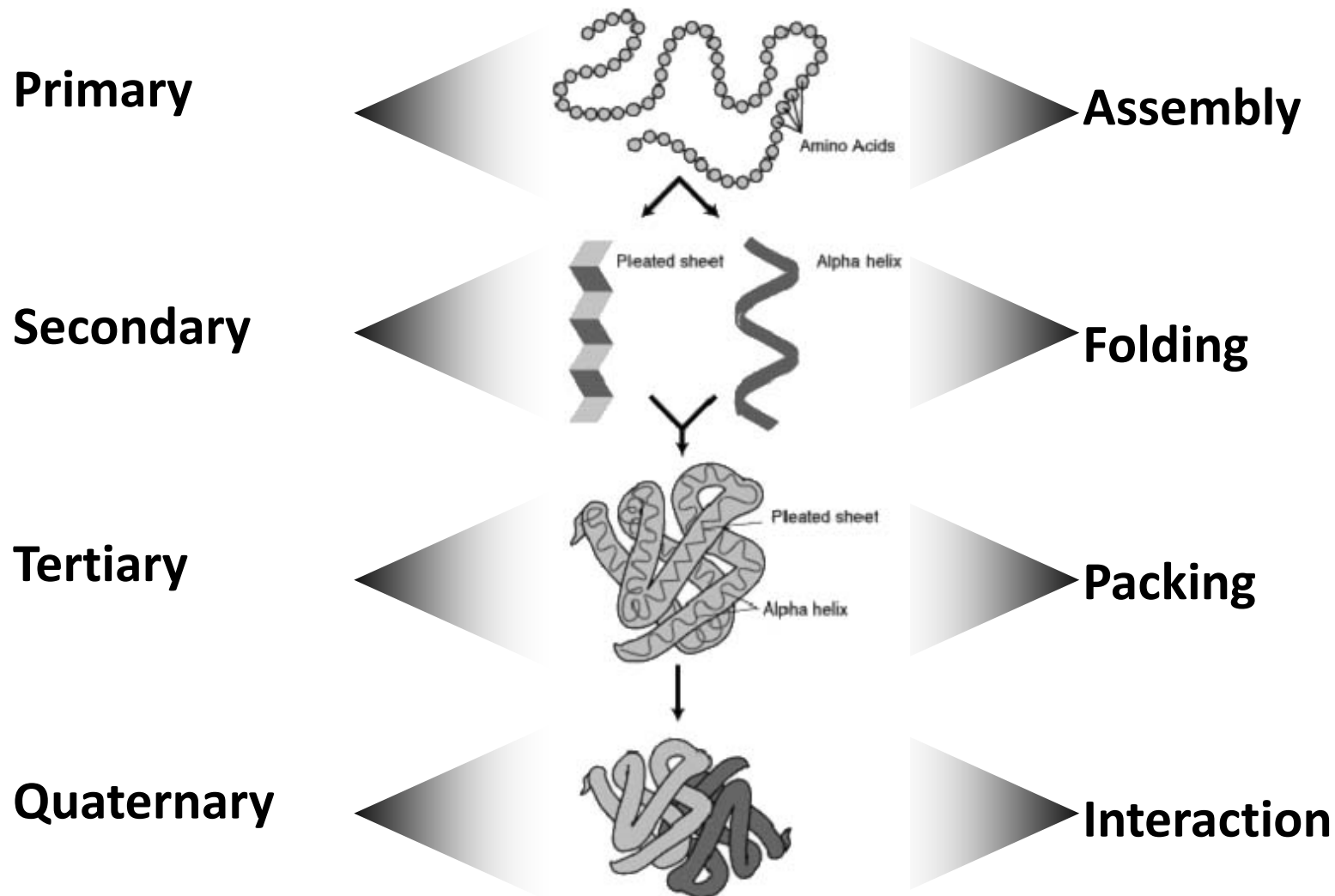
- Cells have developed molecules that catalyze protein folding called chaperones.
 - ☞ Help protein to fold correctly
 - ☞ Can limit regions available for intermolecular interaction, by masking certain regions of the protein.
 - ☞ Prevent folding until the entire protein is synthesized.
 - ☞ Increase during heat stress, are also known as heat shocked protein.
 - ☞ Act catalytically to speed up the process of protein folding by lowering the activation barrier between misfolding and native state.

Chaperone-assisted Protein Folding

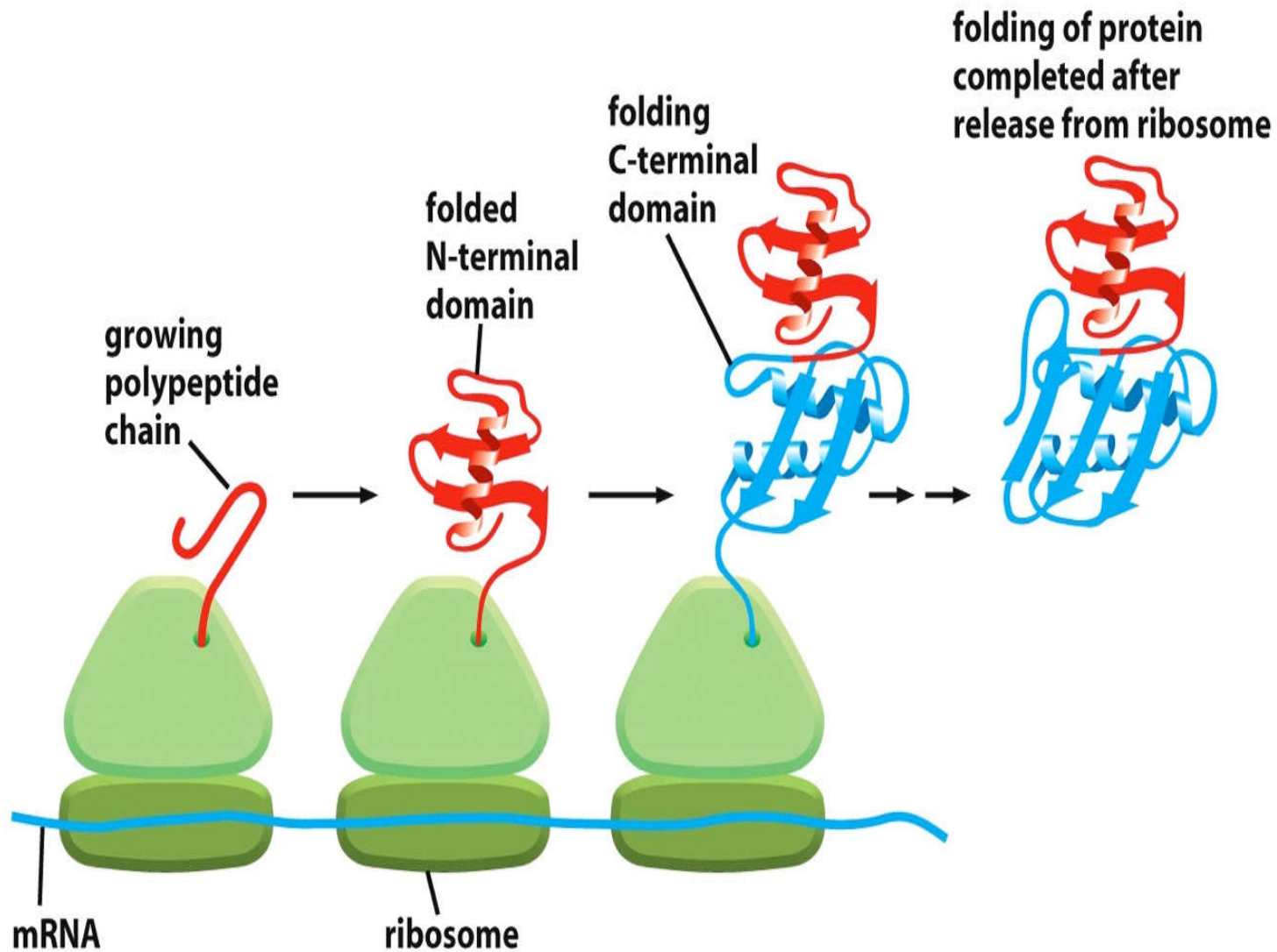
- ☞ The folding of many proteins is kinetically slow and is assisted by folding agents known as **chaperones**.
- ☞ These proteins are found in all organisms and even in different organelles of eukaryotic cells.
- ☞ Chaperones assist in
 - ① folding of nascent polypeptides made by translation, and
 - ② re-folding of proteins denatured by environmental damage, such as heat shock.
- ☞ **Molecular chaperones** bind to unfolded **nascent**

- ✈ polypeptide chains as they emerge from the ribosome, and prevent aggregation, misfolding, and degradation.
- ✈ The hydrolysis of ATP by the chaperone drives conformational changes that prevent aggregation and help drive protein folding.
- ✈ Eukaryotic molecular chaperones such as Hsp 70 (cytosol & mito matrix) and BiP (ER) are related to the bacterial protein DnaK.





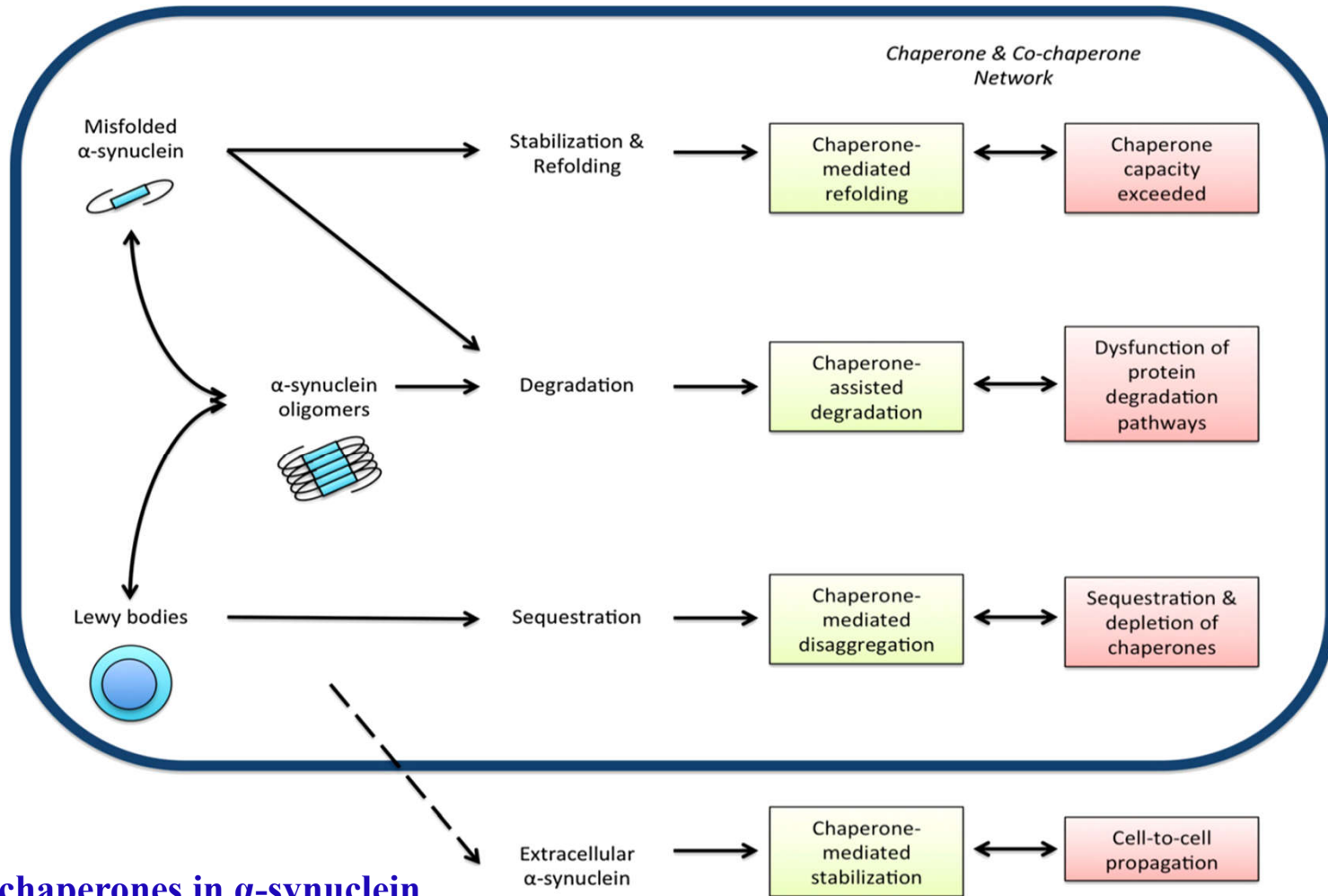
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Functions of a chaperone:

- Assist folding and assembly
- Modulation of conformation
- Transport
- Disaggregation of protein aggregates
- Unfolding of proteins

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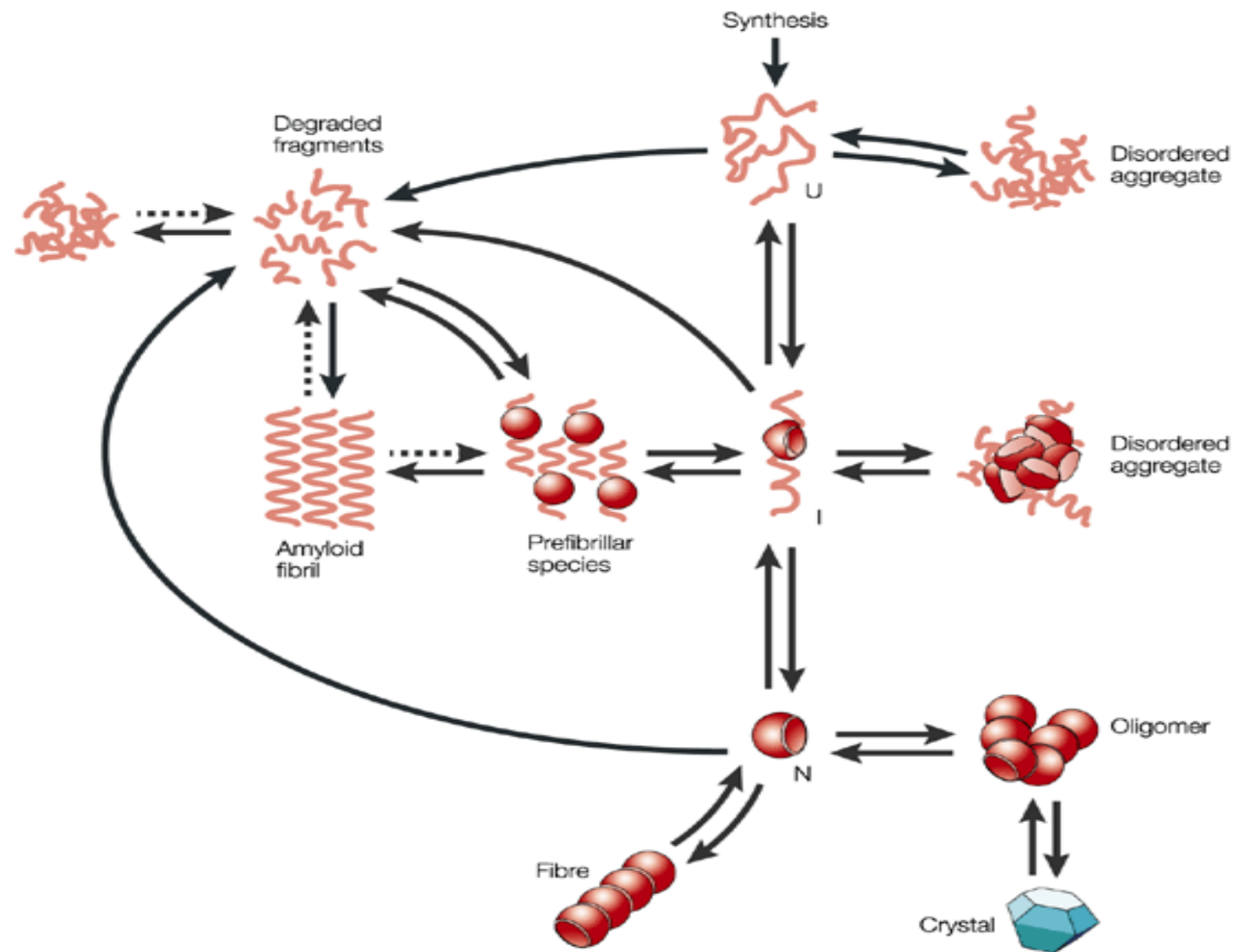
The role of chaperones in α -synuclein metabolism and pathology.

Protein Misfolding

- ✓ The overall **protein folding problem** is to understand the relationship between amino-acid sequence and protein structure.
- ✓ Proteins do not always fold correctly.
- ✓ Protein folding is a complex, and sometimes result in improperly folded molecules.
- ✓ Quality controls exist to identify misfolded proteins, and either refold or destroy them.
- ✓ Failure of misfolding can results in accumulation of prion deposits both inside and outside of the cell.

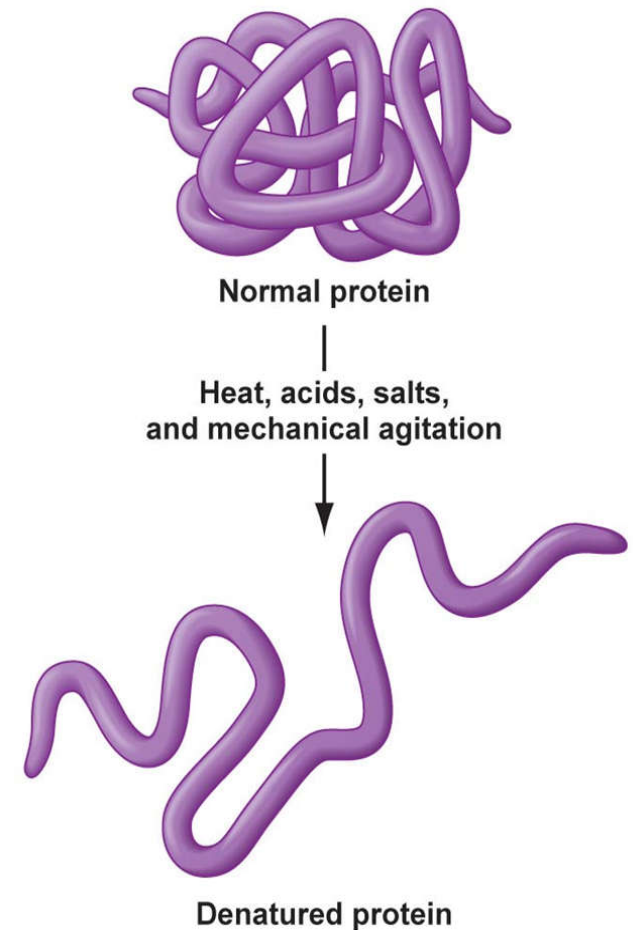
- ☞ Misfolded proteins, aggregates, and inclusion bodies are hallmarks of a range of neurodegenerative disorders including
- ▶ Alzheimer's disease (AD)
 - ▶ Parkinson's disease (PD)
 - ▶ prion disorders
 - ▶ amyotrophic lateral sclerosis (ALS)
 - ▶ polyglutamine (polyQ) diseases that include Huntington's disease (HD) and related ataxia.

...Cont'd



Protein Denaturation

- ✓ Denaturation is loss of native conformation
- ✓ On denaturation, physical, chemical and biological properties of a protein are altered
- ✓ Some of the changes in properties are
 - ☞ Decreased solubility
 - ☞ Unfolding of polypeptide chain,
 - ☞ Loss of helical structure,
 - ☞ Decreased or loss of biological activity,
 - ☞ More susceptible to action of enzymes,
 - ☞ Increased chemical reactivity and
 - ☞ Dissociation of subunits in case of oligomeric proteins.



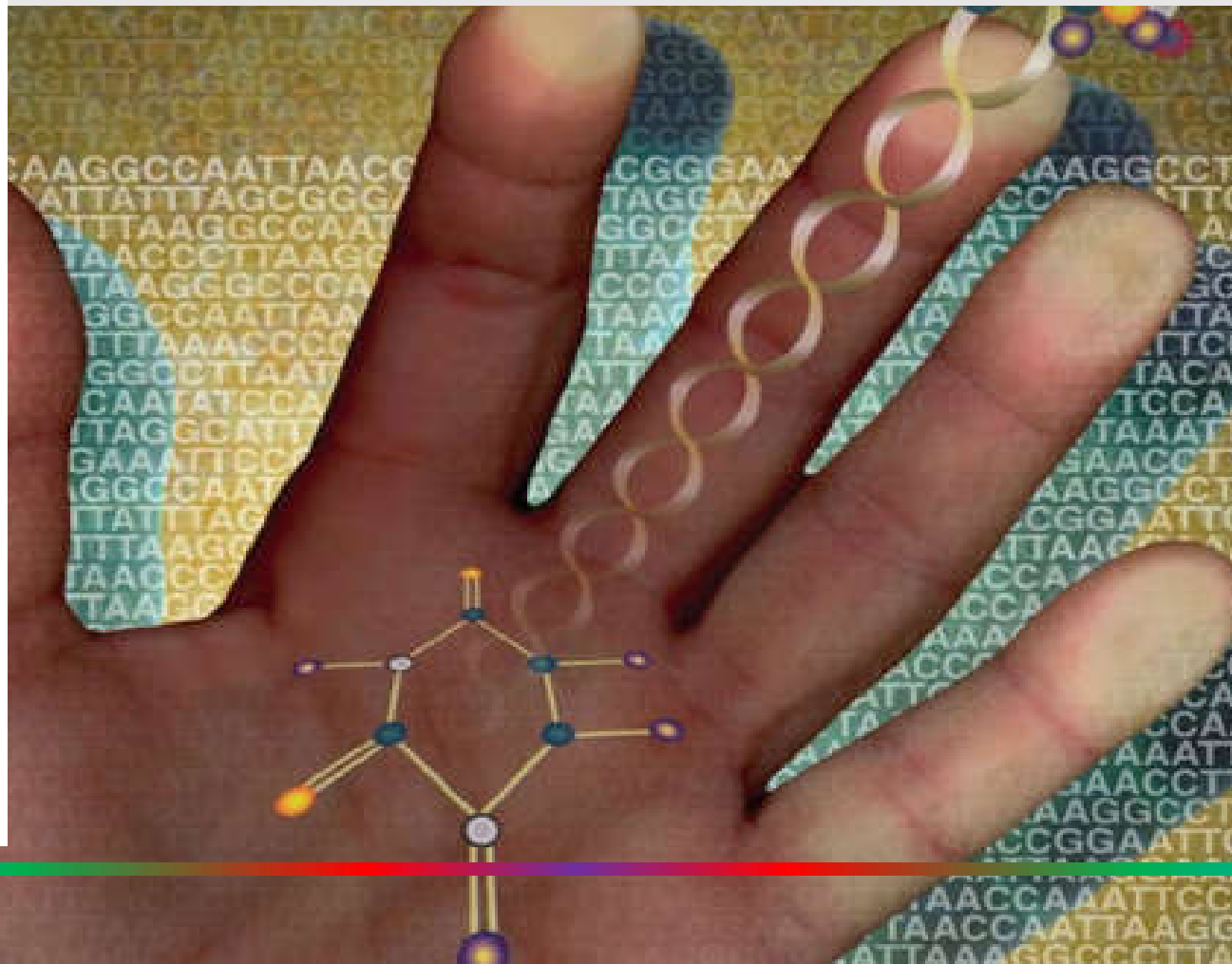
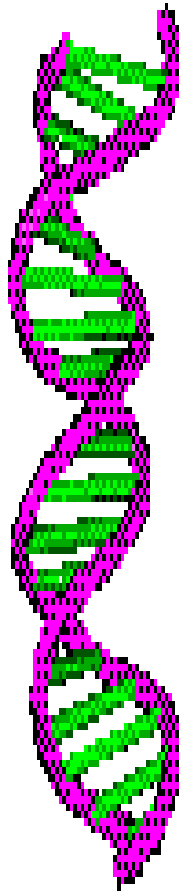
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Causes of Denaturation

- ① High temperature
- ② Extreme alkaline or acidic pH
- ③ Use of urea and guanidine at high concentration
- ④ UV radiation
- ⑤ Sonication and vigorous shaking
- ⑥ Detergent like sodium dodecylsulfate
- ⑦ Treatment with organic solvents like ethanol, acetone etc.
- ⑧ Treatment with strong acids like trichloro acetic acid, picric acid and tungstic acid
- ⑨ Exposure to heavy metals like Pb^{2+} , Ag^{2+} and Cu^{2+}

- ✓ Primary structure is unchanged by protein denaturation.
- ✓ It may be reversible by protein refolding.
- ✓ Example: Ribonuclease denatures on exposure to heat but come back to its native conformation when temperature is lowered.

Nucleic Acids



Introduction to Nucleic Acids

- ♣ Frederic Miesher in 1869, isolated an acidic compound from the nuclear material of SALMON sperms, and named it as NUCLIEN which is now called **NUCLEIC ACID**.
 - ♣ **Jones in 1920** proved the fact there are two types of nucleic acids, i.e., Deoxy ribo nucleic acid (DNA) and Ribonucleic acid (RNA).
 - ♣ In 1935 J. D. Watson and F. H. C Crick, on the basis of information's available not only proposed the “Double helical” structure of DNA but also suggested what Crick termed “central dogma of molecular genetics”, which states that genetic information flows from DNA to RNA to protein.
-

Introduction

- ➡ **Nucleic acids** are present in all living organisms (plants, animal, bacteria and even Virus and virioids) and therefore origin of life is suspected with the formation of nucleic acids.
- ➡ All living cells contain both **DNA and RNA**, while viruses contain either DNA or RNA, usually not both.

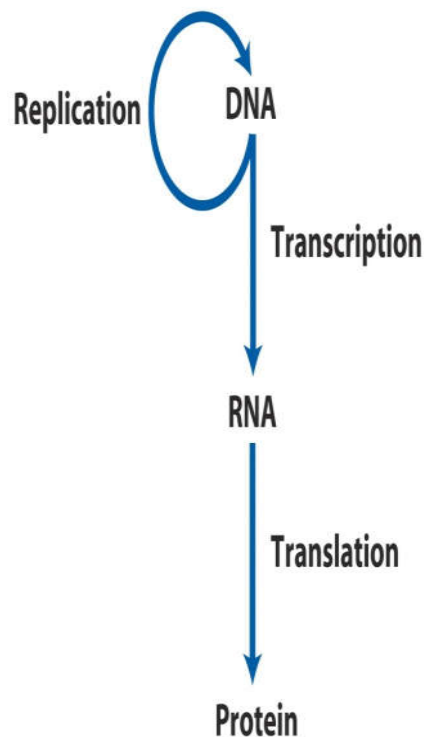
Introduction

- ♣ The nucleic acids are **most important biological macromolecules** responsible for transmission and storage of heredity or genetic information from one generation to next by means of Replication, Transcription and Translation.
- ♣ **Nucleic acids are fibre like molecules having much greater length in comparision to diameter**
- ♣ Nucleic acids have function in encoding, transmitting and expressing genetic information
- ♣ The nucleic acids are generally **associated with proteins** to form nucleoproteins

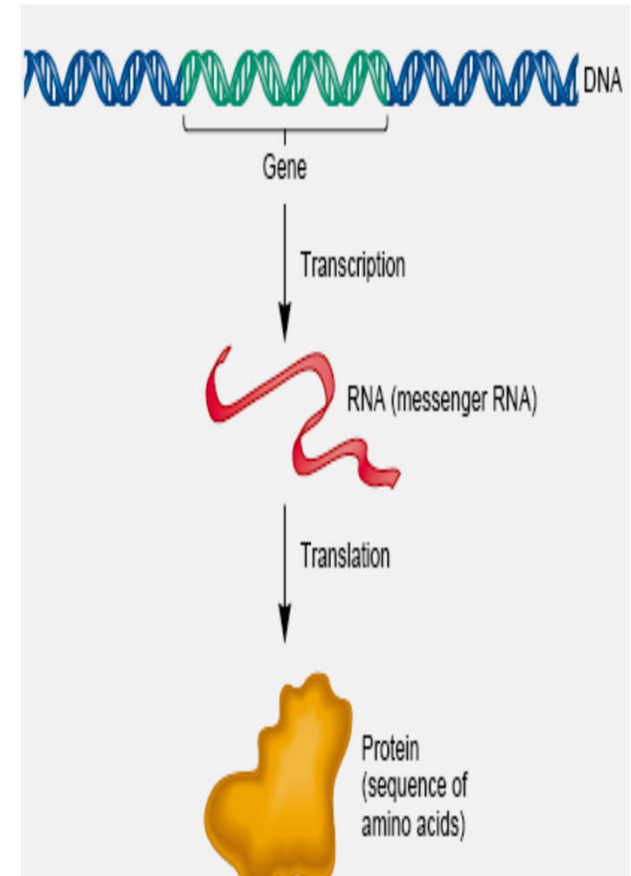
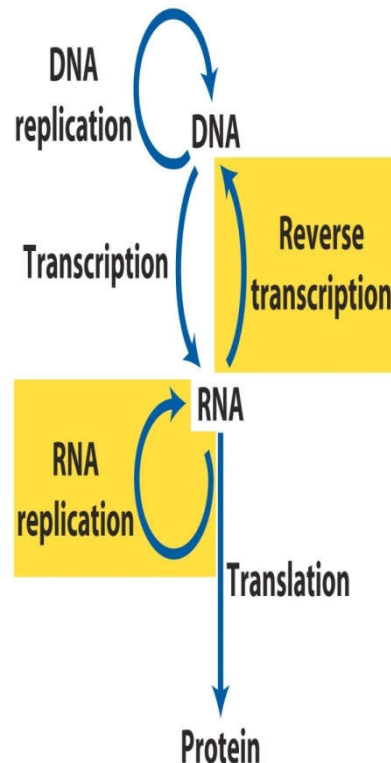
Introduction

- ♪ The genetic information is stored in **DNA** or **RNA** (RNA viruses).
- ♪ **The central dogma**; that is the flow of genetic information from DNA to DNA from DNA to RNA, and then to protein.

Traditional dogma



Extended dogma

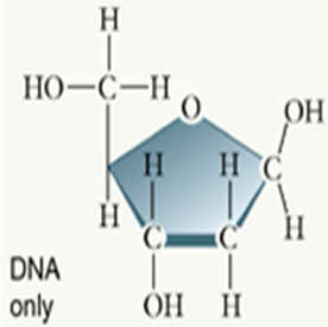
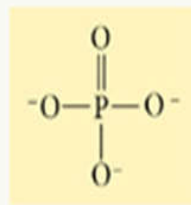
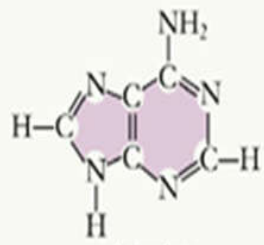
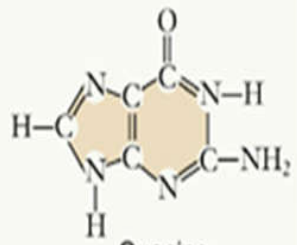
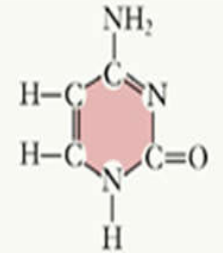
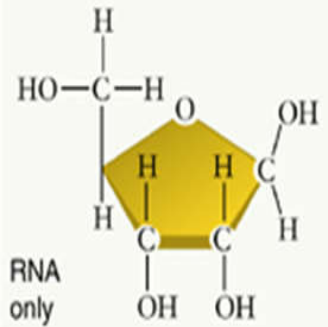


Nucleic Acids

- ♪ They are found in two basic structural forms, deoxyribonucleic acid (DNA) and ribonucleic acid (RNA).
- ♪ DNA is present in nucleus and mitochondria.
- ♪ RNA is present in nucleus and cytoplasm.
- ♪ DNA and RNA are long chain polymers of nucleotides.
- ♪ Each nucleotide is composed of a **base, sugar** (ribose in RNA or deoxyribose in DNA) and a **phosphate group**.
- ♪ The phosphate joins the sugars in a DNA or RNA chain through **3' → 5' phosphodiester bonds**.

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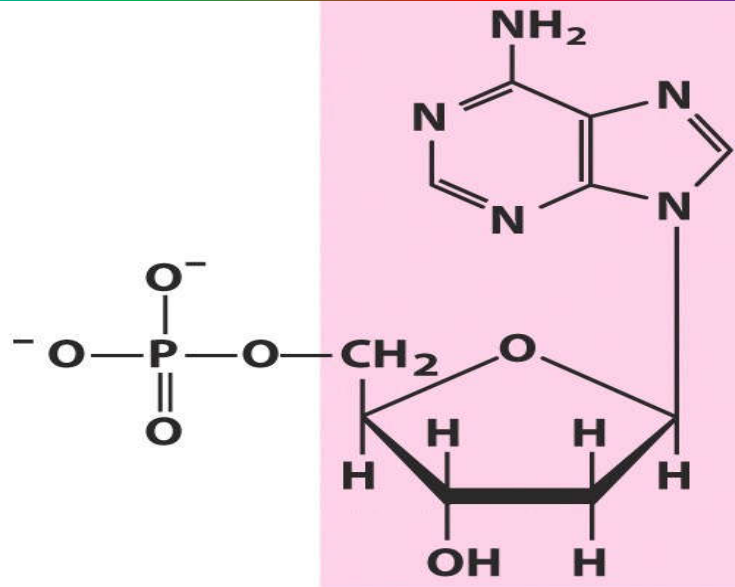
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Five-carbon Sugar		Phosphate Group	Nitrogenous Base		
Deoxyribose	 DNA only		Purines	 Adenine	 Guanine
				Pyrimidines	 Cytosine (both DNA and RNA)
Ribose	 RNA only				

Structure of Nucleotides

- Nucleotides are the fundamental units of nucleic acids.
- Each nucleotide is comprising of a:
 - I. Phosphate group
 - II. Pentose Sugar and
 - III. Nitrogenous base.
- ✓ Sugar + Base = Nucleoside
- ✓ Nucleoside + Phosphate = Nucleotide

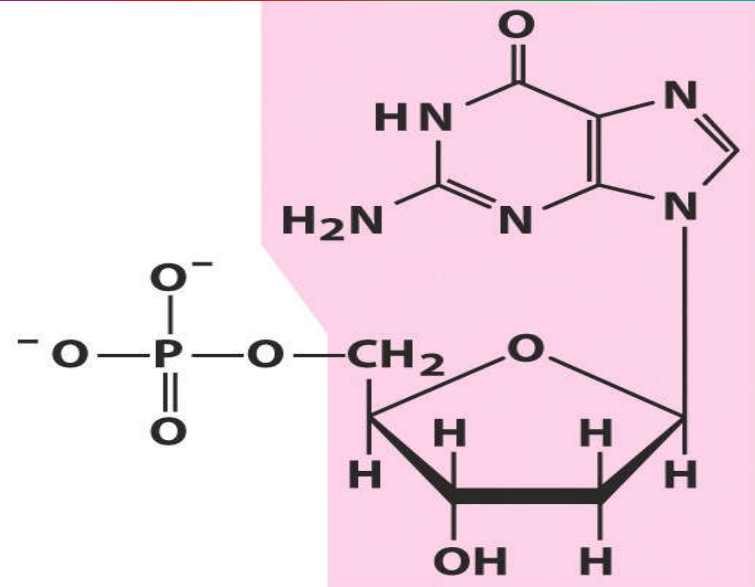
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Nucleotide: Deoxyadenylate
(deoxyadenosine
5'-monophosphate)

Symbols: A, dA, dAMP

Nucleoside: Deoxyadenosine



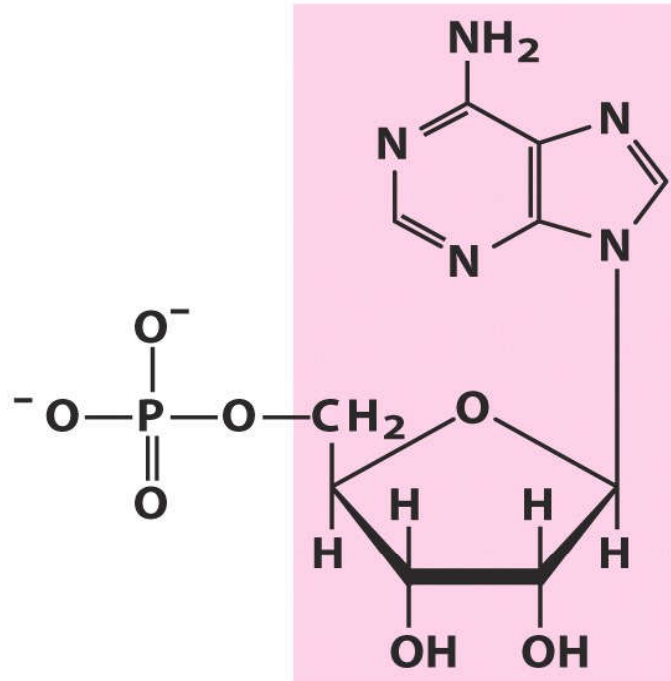
Nucleotide: Deoxyguanylate
(deoxyguanosine
5'-monophosphate)

Symbols: G, dG, dGMP

Nucleoside: Deoxyguanosine

(a) Deoxyribonucleotides

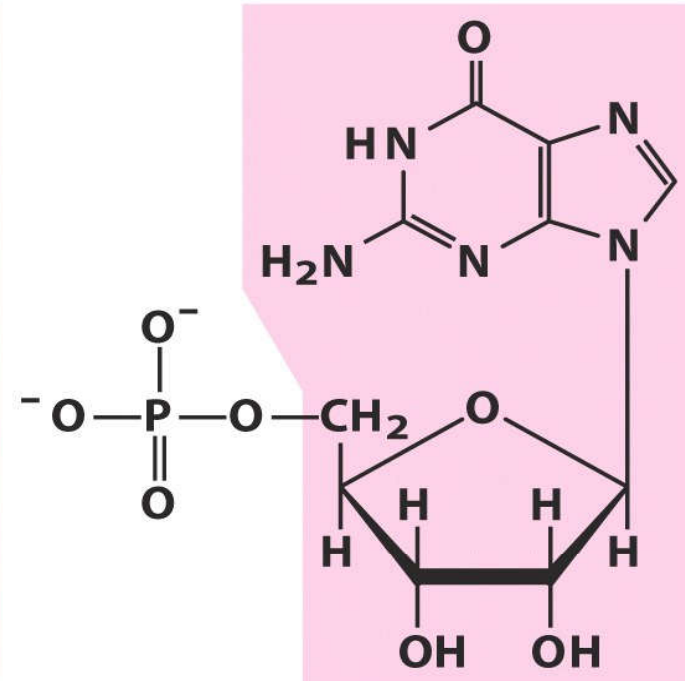
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Nucleotide: Adenylate (adenosine 5'-monophosphate)

Symbols: A, AMP

Nucleoside: Adenosine



Nucleotide: Guanylate (guanosine 5'-monophosphate)

Symbols: G, GMP

Nucleoside: Guanosine

(b) Ribonucleotides

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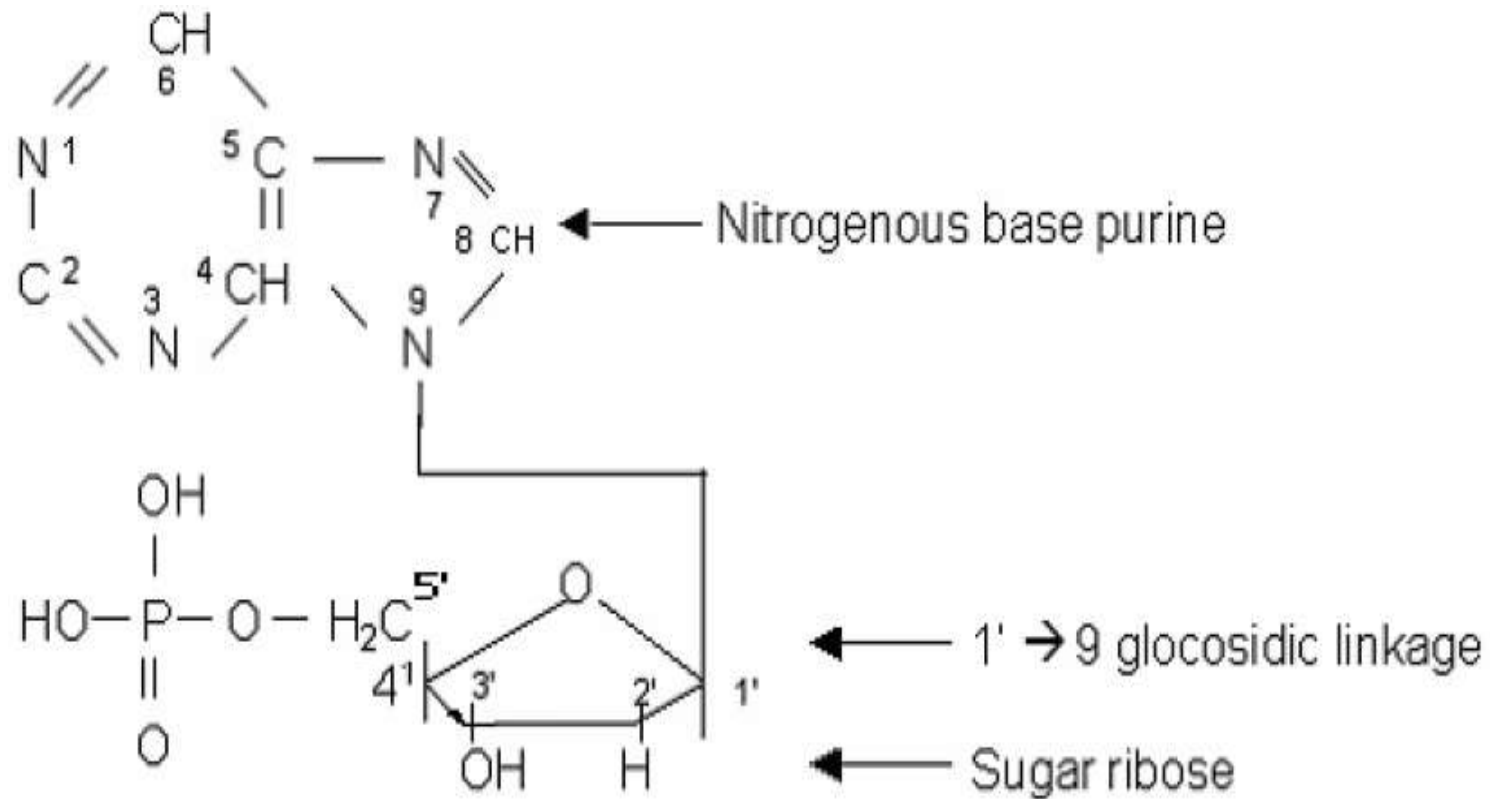


Fig . Purine deoxyribonucleotide

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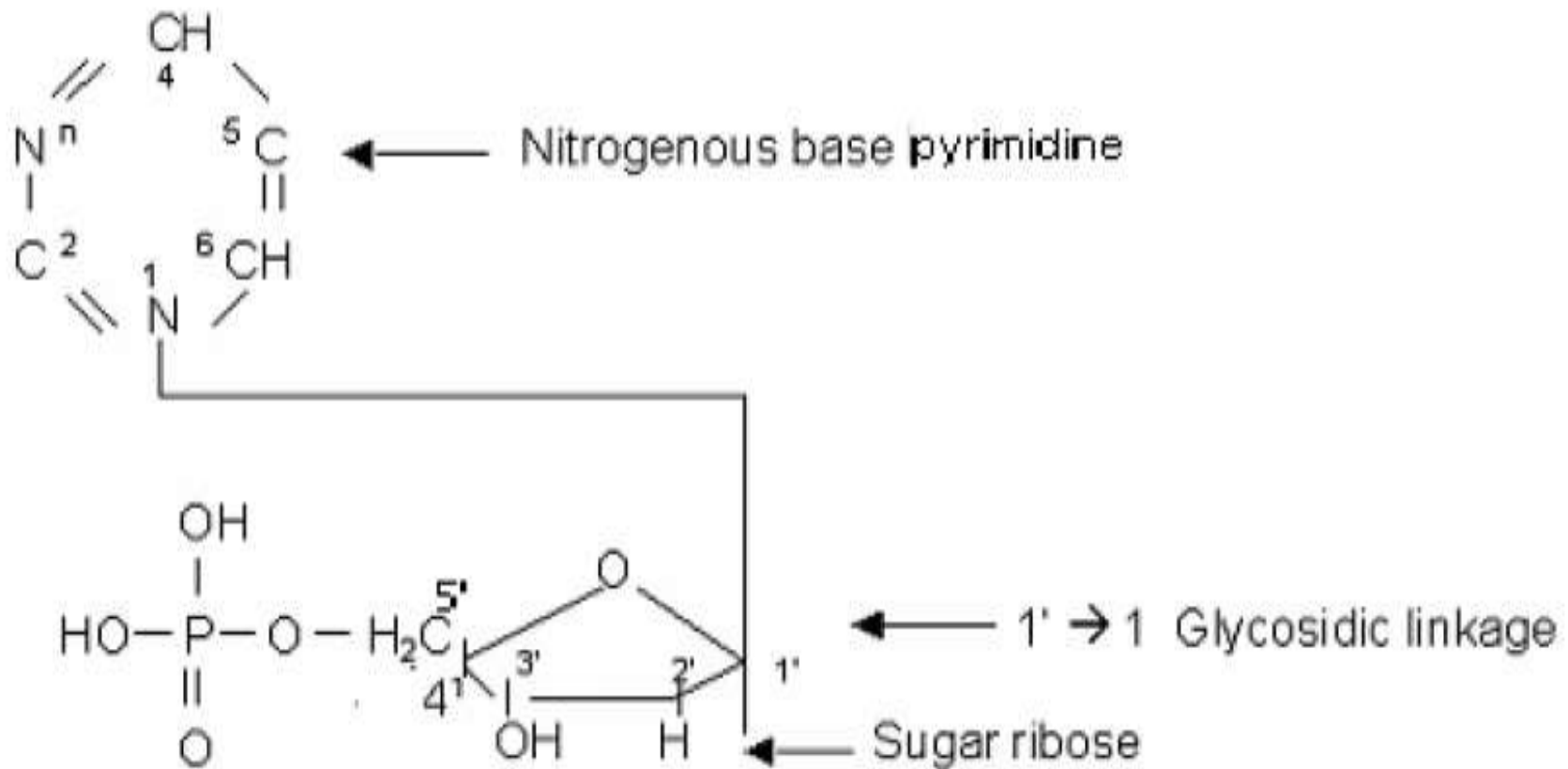


Fig. Pyrmidine deoxyribonucleotide

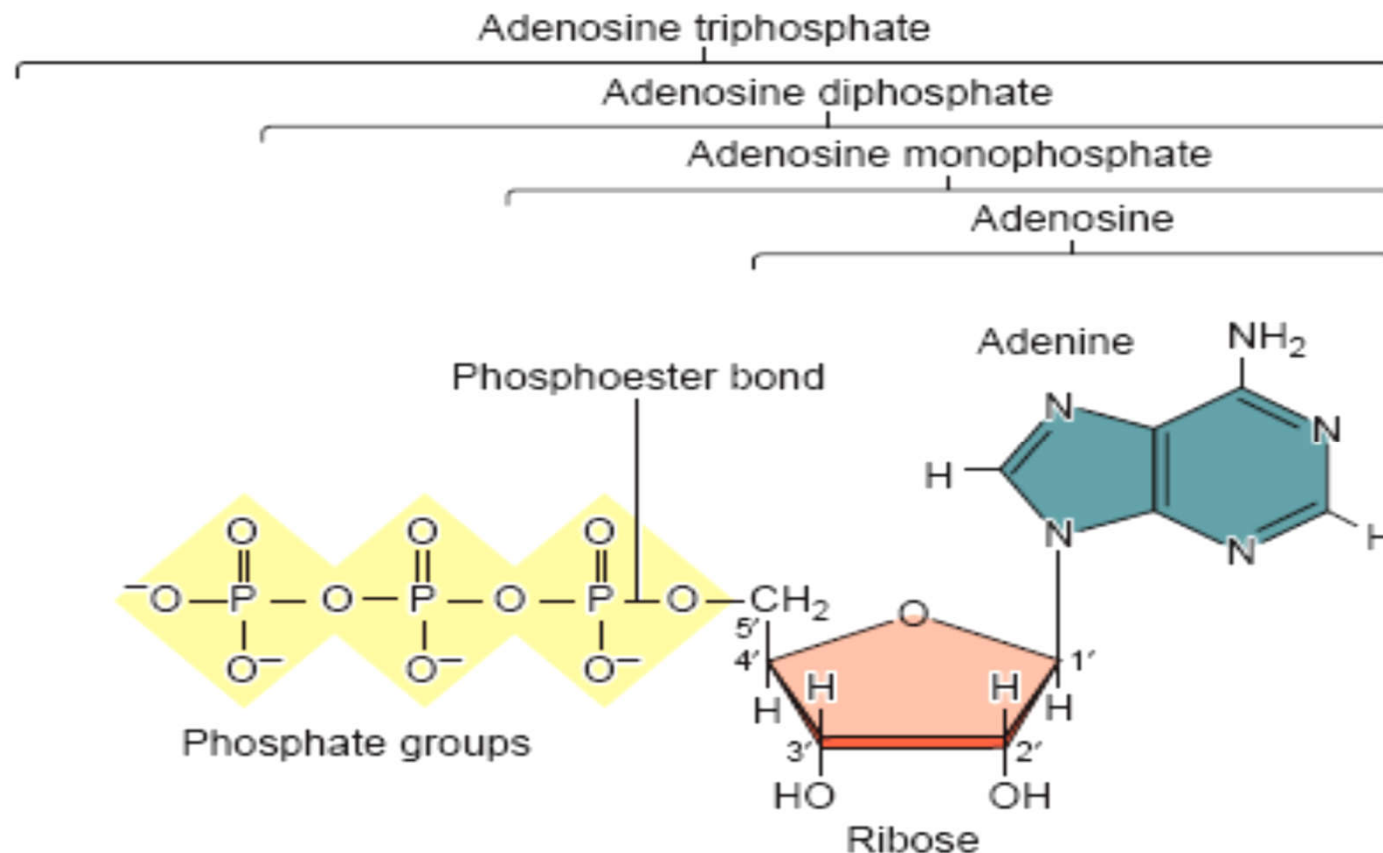
Nomenclature for components of RNA

Base	Ribonucleoside	Ribonucleotide (5'-monophosphate)
Adenine (A)	Adenosine	Adenylate (AMP)
Guanine (G)	Guanosine	Guanylate (GMP)
Uracil (U)	Uridine	Uridylate (UMP)
Cytosine (C)	Cytidine	Cytidylate (CMP)

Nomenclature for components of DNA

Base	Deoxyribonucleoside	Deoxyribonucleotide (5'-monophosphate)
Adenine (A)	Deoxyadenosine	Deoxyadenylate (dAMP)
Guanine (G)	Deoxyguanosine	Deoxyguanylate (dGMP)
Thymine (T)	Thymidine	Thymidylate (TMP)
Cytosine (C)	Deoxycytidine	Deoxycytidylate (dCMP)

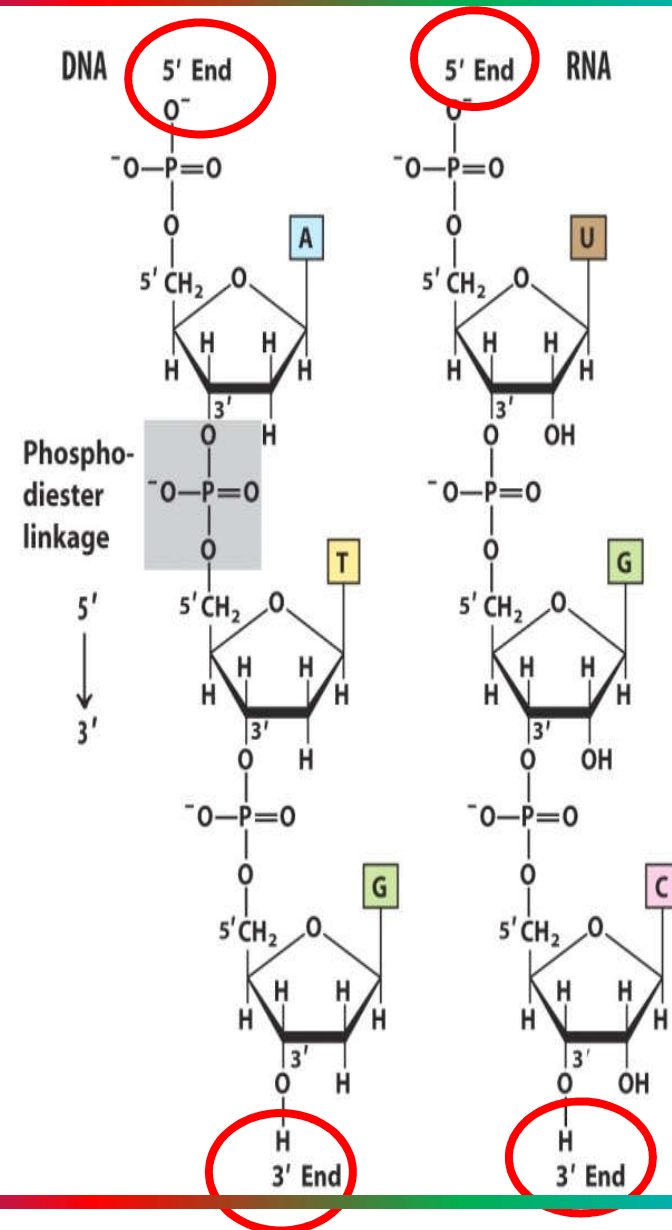
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- ➡ Nucleotides are linked together to form a strand
- ➡ A strand of DNA or RNA has nucleotides that are covalently attached to each other in a linear fashion.

Primary structure of nucleic acids

- ✓ Linear sequence of a polynucleotide is known as primary structure of nucleic acid.
- ✓ Polynucleotide chain has direction ($5' \rightarrow 3'$ direction only).
- ✓ However, the phosphodiester linkage runs in $3' \rightarrow 5'$ direction.
- ✓ Primary structures of DNA and RNA exist in single stranded DNA and RNA organisms.
- ✓ In primary structure, letters A, G, C, T stands for nucleotides and sequence is written from left to right.



Significances of Nucleotides

1. Nucleotides are **precursors of nucleic acids (DNA and RNA)** and are needed for:
 - ✓ **Replication** of the genome
 - ✓ **Transcription** of the genetic material
2. They are essential carriers of chemical energy (ATP and GTP).
3. They are used for **biosynthesis of nucleotide derivatives**
 - ✓ **UDP-glucose** participate in biosynthetic processes ~ the formation of glycogen.
4. Nucleotides are essential components of **signal-transduction pathways**.
 - ✓ **cAMP** and **cGMP** are second messengers that transmit signals both within and between cells.
5. They are components of the coenzymes (NAD, FAD, and coenzyme A)
6. ATP acts as the **donor of phosphoryl** groups transferred by protein kinases.

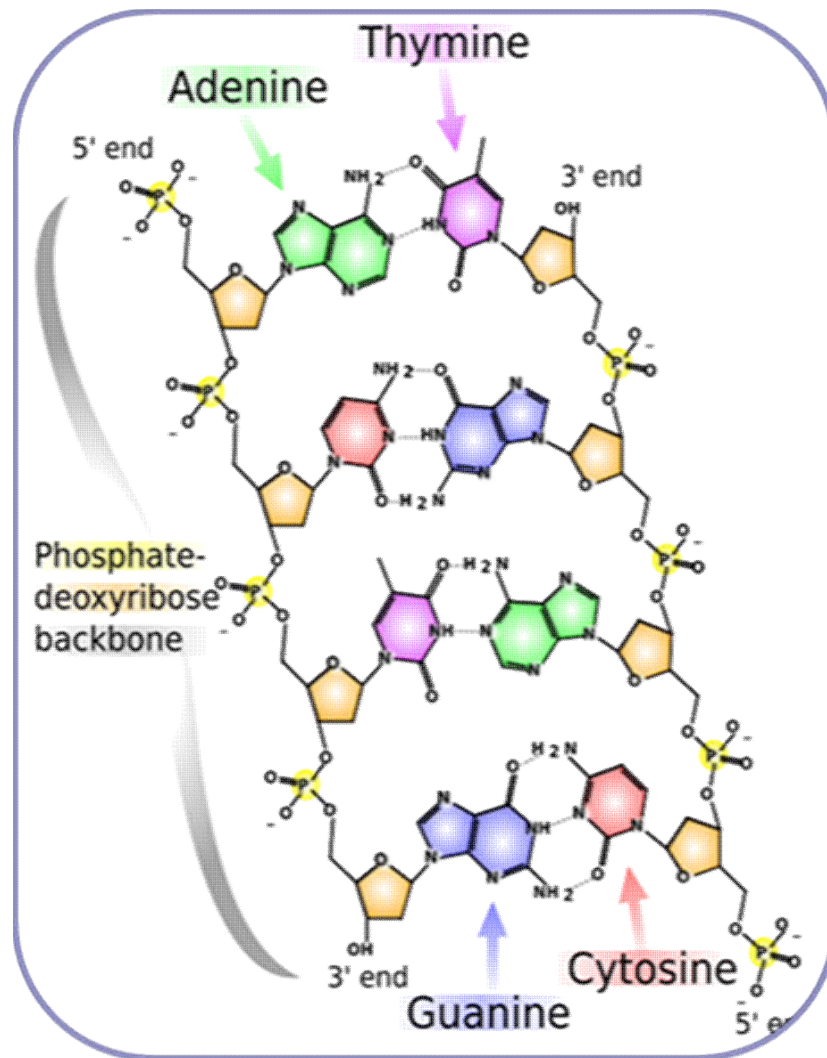
DNA

- ♣ **Deoxyribonucleic Acid (DNA)**; the genetic material of all cellular organisms and most viruses.
- ♣ **DNA**; the gigantic molecule which is used to encode genetic information for all life on Earth.
- ♣ A human cell contains about 2 meters of **DNA**.
- ♣ **DNA** in the body could stretch to the sun and back almost 100 times. So it is tightly packed.
- ♣ **DNA** responsible for preserving, copying and transmitting information within cells and from generation to generation.

DNA Double Helix

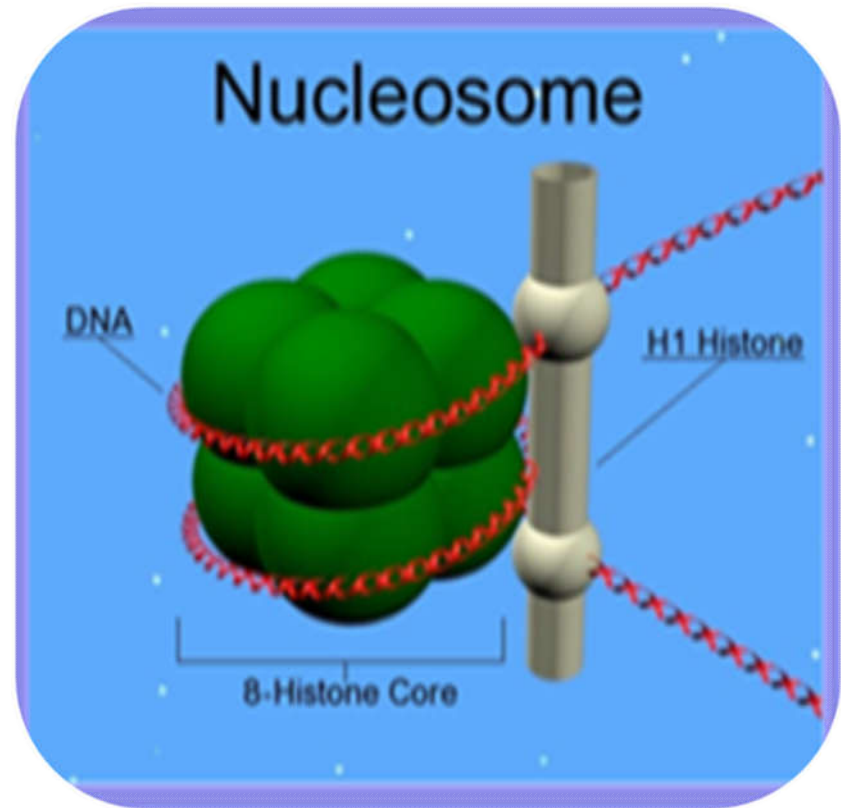
- Linked as a twisted ladder.
- The curving sides of the ladder represent the **sugar-phosphate** backbone of the two DNA strands; the rungs are the **base pairs**.
- Possess **antiparallel** polarity.
- Stabilized by **hydrogen bonds** between the bases.

DNA Double Helix



DNA Organization

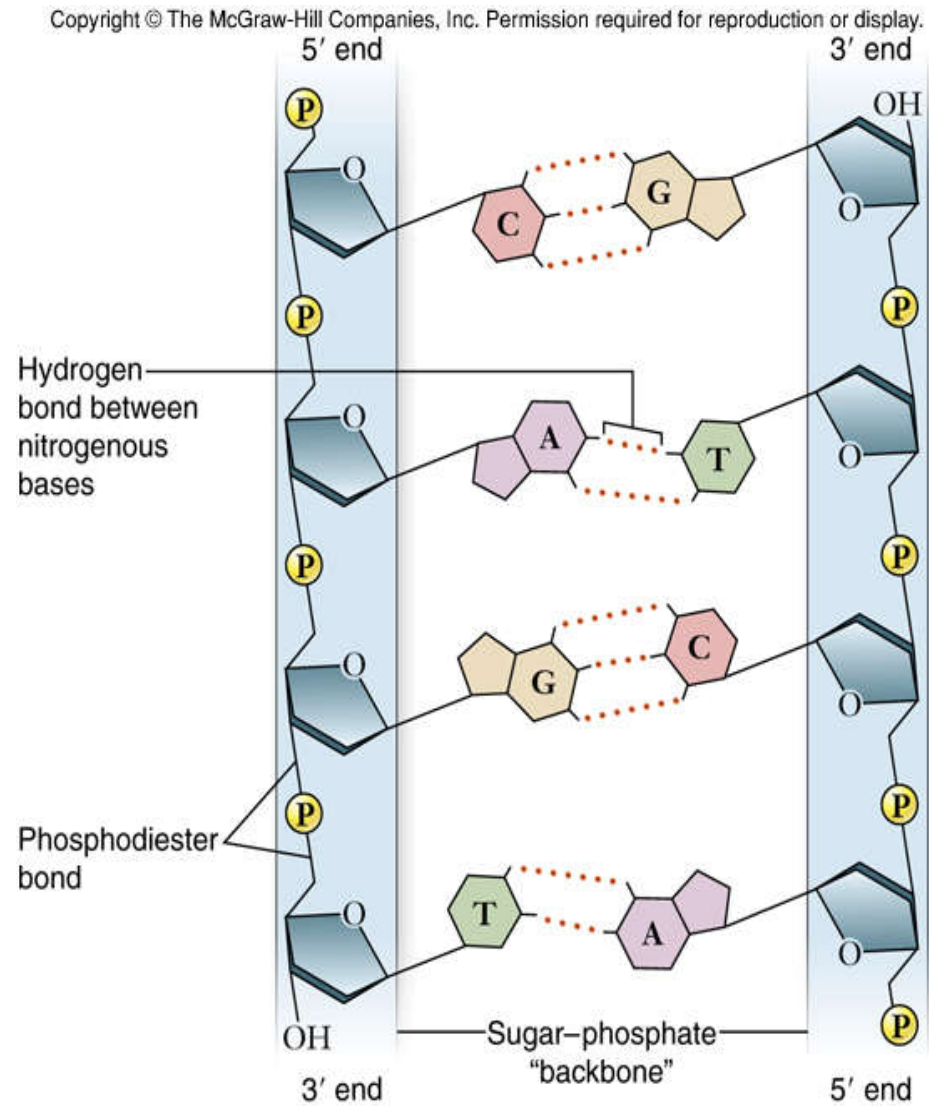
DNA molecules complexed with other proteins, especially basic proteins called **histones** to form a substance known as **chromatin**.



Structure of DNA

- ✓ DNA is polymers of deoxyribonucleotides attached to each other by phosphodiester linkages.
- ✓ Built up of a sugar-phosphate backbone and a sequence of **nucleotides**: **adenine (A)**, **guanine (G)**, **cytosine (C)** and **thymine (T)**
- ✓ The bases pair are in a specific way:
 - ❖ A - T (two hydrogen bonds) and
 - ❖ G – C (three hydrogen bonds)
- ✓ Within the structure of DNA, the number of thymine is always equal to the number of adenine and the number of cytosine is always equal to guanine. This is known as Chargaff's Rules
- ✓ Human DNA consists of approximately 3×10^9 such **“base pairs”**.

...Cont'd



Structure of DNA

Example: A sample of DNA has 10% G; what is the % T?

$$10\% \text{ G} + 10\% \text{ C} = 20\%$$

therefore, % A + % T must total 80%

40% A and 40% T

Ans: 40% T

Homework: A double-stranded RNA genome isolated from a virus in the stool of a child with gastroenteritis was found to contain 15% uracil. What is the percentage of guanine in this genome?

✓ Human DNA consists of approximately 3×10^9 such “base pairs”.

Primary Structure of DNA

- Polynucleotide chain has direction.
 - They are represented in $5' \rightarrow 3'$ *direction only*.
 - However, the *phosphodiester* linkage runs in $3' \rightarrow 5'$ *direction*.
- *Each poly nucleotide chain has two ends.*
 - The *5' end* carrying phosphate is shown on the left hand side and
 - *3' end carrying unreacted hydroxyl* is shown on the right hand side (next Fig.).
- Primary structures of DNA and RNA exist in single stranded DNA and RNA organisms.

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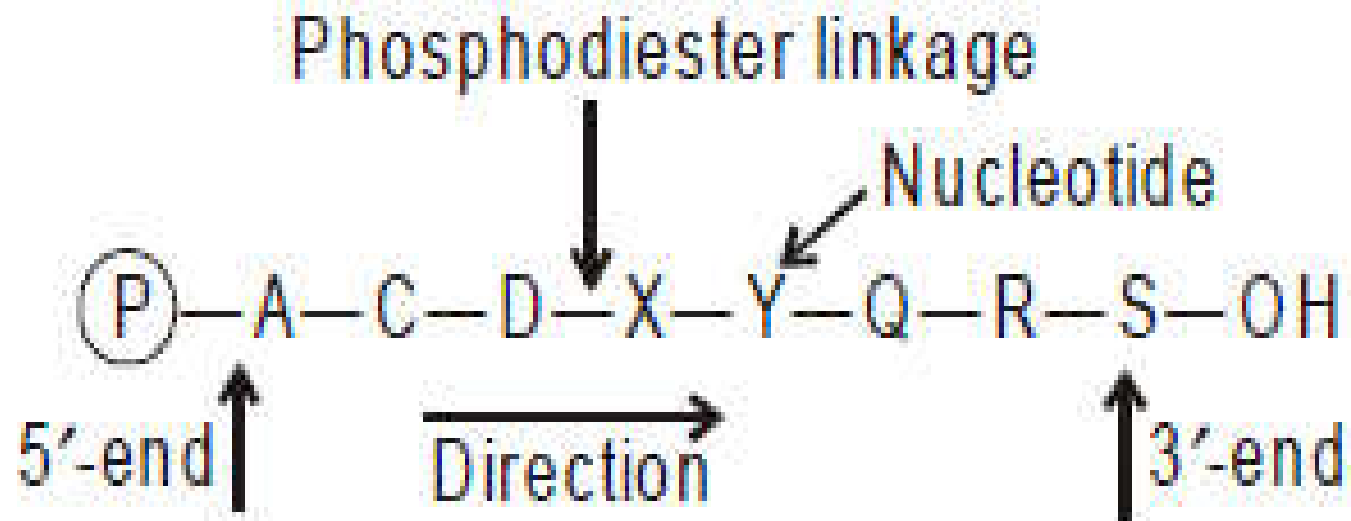
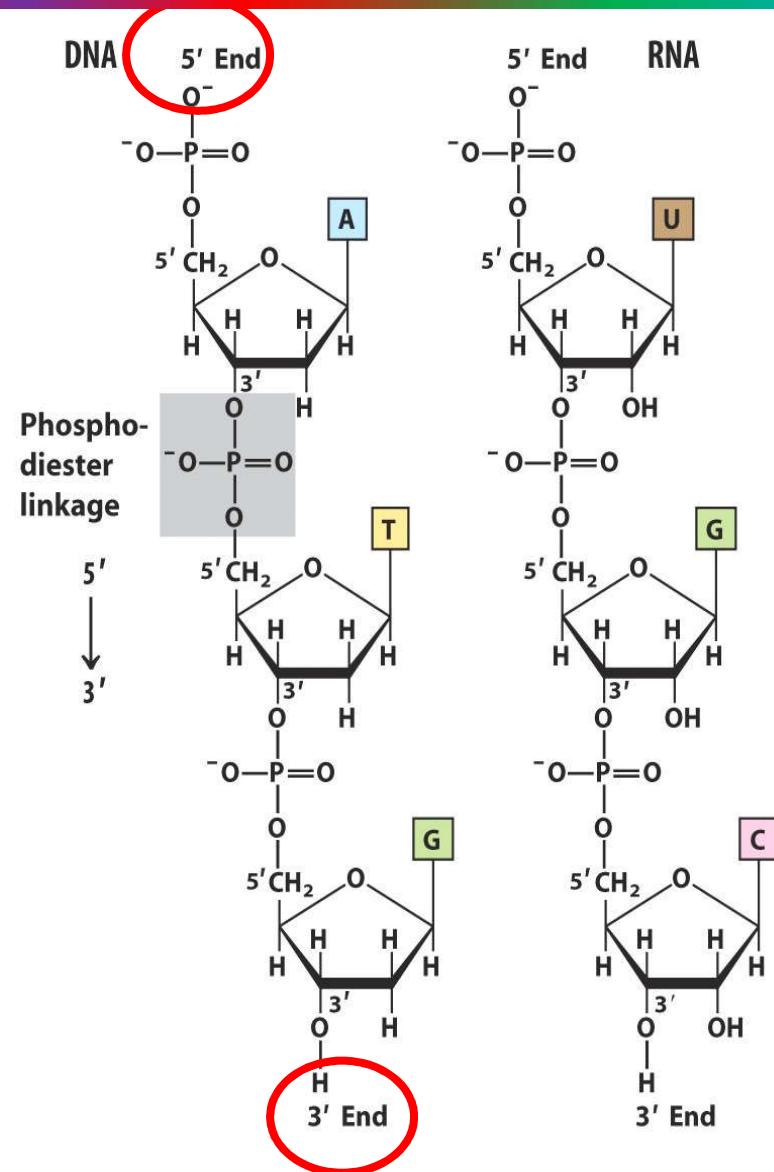


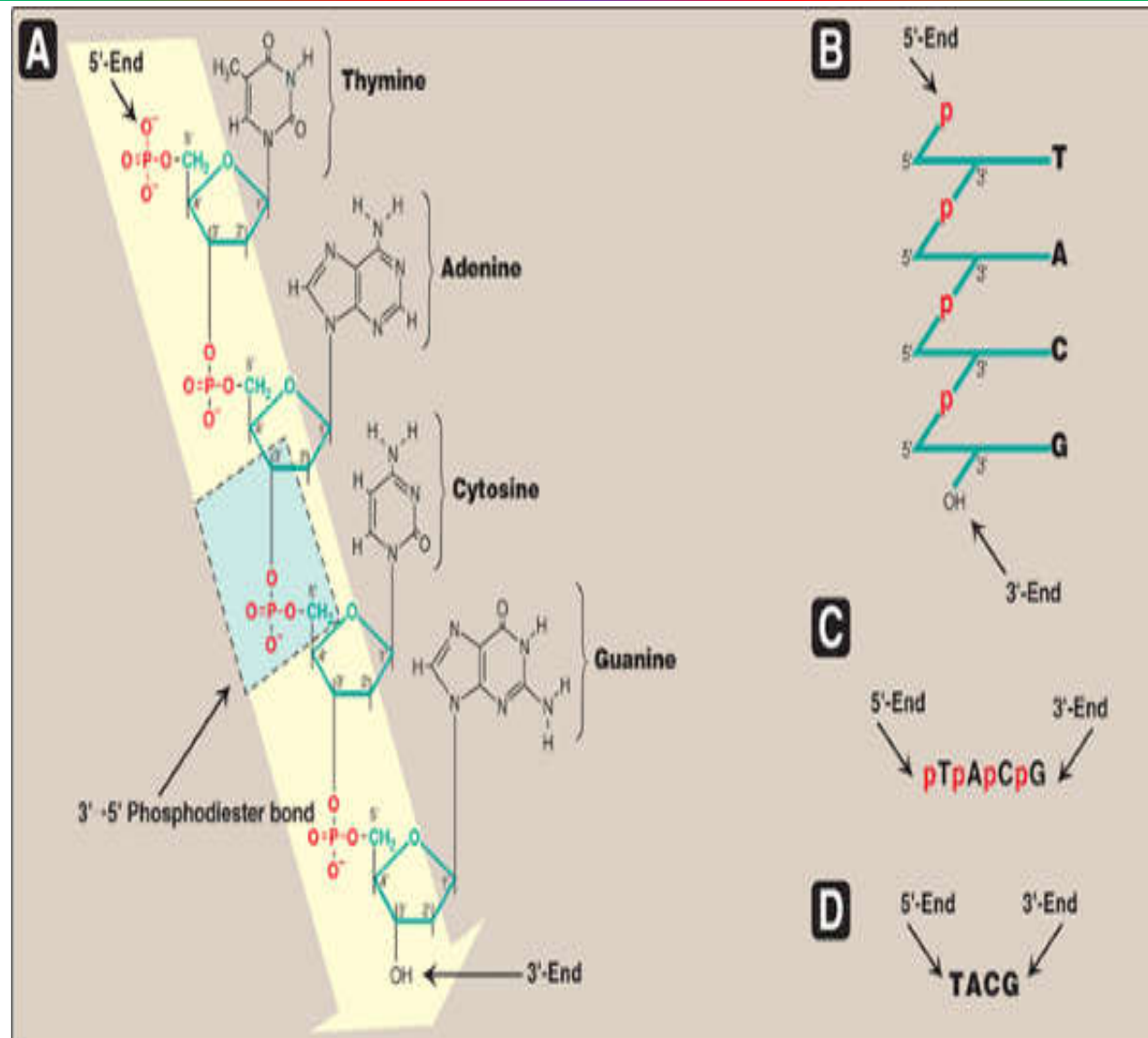
Fig. *Primary structure of nucleic acid. Letters A, C, D, X, Y, Q, R, S are nucleotides*

The Anatomy of DNA

- ✓ A phosphodiester bond joins one nucleotide to another from the 3' ribose oxygen to a phosphate attached to the 5' oxygen (3'-5' phosphodiester bond) : a string of bases is read **5'→3'**
- ✓ Specific long sequences of A, T, G, C nucleotides in DNA are the repository of genetic information.



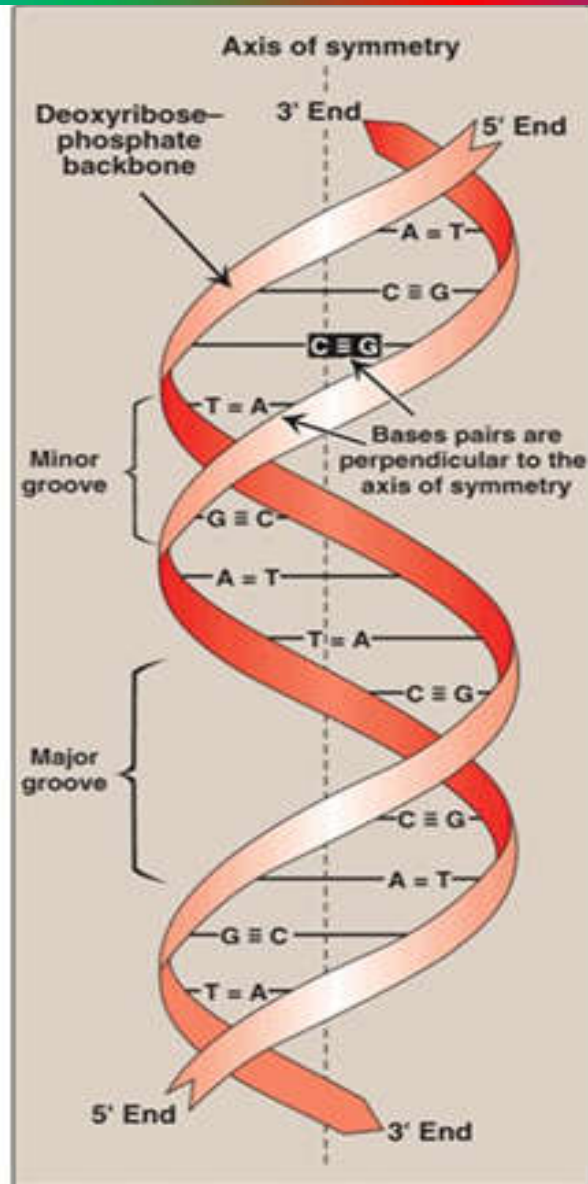
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Secondary Structure of DNA

- ✓ It is performed when the **two strands of DNA are paired** together.
- ✓ The chains are paired in an **antiparallel** manner,
 - i.e. the 5'-end of one strand is paired with the 3'-end of the other strand .
- ✓ The **bases are stacked in the inside** of the two strands.
- ✓ The negatively charged **phosphate group** and the sugar units **expose themselves to the outside of the chain**.
- ✓ The two strands of DNA coil around a single axis forming right handed double helix.
- ✓ The overall structure resembles a twisted ladder.

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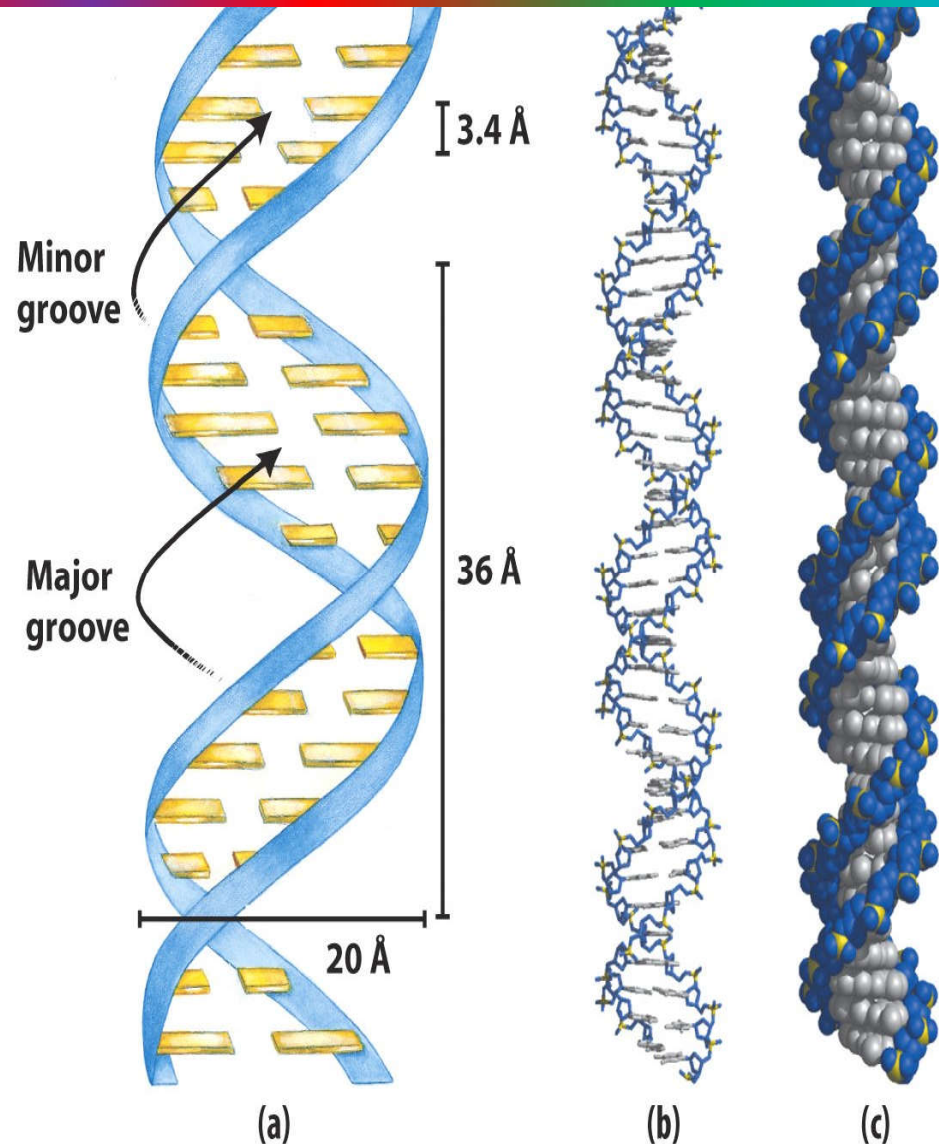


Watson, Crick (1953)
– Right handed, antiparallel double helix (B-DNA)



Right handed, antiparallel double helix (B-DNA)

- ✓ Double helix – 3.4 Å/base
- ✓ The hydrophilic phosphodeoxyribose backbones face out
- ✓ The bases are found on the inside.
- ✓ 10 residues per turn (36 Å)



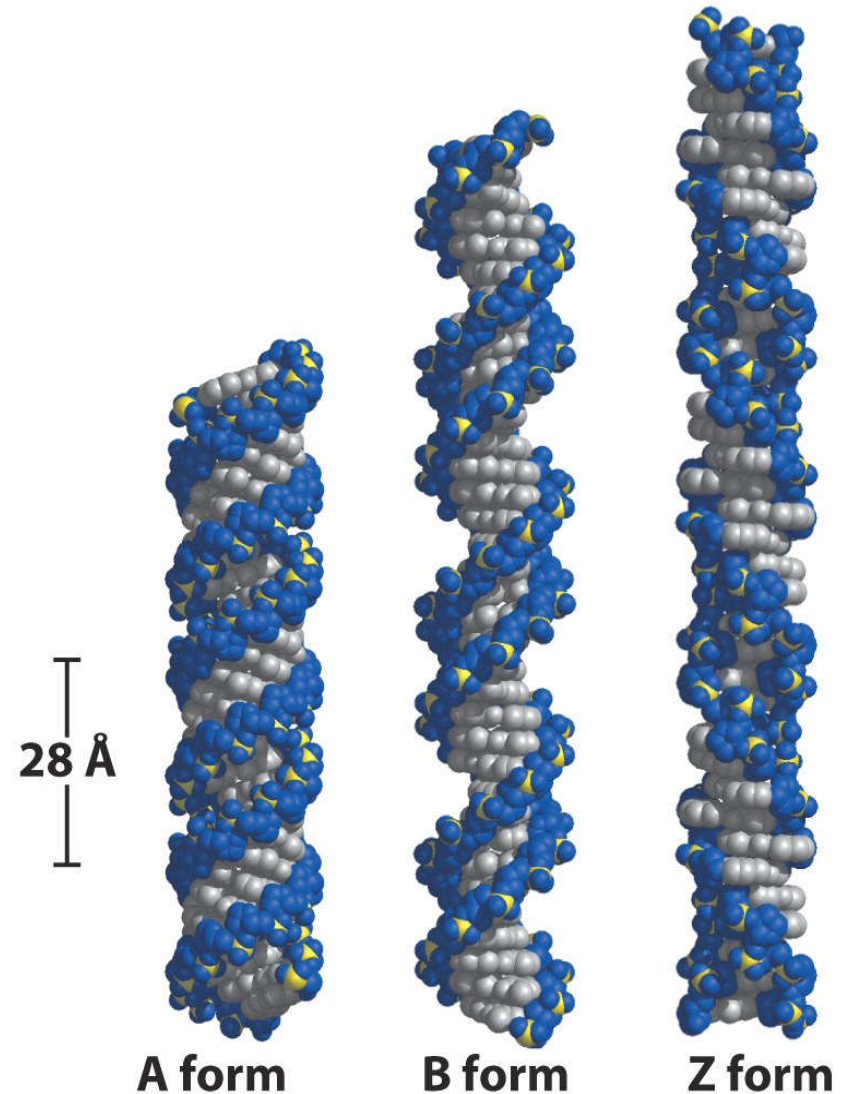
Watson - Crick have proposed a double helical model of DNA, having the following important characteristic features.

1. Two helical polynucleotide chains are coiled around a common axis. The chains run in *opposite directions, (anti parallel)*
2. The two antiparallel polynucleotide chains are **not identical**, but *they are complimentary*.
3. The *purine, pyrimidine bases are on the inside* of the helix, the *phosphate and deoxyribose groups are on the outside*. The planes of the sugars are at right angles to that of the bases.
4. The diameter of the helix is 20 Å, adjacent bases are separated by 3.4 Å.
5. The helical structure repeats after *10 residues* on each chain.

6. The two chains are held together by *hydrogen bonds* between pairs of bases.
 - Adenine is always paired with thymine, Guanine always paired with cytosine.
 - A to T is bonded by *two hydrogen bonds* (A= T), Guanine is bonded to cytosine by *three hydrogen bonds* . . .
7. The double helix is stabilized by interaction between **stacked bases** of the same strand.
8. Watson - Crick Model of DNA is also referred as *B-DNA, which is the most stable* one under physiological conditions.
 - The mitochondrial DNA is circular and there can be formation of Z-DNA and C-DNA which can be performed during either replication or transcription.

DNA duplex can exist in different 3-D forms.

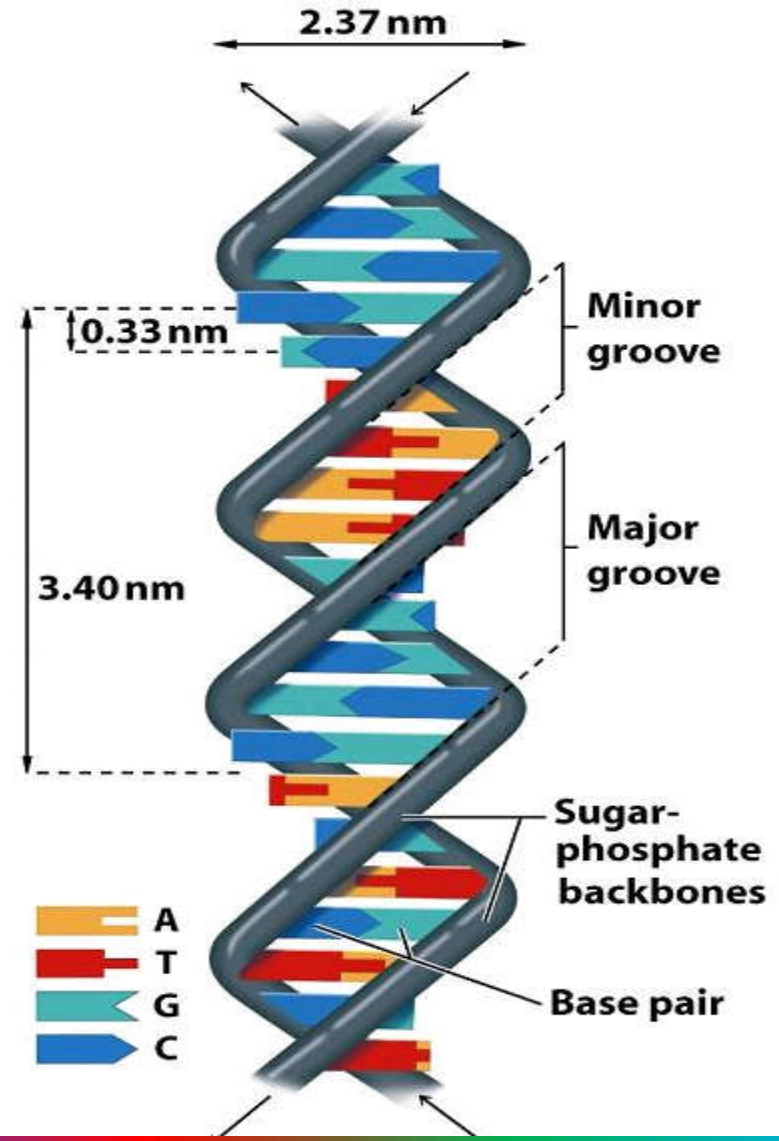
- ✓ **B-form** : the Watson-Crick structure, standard form, most stable under physiological conditions
- ✓ **A-form** : a dehydrated form
- ✓ **Z-form** : left handed A short Z-DNA tracts may play a role in gene regulation.



What holds DNA strands together?

✓ The two strands of ds DNA are held together by a number of weak interactions:

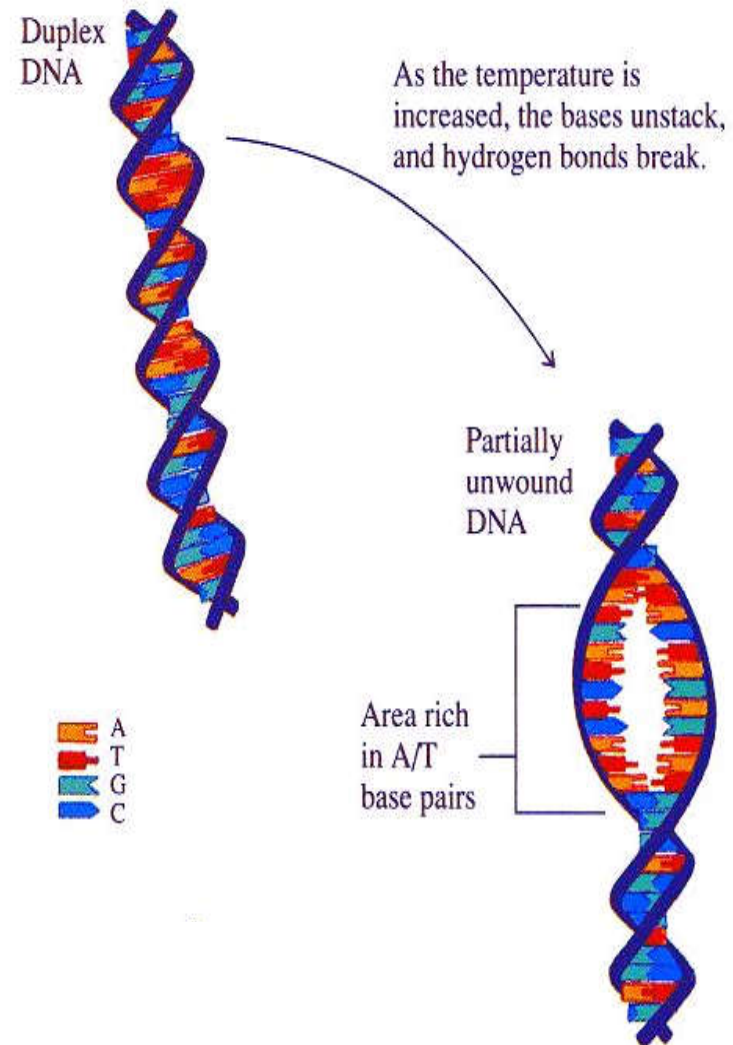
- Hydrogen bonds,
- Stacking interactions, and
- Hydrophobic effects



What happens to ds DNA when temperature is raised?

1. Local unwinding of the ds DNA.
2. This occurs preferentially in regions where the two strands are held together less strongly.
3. In these regions the strands separate to form bubbles of ss regions.
4. The DNA sequence in these regions is enriched in A/T base pairs because the interactions between the two strands are weaker in A/T rich regions.
5. In G/C rich regions strands are held together more strongly so they don't unwind until higher temperatures.

- ❖ A/T rich regions are more likely to open up than G/C rich regions.
- ❖ This is one of the reasons why transcription initiation bubbles and DNA replication origins are often A/T rich.
- ❖ It's easier for the proteins (RNA polymerase, and origin binding proteins) to create the locally unwound regions.



The Structure of RNA

- ❖ They are involved in the transfer and expression of genetic information.
- ❖ They act as primers for DNA formation.
- ❖ Some RNAs act as enzymes as well as coenzymes.
- ❖ RNA also functions as genetic material for viruses.
- ❖ The building unit of RNA is ribonucleotide.
- ❖ Unlike DNA, RNA is synthesized as a single strand
- ❖ Uracil is found in RNA while Thymine is found in DNA.

- ❖ The three RNAs that have important role in protein synthesis are:
 1. Ribosomal RNA (rRNA)
 2. Transfer RNA (tRNA)
 3. Messenger RNA (mRNA)
- ❖ The 3 types of RNA molecules differ from each other by size, function and general stability.
- ❖ Only t-RNA after post transcriptional process can be changed to tertiary structure.

TYPES OF RNAs

S N	Coding RNA	Non-coding RNAs
1	mRNA	
2		tRNA (Transfer RNA)
3		rRNA (Ribosomal RNA
4		snRNA (Small nuclear RNA)
5		snoRNA (Small nucleolar RNA)
6		scRNA (Small cytoplasmic RNA)
7		aRNA (Antisense RNA)
8		miRNA (microRNA)
9		siRNA (small interference RNA)
10		Ribozyme (catalytic RNA)
11		cRNA (Complementary RNA)
12		gRNA (guide RNA)

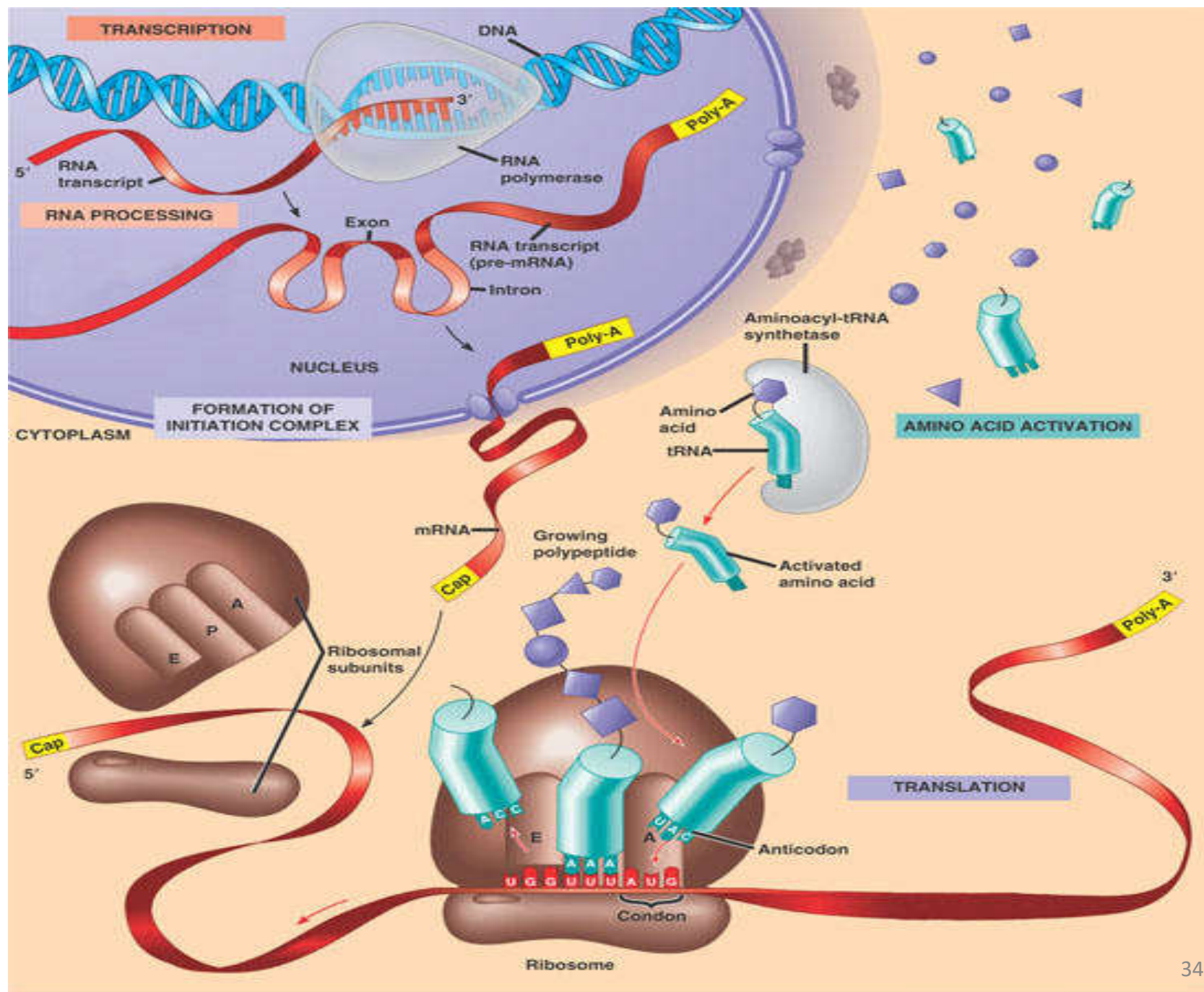
Messenger RNA

- It accounts for 1-5% of cellular RNA.

Structure

- ✓ Majority of mRNA has primary structure.
 - They are single-stranded linear molecules.
 - They consist of 1000-10,000 nucleotides.
- ✓ mRNA molecules have free or phosphorylated 3' and 5' end.
- ✓ mRNA molecules have different life spans.
- ✓ Eukaryotic mRNA are more stable than prokaryotic mRNA.
- ✓ The mRNA nucleotide sequence is complementary from which it is synthesized or copied.

- ✓ Some **eukaryotic mRNA** molecules are **capped at 5' end**.
 - The cap is **methyated GTP (m7 GTP)**.
 - Some mRNA contain internal methylated nucleotides.
 - Capping protects mRNA from nuclease attack.
- ✓ At *3' end of most of eukaryotic mRNA, a polymer of adenylate (poly A) is found as tail.*
 - Poly A tail protects mRNA from nucleases attack.
- ✓ In prokaryotes *5' end of mRNA contains a sequence rich in A and G.*
 - Such sequence is known as ***Shine-Dalgarno sequence***.
 - It helps attachment of ***mRNA with ribosome*** during protein synthesis.



Functions

1. mRNA is *direct carrier* of genetic information from the nucleus to the cytoplasm.
2. Usually a molecule of mRNA contains information required for the formation of one protein molecule.
3. Genetic information is present in mRNA in the form of *genetic code*.
4. Some times single mRNA may contain information for the formation of more than one protein.

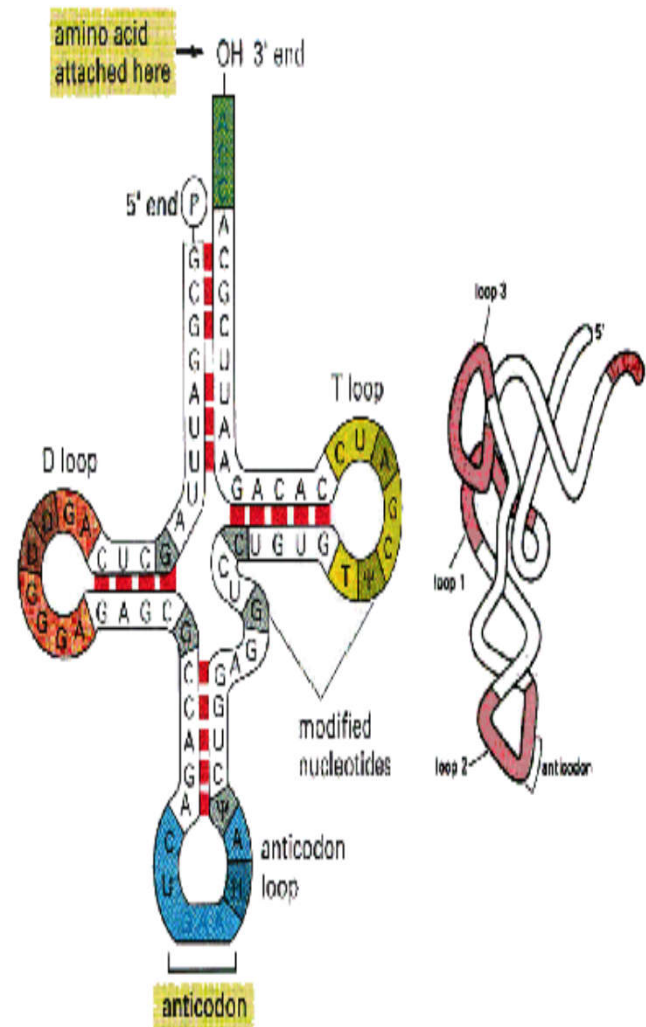
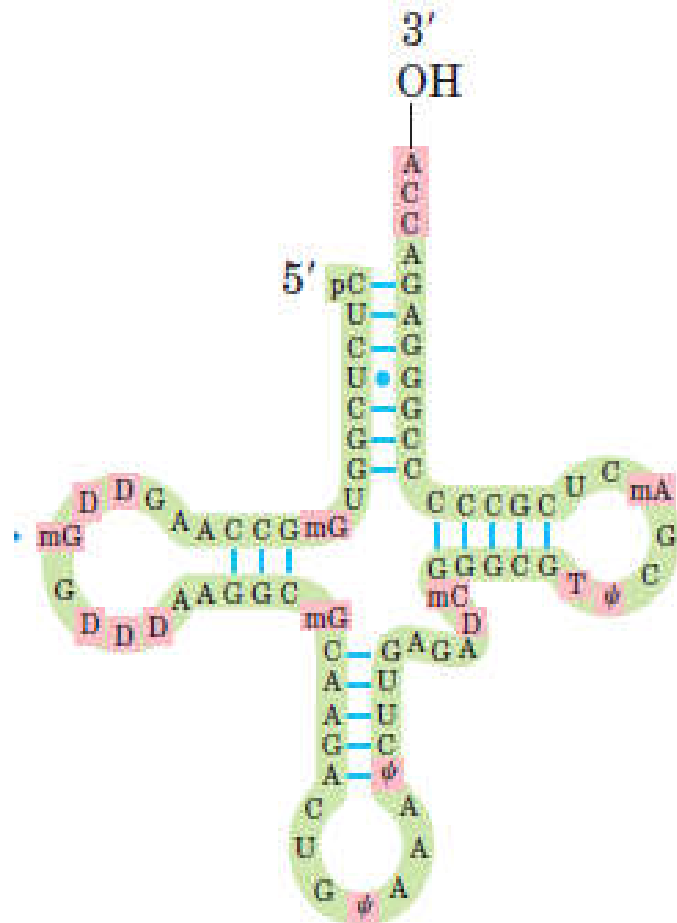
Transfer RNA

- t-RNA accounts for 10-15% of total cell RNA.

Structure

- ✓ They are the **smallest of all the RNAs**.
 - Usually they consist of **50-100 nucleotides**.
- ✓ They are single strand molecules.
 - They t-RNA molecules contain many **unusual bases** 7-15 per molecule.
 - are methylated adenine, guanine, cytosine and thymine, dihydrouracil, pseudo uridine, isopentenyl adenine etc.
- ✓ These unusual bases are important for binding of t-RNA to ribosomes and interaction of t-RNA with aminoacyl-t-RNA synthetases.

Mature tRNA^{Tyr}



Tertiary structure of t-RNA

- ✓ X-ray diffraction analysis indicated complex three-dimensional structure for t-RNA molecule.
- ✓ Three-dimensional structure of t-RNA looks like inverted or tilted L.
- ✓ The anti-codon arm is at the tip of the vertical arm of tilted L.
- ✓ The acceptor arm is at the tip of horizontal arm of tilted L.
- ✓ The D loop and TΨC loop are pushed into corner of tilted L.

Function

1. It is the carrier of amino acids to the site of protein synthesis.
2. There is at least one t-RNA molecule to each of 20 amino acids required for protein synthesis.
3. Eukaryotic t-RNAs are less stable where as prokaryotic RNAs are more stable.

Ribosomal RNA

- ✓ r-RNA accounts for *80% of total cellular RNA*.
 - It is present in ribosomes.
- ✓ In ribosomes, r-RNA is found in combination with protein. It is known as *ribonucleoprotein*.
- ✓ The length of r-RNA ranges from 100-600 nucleotides.
 - Both prokaryotic and eukaryotic ribosomes contain r-RNA molecules.
- ✓ r-RNAs differ in sedimentation coefficients (S).
 - There are **four types of r-RNAs in eukaryotes**. They are **5, 5.8, 18 and 28S r-RNA molecules**.
 - **Prokaryotes** contains 3 types of r-RNA molecules. They are **5, 16 and 23S r-RNA molecules**.

- ❖ The number of proteins in the eukaryotic ribosomal subunits varies somewhat from species to species.

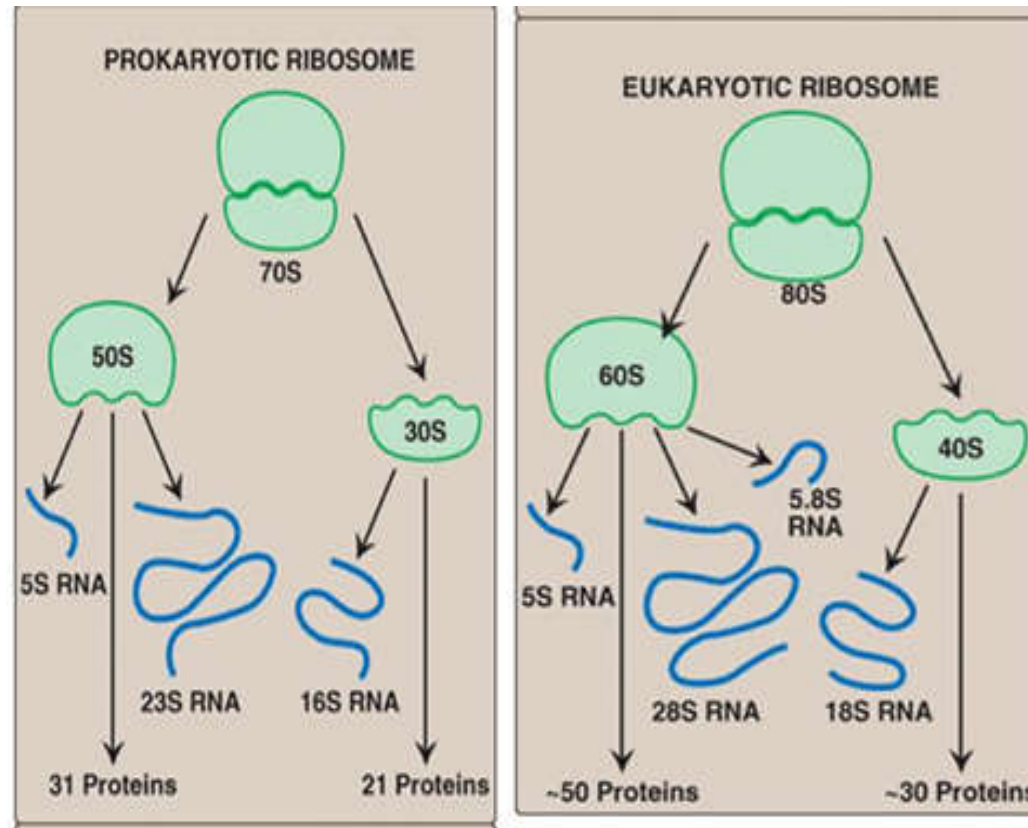


Fig. Ribosomal composition.

- ✓ **The large ribosomal subunit** **catalyzes** formation of the peptide bonds that link amino acid residues in a protein.
- ✓ **The small subunit** **binds mRNA** and
 - is responsible for the accuracy of translation
 - by ensuring correct base-pairing between the **codon in the mRNA** and the **anticodon of the tRNA**.

Functions of rRNA

1. r-RNAs are required for the formation of ribosomes.
2. 16S RNA is involved in initiation of protein synthesis.
3. The function of the rRNA is to provide a mechanism for decoding mRNA into amino acids and to interact with the tRNAs during translation by providing peptidyl transferase activity.

Differences between RNA and DNA

DNA	RNA
Sugar moiety is deoxy ribose	Sugar moiety is ribose
Uracil, a pyrimidine base is usually absent	Thymine, a pyrimidine base is usually absent
Double-stranded molecules	Single stranded molecules
Sum of purine bases is equal to sum of pyrimidine bases ($A + G = C + T$)	Sum of purine bases is not equal to sum pyrimidine bases ($A + G \neq C + T$)
More stable due to absence of hydroxyl group on 2 carbon atom of deoxyribose	Less stable due to presence of hydroxyl group on 2 carbon atom of ribose
Bases are not modified	Bases are modified
No catalytic activity	Some RNA are catalytically active
Not subjected to degradation in cell	Degraded in the cell by nucleases

